STRATA-AdamW: A Cautionary Study of Flag-Manifold Optimizer State

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Abstract

STRATA-AdamW stores adaptive optimizer state on a nested flag manifold, using multi-timescale covariance sketches, transported first moments, and blockwise adaptive updates. Separate moment counters and a projected-statistic epsilon floor make the resulting update numerically stable. Completed seed-42 comparisons show gains over same-rank Single-Grassmann in three of twelve citation-graph settings and five of six newly tested forecasting, recommendation, and sensor-drift settings. On ETTh2, STRATA lowers MSE by 19.6% relative to AdamW and 2.7% relative to GaLore; sensor-drift replay reaches 96.506% accuracy using 55.1% of AdamW's stored state, versus AdamW's 96.536%. On PubMed/SAGE, the full method exceeds all ten tested interventions, while a state-matched six-candidate comparison gives a smaller 0.460-percentage-point gain over Single-Grassmann. We develop detailed conditional guarantees for flag optimality, statistical tracking, adaptive-state perturbations, nonconvex and PL convergence, and generalization through coupled optimizer states and an inverse-KL, flag-aligned PAC–Bayes bound. Complete comparisons retain stronger baselines and identify the settings in which the combined design is effective.

1. Introduction

Low-rank training methods such as GaLore, OSD, SubTrack++, and related projection optimizers reduce optimizer state by representing gradients in a learned subspace rather than maintaining full Adam moments for every matrix entry (Zhao et al., 2024; Su et al., 2025; Liang et al., 2024; Rajabi et al., 2025; Rajabi & Rambhatla, 2024; Refael et al., 2025; Zhu et al., 2025; Nikdan et al., 2024; Zhang et al., 2026). They are motivated by empirical evidence that neural gradients have persistent low-dimensional structure (Thomas, 2026; Huang et al., 2026; others, 2023). However, a single subspace is a strong modeling choice: for example, a PINN layer may contain a slowly moving core induced by the PDE residual while also needing fast directions near transient boundary corrections; a graph model may begin with highly aligned neighborhood-aggregation gradients and lose that alignment as representations specialize; and a vision classifier may show low-rank structure that is stable enough for compression but not predictive of optimization gains.

This paper studies whether a **flag-manifold optimizer state** is a useful refinement of the single-subspace assumption. A flag is a nested chain $U_1 \subset U_2 \subset \cdots \subset U_K$, so it can represent a stable inner core and progressively larger outer directions. STRATA-AdamW applies this idea to optimizer state: it maintains multi-timescale covariance sketches, updates a flag by Riemannian steps, aligns blocks by Procrustes transport, and applies Adam-style moments in the incremental flag blocks. This construction is related to mathematical work on flag optimization (Ye & Wong, 2022; Lai et al., 2022; Szwagier & Pennec, 2025; Jin et al., 2025; Marrinan et al., 2024; Xiao et al., 2026; Ye & Lim, 2021; others, 2025), but the object being optimized is different: STRATA uses a flag inside a stochastic optimizer for changing neural-network gradients.

Our study is intentionally not written as a state-of-the-art claim. The original implementation exposed two numerical specification bugs that are easy to miss because they arise only after projecting gradients into low-dimensional flag blocks. First, transporting a mature first moment while resetting the second moment but sharing a single Adam bias-correction counter over-corrects the first moment after each flag update. Second, using the usual 10^{-8} Adam epsilon in near-zero projected blocks can divide by a vanishing second moment, producing huge block updates. Fixing these issues makes the optimizer much more stable, but it does not turn STRATA into a universal winner.

The completed follow-ups identify targeted benefits in three of twelve citation-graph settings (Table 4), with every remaining setting retained in Appendix J. On PubMed/SAGE, independently validation-selected STRATA improves on both same-rank Single-Grassmann and AdamW, while its measured optimizer tensor state is smaller than AdamW's (Table 3). This does not establish a family-wide ranking:

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Preprint. November 5, 2026.

Table 1. **One STRATA-AdamW iteration.** The parameter branch uses the current flag; refreshed coordinates are used at the next iteration. Block alignment rotates first-moment coordinates but need not preserve ambient momentum exactly.

Stage	Operation	State or output
Project	Decompose the gradient into orthogonal incremental flag blocks.	$Z_{k,t} = G_t U_t^{[k]}, \quad G_{\perp,t} = G_t(I - P_{K,t})$.
Adapt	Update block first and second moments with separate ages.	$\widehat{M}_{k,t}$ and $\widehat{V}_{k,t}$ use counters c_k^M and c_k^V.
Lift	Lift the blockwise adaptive directions; add the normalized residual.	$D_t = \sum_k a_k H_{k,t} U_t^{[k]\top} + a_\perp G_{\perp,t}/(\|G_{\perp,t}\|_F/\sqrt{md} + \epsilon)$.
Update	Apply the direction and decoupled weight decay.	Unconstrained matrix parameter W_{t+1}.
Track	Update covariance sketches at every step.	$Y_{k,t} = \rho_k Y_{k,t-1} + (1-\rho_k) G_t^\top (G_t \Omega_k)$.
Refresh	Every τ steps, take a flag step, retract, and align blocks.	Procrustes rotations transport M_k; reset V_k and c_k^V, retain mature c_k^M.

GaLore and Independent-Grassmann remain stronger in that cohort, several other graph and tabular settings favor the baselines, and the medium-scale image searches are still running. An approximately state-matched Single-Grassmann control wins at the default flag step; equal six-candidate flag-step/rate searches yield a smaller STRATA gain of 0.460 percentage points (Table 21). We therefore retain budget and hyperparameter matching when testing the positive validation signal across a balanced architecture matrix and mechanism controls. Equalizing flag-layer coefficients barely changes the best archived Burgers or Poisson cells. In the new PubMed/SAGE grid, by contrast, all ten validation-selected interventions reduce test accuracy relative to full STRATA; these setting-dependent results support testing the combined design without identifying a universal causal mechanism. Our contributions are: (i) we identify and fix two flag-projection-specific numerical stability bugs; (ii) we introduce δ_{nest} as a descriptive diagnostic of multi-timescale gradient nesting, while explicitly showing that it is not a performance predictor; (iii) we provide a fully proved conditional analysis connecting flag optimality, tracking, perturbation, nonconvex and PL convergence, and full-state generalization stability, with a transparent accounting of which quantities are measured in this project and which remain out of scope.

2. Related Work

Compressed adaptive state and online tracking. AdamW maintains coordinatewise adaptive statistics with decoupled weight decay (Loshchilov & Hutter, 2019). GaLore reduces state through low-rank gradient projection and periodic SVD refresh (Zhao et al., 2024). OSD replaces spectral refresh with online PCA (Liang et al., 2024), while SubTrack++ uses Grassmannian tracking together with projection-aware adaptive statistics (Rajabi et al., 2025). APOLLO uses random projections to estimate structured gradient scaling (Zhu et al., 2025). STRATA shares the goal of compressing optimizer state, but maintains a nested chain associated with several covariance decay rates instead of one tracked projection. Table 2 distinguishes these mechanisms. In particular, tracking a subspace and correcting moments as its coordinates change are established ideas; the proposed combination is a multi-timescale flag state, not the introduction of either idea alone.

Matrix-aware update geometry. Shampoo uses matrix preconditioning (Gupta et al., 2018); SOAP applies Adam in the preconditioner's evolving eigenbasis (Vyas et al., 2024). Muon approximately orthogonalizes matrix-valued update directions (Jordan, 2024). These methods concern update geometry and need not constrain the trained weights. STRATA likewise leaves W unconstrained, but allocates low-dimensional adaptive statistics to nested gradient-coordinate blocks. SOAP's basis adaptation and SubTrack++'s projection-aware moments are particularly relevant precedents for our alignment and state-consistency audit. The feature table does not imply a universal memory or convergence advantage: stored probes, residual updates, and state refresh all contribute costs.

Flag geometry and nested representations. Flag optimization supplies the differential geometry of nested subspaces (Lai et al., 2022); nested-subspace learning lifts Grassmannian criteria through nested projectors (Szwagier & Pennec, 2025). Thus nestedness itself is not new. Our use is optimizer-internal: multiple moving gradient covariance estimates drive a flag, and each incremental block carries

Table 2. **Where STRATA sits among related methods.** Keywords describe the mechanism; symbols describe structural properties, not empirical superiority. The final row combines nested subspaces with separate covariance timescales inside the optimizer.

Method / family	Core mechanism	Low-rank moments	Online tracking	Nested subspaces	Multiple covariance EMAs
AdamW (Loshchilov & Hutter, 2019)	Diagonal adaptivity	✗	✗	✗	✗
GaLore (Zhao et al., 2024)	Periodic SVD	✓	✗	✗	✗
OSD (Liang et al., 2024)	Online PCA	✓	✓	✗	✗
SubTrack++ (Rajabi et al., 2025)	Grassmann tracking	✓	✓	✗	✗
APOLLO (Zhu et al., 2025)	Random-projection scaling	✓	✗	✗	✗
SOAP (Vyas et al., 2024)	Eigenbasis Adam	✗	✗	✗	✗
Muon (Jordan, 2024)	Polar update direction	✗	✗	✗	✗
Flag methods (Lai et al., 2022; Szwagier & Pennec, 2025)	Nested projectors	—	—	✓	—
Single-Grassmann (our control)	One tracked subspace	✓	✓	✗	✗
STRATA (ours)	**Multi-timescale flag state**	✓	✓	✓	✓

✓ present; ✗ absent in the listed formulation; — not an adaptive-optimizer property. *Low-rank moments*: adaptive statistics are stored in compressed coordinates (including APOLLO's auxiliary scaling state). *Online tracking*: incremental data-driven tracking of a compressed gradient subspace, excluding periodic SVD/random refresh and full eigenbasis changes. *Nested*: a nontrivial chain with $K > 1$; a lone subspace is not counted as a multi-level flag. *Multiple covariance EMAs*: distinct decays for the same gradient covariance, not merely Adam's different first-/second-moment decays or SOAP's row/column factors.

adaptive moments. This adds a stochastic state-management problem to a geometric representation problem. The complete analysis explicitly separates quotient-invariant projector identities from basis-dependent coordinatewise Adam updates, and the experiments test whether this additional structure yields a benefit rather than assuming that it must.

Adaptive convergence and generalization. Adam-type convergence can fail even for convex objectives (Reddi et al., 2018). Algorithmic stability links coupled training trajectories to expected generalization (Hardt et al., 2016). Our analysis includes the complete STRATA state and treats current-gradient normalization, transport/reset variation, and flag-aligned posterior families under explicit assumptions.

3. Preliminaries and Problem Setup

For a matrix parameter $W \in \mathbb{R}^{m\times n}$, let $d = \min(m, n)$ and let R be the maximum low-rank budget. A Single-Grassmann optimizer maintains one rank-R subspace. STRATA maintains a flag $U_1 \subset \cdots \subset U_K$ with ranks $r_1 < \cdots < r_K = R$ and incremental projectors $\Pi_k = P_k - P_{k-1}$, $P_k = U_k U_k^\top$. For an exponential moving average timescale ρ_k, the population covariance is modeled as $\Sigma_{k,t} = \mathbb{E}C_{k,t}$ with $C_{k,t} = \rho_k C_{k,t-1} + (1-\rho_k) g_t g_t^\top$.

The descriptive nesting diagnostic compares independently optimal timescale subspaces before imposing a flag:

$$\delta_{\mathrm{nest},t} = \sum_{k=1}^{K-1} \|(I - \hat{P}_{k+1,t})\hat{P}_{k,t}\|_F^2, \tag{1}$$

where $\hat{P}_{k,t}$ projects onto the independently estimated top-r_k subspace. Small values mean that the independent optima already look nested; rising values mean that this structure is being lost. In this paper, δ_{nest} is a structural measurement, not a performance predictor.

The main theoretical object is a flag objective

$$\Phi_t(U) = \sum_{k=1}^{K} \omega_k \mathrm{tr}(P_k(U)\Sigma_{k,t}), \tag{2}$$

which assigns the nested blocks to multi-timescale covariance mass. The theory analyzes when this objective has an optimal flag, how sketch/statistical/tracking errors perturb it, and how flag error enters parameter optimization through layer coefficient differences in a linear block-isotropic surrogate, with separate discrepancy terms for the practical Adam rule. The surrogate coefficient variation is $\Gamma_t^a = \sum_k |a_{k,t} - a_{k+1,t}|$.

4. STRATA-AdamW

Table 1 gives the parameter and optimizer-state branches of one iteration. If long and short gradient timescales share a

core but differ in outer directions, a nested state can store the core once and allocate extra rank where the faster timescale needs it. Each projection creates low-dimensional statistics whose numerical scale can differ from full-gradient AdamW.

The two most important implementation details are precisely the ones exposed by the experiments. First, the first and second moment counters cannot be shared after a flag update: the first moment is transported and remains mature, while the second moment is reinitialized in the new projected coordinates. Second, projected block gradients may be genuinely near zero; using the standard AdamW epsilon can produce enormous normalized updates, so the projected optimizers use an epsilon floor of 10^{-6}. Archived full-AdamW references retain their original settings; the new controlled revision cohorts use 10^{-6} for both full AdamW and projected Adam controls.

For a matrix oriented as $m \times d$ with $d \le m$ and sketch widths $s_k = \min(d, r_k + p)$, the complete persistent tensor count in the flag branch is

$$\text{state}_{\text{STRATA}} = 2mR + dR + 2d\sum_k s_k, \tag{3}$$

$$\text{state}_{\text{AdamW}} = 2md. \tag{4}$$

The terms are block moments (M, V), the flag U, sketches Y_k, and the fixed probes Ω_k. Earlier accounting omitted the probes, although the implementation stores them; its smaller formula is retained only as a historical partial count. The corrected ratio for 128×128, ranks $(16, 32, 64)$ and $p = 8$ is 1.8125, so low-rank state is not automatically smaller. Noneligible parameters use full Adam state; workspaces, gradients and activations are additional costs.

Theory summary. Appendix A gives complete statements and proofs. Commuting covariance optimization reduces to a capacity-constrained assignment; commutation alone does not eliminate strict nonglobal local maxima. Under explicit local curvature, statistical, sketch and drift errors yield a filtered flag-tracking bound. For predictable bounded preconditioners, nonconvex stationarity and PL contraction hold with an explicit discrepancy term for practical momentum, current-gradient normalization and transport. Generalization is controlled through a coupled recursion for parameters and the complete optimizer state. Equal scalar layer coefficients cancel flag dependence only in the linear projector surrogate, not in coordinatewise Adam. A5 therefore tests layer weighting rather than removal of all flag effects.

5. Convergence and Generalization

We analyze the chain from covariance estimation to learned flags, evolving adaptive states, parameter descent, and generalization. Complete hypotheses and proofs are in Appendix A; Appendix B retains the expanded mathematical bridge.

Flag geometry and tracking. For $P_k = U_{:r_k} U_{:r_k}^\top$, put $\Pi_k = P_k - P_{k-1}$ and $\Phi(U) = \sum_k \omega_k \operatorname{tr}(P_k C_k)$. The overlap coupling $T_{ja} = \operatorname{tr}(\Pi_j(U)\Pi_a(V))$ gives the exact distance identity

$$d_P^2(U, V) = \sum_{j,a} |j - a| T_{ja}. \tag{5}$$

Charging each misplaced direction only for the prefix boundaries it crosses gives the sharp factor K in the assignment-distance comparison. Hoffman–Kruskal integrality characterizes the commuting-covariance global optimum as a capacity-constrained assignment (Theorem 1). A commuting counterexample still has a strict nonglobal local maximum (Proposition 6). Under the regular local-chart hypotheses, the oracle perturbation and tracking estimates are

$$\|x^\star_{\widehat{C}} - x^\star_C\| \le \frac{2J_\psi}{\mu_\phi} \sum_k \omega_k \sqrt{r_k}\epsilon_k, \tag{6}$$

$$\sqrt{E_s} \le q^s \sqrt{E_0} + \sum_{j<s} q^{s-1-j} b_j. \tag{7}$$

Here ϵ_k bounds covariance error and b_j combines statistical, sketch, and drift forcing. The sharp quadratic contraction is $q_\star = (L_\phi - \mu_\phi)/(L_\phi + \mu_\phi)$. Centered noise and persistent drift have different gains, $\xi^2/(1-q^2)$ and $1/(1-q)$, respectively (Theorem 3).

Convergence with evolving adaptive state. For a predictable preconditioner $mI \preceq B_t \preceq MI$, write $w_{t+1} = w_t - \eta_t(B_t g_t + e_t)$ and retain the exact discrepancy e_t from the practical update. With $S_T = \sum_{t<T} \eta_t$, the nonconvex estimate is

$$\frac{\sum_{t<T} \eta_t \mathbb{E}\|\nabla F(w_t)\|^2}{S_T} \le \frac{2\Delta_0}{mS_T} + \frac{2L\sum_{t<T}\eta_t^2 \mathbb{E}\mathsf{v}_t}{mS_T} + \mathcal{E}_T, \tag{8}$$

where $\mathcal{E}_T$ is the explicit bias, transport, and normalization discrepancy term in Theorem 5; $\mathsf{v}_t = \operatorname{tr}(B_t^2 \Sigma_t)$ retains directional noise rather than replacing it by $M^2\sigma_t^2$. Under the PL condition, the initial gap contracts by $(1 - \mu m \eta)^T$ toward a stated noise/discrepancy floor (Theorem 6). The implemented current-gradient normalizer requires separate analysis: Theorem 14 controls its conditional symmetry defect, and Theorem 15 propagates observable moment/reset/transport budgets. The explicit asymmetric-noise stationary bias in Proposition 23 explains why gradient unbiasedness alone is insufficient.

Generalization and directional complexity. Coupling neighboring samples gives a five-component recursion for

Algorithm 1 STRATA-AdamW for one matrix parameter

Maintain flag $U_1 \subset \cdots \subset U_K$, block moments (M_k, V_k), counters (c_k^M, c_k^V), and a stateless residual direction.
for training step t **do**
  Receive gradient G_t and compute projected block gradients $Z_{k,t} = G_t U^{[k]}$ plus residual component.
  Update first moments $M_k \leftarrow \beta_{1,k} M_k + (1-\beta_{1,k}) Z_{k,t}$ and second moments $V_k \leftarrow \beta_{2,k} V_k + (1-\beta_{2,k}) Z_{k,t}^{\odot 2}$.
  Form Adam-style block updates with epsilon 10^{-6} for projected/low-rank statistics.
  Combine updates as $D_t = \sum_k a_k H_{k,t} U_t^{[k]\top} + a_\perp G_{\perp,t}/(\|G_{\perp,t}\|_F/\sqrt{md}+\epsilon)$ and update W with decoupled weight decay.
  Update all covariance sketches using the current gradient.
  if t is a flag-update step **then**
    Take a Riemannian flag step and retract; align old and new blocks by Procrustes.
    Transport M_k through the block rotations, reset V_k, keep c_k^M mature, and reset only c_k^V.
  end if
end for

parameters, flags, sketches, first moments, and square-root second moments. Its nonnegative transition matrices $\mathcal{M}_t$ yield

$$\epsilon_T^{\mathrm{stab}} \le \frac{L_0}{n} e_w \sum_{t<T} b_t (\mathcal{M}_{T-1} \cdots \mathcal{M}_{t+1}) d_t. \tag{9}$$

This accounts for the whole optimizer state (Theorem 10). Independently, the Seeger–Maurer route keeps inverse binary KL and the exact moment constant Ξ_n, avoiding a Pinsker relaxation (Theorem 13). Under the stated bounded-loss curvature hypothesis, optimizing a full-rank Gaussian posterior aligned with the flag gives

$$\mathcal{C}_\lambda(U) = \frac{1}{2\lambda} \sum_j m q_j \log(1 + 2\lambda \sigma_0^2 \overline{h}_j(U)), \tag{10}$$

$$(\sigma_j^2)^\star = \frac{\sigma_0^2}{1 + 2\lambda \sigma_0^2 \overline{h}_j(U)}. \tag{11}$$

Refining a block with unequal average curvatures strictly decreases this optimized complexity; upper and lower gains depend on their weighted variance (Proposition 25). Tracking sensitivity depends on spectral spread, with isotropic curvature canceling exactly. These are conditional guarantees with explicit hypotheses, rather than empirical certificates of those hypotheses on every benchmark.

6. Experiments

Execution snapshot. At 2026-11-05 04:08 UTC, the original 856 configurations include 556 complete, 21 retained unfinished, 278 deferred, and 1 failed. The reduced scope is explicit in Appendix I; deferred runs supply no evidence.

Protocol. All runs use a single seed, seed 42. We therefore report long-run endpoints and curves as deterministic case studies, not as high-confidence statistical estimates. The revision recomputes configuration selection on validation criteria within each method (Appendix D); and the main comparison is against same-maximum-rank Single-Grassmann. PINN metrics are relative L_2 error where lower is better; classification and graph metrics are accuracies where higher is better.

Completed follow-up evidence. The completed PubMed/SAGE two-rate cohort gives STRATA test accuracy 0.71623, versus 0.67902 for Single-Grassmann and 0.69061 for AdamW: descriptive gains of 3.721 and 2.562 percentage points. Each method selects its rate using terminal-100 validation accuracy. Its archived optimizer tensor state, including fixed probes, is 58.1% of AdamW's. Table 3 gives all five methods, including stronger competitors; this is a comparison-specific benefit, not a graph-family or significance claim. The completed default-protocol PubMed architectures are gat, gcn, gcnii, sage; positive STRATA–Single-G test differences occur in gcn, sage. Every completed architecture remains in Table 28. Further controlled ablations and an equal-budget flag-step sweep are reported in Appendix I; experimental variants do not overwrite default-protocol results.

Mechanism evidence in the positive setting. Both rates for all ten PubMed/SAGE interventions are complete (22 configurations including the full controls). Full STRATA reaches 71.623%, exceeding every validation-selected ablation by 0.435–3.984 percentage points (Table 18). This supports the combined configuration in this setting; interventions that also alter resets or the covariance objective do not isolate a single geometric cause.

Matched flag-step search. In the completed exploratory six-candidate grid for each geometric method, validation selects STRATA at $\xi = 10$ and $\eta = 10^{-3}$, with test accuracy 73.383%, versus 67.902% for Single-Grassmann

Table 3. PubMed/SAGE: validation-selected two-rate, seed-42, 1000-epoch comparisons. State is unique optimizer tensor storage read from the final checkpoint, including probes. Peak is maximum recorded CUDA allocation; wall time is shared-server epoch timing, not isolated throughput.

Method	Val acc	Test acc	State MiB	Peak MiB	ms/epoch
STRATA	0.7179	0.7162	2.32	391.1	29.6
Single-G	0.7067	0.6790	1.82	390.6	24.2
AdamW	0.7041	0.6906	3.99	392.7	**16.6**
GaLore	**0.7590**	**0.7580**	**1.26**	**390.0**	22.3
Independent	0.7419	0.7415	2.20	391.0	27.3

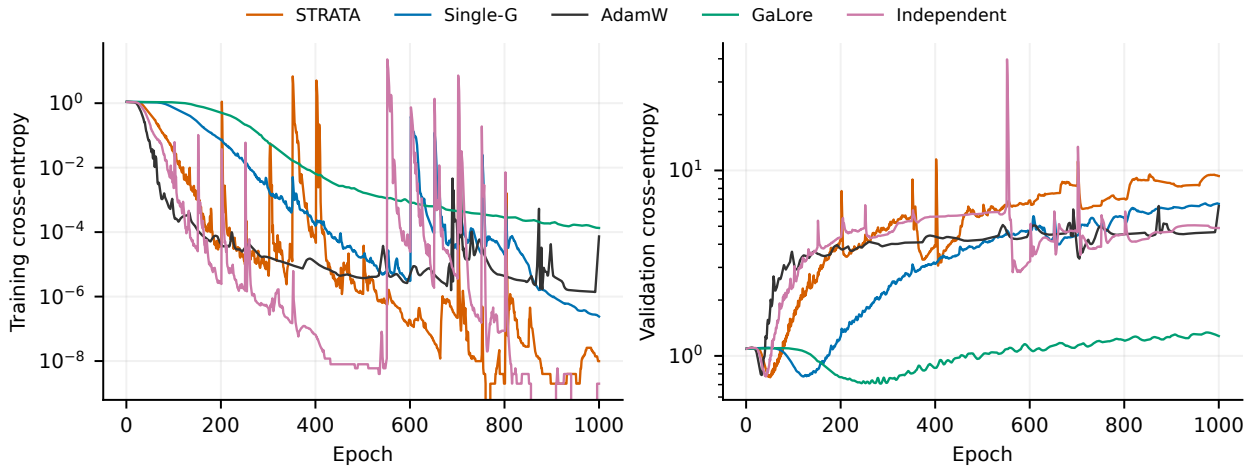


Figure 1. Completed PubMed/SAGE training and validation trajectories. Each of five methods selects its learning rate using terminal-100 validation accuracy after both full-budget candidates finish. All trajectories are raw; held-out test outcomes are reported in Table 3.

and 74.151% for Independent. The STRATA–Single-G difference is 5.481 percentage points; STRATA improves by 1.760 points over its default-step result. Table 19 and the full profiles retain every setting. This is a separate sensitivity result, and Independent remains stronger on held-out test accuracy.

Equal-state follow-up. A resource-only calibration selects Single-Grassmann rank 82, whose stored optimizer tensor state differs from STRATA R64 by -0.408%. After the same six-candidate validation search for both methods, test accuracies are 73.383% and 72.923%, respectively (signed STRATA difference +0.460 percentage points; Table 21). At the default flag step, the two-rate validation selections instead give STRATA 71.623% and Single-G R82 72.923%. Thus extra state is a material confound in the equal-rank comparison, and the retained six-grid gain is smaller and protocol-dependent.

Scope of the positive cases. Table 4 collects three illustrative comparisons from the completed default graph matrix. These rows are chosen for exposition after observing the matrix; they are not a preselected confirmatory subset. Within every row, all methods finish both rates before validation selects a configuration. STRATA improves on Single-Grassmann in three of the twelve dataset–architecture settings, and does not in the other nine. The complete matrix and adverse cases appear in Appendix J. GaLore is stronger in these three cases, so they support restricted comparisons rather than a universal graph-optimizer ranking.

Other restricted benefits. The archived MNIST/MLP case gives STRATA accuracy 0.98273 versus Single-Grassmann 0.98011, while the available full-AdamW reference is about 0.987. Both completed five-method Rotated-MNIST grids below improve on Single-Grassmann after two-rate validation selection. In Poisson, STRATA has lower error than same-rank Single-Grassmann in an unstable baseline grid, while full AdamW is substantially better. These are specific vision, continual-learning and numerical-stability observations. They do not establish superiority over the strongest baseline across those families, and their complete adverse comparisons remain in the appendix.

Numerical stability findings. Tables 5 and 6 summarize the engineering contribution. The counter bug was diagnosed by observing gradient-norm oscillations synchronized with flag updates; transporting M while resetting a shared

Table 4. Illustrative positive same-rank comparisons, seed42, terminal-100 test accuracy after full two-rate validation selection. Every method is shown; higher is better. The full twelve-setting graph matrix is retained in the appendix.

Setting	STRATA	Single-G	AdamW	GaLore	Indep.
PubMed/SAGE	0.71623	0.67902	0.69061	**0.75801**	0.74151
PubMed/GCN	0.71742	0.71208	0.70445	**0.75915**	0.68782
CiteSeer/SAGE	0.55747	0.53980	0.55302	**0.65124**	0.55836

Table 5. **Numerical specification fixes.** The MNIST epsilon check is a diagnostic configuration; gradient norm and accuracy are the recorded values for that check.

Projected ϵ	Max. $\|g\|$	Test acc. ↑	Separate counters
10^{-8}	1.87×10^5	0.686	–
10^{-6}	14.2	**0.969**	–

Table 6. **Counter handling at a flag refresh.** Transported mature first moments and reinitialized second moments have different statistical ages.

Rule	First-moment age	Second-moment age
Shared reset	Reset despite transport	Reset
Separate counters	Retain mature age	Reset

counter made $\hat{M}$ roughly ten times too large after reset. The epsilon bug was diagnosed by decomposing update components: residual updates stayed bounded, while a flag-block Adam branch exploded because $V \approx 0$ and epsilon was 10^{-8}. On MNIST at the diagnostic setting, $\epsilon = 10^{-8}$ produced max gradient norm 1.87×10^5 and test accuracy 0.686, while $\epsilon = 10^{-6}$ produced max gradient norm 14.2 and test accuracy 0.969.

Memory/state accounting. The revised tensor count includes both fixed probes and EMA sketches. For ranks $(16, 32, 64)$ and $p = 8$, the STRATA/full-AdamW ratios for individual eligible matrices are 1.8125 for 128×128, 0.9063 for 256×256, 0.3393 for 784×512, 0.2070 for 512×2048, and 0.1035 for 4096×1024. Earlier reported model ratios excluded probes and are not treated as complete measured state. Archived peak-allocation logs show about 0.709 versus 0.772 GB on CIFAR, while STRATA takes about 24.2 versus 15.5 seconds per epoch; these small-model observations do not establish a large-model memory advantage. New runs retain actual tensor-state and peak-allocation measurements.

Completed continual-learning comparisons. The Rotated-MNIST fine-tuning and replay grids each complete all ten candidates (five methods, two rates), with 1000 epochs across ten rotations, a two-layer 256-wide MLP, rank 64 and seed 42. Each method selects its rate by terminal-100 seen-task validation accuracy. In fine-tuning, STRATA attains 57.655% test accuracy versus Single-Grassmann 57.109% (+0.546 percentage points) and Independent 57.278%; AdamW (62.193%) and GaLore (59.691%) remain stronger. With replay, STRATA reaches 88.582%, above Single-Grassmann 87.170% (+1.412 points) and Independent 87.140%, while AdamW (90.720%) and GaLore (88.880%) remain higher. Thus both completed settings support a gain over Single-Grassmann, not an overall baseline ranking. Table 28 retains the full comparisons and replay protocol; Appendix G.1 gives every selected loss trajectory. EWC remains deferred.

Tabular classification. Completed 512-wide MLP grids give comparison-specific gains after all ten trials finish and each method selects its rate on validation. Red Wine Quality yields STRATA 71.500% versus Single-Grassmann 69.702% (+1.798 percentage points), with AdamW at 71.491%. Adult gives 81.509% versus Single-Grassmann 81.281% (+0.227 points); AdamW and GaLore remain higher at 81.909% and 81.838%. Covertype gives 96.644%, above AdamW's 96.448% (+0.196 points) and GaLore's 96.597%. Single-Grassmann is effectively tied at 96.642%, while Independent is higher at 96.755%. These are single-seed descriptive differences. The corresponding 256-wide models do not improve on Single-Grassmann. Table 28 retains all methods and completed settings; Appendix G.1 gives the selected training and validation losses.

Token-budget language modeling. The completed WikiText-103/GPT-2-small rank-16 grid gives STRATA test perplexity 144.54 versus GaLore 192.20, a 24.8% reduction after two-rate validation selection. AdamW (80.00), Single-Grassmann (139.67), and Independent (140.81) remain stronger. Every candidate trains from scratch for 10000 updates and 20.48 million tokens, comprising 1000 ten-step evaluation units rather than corpus epochs. Table 23 reports the full comparison and measured resources; Appendix G.1 retains all selected loss trajectories.

Transfer to forecasting, recommendation, and sensor drift. All 180 configurations across six settings have completed their 1000-epoch budgets. Each of five methods selects among six candidates using terminal-100 validation metrics (Appendix K). STRATA improves on same-rank Single-Grassmann in 5/6 completed settings. Table 7 high-

Table 7. **Completed new-task positive cases.** Terminal-100 test metrics after equal-count validation search. All five methods are shown; bold marks the best value. MSE/RMSE are lower-is-better and accuracy is higher-is-better. The full six-setting matrix and selected losses are in Appendix K.1.

Setting	Metric	STRATA	Single-G	AdamW	GaLore	Indep.
ETTh2 / H24	MSE	**1.74404**	2.37970	2.16884	1.79170	2.18874
MovieLens100K / ratings	RMSE	0.96949	1.00529	**0.93295**	0.95778	1.01755
MovieLens1M / ratings	RMSE	1.05052	1.09444	0.98066	**0.91712**	1.08146
Gas sensors / fine-tuning	acc	0.85453	0.85375	0.82335	**0.90656**	0.80659
Gas sensors / replay	acc	0.96506	0.96409	**0.96536**	0.96232	0.96343

lights these positive cases after observing the complete matrix; all settings, including ETTh1, appear in Table 30. The results use seed 42 and describe the tested configurations.

Forecasting. On ETTh2/H24, STRATA attains MSE 1.74404, the lowest among the five methods, reducing error by 26.7%, 19.6%, and 2.7% relative to Single-Grassmann, AdamW, and GaLore, respectively. It stores 36.9% of AdamW's optimizer tensor state. The common forecaster is an MLP; this comparison tests the optimizer under the specified chronological split.

Recommendation. STRATA reduces RMSE relative to Single-Grassmann by 3.56% on MovieLens100K and 4.01% on MovieLens1M. AdamW and GaLore remain stronger on both datasets.

Sensor drift. Fine-tuning gives 85.453% accuracy, 3.118 percentage points above AdamW, while GaLore is stronger. With replay, STRATA reaches 96.506%, versus AdamW's 96.536%, using 55.1% of its stored optimizer state. State counts include probes and buffers. These same-rank comparisons do not equalize geometric-method state, and stored-state savings do not imply peak-memory or throughput gains.

7. Conclusion

STRATA-AdamW combines nested gradient subspaces, multi-timescale covariance sketches, and transported adaptive state. The analysis connects flag geometry and covariance tracking to explicit update perturbations, nonconvex stationarity, PL contraction, and full-state stability and PAC–Bayes generalization. Separate moment counters and a projected-statistic epsilon floor resolve the two observed numerical failure modes.

Completed comparisons identify benefits on specific graph, forecasting, recommendation, and continual-learning settings. The completed 180-configuration extension favors STRATA over same-rank Single-Grassmann in five of six settings; ETTh2 achieves the lowest MSE among all five tested methods, and both MovieLens datasets improve on Single-Grassmann. PubMed/SAGE favors the full design over all ten tested interventions; its equal-state, equal-count search retains a 0.460-percentage-point advantage over Single-Grassmann. Real sensor-drift replay provides an accuracy–state tradeoff, reaching 96.506% with 55.1% of AdamW's stored optimizer state. The complete single-seed comparison matrices and validation-based selection records give the empirical context for these results and the conditional theoretical guarantees.

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A. Mathematical Analysis

The analysis follows the two interacting parts of STRATA: learning a flag from gradient statistics and updating parameters with state expressed in that flag. We first establish the geometry and covariance-to-tracking estimates, then propagate state errors into descent and sample sensitivity. The final two subsections treat current-gradient normalization directly and optimize a directional generalization bound. The linear surrogate, the implemented adaptive map, and the Gaussian posterior each serve a different role; their assumptions are stated at the point where each is used. Three structural reductions sharpen the estimates: prefix overlap becomes a transportation cost, spectral centering removes isotropic curvature, and inverse binary KL preserves the empirical-risk dependence. We distinguish these exact identities and sharp comparison constants from conditional envelopes for covariance approximation and optimizer-state variation.

A.1. Flag quotient, tangent decomposition, and retractions

The geometric variable is a chain of nested subspaces. Rotating a basis inside one block leaves that chain unchanged, which motivates the quotient and its horizontal tangent space. The orthogonal block decomposition also separates captured gradient energy from the residual exactly. Coordinatewise adaptive moments require additional information about the chosen basis; their behavior is examined in Proposition 12.

Definition 1 (Ambient spaces and block selectors)**.**

$$\langle X, Y\rangle_F := \operatorname{tr}(X^\top Y), \quad \|X\|_F^2 := \langle X, X\rangle_F, \quad \|X\|_{\mathrm{op}} := \sup_{\|v\|_2=1} \|Xv\|_2, \tag{12}$$

$$\operatorname{sym} A := \tfrac{1}{2}(A + A^\top), \qquad \operatorname{skew} A := \tfrac{1}{2}(A - A^\top), \tag{13}$$

$$0 = r_0 < r_1 < \cdots < r_K = R < d \le m, \quad q_j := r_j - r_{j-1}, \quad q_{K+1} := d - R, \tag{14}$$

$$I_{K+1} := \{R+1, \dots, d\}, \tag{15}$$

$$I_j := \{r_{j-1}+1, \dots, r_j\}, \quad E_j := \operatorname{diag}(\mathbf{1}_{I_j}) \in \mathbb{R}^{R\times R}, \quad F_k := \sum_{j\le k} E_j, \tag{16}$$

$$\operatorname{St}(d, R) := \{U \in \mathbb{R}^{d\times R} : U^\top U = I_R\}, \tag{17}$$

$$\mathcal{G} := \prod_{j=1}^{K} \mathrm{O}(q_j), \qquad U \sim V \iff V = U \operatorname{diag}(O_1, \dots, O_K), \quad O_j \in \mathrm{O}(q_j), \tag{18}$$

$$\mathfrak{F}(q_1, \dots, q_{K+1}) := \operatorname{St}(d, R)/\mathcal{G} \simeq \mathrm{O}(d)/\prod_{j=1}^{K+1} \mathrm{O}(q_j), \tag{19}$$

$$U_j := U_{:,I_j}, \quad P_k(U) := UF_kU^\top, \quad \Pi_j(U) := UE_jU^\top, \quad \Pi_{K+1}(U) := I_d - UU^\top. \tag{20}$$

Proposition 1 (Projector algebra)**.**

$$P_k^\top = P_k = P_k^2, \quad \operatorname{tr} P_k = r_k, \quad P_kP_l = P_{\min(k,l)}, \tag{21}$$

$$\Pi_i^\top = \Pi_i, \quad \Pi_i\Pi_j = \delta_{ij}\Pi_i, \quad \sum_{j=1}^{K+1} \Pi_j = I_d, \tag{22}$$

$$G = \sum_{j=1}^{K+1} G\Pi_j, \quad \langle G\Pi_i, G\Pi_j\rangle_F = 0 \quad (i \ne j), \tag{23}$$

$$\|G\|_F^2 = \sum_{j=1}^{K} \|GU_j\|_F^2 + \|G\Pi_{K+1}\|_F^2, \tag{24}$$

$$\|G - GP_k\|_F^2 = \operatorname{tr}(G^\top G) - \operatorname{tr}(P_kG^\top G). \tag{25}$$

Proof.

$$P_k P_l = U F_k (U^\top U) F_l U^\top = U F_{\min(k,l)} U^\top, \tag{26}$$

$$\Pi_i \Pi_j = U E_i E_j U^\top = \delta_{ij} U E_i U^\top, \quad i,j \le K, \tag{27}$$

$$(I - UU^\top) U E_j = 0, \qquad \sum_{j \le K} U E_j U^\top = UU^\top, \tag{28}$$

$$\langle G\Pi_i, G\Pi_j \rangle_F = \operatorname{tr}(\Pi_i G^\top G \Pi_j) = \operatorname{tr}(\Pi_j \Pi_i G^\top G) = 0, \tag{29}$$

$$\|G U_j U_j^\top\|_F^2 = \operatorname{tr}(U_j U_j^\top G^\top G U_j U_j^\top) = \operatorname{tr}(U_j^\top G^\top G U_j), \tag{30}$$

$$\|G(I - P_k)\|_F^2 = \operatorname{tr}((I - P_k) G^\top G (I - P_k)) = \operatorname{tr}(G^\top G) - \operatorname{tr}(P_k G^\top G). \tag{31}$$

□

Proposition 2 (Vertical and horizontal spaces)**.**

$$T_U \operatorname{St}(d,R) = \{Z : U^\top Z + Z^\top U = 0\}, \tag{32}$$

$$\mathcal{V}_U = \{U\Omega : \Omega^\top = -\Omega,\ \Omega = \operatorname{blkdiag}(\Omega_{11}, \dots, \Omega_{KK})\}, \tag{33}$$

$$\mathcal{H}_U = \{Z \in T_U \operatorname{St}(d,R) : E_j U^\top Z E_j = 0\ (1 \le j \le K)\}, \tag{34}$$

$$T_U \operatorname{St}(d,R) = \mathcal{H}_U \perp \mathcal{V}_U, \tag{35}$$

$$\mathsf{H}_U(A) = A - U \operatorname{sym}(U^\top A) - U \operatorname{blkdiag}\big(\operatorname{skew}(U^\top A)\big),\ \dim \mathfrak{F} \qquad = dR - \frac{R^2 + \sum_{j=1}^K q_j^2}{2}. \tag{36}$$

Proof.

$$0 = \frac{\mathrm{d}}{\mathrm{d}s}(U + sZ)^\top (U + sZ)\Big|_{s=0} = U^\top Z + Z^\top U, \tag{37}$$

$$Z = U\Omega + U_\perp B, \quad \Omega = U^\top Z = -\Omega^\top, \quad B = U_\perp^\top Z, \tag{38}$$

$$\langle Z, U\Lambda \rangle_F = \operatorname{tr}(\Omega^\top \Lambda) = \sum_j \operatorname{tr}(\Omega_{jj}^\top \Lambda_{jj}), \tag{39}$$

$$Z \perp \mathcal{V}_U \iff \Omega_{jj} = 0 \quad (\forall j), \tag{40}$$

$$A = \underbrace{U \operatorname{sym}(U^\top A)}_{(T_U \operatorname{St})^\perp} + \underbrace{U \operatorname{blkdiag}(\operatorname{skew}(U^\top A))}_{\mathcal{V}_U} + \underbrace{\mathsf{H}_U(A)}_{\mathcal{H}_U}, \tag{41}$$

$$\langle US, U\Omega \rangle_F = \operatorname{tr}(S\Omega) = 0 \quad (S^\top = S,\ \Omega^\top = -\Omega), \tag{42}$$

$$\dim \mathcal{H}_U = dR - \frac{R(R+1)}{2} - \sum_j \frac{q_j(q_j - 1)}{2}. \tag{43}$$

□

We use projector distances to compare subspaces independently of their basis representatives. The aligned-basis distance is convenient for moment transport; the following comparisons allow estimates in these two representations to be combined with explicit constants.

Definition 2 (Quotient metric and projector distances)**.**

$$g_{[U]}(\xi,\eta) := \langle \xi_U^{\mathcal{H}}, \eta_U^{\mathcal{H}} \rangle_F, \tag{44}$$

$$d_P(U,V)^2 := \sum_{k=1}^{K} \|P_k(U) - P_k(V)\|_F^2, \tag{45}$$

$$d_B(U,V)^2 := \sum_{j=1}^{K} \min_{O_j \in \mathrm{O}(q_j)} \|U_j - V_j O_j\|_F^2, \tag{46}$$

$$d_\Pi(U,V)^2 := \sum_{j=1}^{K} \|\Pi_j(U) - \Pi_j(V)\|_F^2, \tag{47}$$

$$\overline{d}_\Pi(U,V)^2 := \sum_{j=1}^{K+1} \|\Pi_j(U) - \Pi_j(V)\|_F^2. \tag{48}$$

Lemma 1 (Overlap transport identity and sharp metric comparisons)**.** *For $T_{ja} := \mathrm{tr}(\Pi_j(U)\Pi_a(V))$, $1 \le j, a \le K+1$, set $W := \sum_{j \neq a} T_{ja}$. Then*

$$d_P^2 = \sum_{j,a} |j-a| T_{ja}, \quad \overline{d}_\Pi^2 = 2W, \tag{49}$$

$$d_B^2 \le d_\Pi^2 \le 2d_B^2, \qquad d_\Pi^2 \le \overline{d}_\Pi^2 \le 2d_P^2, \tag{50}$$

$$d_P^2 \le \frac{K}{2}\overline{d}_\Pi^2 \le K d_\Pi^2 \le 2K d_B^2. \tag{51}$$

The constants in $\overline{d}_\Pi^2 \le 2d_P^2$ and $d_P^2 \le (K/2)\overline{d}_\Pi^2$ are attained by adjacent-block and first-to-residual rotations, respectively.

Proof.

$$U_j^\top V_j = L_j \,\mathrm{diag}(\sigma_{j1}, \dots, \sigma_{jq_j}) R_j^\top, \quad 0 \le \sigma_{ji} \le 1, \tag{52}$$

$$\min_{O_j} \|U_j - V_j O_j\|_F^2 = 2\sum_i (1 - \sigma_{ji}), \tag{53}$$

$$\|\Pi_j(U) - \Pi_j(V)\|_F^2 = 2\sum_i (1 - \sigma_{ji}^2), \tag{54}$$

$$1 - \sigma \le 1 - \sigma^2 \le 2(1-\sigma), \quad \sigma \in [0,1], \tag{55}$$

$$\sum_a T_{ja} = q_j, \quad \sum_j T_{ja} = q_a, \tag{56}$$

$$\|P_k(U) - P_k(V)\|_F^2 = \sum_{j \le k < a} T_{ja} + \sum_{a \le k < j} T_{ja}, \tag{57}$$

$$\sum_k \|P_k(U) - P_k(V)\|_F^2 = \sum_{j,a} |j-a| T_{ja}, \tag{58}$$

$$\sum_j \|\Pi_j(U) - \Pi_j(V)\|_F^2 = 2\sum_{j \neq a} T_{ja} = 2W, \tag{59}$$

$$\|\Pi_{K+1}(U) - \Pi_{K+1}(V)\|_F^2 = 2\sum_{a \le K} T_{K+1,a} \le W. \tag{60}$$

Thus $W \le d_P^2 \le KW$ and $d_\Pi^2 \ge W$. A two-plane rotation between blocks $a < b$ has $W = 2\sin^2\theta$ and $d_P^2 = (b-a)W$, proving the equality claims. Keeping the overlap flow avoids repeated triangle inequalities for the nested prefixes. □

Lemma 2 (Infinitesimal projector metric)**.** *For $Q = [U, U_\perp] \in \mathrm{O}(d)$, $\Omega^\top = -\Omega$, $\Omega_{jj} = 0$, and $U(s) = (Qe^{s\Omega})_{:R}$,*

$$\dot{P}_k(0) = Q[\Omega, \widehat{F}_k]Q^\top, \quad \widehat{F}_k := \mathrm{diag}(I_{r_k}, 0_{d-r_k}), \tag{61}$$

$$\sum_k \|\dot{P}_k(0)\|_F^2 = 2 \sum_{1 \le a < b \le K+1} (b-a)\|\Omega_{ab}\|_F^2, \tag{62}$$

$$\|\dot{U}(0)\|_F^2 = 2 \sum_{a<b\le K} \|\Omega_{ab}\|_F^2 + \sum_{a \le K} \|\Omega_{a,K+1}\|_F^2, \tag{63}$$

$$\|\dot{U}(0)\|_F^2 \le \sum_k \|\dot{P}_k(0)\|_F^2 \le 2K\|\dot{U}(0)\|_F^2. \tag{64}$$

Proof.

$$P_k(s) = Qe^{s\Omega}\widehat{F}_k e^{-s\Omega}Q^\top, \tag{65}$$

$$[\Omega, \widehat{F}_k]_{ab} = (\mathbf{1}_{b\le k} - \mathbf{1}_{a \le k})\Omega_{ab}, \tag{66}$$

$$\|[\Omega, \widehat{F}_k]\|_F^2 = 2 \sum_{a \le k < b} \|\Omega_{ab}\|_F^2, \tag{67}$$

$$\sum_{k=1}^{K} \mathbf{1}_{a \le k < b} = b - a, \tag{68}$$

$$\|(Q\Omega)_{:R}\|_F^2 = \|\Omega_{1:K,1:K}\|_F^2 + \|\Omega_{K+1,1:K}\|_F^2. \tag{69}$$

□

Proposition 3 (Polar and QR retractions)**.** *For $Z \in T_U \,\mathrm{St}(d, R)$,*

$$X := U + Z, \quad X^\top X = I + Z^\top Z, \quad \sigma_{\min}(X) \ge 1, \tag{70}$$

$$R_U^{\mathrm{pol}}(Z) := (U+Z)(I + Z^\top Z)^{-1/2}, \tag{71}$$

$$R_U^{\mathrm{qr}}(Z) := (U+Z)\,\mathrm{chol}(I + Z^\top Z)^{-\top}, \tag{72}$$

$$R_U^{\mathrm{pol}}(0) = R_U^{\mathrm{qr}}(0) = U, \quad \mathrm{D}R_U^{\mathrm{pol}}(0)[Z] = \mathrm{D}R_U^{\mathrm{qr}}(0)[Z] = Z, \tag{73}$$

$$\|R_U^{\mathrm{pol}}(Z) - U - Z\|_F \le \tfrac{1}{2}\sqrt{1 + \|Z\|_{\mathrm{op}}^2}\,\|Z\|_F^2, \tag{74}$$

$$\|R_U^{\mathrm{qr}}(Z) - U - Z\|_F \le (1 + \|Z\|_{\mathrm{op}}^2)\|Z\|_F^2. \tag{75}$$

Proof.

$$X^\top X = I + U^\top Z + Z^\top U + Z^\top Z = I + Z^\top Z, \tag{76}$$

$$0 \preceq I - (I+E)^{-1/2} \preceq \tfrac{1}{2}E, \quad E := Z^\top Z, \tag{77}$$

$$\|R_U^{\mathrm{pol}}(Z) - X\|_F \le \|X\|_{\mathrm{op}}\|(I+E)^{-1/2} - I\|_F \le \tfrac{1}{2}\sqrt{1 + \|Z\|_{\mathrm{op}}^2}\|E\|_F, \tag{78}$$

$$L(s)L(s)^\top = I + sE, \quad L(0) = I, \tag{79}$$

$$L'(s) = L(s)\mathcal{T}\big(L(s)^{-1}EL(s)^{-\top}\big), \tag{80}$$

$$\mathcal{T}(A) := \mathrm{tril}(A, -1) + \tfrac{1}{2}\,\mathrm{diag}(A), \tag{81}$$

$$\|\mathcal{T}(A)\|_F \le \|A\|_F, \quad \|L(s)^{-1}\|_{\mathrm{op}} \le 1, \tag{82}$$

$$\|L(1) - I\|_F \le \int_0^1 \|L'(s)\|_F \,\mathrm{d}s \le \sqrt{1 + \|E\|_{\mathrm{op}}}\|E\|_F, \tag{83}$$

$$\|XL(1)^{-\top} - X\|_F \le \|X\|_{\mathrm{op}}\|L(1)^{-1}\|_{\mathrm{op}}\|L(1) - I\|_F \tag{84}$$

$$\le (1 + \|E\|_{\mathrm{op}})\|E\|_F \le (1 + \|Z\|_{\mathrm{op}}^2)\|Z\|_F^2. \tag{85}$$

□

The retraction error is quadratic in the tangent step, so a small horizontal update preserves its intended first-order motion. Comparing two different runs additionally requires control of the QR map between their inputs. The next lemma makes the necessary singular-value condition on that connecting segment explicit.

Lemma 3 (Differential of the positive-diagonal QR map)**.** *For $X = QR$, $R_{ii} > 0$, $\sigma_{\min}(X) \geq a > 0$,*

$$A := Q^\top(\mathrm{d}X)R^{-1}, \tag{86}$$

$$\mathrm{d}Q = (I - QQ^\top)(\mathrm{d}X)R^{-1} + Q\big(\mathrm{tril}(A,-1) - \mathrm{tril}(A,-1)^\top\big), \tag{87}$$

$$\|\mathrm{d}Q\|_F^2 \leq 2\|(\mathrm{d}X)R^{-1}\|_F^2 \leq \frac{2}{a^2}\|\mathrm{d}X\|_F^2. \tag{88}$$

If $\inf_{s\in[0,1]} \sigma_{\min}(X + s(Y - X)) \geq a$,

$$\|\operatorname{qf}(X) - \operatorname{qf}(Y)\|_F \leq \frac{\sqrt{2}}{a}\|X - Y\|_F. \tag{89}$$

Proof.

$$\mathrm{d}X = (\mathrm{d}Q)R + Q\,\mathrm{d}R, \tag{90}$$

$$Q^\top(\mathrm{d}X)R^{-1} = Q^\top\mathrm{d}Q + (\mathrm{d}R)R^{-1}, \tag{91}$$

$$(Q^\top\mathrm{d}Q)^\top = -Q^\top\mathrm{d}Q, \qquad \mathrm{tril}((\mathrm{d}R)R^{-1}, -1) = 0, \tag{92}$$

$$\|\mathrm{tril}(A,-1) - \mathrm{tril}(A,-1)^\top\|_F^2 = 2\|\mathrm{tril}(A,-1)\|_F^2 \leq 2\|A\|_F^2, \tag{93}$$

$$\operatorname{qf}(Y) - \operatorname{qf}(X) = \int_0^1 \mathrm{D}\operatorname{qf}(X + s(Y - X))[Y - X]\,\mathrm{d}s. \tag{94}$$

□

A.2. Variational structure, spectral assignment, and local curvature

Each prefix of the flag is evaluated against a covariance with its own memory scale. Consequently a direction assigned to an inner block contributes to several prefix objectives, as encoded by A_j. The assignment theorem below resolves these competing contributions under simultaneous diagonalization, while respecting the prescribed capacity of every block.

Definition 3 (Flag energy)**.**

$$C_k = C_k^\top \succeq 0, \quad \omega_k > 0, \quad A_j := \sum_{k=j}^{K} \omega_k C_k, \quad A_{K+1} := 0, \tag{95}$$

$$\Phi(U; C) := \sum_{k=1}^{K} \omega_k \operatorname{tr}(P_k(U)C_k) = \sum_{j=1}^{K} \operatorname{tr}(U_j^\top A_j U_j), \tag{96}$$

$$H(U; C) := 2\sum_k \omega_k C_k U F_k, \quad \operatorname{grad}\Phi(U; C) = \mathsf{H}_U(H(U; C)). \tag{97}$$

Proposition 4 (First and second variations)**.** *For $Z \in \mathcal{H}_U$, $Q = [U, U_\perp]$, and horizontal Ω with $Z = (Q\Omega)_{:R}$,*

$$\mathrm{D}\Phi(U)[Z] = 2\sum_k \omega_k \operatorname{tr}(Z^\top C_k U F_k) = \langle \operatorname{grad}\Phi, Z\rangle_F, \tag{98}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}\Phi((Qe^{s\Omega})_{:R}; C)\Big|_{s=0} = \sum_k \omega_k \operatorname{tr}\big(Q^\top C_k Q[\Omega, [\Omega, \widehat{F}_k]]\big), \tag{99}$$

$$\operatorname{grad}\Phi(U; C) = 0 \Longrightarrow \quad \langle Z, \operatorname{Hess}\Phi(U)[Z]\rangle_F = \sum_k \omega_k \operatorname{tr}\big(Q^\top C_k Q[\Omega, [\Omega, \widehat{F}_k]]\big). \tag{100}$$

Proof.

$$\mathrm{D}(UF_kU^\top)[Z] = ZF_kU^\top + UF_kZ^\top, \tag{101}$$

$$\frac{\mathrm{d}}{\mathrm{d}s}(e^{s\Omega}\widehat{F}_k e^{-s\Omega}) = e^{s\Omega}[\Omega, \widehat{F}_k]e^{-s\Omega}, \tag{102}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}(e^{s\Omega}\widehat{F}_k e^{-s\Omega}) = e^{s\Omega}[\Omega, [\Omega, \widehat{F}_k]]e^{-s\Omega}, \tag{103}$$

$$(\Phi \circ \gamma)''(0) = \operatorname{Hess}\Phi[\dot\gamma(0), \dot\gamma(0)] + \langle \operatorname{grad}\Phi, \nabla_{\dot\gamma}\dot\gamma\rangle_F. \tag{104}$$

□

Theorem 1 (Commuting case: integral assignment)**.** *Suppose*

$$C_k = Q\,\mathrm{diag}(\lambda_{k1}, \dots, \lambda_{kd})Q^\top, \quad b_{ji} := \sum_{k=j}^{K} \omega_k\lambda_{ki}, \quad b_{K+1,i} := 0, \tag{105}$$

$$\mathcal{X} := \{x \geq 0 : \sum_{j=1}^{K+1} x_{ji} = 1,\ \sum_{i=1}^{d} x_{ji} = q_j\}. \tag{106}$$

Then

$$\max_{[U]\in\mathfrak{F}} \Phi(U; C) = \max_{x\in\mathcal{X}} \sum_{j,i} b_{ji}x_{ji} \tag{107}$$

$$= \max_{\substack{\{S_j\}\ \text{partition}\ [d] \\ |S_j| = q_j}} \sum_{j=1}^{K+1} \sum_{i\in S_j} b_{ji} \tag{108}$$

$$= \min_{\alpha\in\mathbb{R}^d,\, \beta\in\mathbb{R}^{K+1}} \left\{ \sum_i \alpha_i + \sum_j q_j\beta_j : \alpha_i + \beta_j \geq b_{ji} \right\}. \tag{109}$$

Proof.

$$x_{ji}(U) := e_i^\top Q^\top \Pi_j(U) Q e_i \geq 0, \tag{110}$$

$$\sum_j x_{ji}(U) = 1, \qquad \sum_i x_{ji}(U) = \operatorname{tr}\Pi_j = q_j, \tag{111}$$

$$\Phi(U; C) = \sum_{j,i} b_{ji}x_{ji}(U) \leq \max_{x\in\mathcal{X}} \langle b, x\rangle. \tag{112}$$

Changing the signs of the block-sum rows turns the constraint matrix into a directed bipartite incidence matrix. It is totally unimodular, so the Hoffman–Kruskal integrality theorem (Hoffman & Kruskal, 1956), with integer margins $(q_j, 1)$, gives integral vertices. Column sums of one then give $x_{ji} \in \{0, 1\}$. For $S_j := \{i : x_{ji} = 1\}$ choose $U_j = Q_{:,S_j}$. Then $x(U) = x$. Finally,

$$\mathcal{L}(x, \alpha, \beta) = \sum_i \alpha_i + \sum_j q_j\beta_j + \sum_{j,i} (b_{ji} - \alpha_i - \beta_j)x_{ji}, \tag{113}$$

$$\sup_{x\geq 0} \mathcal{L} < \infty \iff \alpha_i + \beta_j \geq b_{ji}, \tag{114}$$

$$\varnothing \neq \mathcal{X}\ \text{compact} \implies \max\langle b, x\rangle = \min \mathcal{L}_{\mathrm{dual}}. \tag{115}$$

□

Corollary 1 (Common order and strict assignment margin)**.** *If $\lambda_{k1} \geq \cdots \geq \lambda_{kd}$ for every k, then $U^\star = Q_{:,1:R}$ maximizes Φ. More generally, let an optimal partition $S^\star$ admit a dual optimum satisfying*

$$s_{ji} := \alpha_i + \beta_j - b_{ji}, \quad s_{ji} = 0\ (i \in S_j^\star), \quad s_{ji} \geq \delta > 0\ (i \notin S_j^\star), \tag{116}$$

$$W(U) := \sum_j \sum_{i\notin S_j^\star} x_{ji}(U). \tag{117}$$

Then

$$\Phi(U^\star) - \Phi(U) = \sum_{j,i} s_{ji} x_{ji}(U) \geq \delta W(U), \tag{118}$$

$$d_P(U, U^\star)^2 \leq K W(U), \tag{119}$$

$$\Phi(U^\star) - \Phi(U) \geq \frac{\delta}{K} d_P(U, U^\star)^2. \tag{120}$$

The factor K in the distance comparison is attained by a rotation between the first and residual blocks.

Proof. Ky Fan's maximum principle (Fan, 1949) gives the common-order optimum. For the quantitative bound let $T_{ja} := \operatorname{tr}(\Pi_j(U)\Pi_a(U^\star))$, including the residual block. Its row and column sums both equal the block capacities.

$$\operatorname{tr}(P_k C_k) \leq \sum_{i=1}^{r_k} \lambda_{ki} = \operatorname{tr}(P_k(U^\star) C_k), \tag{121}$$

$$\|\Pi_j(U) - \Pi_j(U^\star)\|_F^2 = 2q_j - 2\sum_{i \in S_j^\star} x_{ji} = 2\sum_{i \notin S_j^\star} x_{ji}, \tag{122}$$

$$\|P_k(U) - P_k(U^\star)\|_F^2 = 2\sum_{j \leq k < a} T_{ja} = \sum_{j \leq k < a} T_{ja} + \sum_{a \leq k < j} T_{ja}, \tag{123}$$

$$d_P^2 = \sum_{j,a} |j - a| T_{ja} \leq K \sum_{j \neq a} T_{ja} = K W(U). \tag{124}$$

For a rotation of angle θ between a vector of block 1 and one of block $K+1$, $W = 2\sin^2\theta$ and $d_P^2 = 2K\sin^2\theta$. Thus the geometric constant cannot be reduced. The dual-slack identity gives the objective bound. □

Proposition 5 (Perturbation and matched-rank reconstruction)**.** *For $\|\widehat{C}_k - C_k\|_{\mathrm{op}} \leq \epsilon_k$, define $D_C := \sum_k \omega_k r_k \epsilon_k$. If $U^\star, \widehat{U}$ maximize their respective objectives, then*

$$\sup_U |\Phi(U; \widehat{C}) - \Phi(U; C)| \leq D_C, \qquad 0 \leq \Phi(U^\star; C) - \Phi(\widehat{U}; C) \leq 2D_C. \tag{125}$$

For every rank-R projector Q_R containing $\operatorname{range} P_k$*,*

$$\|G - GQ_R\|_F^2 \leq \|G - GP_k\|_F^2. \tag{126}$$

Proof.

$$|\operatorname{tr}(P_k \Delta C_k)| \leq \|P_k\|_* \|\Delta C_k\|_{\mathrm{op}} = r_k \epsilon_k, \tag{127}$$

$$\Phi(U^\star; C) - \Phi(\widehat{U}; C) \leq \Phi(U^\star; C) - \Phi(U^\star; \widehat{C}) + \Phi(\widehat{U}; \widehat{C}) - \Phi(\widehat{U}; C), \tag{128}$$

$$Q_R - P_k \succeq 0, \quad \|G - GP_k\|_F^2 - \|G - GQ_R\|_F^2 = \operatorname{tr}((Q_R - P_k)G^\top G) \geq 0. \tag{129}$$

□

Proposition 6 (A strict nonglobal local maximum)**.** *Let*

$$d = 3, \quad K = 2, \quad (r_1, r_2) = (1, 2), \quad \omega_1 = \omega_2 = 1, \tag{130}$$

$$C_1 = \operatorname{diag}(3, \tfrac{5}{2}, \tfrac{1}{2}), \quad C_2 = \operatorname{diag}(0, 2, \tfrac{3}{2}), \quad (b_{ji}) = \begin{pmatrix} 3 & 9/2 & 2 \\ 0 & 2 & 3/2 \\ 0 & 0 & 0 \end{pmatrix}, \tag{131}$$

$$U_0 = (e_1, e_2), \qquad U_\star = (e_2, e_3). \tag{132}$$

Then $[C_1, C_2] = 0$, $\Phi(U_0) = 5 < 6 = \Phi(U_\star)$, and

$$\Phi((e^\Omega)_{:2}) = 5 - \tfrac{1}{2}\Omega_{12}^2 - \Omega_{13}^2 - \tfrac{1}{2}\Omega_{23}^2 + O(\|\Omega\|_F^3). \tag{133}$$

Proof. For eigen-indices $i \in S_a$, $l \in S_b$, $a < b$, put

$$\Delta_{ai,bl} := b_{ai} + b_{bl} - b_{al} - b_{bi}, \tag{134}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}s^2}\Phi((Qe^{s\Omega})_{:R})\Big|_0 = -2\sum_{a<b}\sum_{i\in S_a, l\in S_b}\Delta_{ai,bl}\Omega_{il}^2. \tag{135}$$

At U_0,

$$\Delta_{11,22} = \tfrac{1}{2}, \quad \Delta_{11,33} = 1, \quad \Delta_{22,33} = \tfrac{1}{2}, \quad \operatorname{grad}\Phi(U_0) = 0, \quad \operatorname{Hess}\Phi(U_0) \prec 0. \tag{136}$$

The six assignment scores are $5, 9/2, 9/2, 6, 2, 4$. □

The counterexample distinguishes global allocation from local improvement: every infinitesimal pairwise rotation at U_0 decreases the objective, although a different assignment has greater value. Commutativity therefore provides an exact global characterization without ensuring that ascent finds it. The oracle used subsequently is an interior stationary branch with uniform local curvature; the analysis requires the iterates to remain in its regular chart.

Assumption 1 (A regular oracle chart)**.** *Let $\psi : \mathcal{D} \subset \mathbb{R}^p \to \mathfrak{F}$ be a smooth chart on a convex set, $f_C := \Phi(\psi(\cdot); C)$, and*

$$\nabla f_C(x_C^\star) = 0, \quad -L_\phi I \preceq \nabla^2 f_C(x) \preceq -\mu_\phi I \quad (x \in \mathcal{D}), \tag{137}$$

$$0 < \mu_\phi \le L_\phi, \quad \|\mathrm{D}\psi(x)\|_{2\to F} \le J_\psi, \quad \|\nabla^3 f_C(x)\| \le T_\phi. \tag{138}$$

Lemma 4 (Oracle sensitivity and differentiability)**.** *If $C, \widehat{C}$ satisfy Assumption 1 in the same chart, then, with $E_C := 2J_\psi \sum_k \omega_k \sqrt{r_k}\epsilon_k$,*

$$\sup_x \|\nabla f_{\widehat{C}}(x) - \nabla f_C(x)\| \le E_C, \tag{139}$$

$$\|x_{\widehat{C}}^\star - x_C^\star\| \le E_C/\mu_\phi, \tag{140}$$

$$\mathrm{D}x_C^\star[\Delta C] = -[\nabla^2 f_C(x_C^\star)]^{-1}\nabla f_{\Delta C}(x_C^\star). \tag{141}$$

If $\|\nabla^2 f_{\Delta C}\| \le E_H$, and $v = x^\star_{C+\Delta C} - x^\star_C$,

$$\|v - \mathrm{D}x_C^\star[\Delta C]\| \le \frac{T_\phi}{2\mu_\phi^3}E_C^2 + \frac{E_H}{\mu_\phi^2}E_C. \tag{142}$$

Proof.

$$\|H(U;\Delta C)\|_F \le 2\sum_k \omega_k\epsilon_k\|UF_k\|_F = 2\sum_k \omega_k\sqrt{r_k}\epsilon_k, \tag{143}$$

$$\mu_\phi\|v\|^2 \le -\langle \nabla f_C(x_C^\star + v) - \nabla f_C(x_C^\star), v\rangle \le E_C\|v\|, \tag{144}$$

$$0 = \nabla f_{C+\Delta C}(x_C^\star + v) = \nabla^2 f_C(x_C^\star)v + \nabla f_{\Delta C}(x_C^\star) + R, \tag{145}$$

$$\|R\| \le \tfrac{1}{2}T_\phi\|v\|^2 + E_H\|v\|, \tag{146}$$

$$v = -[\nabla^2 f_C(x_C^\star)]^{-1}(\nabla f_{\Delta C}(x_C^\star) + R). \tag{147}$$

□

The factor μ_ϕ^{-1} measures how strongly estimation errors can move the local flag optimum. A nearly flat objective can change its preferred flag substantially under a small covariance perturbation. The next spectral bound exhibits the corresponding gap dependence for a single projector; a general multi-covariance flag uses the chart curvature above.

Proposition 7 (Residual Davis–Kahan bound)**.** *Let $C = Q_1\Lambda_1Q_1^\top + Q_2\Lambda_2Q_2^\top$, $\dim Q_1 = r$, and $g := \lambda_r(C) - \lambda_{r+1}(C) > 0$. If $\widehat{C} = C + E$, $\|E\|_{\mathrm{op}} \le \epsilon < g$, and $\widehat{P}$ is its leading rank-r projector, set $s_r := \min(r, d-r)$ and $g_{\mathrm{sep}} := \lambda_{\min}(\widehat{\Lambda}_1) - \lambda_{\max}(\Lambda_2)$. Then $g_{\mathrm{sep}} \ge g - \epsilon > 0$ and*

$$\|\widehat{P} - Q_1Q_1^\top\|_F \le \frac{\sqrt{2}\|Q_2^\top E\widehat{Q}_1\|_F}{g_{\mathrm{sep}}} \le \frac{\sqrt{2s_r}\,\epsilon}{g-\epsilon}. \tag{148}$$

The left side is also at most $\sqrt{2s_r}$. Keeping the off-diagonal residual can be substantially sharper than replacing it by $\sqrt{s_r}\epsilon$.

Proof. This is the Sylvester-equation form of the Davis–Kahan $\sin\Theta$ theorem (Davis & Kahan, 1970); Weyl's inequality supplies the separation from the unperturbed gap.

$$\widehat{C}\widehat{Q}_1 = \widehat{Q}_1\widehat{\Lambda}_1, \quad X := Q_2^\top \widehat{Q}_1, \quad X\widehat{\Lambda}_1 - \Lambda_2 X = Q_2^\top E\widehat{Q}_1 =: B, \tag{149}$$

$$\lambda_{\min}(\widehat{\Lambda}_1) - \lambda_{\max}(\Lambda_2) \geq g - \epsilon, \tag{150}$$

$$X_{ij} = B_{ij}/(\widehat{\lambda}_j - \lambda_{r+i}), \quad \|X\|_F \leq \|B\|_F/g_{\text{sep}}, \quad \|B\|_F \leq \sqrt{s_r}\|E\|_{\text{op}}, \tag{151}$$

$$\|\widehat{P} - P\|_F^2 = 2r - 2\,\mathrm{tr}(\widehat{P}P) = 2\|Q_2^\top \widehat{Q}_1\|_F^2. \tag{152}$$

□

Corollary 2 (Local curvature at a common-order flag)**.** *If $\lambda_{k1} \geq \cdots \geq \lambda_{kd}$, then for $i \in I_a$, $l \in I_b$,*

$$\Delta_{ai,bl} = \sum_{k=a}^{b-1} \omega_k(\lambda_{ki} - \lambda_{kl}), \quad \lambda_{k\cdot} := 0 \ (k > K), \tag{153}$$

$$\delta_\star := \min_{a<b,\, i\in I_a,\, l\in I_b} \Delta_{ai,bl} > 0 \implies -\mathrm{Hess}\,\Phi(U^\star)[Z,Z] \geq \delta_\star \|Z\|_F^2. \tag{154}$$

For orthonormal chart coordinates at $U^\star$, and $\|\nabla^2 f(x) - \nabla^2 f(0)\| \leq T_\phi\|x\|$,

$$\|x\| \leq \delta_\star/(2T_\phi) \implies -\nabla^2 f(x) \succeq (\delta_\star/2)I. \tag{155}$$

A.3. Adapted covariance statistics, sketches, and tracking

Gradients depend on previous updates, so their second moments form an adapted process. We separate fluctuations around each conditional covariance from the lag introduced by averaging a changing target. Normalized and raw EMAs are distinguished throughout because their early-time scaling changes the relative weights in the flag objective.

Definition 4 (Filtration and normalized EMA)**.** *For a filtration $(\mathcal{F}_t)$, let G_t be $\mathcal{F}_t$-measurable and*

$$\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}], \tag{156}$$

$$X_t := G_t^\top G_t, \quad C_t := \mathbb{E}[X_t \mid \mathcal{F}_{t-1}], \tag{157}$$

$$Z_t := X_t - C_t, \quad \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}] = 0, \tag{158}$$

$$\overline{C}_{k,t} := (1-\rho_k)\sum_{i=1}^{t} \rho_k^{t-i} X_i, \quad \widehat{C}_{k,t} := \overline{C}_{k,t}/(1-\rho_k^t), \quad 0 < \rho_k < 1, \tag{159}$$

$$\alpha_{k,t,i} := \frac{(1-\rho_k)\rho_k^{t-i}}{1-\rho_k^t}, \qquad C^\circ_{k,t} := \sum_{i=1}^{t} \alpha_{k,t,i} C_i. \tag{160}$$

Lemma 5 (Exact EMA identities)**.**

$$\sum_{i=1}^{t} \alpha_{k,t,i} = 1, \quad \max_i \alpha_{k,t,i} = \frac{1-\rho_k}{1-\rho_k^t}, \tag{161}$$

$$\sum_{i=1}^{t} \alpha_{k,t,i}^2 = \frac{1-\rho_k}{1+\rho_k}\frac{1+\rho_k^t}{1-\rho_k^t}, \tag{162}$$

$$A_{k,t} := \sum_i (t-i)\alpha_{k,t,i} = \frac{\rho_k}{1-\rho_k} - \frac{t\rho_k^t}{1-\rho_k^t}, \tag{163}$$

$$\|C_i - C_{i-1}\|_{\text{op}} \leq \nu \implies \|C^\circ_{k,t} - C_t\|_{\text{op}} \leq \nu A_{k,t}. \tag{164}$$

Proof.

$$\sum_{j=0}^{t-1}\rho^j=(1-\rho^t)/(1-\rho),\quad \sum_{j=0}^{t-1}\rho^{2j}=(1-\rho^{2t})/(1-\rho^2), \tag{165}$$

$$\sum_{j=0}^{t-1}j\rho^j=\rho\frac{\mathrm{d}}{\mathrm{d}\rho}\frac{1-\rho^t}{1-\rho}, \tag{166}$$

$$\|C^\circ_{k,t}-C_t\|_{\mathrm{op}}\le\sum_i\alpha_{k,t,i}\sum_{j=i+1}^{t}\|C_j-C_{j-1}\|_{\mathrm{op}}\le\nu\sum_i(t-i)\alpha_{k,t,i}. \tag{167}$$

□

Theorem 2 (Finite-horizon adapted covariance bound)**.** *Suppose, almost surely,*

$$\|Z_i\|_{\mathrm{op}}\le L_Z,\quad \mathbb{E}[Z_i^2\mid\mathcal{F}_{i-1}]\preceq v_iI,\quad v_i \textit{ deterministic}, \tag{168}$$

$$V_{k,t}:=\sum_i\alpha_{k,t,i}^2v_i,\quad R_{k,t}:=L_Z\max_i\alpha_{k,t,i}, \tag{169}$$

$$u:=\log(2dKT/\delta),\quad \mathfrak{r}(V,R,u):=\frac{Ru}{3}+\sqrt{2Vu+\frac{R^2u^2}{9}}. \tag{170}$$

With probability at least $1-\delta$*, simultaneously for* $k\le K, t\le T$*,*

$$\|\widehat{C}_{k,t}-C_t\|_{\mathrm{op}}\le\mathfrak{r}(V_{k,t},R_{k,t},u)+\nu A_{k,t}, \tag{171}$$

$$\|\overline{C}_{k,t}-C_t\|_{\mathrm{op}}\le(1-\rho_k^t)\left(\mathfrak{r}(V_{k,t},R_{k,t},u)+\nu A_{k,t}\right)+\rho_k^t\|C_t\|_{\mathrm{op}}. \tag{172}$$

Proof. For fixed (k,t), $D_i:=\alpha_{k,t,i}Z_i$ satisfies

$$\mathbb{E}[D_i\mid\mathcal{F}_{i-1}]=0,\quad \lambda_{\max}(\pm D_i)\le R_{k,t},\quad \left\|\sum_i\mathbb{E}[D_i^2\mid\mathcal{F}_{i-1}]\right\|_{\mathrm{op}}\le V_{k,t}. \tag{173}$$

The matrix Freedman inequality (Tropp, 2011) gives

$$\mathbb{P}\left(\left\|\sum_iD_i\right\|_{\mathrm{op}}\ge s\right)\le2d\exp\left(-\frac{s^2}{2(V_{k,t}+R_{k,t}s/3)}\right), \tag{174}$$

$$s=\mathfrak{r}(V_{k,t},R_{k,t},u)\Longrightarrow s^2=2u(V_{k,t}+R_{k,t}s/3), \tag{175}$$

$$\mathbb{P}\left(\exists k,t:\|\widehat{C}_{k,t}-C^\circ_{k,t}\|_{\mathrm{op}}>s\right)\le2dKTe^{-u}=\delta, \tag{176}$$

$$\overline{C}_{k,t}-C_t=(1-\rho_k^t)(\widehat{C}_{k,t}-C_t)-\rho_k^tC_t. \tag{177}$$

Thus the radius exactly inverts the Bernstein exponent in matrix Freedman; it avoids the further relaxation $\mathfrak{r}(V,R,u)\le\sqrt{2Vu}+2Ru/3$. A zero conditional-variance budget means the martingale sum vanishes almost surely and its radius can be set to zero. □

Corollary 3 (Memory–drift balance)**.** *For the limiting EMA weights,* $v_i\le v$*,* $h=1-\rho$*, and fixed* u*,*

$$\epsilon(h)\le\mathfrak{r}\left(\frac{vh}{2-h},L_Zh,u\right)+\frac{\nu(1-h)}{h}, \tag{178}$$

$$\mathfrak{r}\left(\frac{vh}{2-h},L_Zh,u\right)=\sqrt{vu}\sqrt{h}+\frac{L_Zu}{3}h+O(h^{3/2}),\quad h\downarrow0, \tag{179}$$

$$a:=\sqrt{vu}>0,\quad \nu>0,\quad h_0:=(2\nu/a)^{2/3}\le1,\quad a\sqrt{h_0}+\nu/h_0=3\,2^{-2/3}a^{2/3}\nu^{1/3}. \tag{180}$$

The choice h_0 *minimizes the leading stochastic–drift expression* $a\sqrt{h}+\nu/h$*; the first display retains the finite-*h *constants.*

Longer memory reduces the stochastic term by spreading mass over more observations, but increases the age of the information used to track the current covariance. The displayed balance quantifies this tradeoff. Different timescales can address different drift regimes; their joint benefit still depends on the flag objective and the resulting update.

Definition 5 (Fixed-probe Nyström reconstruction)**.**

$$\Omega_k \in \mathbb{R}^{d\times s_k}, \quad s_k = \min(d, r_k + p), \quad Y_k = C_k\Omega_k, \quad B_k = \mathrm{sym}(\Omega_k^\top Y_k) + \gamma I, \quad \gamma > 0, \tag{181}$$

$$\mathcal{N}_\gamma(Y_k) := Y_k B_k^{-1} Y_k^\top. \tag{182}$$

Proposition 8 (PSD order, regularization, and probe blindness)**.** *For $C \succeq 0$, $Y = C\Omega$, and $Z = C^{1/2}\Omega$,*

$$\mathcal{N}_\gamma(C\Omega) = C^{1/2} Z (Z^\top Z + \gamma I)^{-1} Z^\top C^{1/2}, \tag{183}$$

$$0 \preceq \mathcal{N}_\gamma(C\Omega) \preceq C, \tag{184}$$

$$\mathcal{N}_0(C\Omega) - \mathcal{N}_\gamma(C\Omega) = C^{1/2} L \,\mathrm{diag}\left(\frac{\gamma}{\sigma_i^2 + \gamma}\right) L^\top C^{1/2}, \quad Z = L\,\mathrm{diag}(\sigma_i) R^\top \quad (\sigma_i > 0). \tag{185}$$

If $s < d$, $v \in \ker \Omega^\top$, and $\|v\| = 1$, then

$$C = avv^\top \quad \Longrightarrow \quad Y = 0, \quad \mathcal{N}_\gamma(Y) = 0, \quad \|C - \mathcal{N}_\gamma(Y)\|_{\mathrm{op}} = a. \tag{186}$$

Here $\mathcal{N}_0$ uses the Moore–Penrose inverse.

Proof.

$$Z(Z^\top Z + \gamma I)^{-1} Z^\top = L\,\mathrm{diag}\left(\frac{\sigma_i^2}{\sigma_i^2 + \gamma}\right) L^\top, \tag{187}$$

$$0 \le \frac{\sigma_i^2}{\sigma_i^2 + \gamma} \le 1, \quad Z(Z^\top Z)^\dagger Z^\top = LL^\top, \tag{188}$$

$$C\Omega = av(v^\top \Omega) = 0. \tag{189}$$

□

Lemma 6 (Sketch sensitivity)**.** *Suppose $B = \mathrm{sym}(\Omega^\top Y) + \gamma I \succeq \gamma I$ and $B' = \mathrm{sym}(\Omega^\top Y') + \gamma I \succeq \gamma I$, with $\max(\|Y\|_{\mathrm{op}}, \|Y'\|_{\mathrm{op}}) \le Y_\star$. Then*

$$\|\mathcal{N}_\gamma(Y) - \mathcal{N}_\gamma(Y')\|_F \le L_N \|Y - Y'\|_F, \quad L_N := 2Y_\star/\gamma + Y_\star^2 \|\Omega\|_{\mathrm{op}}/\gamma^2, \tag{190}$$

$$\mathrm{D}\mathcal{N}_\gamma(Y)[E] = EB^{-1}Y^\top + YB^{-1}E^\top - YB^{-1}\,\mathrm{sym}(\Omega^\top E)B^{-1}Y^\top. \tag{191}$$

Proof.

$$B^{-1} - B'^{-1} = B^{-1}(B' - B)B'^{-1}, \tag{192}$$

$$YB^{-1}Y^\top - Y'B'^{-1}Y'^\top = (Y - Y')B^{-1}Y^\top + Y'B^{-1}(Y - Y')^\top + Y'(B^{-1} - B'^{-1})Y'^\top, \tag{193}$$

$$\|B^{-1} - B'^{-1}\|_F \le \gamma^{-2}\|\Omega\|_{\mathrm{op}}\|Y - Y'\|_F, \qquad \mathrm{D}(B^{-1})[E] = -B^{-1}\,\mathrm{sym}(\Omega^\top E)B^{-1}. \tag{194}$$

□

The probe-blindness example explains why sketch error must remain a separate quantity: a fixed probe can miss an entire energetic direction. We now combine this approximation error with statistical error and target motion. The tracking recursion measures distance to a local oracle at refresh times, rather than assuming exact covariance information at every parameter update.

Definition 6 (Refresh-time chart recursion)**.** *At refresh index s, let*

$$f_s(x) := \Phi(\psi(x); C_s), \quad \nabla f_s(x_s^\star) = 0, \tag{195}$$

$$x_{s+1} = x_s + \xi\nabla f_s(x_s) + \xi e_s + \xi\zeta_s + r_s, \tag{196}$$

$$\Delta_s := x_{s+1}^\star - x_s^\star, \quad E_s := \mathbb{E}\|x_s - x_s^\star\|^2, \tag{197}$$

$$q := \max\{|1 - \xi\mu_\phi|, |1 - \xi L_\phi|\} < 1, \quad 0 < \xi < 2/L_\phi. \tag{198}$$

Theorem 3 (Cocoercive tracking with exact forcing gains)**.** *Under Assumption 1, assume the relevant iterates and connecting segments remain in its convex chart. Define*

$$v_s := \xi e_s + \xi\zeta_s + r_s - \Delta_s, \quad b_s := \sqrt{\mathbb{E}\|v_s\|^2}. \tag{199}$$

Then the root-mean-square error obeys

$$\sqrt{E_{s+1}} \le q\sqrt{E_s} + b_s, \tag{200}$$

$$\sqrt{E_s} \le q^s\sqrt{E_0} + \sum_{j<s} q^{s-1-j} b_j. \tag{201}$$

The contraction is minimized at $\xi_\star = 2/(L_\phi + \mu_\phi)$, giving $q_\star = (L_\phi - \mu_\phi)/(L_\phi + \mu_\phi)$. For comparison, a squared-error form remains valid: for $\alpha > 0$ with $\kappa := (1+\alpha)q^2 < 1$,

$$E_{s+1} \le \kappa E_s + B_s, \tag{202}$$

$$B_s := 4(1+\alpha^{-1})\mathbb{E}\left[\xi^2\|e_s\|^2 + \xi^2\|\zeta_s\|^2 + \|r_s\|^2 + \|\Delta_s\|^2\right], \tag{203}$$

$$E_s \le \kappa^s E_0 + \sum_{j=0}^{s-1} \kappa^{s-1-j} B_j, \tag{204}$$

$$\sum_{s=0}^{S-1} E_s \le \frac{E_0 + \sum_{s=0}^{S-2} B_s}{1-\kappa}. \tag{205}$$

If $\mathbb{E}[\zeta_s \mid \mathcal{H}_s] = 0$, where $\mathcal{H}_s$ contains $(f_s, x_s, x_s^\star, e_s, r_s, \Delta_s)$, set $d_s^2 := \mathbb{E}\|\xi e_s + r_s - \Delta_s\|^2$ and $\sigma_s^2 := \mathbb{E}\|\zeta_s\|^2$. Then

$$E_{s+1} \le (q\sqrt{E_s} + d_s)^2 + \xi^2\sigma_s^2. \tag{206}$$

For uniform envelopes $d_s \le \overline{d}$, $\sigma_s \le \overline{\sigma}$, define

$$x_\infty := \frac{q\overline{d} + \sqrt{\overline{d}^2 + (1-q^2)\xi^2\overline{\sigma}^2}}{1-q^2}. \tag{207}$$

Then $\sqrt{E_s} \le x_\infty + q^s(\sqrt{E_0} - x_\infty)_+$. In particular the pure-drift RMS gain is $1/(1-q)$, whereas the centered-noise variance gain is $\xi^2/(1-q^2)$; scalar quadratic recursions attain both.

Proof. The smooth strongly convex interpolation inequality associated with Baillon–Haddad cocoercivity (Bauschke & Combettes, 2010) gives the optimal-step contraction. Here its local Hessian proof also gives every admissible step: with $v = x - y$ and $g = \nabla(-f_s)(x) - \nabla(-f_s)(y)$,

$$\langle v, g\rangle \ge \frac{\mu_\phi L_\phi}{\mu_\phi + L_\phi}\|v\|^2 + \frac{\|g\|^2}{\mu_\phi + L_\phi}, \tag{208}$$

$$T_s(x) - T_s(y) = \left[I + \xi\int_0^1 \nabla^2 f_s(y + u(x-y))\,\mathrm{d}u\right](x-y), \tag{209}$$

$$\|T_s(x) - T_s(y)\| \le q\|x-y\|, \quad T_s(x_s^\star) = x_s^\star, \tag{210}$$

$$x_{s+1} - x_{s+1}^\star = T_s(x_s) - T_s(x_s^\star) + \xi e_s + \xi\zeta_s + r_s - \Delta_s, \tag{211}$$

$$\|a+b\|^2 \le (1+\alpha)\|a\|^2 + (1+\alpha^{-1})\|b\|^2, \tag{212}$$

$$\|b_1+b_2+b_3+b_4\|^2 \le 4\sum_{i=1}^4 \|b_i\|^2, \tag{213}$$

$$\sum_{s=0}^{S-1}\sum_{j<s} \kappa^{s-1-j} B_j \le (1-\kappa)^{-1}\sum_{j=0}^{S-2} B_j. \tag{214}$$

The averaged Hessian in the first map identity has spectrum in $[-L_\phi, -\mu_\phi]$, so the endpoint formula for q is exact for the quadratic class. Minkowski's inequality in L^2 gives (201) directly. Conditional centering removes the cross term with ζ_s.

The scalar map $x \mapsto \sqrt{(qx+\overline{d})^2+\xi^2\overline{\sigma}^2}$ has Lipschitz constant at most q and fixed point x_∞, proving the last bound. Constant scalar forcing and a scalar AR(1) process respectively attain the two stated gains. Sharpness here concerns the comparison recursion; applicability to a flag trajectory still requires the stated chart invariance. □

Corollary 4 (End-to-end flag forcing)**.** *Suppose* $\|\widetilde{C}_{k,s}-\widehat{C}_{k,s}\|_{\mathrm{op}} \le a_{k,s}$ *and* $\|\widehat{C}_{k,s}-C_{k,s}\|_{\mathrm{op}} \le \epsilon_{k,s}$*. Then*

$$\|e_s\| \le 2J_\psi \sum_k \omega_k \sqrt{r_k}(a_{k,s}+\epsilon_{k,s}), \tag{215}$$

$$\|\Delta_s\| \le \frac{2J_\psi}{\mu_\phi} \sum_k \omega_k \sqrt{r_k}\|C_{k,s+1}-C_{k,s}\|_{\mathrm{op}}. \tag{216}$$

If $\|C_k(w)-C_k(w')\|_{\mathrm{op}} \le L_{C,k}\|w-w'\|$ *and* $w_{t+1}=w_t-\eta_t d_t$*, with* τ *steps per refresh, then*

$$\|\Delta_s\|^2 \le L_\star^2 \tau \sum_{t=s\tau}^{(s+1)\tau-1} \eta_t^2\|d_t\|^2, \quad L_\star := \frac{2J_\psi}{\mu_\phi}\sum_k \omega_k\sqrt{r_k}L_{C,k}. \tag{217}$$

Proposition 9 (Bounded-gradient constants and raw-EMA weights)**.** *If* $\|G_t\|_F \le G_\star$ *almost surely, then*

$$0 \preceq X_t, C_t \preceq G_\star^2 I, \quad \|Z_t\|_{\mathrm{op}} \le G_\star^2, \quad \mathbb{E}_t Z_t^2 = \mathbb{E}_t X_t^2 - C_t^2 \preceq G_\star^4 I, \tag{218}$$

$$L_Z = G_\star^2, \qquad v_t = G_\star^4 \quad \textit{are admissible in Theorem 2}, \tag{219}$$

$$\Phi(U;\overline{C}_{1:K,t}) = \sum_k \omega_k(1-\rho_k^t)\,\mathrm{tr}(P_k(U)\widehat{C}_{k,t}). \tag{220}$$

Thus replacing raw EMAs by normalized EMAs changes the relative objective weights unless all factors $1-\rho_k^t$ *coincide.*

Proof.

$$-G_\star^2 I \preceq X_t - C_t \preceq G_\star^2 I, \quad X_t^2 \preceq G_\star^2 X_t, \tag{221}$$

$$\mathbb{E}_t(X_t-C_t)^2 = \mathbb{E}_t X_t^2 - C_t^2 \preceq G_\star^2 C_t \preceq G_\star^4 I. \tag{222}$$

□

The preceding chart recursion uses a Euclidean gradient for clarity. Actual manifold ascent uses the inverse pullback metric, and QR introduces a retraction remainder. The next theorem controls both effects so that the same tracking argument applies under explicit local metric and step-size conditions.

Theorem 4 (Actual Riemannian ascent in a non-Euclidean chart)**.** *Under Assumption 1, let* $\mathcal{D}$ *lie in the radius-*r *ball about* $x^\star$*, and define the pullback metric* $\mathsf{G}(x)$ *and natural gradient*

$$\mathsf{G}(x)_{ab} := g_{\psi(x)}(\mathrm{D}\psi(x)e_a, \mathrm{D}\psi(x)e_b), \quad A(x) := \mathsf{G}(x)^{-1}\nabla f(x), \tag{223}$$

$$\|\mathsf{G}^{-1}(x)-I\|_{\mathrm{op}} \le \delta_M, \quad \|\mathrm{D}(\mathsf{G}^{-1})(x)\| \le L_M, \tag{224}$$

$$\mu_A := \mu_\phi - L_\phi(\delta_M + L_M r) > 0, \quad L_A := L_\phi(1+\delta_M+L_M r), \tag{225}$$

$$0 < \xi < 2\mu_A/L_A^2, \quad q_A := \sqrt{1-2\mu_A\xi+L_A^2\xi^2} < 1. \tag{226}$$

Then

$$\langle x-y, A(x)-A(y)\rangle \le -\mu_A\|x-y\|^2, \tag{227}$$

$$\|A(x)-A(y)\| \le L_A\|x-y\|, \tag{228}$$

$$\|x+\xi A(x)-y-\xi A(y)\| \le q_A\|x-y\|. \tag{229}$$

Let $\mathcal{R}_x(v)$ *be the chart representation of the quotient QR retraction, with* $\mathcal{R}_x(0)=x$, $\mathrm{D}\mathcal{R}_x(0)=I$*, and* $\|\mathrm{D}^2\mathcal{R}_x(v)\| \le C_\psi$ *on the relevant segment. For an estimated covariance* $\widehat{C}$*, the actual ascent is*

$$x^+ = \mathcal{R}_x\left(\xi\mathsf{G}(x)^{-1}\nabla f_{\widehat{C}}(x)\right) = x + \xi A(x) + \xi e(x) + r(x), \tag{230}$$

$$\|e(x)\| \le (1+\delta_M)E_C, \qquad \|r(x)\| \le \tfrac{1}{2}C_\psi\xi^2\|A(x)+e(x)\|^2. \tag{231}$$

Theorem 3 therefore applies with q *replaced by* q_A*,* $\kappa = (1+\alpha)q_A^2 < 1$*, and the displayed error and remainder bounds.*

Proof. For $M(x) := \mathsf{G}(x)^{-1}$,

$$\mathrm{D}A(x)[v] = \nabla^2 f(x)v + (M(x) - I)\nabla^2 f(x)v + (\mathrm{D}M(x)[v])\nabla f(x), \tag{232}$$

$$\|\nabla f(x)\| = \|\nabla f(x) - \nabla f(x^\star)\| \le L_\phi r, \tag{233}$$

$$v^\top \mathrm{D}A(x)v \le -\mu_\phi \|v\|^2 + \delta_M L_\phi \|v\|^2 + L_M L_\phi r \|v\|^2 = -\mu_A \|v\|^2, \tag{234}$$

$$\|\mathrm{D}A(x)\|_{\mathrm{op}} \le L_A, \tag{235}$$

$$\|x - y + \xi(A(x) - A(y))\|^2 \le (1 - 2\mu_A \xi + L_A^2 \xi^2)\|x - y\|^2, \tag{236}$$

$$e(x) := M(x)(\nabla f_{\widehat{C}}(x) - \nabla f_C(x)), \tag{237}$$

$$r(x) = \int_0^1 (1 - s)\mathrm{D}^2 \mathcal{R}_x(sv)[v, v]\,\mathrm{d}s, \quad v = \xi(A(x) + e(x)). \tag{238}$$

□

A.4. The implemented adaptive map and its coordinate errors

This subsection fixes the order of operations in the implemented optimizer. The parameter step uses the current flag; a scheduled refresh then aligns the new basis, transports first moments, and resets second moments. Keeping their ages separate is essential because the two statistics retain different amounts of history after a refresh.

Definition 7 (One exact matrix-state transition)**.** *Let $W_t, G_t \in \mathbb{R}^{m\times d}$, $U_t \in \mathrm{St}(d, R)$, and $\chi_t := \mathbf{1}_{t\equiv 0 \pmod\tau}$. Operations $\odot$, division, and square roots below are entrywise. Assume $0 \le \beta_{jk} < 1$, $\epsilon, \gamma, \xi > 0$, $\tau \in \mathbb{N}$. Before the refresh,*

$$Y^-_{k,t} = \rho_k Y_{k,t-1} + (1 - \rho_k) G_t^\top G_t \Omega_k, \tag{239}$$

$$Z_{k,t} = G_t U_{k,t}, \tag{240}$$

$$M^-_{k,t} = \beta_{1k} M^+_{k,t-1} + (1 - \beta_{1k}) Z_{k,t}, \tag{241}$$

$$V^-_{k,t} = \beta_{2k} V^+_{k,t-1} + (1 - \beta_{2k}) Z^{\odot 2}_{k,t}, \tag{242}$$

$$c^-_{M,k,t} = c^+_{M,k,t-1} + 1, \quad c^-_{V,k,t} = c^+_{V,k,t-1} + 1, \tag{243}$$

$$\widehat{M}_{k,t} = M^-_{k,t}/(1 - \beta_{1k}^{c^-_{M,k,t}}), \quad \widehat{V}_{k,t} = V^-_{k,t}/(1 - \beta_{2k}^{c^-_{V,k,t}}), \tag{244}$$

$$H_{k,t} = \widehat{M}_{k,t}/(\sqrt{\widehat{V}_{k,t}} + \epsilon), \tag{245}$$

$$D_t = \sum_k a_k H_{k,t} U_{k,t}^\top + a_\perp \mathcal{R}_\epsilon(G_t(I - U_t U_t^\top)), \tag{246}$$

$$\mathcal{R}_\epsilon(X) := X/(\|X\|_F/\sqrt{md} + \epsilon), \tag{247}$$

$$W_{t+1} = (1 - \eta_t \lambda) W_t - \eta_t D_t. \tag{248}$$

If $\chi_t = 1$,

$$\widetilde{C}_{k,t} := \mathcal{N}_\gamma(Y^-_{k,t}), \quad \overline{U}_t := \mathrm{qf}(U_t + \xi \operatorname{grad}\Phi(U_t; \widetilde{C}_t)), \tag{249}$$

$$O_{k,t} := \mathrm{polar}(\overline{U}_{k,t}^\top U_{k,t}), \quad U_{k,t+1} := \overline{U}_{k,t} O_{k,t}, \tag{250}$$

$$M^+_{k,t} := M^-_{k,t} O_{k,t}^\top, \quad V^+_{k,t} := 0, \quad c^+_{V,k,t} := 0, \quad c^+_{M,k,t} := c^-_{M,k,t}. \tag{251}$$

If $\chi_t = 0$, set $(U_{t+1}, M^+_t, V^+_t, c^+_t) = (U_t, M^-_t, V^-_t, c^-_t)$. In both cases $Y_{k,t} = Y^-_{k,t}$. Thus D_t uses U_t, not U_{t+1}.

Lemma 7 (Procrustes optimization and polar sensitivity)**.** *For orthonormal $A, B \in \mathbb{R}^{d\times q}$ and nonsingular $A^\top B = L\Sigma R^\top$,*

$$\arg\min_{O\in \mathrm{O}(q)} \|AO - B\|_F^2 = LR^\top, \qquad \min_O \|AO - B\|_F^2 = 2q - 2\|A^\top B\|_*, \tag{252}$$

$$\|\operatorname{polar}(X) - \operatorname{polar}(Y)\|_F \le \frac{2\|X - Y\|_F}{\sigma_{\min}(X) + \sigma_{\min}(Y)}. \tag{253}$$

Proof.

$$\|AO - B\|_F^2 = 2q - 2\,\mathrm{tr}(O^\top A^\top B), \tag{254}$$

$$\mathrm{tr}(O^\top L\Sigma R^\top) = \mathrm{tr}(\Sigma R^\top O^\top L) \le \mathrm{tr}\,\Sigma, \tag{255}$$

$$X = PH, \quad Y = P'H', \quad Q = P^\top P', \quad E = X - Y, \tag{256}$$

$$P^\top E - E^\top P' = H(I - Q) + (I - Q)H', \tag{257}$$

$$\langle T, HT + TH'\rangle_F \ge (\lambda_{\min}(H) + \lambda_{\min}(H'))\|T\|_F^2, \tag{258}$$

$$\|I - Q\|_F \le \frac{\|P^\top E - E^\top P'\|_F}{\lambda_{\min}(H) + \lambda_{\min}(H')} \le \frac{2\|E\|_F}{\sigma_{\min}(X) + \sigma_{\min}(Y)}. \tag{259}$$

□

Proposition 10 (Exact least-squares transport and implemented discrepancy)**.** *For old and new block bases $U, V \in \mathrm{St}(d, q)$, $A := MU^\top$, the unique minimizer is*

$$M_{\mathrm{LS}} := \arg\min_N \|NV^\top - A\|_F^2 = MU^\top V, \tag{260}$$

$$\|NV^\top - A\|_F^2 = \|N - MU^\top V\|_F^2 + \|MU^\top(I - VV^\top)\|_F^2. \tag{261}$$

For $V = \overline{U}O$ and the implemented $M^+ = MO^\top$,

$$\|M^+V^\top - MU^\top\|_F^2 = \|M(O^\top - U^\top V)\|_F^2 + \|MU^\top(I - VV^\top)\|_F^2, \tag{262}$$

$$\|MU^\top(I - VV^\top)\|_F \le \|M\|_{\mathrm{op}}\|UU^\top - VV^\top\|_F/\sqrt{2}. \tag{263}$$

In particular,

$$\overline{U} = UR, \quad O = R^\top, \quad V = U, \quad M^+ = MR, \tag{264}$$

$$d_P(U, V) = 0, \qquad \|M^+V^\top - MU^\top\|_F = \|M(R - I)\|_F. \tag{265}$$

Proof.

$$NV^\top - A = (N - AV)V^\top - A(I - VV^\top), \tag{266}$$

$$\langle (N - AV)V^\top, A(I - VV^\top)\rangle_F = 0, \tag{267}$$

$$\|U^\top(I - VV^\top)\|_F^2 = q - \mathrm{tr}(UU^\top VV^\top) = \tfrac{1}{2}\|UU^\top - VV^\top\|_F^2, \tag{268}$$

$$\mathrm{polar}((UR)^\top U) = R^\top. \tag{269}$$

□

The transport identity separates two losses of information. Projection onto a changed subspace discards the component outside it, while a coordinate mismatch can perturb the represented moment even inside the new subspace. Least-squares transport minimizes their total squared error. The implemented rotation has an additional, explicitly retained discrepancy, including in the pure-basis example above.

Lemma 8 (First-moment unrolling and boundedness)**.** *If $M_0 = 0$, $M_t^- = \beta M_{t-1}^+ + (1 - \beta)Z_t$, $M_t^+ = M_t^- T_t$, and $\|T_t\|_{\mathrm{op}} \le 1$, then*

$$M_t^- = (1 - \beta)\sum_{i=1}^{t} \beta^{t-i} Z_i T_i T_{i+1} \cdots T_{t-1}, \tag{270}$$

$$\|Z_i\|_F \le G_\star \Longrightarrow \|M_t^-\|_F \le (1 - \beta^t)G_\star, \quad \|\widehat{M}_t\|_F \le G_\star. \tag{271}$$

For $T_t = O_t^\top$ or $T_t = U_t^\top U_{t+1}$, $\|T_t\|_{\mathrm{op}} \le 1$. If the second moment age is $a_t \ge 1$ and $\|Z_i\|_F \le G_\star$, then

$$\|\sqrt{\widehat{V}_t}\|_F^2 \le G_\star^2, \quad \|H_t\|_F \le G_\star/\epsilon, \quad \|D_t\|_F \le \epsilon^{-1}G_\star \sum_k |a_k| + |a_\perp|\sqrt{md}. \tag{272}$$

Proof.

$$\|M_t^-\|_F \le (1-\beta)G_\star \sum_{i=1}^{t} \beta^{t-i} = (1-\beta^t)G_\star, \tag{273}$$

$$\operatorname{tr} \widehat{V}_t = \frac{1-\beta_2}{1-\beta_2^{a_t}} \sum_{j=0}^{a_t-1} \beta_2^j \|Z_{t-j}\|_F^2 \le G_\star^2, \tag{274}$$

$$\|H_t\|_F \le \|\widehat{M}_t\|_F/\epsilon, \quad \|\mathcal{R}_\epsilon(X)\|_F = \frac{\|X\|_F}{\|X\|_F/\sqrt{md}+\epsilon} \le \sqrt{md}. \tag{275}$$

□

Lemma 9 (Square-root state and normalization)**.** *For $S = \sqrt{V}$, define $\mathcal{S}_\beta(S,Z) := \sqrt{\beta S^{\odot 2} + (1-\beta)Z^{\odot 2}}$. Then*

$$\|\mathcal{S}_\beta(S,Z) - \mathcal{S}_\beta(S',Z')\|_F \le \sqrt{\beta}\|S-S'\|_F + \sqrt{1-\beta}\|Z-Z'\|_F, \tag{276}$$

$$\left\| \frac{M}{S+\epsilon} - \frac{M'}{S'+\epsilon} \right\|_F \le \epsilon^{-1}\|M-M'\|_F + \epsilon^{-2}\|M'\|_{\max}\|S-S'\|_F. \tag{277}$$

For synchronized ages $c \ge 1, a \ge 1$ and $\|M'\|_{\max} \le G_\star$,

$$\|H-H'\|_F \le A_M\|M-M'\|_F + A_S\|S-S'\|_F, \tag{278}$$

$$A_M := \frac{1}{(1-\beta_1)\epsilon}, \quad A_S := \frac{G_\star}{(1-\beta_1)\epsilon^2\sqrt{1-\beta_2}}. \tag{279}$$

Proof.

$$|\mathcal{S}_\beta(s,z) - \mathcal{S}_\beta(s',z')| = \left| \|(\sqrt{\beta}s, \sqrt{1-\beta}z)\|_2 - \|(\sqrt{\beta}s', \sqrt{1-\beta}z')\|_2 \right| \tag{280}$$

$$\le \sqrt{\beta|s-s'|^2 + (1-\beta)|z-z'|^2}, \tag{281}$$

$$\frac{m}{s+\epsilon} - \frac{m'}{s'+\epsilon} = \frac{m-m'}{s+\epsilon} + \frac{m'(s'-s)}{(s+\epsilon)(s'+\epsilon)}. \tag{282}$$

□

Using $\sqrt{V}$ as the state distance avoids differentiating the square-root function at a zero variance entry. The resulting normalization bound remains finite because of ϵ, but its sensitivity can scale as ϵ^{-2}. This identifies a concrete connection between numerical regularization and the later stability constants.

Proposition 11 (Radial residual map)**.** *Writing $x = \operatorname{vec} X$, $N = md$, $r = \|x\|$, the eigenvalues of $\mathrm{D}\mathcal{R}_\epsilon(x)$ are*

$$\lambda_{\mathrm{rad}} = \frac{\epsilon}{(r/\sqrt{N}+\epsilon)^2}, \quad \lambda_{\tan} = \frac{1}{r/\sqrt{N}+\epsilon}, \tag{283}$$

$$0 \preceq \mathrm{D}\mathcal{R}_\epsilon(x) \preceq \epsilon^{-1}I, \quad \|\mathcal{R}_\epsilon(x) - \mathcal{R}_\epsilon(y)\| \le \epsilon^{-1}\|x-y\|, \tag{284}$$

$$\langle x, \mathcal{R}_\epsilon(x)\rangle = \frac{r^2}{r/\sqrt{N}+\epsilon} \ge 0. \tag{285}$$

At $x = 0$, $\mathrm{D}\mathcal{R}_\epsilon(0) = \epsilon^{-1}I$.

Proof.

$$\mathrm{D}\mathcal{R}_\epsilon(x) = \frac{I}{r/\sqrt{N}+\epsilon} - \frac{xx^\top}{\sqrt{N}\, r(r/\sqrt{N}+\epsilon)^2} \quad (r>0). \tag{286}$$

□

Proposition 12 (Coordinatewise Adam is not a quotient map)**.** *For $U = (e_1, e_2) \in \mathrm{St}(3,2)$, $G = (1,0,0)$, zero initial moments, and $R = 2^{-1/2}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$,*

$$UU^\top = (UR)(UR)^\top, \tag{287}$$

$$\frac{GU}{|GU| + \epsilon} U^\top = \frac{e_1^\top}{1+\epsilon}, \qquad \frac{GUR}{|GUR| + \epsilon} R^\top U^\top = \frac{e_1^\top}{2^{-1/2} + \epsilon}. \tag{288}$$

For a signed permutation Q, the transformations

$$U' = UQ, \quad Z' = ZQ, \quad M' = MQ, \quad V' = V|Q|, \tag{289}$$

$$H' = HQ, \quad H'U'^\top = HU^\top \tag{290}$$

are equivariant. The corresponding identities do not hold for general $Q \in \mathrm{O}(q)$.

Proof.

$$(ZQ)^{\odot 2} = Z^{\odot 2}|Q| \quad (Q \text{ signed permutation}), \tag{291}$$

$$(ZR)^{\odot 2} \neq Z^{\odot 2} R \quad \text{in general}, \qquad (|ZR| + \epsilon)^{-1} \odot ZR \neq ((|Z| + \epsilon)^{-1} \odot Z)R. \tag{292}$$

□

Proposition 13 (Linear projector surrogate)**.** *Set $a_{K+1} := a_\perp$ and*

$$B(U) := \sum_{j=1}^{K} a_j \Pi_j(U) + a_\perp \Pi_{K+1}(U) \tag{293}$$

$$= a_\perp I + \sum_{k=1}^{K} (a_k - a_{k+1}) P_k(U), \tag{294}$$

$$a_- I \preceq B(U) \preceq a_+ I, \quad a_- := \min_j a_j > 0, \quad a_+ := \max_j a_j, \tag{295}$$

$$\Gamma^a := \sum_k |a_k - a_{k+1}|, \quad \|B(U) - B(V)\|_{\mathrm{op}} \leq \Gamma^a d_P(U, V). \tag{296}$$

In particular, $a_1 = \cdots = a_K = a_\perp = a$ implies $B(U) = aI$.

Proof.

$$x^\top B(U) x = \sum_{j=1}^{K+1} a_j \|\Pi_j x\|^2, \quad \sum_j \|\Pi_j x\|^2 = \|x\|^2, \tag{297}$$

$$\|B(U) - B(V)\|_{\mathrm{op}} \leq \sum_k |a_k - a_{k+1}| \|P_k(U) - P_k(V)\|_F. \tag{298}$$

□

For the linear surrogate, equal block coefficients remove all dependence on the flag through $\Gamma^a = 0$. Proposition 12 shows why the corresponding Adam ablation has a narrower interpretation: entrywise normalization still depends on the basis. The following decomposition keeps moment lag and changing normalizers explicit when applying surrogate descent results to the implemented update.

Definition 8 (Exact predictable-surrogate decomposition)**.** *For any $\mathcal{F}_{t-1}$-measurable positive operator B_t on $\mathbb{R}^{m \times d}$, define*

$$e_t^{\mathrm{alg}} := D_t - B_t G_t, \qquad e_t := e_t^{\mathrm{alg}} + \lambda W_t, \tag{299}$$

$$W_{t+1} = W_t - \eta_t (B_t G_t + e_t). \tag{300}$$

For a rowwise lagged diagonal surrogate, let

$$h^-_{k,ij,t} := (\sqrt{\widehat{V}_{k,ij,t-1}} + \epsilon)^{-1}, \quad b^-_{i,t} := \sum_k a_k U_{k,t} \operatorname{diag}(h^-_{k,i:,t}) U^\top_{k,t} + a_\perp c^-_t \Pi_{K+1,t}, \tag{301}$$

$$(B_t G)_{i:} := G_{i:} b^-_{i,t}, \qquad c^-_t > 0 \text{ predictable}. \tag{302}$$

Then, exactly,

$$e^{\mathrm{alg}}_t = \sum_k a_k \left[(\widehat{M}_{k,t} - Z_{k,t}) \odot h^-_{k,t} + \widehat{M}_{k,t} \odot (h_{k,t} - h^-_{k,t}) \right] U^\top_{k,t} \tag{303}$$

$$+ a_\perp (c_t - c^-_t) G_t \Pi_{K+1,t}, \quad c_t := (\|G_t \Pi_{K+1,t}\|_F / \sqrt{md} + \epsilon)^{-1}, \tag{304}$$

$$\|e^{\mathrm{alg}}_t\|_F \le \sum_k |a_k| \left[\epsilon^{-1} \|\widehat{M}_{k,t} - Z_{k,t}\|_F + G_\star \|h_{k,t} - h^-_{k,t}\|_{\max} \right] + |a_\perp| G_\star |c_t - c^-_t|. \tag{305}$$

Set $h^-_{k,t} = \epsilon^{-1}\mathbf{1}$ when the lagged second-moment age is zero. The moment-lag term includes the transport discrepancy of Proposition 10.

Lemma 10 (Clipping bias)**.** *Let $\widetilde{g}$ be an unbiased gradient, $g = \widetilde{g} \min(1, c/\|\widetilde{g}\|)$, $h = \mathbb{E}\widetilde{g}$, and $\mathbb{E}\|\widetilde{g}\|^p \le M_p$, $p > 1$. Then*

$$b_c := \mathbb{E}g - h, \quad \|b_c\| \le \mathbb{E}(\|\widetilde{g}\| - c)_+ \le \frac{(p-1)^{p-1}}{p^p} \frac{M_p}{c^{p-1}}. \tag{306}$$

Proof.

$$\|g - \widetilde{g}\| = (\|\widetilde{g}\| - c)_+, \qquad \sup_{x > c} \frac{x - c}{x^p} = \frac{(p-1)^{p-1}}{p^p c^{p-1}}, \quad x_{\max} = pc/(p-1). \tag{307}$$

□

Proposition 14 (Second-order coordinate error for a horizontal QR step)**.** *Let $Z \in \mathcal{H}_U$, $\overline{U} = \mathrm{qf}(U + Z) = U + Z + E$, $e := \|E\|_F < 1$, and, blockwise, $C_j := \overline{U}_j^\top U_j = O_j H_j$. Then*

$$U_j^\top Z_j = 0, \quad \|C_j - I\|_F \le e_j := \|E_j\|_F, \quad H_j = U_j^\top (\overline{U}_j O_j), \tag{308}$$

$$\|O_j - I\|_F \le \frac{2e_j}{2 - e_j}, \quad \|H_j - I\|_F \le e_j, \quad \|O_j^\top - U_j^\top (\overline{U}_j O_j)\|_F \le \left(1 + \frac{2}{2 - e_j}\right) e_j, \tag{309}$$

$$e \le (1 + \|Z\|^2_{\mathrm{op}}) \|Z\|^2_F. \tag{310}$$

Consequently the coordinate part of the transport discrepancy obeys

$$\|M_j (O_j^\top - U_j^\top U_j^+)\|_F \le \|M_j\|_{\mathrm{op}} \left(1 + \frac{2}{2 - e}\right) (1 + \|Z\|^2_{\mathrm{op}}) \|Z\|^2_F. \tag{311}$$

Proof.

$$C_j = I + Z_j^\top U_j + E_j^\top U_j = I + E_j^\top U_j, \tag{312}$$

$$\sigma_{\min}(C_j) \ge 1 - e_j, \qquad C_j^\top O_j = H_j, \tag{313}$$

$$\|O_j - I\|_F \le 2\|C_j - I\|_F / (\sigma_{\min}(C_j) + 1), \tag{314}$$

$$\|C_j - I\|^2_F - \|H_j - I\|^2_F = 2\operatorname{tr}(H_j) - 2\operatorname{tr}(O_j H_j) \ge 0, \tag{315}$$

$$\|O_j^\top - H_j\|_F \le \|O_j - I\|_F + \|I - H_j\|_F. \tag{316}$$

□

A.5. Nonconvex convergence, PL rates, and coupled Lyapunov bounds

We first analyze a predictable positive preconditioner with an explicit discrepancy term. This separates the descent mechanism from the costs of momentum, normalization, clipping, and transport. For STRATA the decomposition is exact, but obtaining a vanishing error requires control of those costs. The later analysis of the actual normalization field provides a second route that retains its dependence on the current gradient directly.

Assumption 2 (Smooth objective and predictable surrogate)**.** *For $F : \mathbb{R}^D \to \mathbb{R}$, $w_t \in \mathcal{F}_{t-1}$, and a self-adjoint B_t, assume*

$$F_{\inf} := \inf_w F(w) > -\infty, \quad \mathbb{E}|F(w_0)| < \infty, \tag{317}$$

$$\|\nabla F(w) - \nabla F(v)\| \le L\|w - v\|, \tag{318}$$

$$h_t := \nabla F(w_t), \quad g_t = h_t + b_t + \zeta_t, \quad \mathbb{E}[\zeta_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[\|\zeta_t\|^2 \mid \mathcal{F}_{t-1}] \le \sigma_t^2, \tag{319}$$

$$B_t \in \mathcal{F}_{t-1}, \quad 0 < mI \preceq B_t \preceq MI, \tag{320}$$

$$w_{t+1} = w_t - \eta_t(B_t g_t + e_t), \quad \varepsilon_t := e_t + B_t b_t. \tag{321}$$

Theorem 5 (Variance-resolved finite-time stationarity)**.** *Under Assumption 2, let $0 < \eta_t \le 1/(4LM)$, $S_T := \sum_{t<T} \eta_t$, and $\Delta_0 := \mathbb{E}F(w_0) - F_{\inf}$. Define*

$$\Sigma_t := \mathbb{E}_t[\zeta_t \zeta_t^\top], \quad \mathsf{v}_t := \operatorname{tr}(B_t^2 \Sigma_t) = \mathbb{E}_t\|B_t\zeta_t\|^2 \le M^2\sigma_t^2. \tag{322}$$

Write $\|x\|_A^2 := x^\top A x$ for $A \succ 0$. Then

$$\mathbb{E}F(w_{t+1}) \le \mathbb{E}F(w_t) - \tfrac{1}{2}\eta_t \mathbb{E}\|h_t\|_{B_t}^2 + L\eta_t^2 \mathbb{E}\mathsf{v}_t + \eta_t \mathbb{E}\|\varepsilon_t\|_{B_t^{-1}}^2 + L\eta_t^2 \mathbb{E}\|\varepsilon_t\|^2, \tag{323}$$

$$\frac{\sum_{t<T} \eta_t \mathbb{E}\|h_t\|^2}{S_T} \le \frac{2\Delta_0}{mS_T} + \frac{2L\sum_{t<T} \eta_t^2 \mathbb{E}\mathsf{v}_t}{mS_T} + \frac{2\sum_{t<T}\left[\eta_t \mathbb{E}\|\varepsilon_t\|_{B_t^{-1}}^2 + L\eta_t^2 \mathbb{E}\|\varepsilon_t\|^2\right]}{mS_T}. \tag{324}$$

For an independent output index J with $\mathbb{P}(J = t) = \eta_t/S_T$, the left side is $\mathbb{E}\|\nabla F(w_J)\|^2$.

Proof. Use the descent lemma (Nesterov, 2004) and the positive-operator inequality $B_t^2 \preceq MB_t$. Weighted Fenchel–Young bounds the defect in the matching dual metric. With $\overline{g}_t := h_t + \zeta_t$ and $Q_t := \langle h_t, B_t h_t\rangle$,

$$F(w_{t+1}) \le F(w_t) - \eta_t\langle h_t, B_t\overline{g}_t + \varepsilon_t\rangle + \tfrac{L}{2}\eta_t^2\|B_t\overline{g}_t + \varepsilon_t\|^2, \tag{325}$$

$$\mathbb{E}_t\langle h_t, B_t\overline{g}_t\rangle = \langle h_t, B_t h_t\rangle, \tag{326}$$

$$|\langle h_t, \varepsilon_t\rangle| \le \tfrac{1}{4}Q_t + \|\varepsilon_t\|_{B_t^{-1}}^2, \tag{327}$$

$$\mathbb{E}_t\|B_t\overline{g}_t + \varepsilon_t\|^2 \le 2\|B_t h_t\|^2 + 2\mathsf{v}_t + 2\mathbb{E}_t\|\varepsilon_t\|^2, \tag{328}$$

$$\|B_t h_t\|^2 \le M\langle h_t, B_t h_t\rangle, \tag{329}$$

$$-\eta_t(1 - LM\eta_t)Q_t + \tfrac{1}{4}\eta_t Q_t \le -\tfrac{1}{2}\eta_t Q_t, \tag{330}$$

$$\tfrac{1}{2}\sum_{t<T} \eta_t \mathbb{E}\|h_t\|_{B_t}^2 \le \mathbb{E}F(w_0) - \mathbb{E}F(w_T) + L\sum_{t<T} \eta_t^2 \mathbb{E}\mathsf{v}_t \tag{331}$$

$$+ \sum_{t<T}\left[\eta_t \mathbb{E}\|\varepsilon_t\|_{B_t^{-1}}^2 + L\eta_t^2 \mathbb{E}\|\varepsilon_t\|^2\right]. \tag{332}$$

Finally $\|h_t\|_{B_t}^2 \ge m\|h_t\|^2$. The defect energy is kept directional; replacing it by $m^{-1}\|\varepsilon_t\|^2$ is an optional later relaxation. □

Corollary 5 (Sharp descent in the zero-discrepancy case)**.** *If $\varepsilon_t = 0$ and $0 < \eta_t < 2/(LM)$, then*

$$\sum_{t<T} \eta_t(1 - LM\eta_t/2)\mathbb{E}\|h_t\|_{B_t}^2 \le \Delta_0 + \frac{L}{2}\sum_{t<T} \eta_t^2 \mathbb{E}\mathsf{v}_t. \tag{333}$$

Proof. Conditional centering gives the exact second-moment identity $\mathbb{E}_t\|B_t(h_t + \zeta_t)\|^2 = \|B_t h_t\|^2 + \mathsf{v}_t$. Apply the descent lemma without a two-term norm relaxation and sum. For $F(w) = L\|w\|^2/2$ and $B_t = MI$, every one-step inequality is an equality: both the noise coefficient $L/2$ and the endpoint $2/(LM)$ are sharp for this class. The positive-operator step also explains the larger step range in Theorem 5. □

The stationarity bound has three sources: the initial objective gap, accumulated stochastic variance, and accumulated update discrepancy. Its left side concerns a step-size-weighted output iterate. Thus a decreasing training loss alone does not verify the conclusion, and persistent discrepancy can leave a nonvanishing stationarity floor even as the stochastic term decreases.

Corollary 6 (Rates and almost-sure stationarity along a subsequence)**.** *If* $\mathbb{E}\mathsf{v}_t \le \overline{v}$, $\eta_t = \eta/\sqrt{T} \le 1/(4LM)$, *then*

$$\mathbb{E}\|\nabla F(w_J)\|^2 \le \frac{2\Delta_0}{m\eta\sqrt{T}} + \frac{2L\eta\overline{v}}{m\sqrt{T}} + \frac{2}{m}\left(\frac{1}{m} + \frac{L\eta}{\sqrt{T}}\right)\frac{1}{T}\sum_{t<T}\mathbb{E}\|\varepsilon_t\|^2. \tag{334}$$

If almost surely

$$\sum_t \eta_t = \infty, \quad \sum_t \eta_t^2 \mathsf{v}_t < \infty, \quad \sum_t (\eta_t/m + L\eta_t^2)\mathbb{E}_t\|\varepsilon_t\|^2 < \infty, \tag{335}$$

then

$$F(w_t) \text{ converges a.s.}, \qquad \sum_t \eta_t \|h_t\|^2 < \infty \text{ a.s.}, \qquad \liminf_{t\to\infty} \|h_t\| = 0 \text{ a.s.} \tag{336}$$

Proof. The finite-time assertion follows by substitution; one may always take $\overline{v} = M^2\sigma^2$ when $\sigma_t^2 \le \sigma^2$, and $\|\varepsilon_t\|^2_{B_t^{-1}} \le m^{-1}\|\varepsilon_t\|^2$. For the almost-sure assertion, apply the nonnegative almost-supermartingale theorem (Robbins & Siegmund, 1971) to $F(w_t) - F_{\inf}$, the summable conditional error, and $\frac{m}{2}\eta_t\|h_t\|^2$; an eventually positive lower bound on $\|h_t\|$ would contradict $\sum_t \eta_t = \infty$. □

Theorem 6 (PL contraction and an explicit floor)**.** *In addition, suppose*

$$\tfrac{1}{2}\|\nabla F(w)\|^2 \ge \mu(F(w) - F^\star), \quad \mu > 0, \tag{337}$$

$$0 < \eta \le \min\{1/(4LM), 1/(\mu m)\}, \quad \mathbb{E}\mathsf{v}_t \le \overline{v}, \quad \mathbb{E}\|\varepsilon_t\|^2 \le E^2. \tag{338}$$

For $a_t := \mathbb{E}[F(w_t) - F^\star]$, $q := 1 - \mu m\eta$,

$$a_{t+1} \le q a_t + L\eta^2\overline{v} + (\eta/m + L\eta^2)E^2, \tag{339}$$

$$a_T \le q^T a_0 + \frac{L\eta\overline{v} + (m^{-1} + L\eta)E^2}{\mu m}(1 - q^T). \tag{340}$$

Proof.

$$-\tfrac{m}{2}\eta\mathbb{E}\|h_t\|^2 \le -\mu m\eta a_t, \qquad \sum_{j=0}^{T-1} q^j = (1 - q^T)/(\mu m\eta). \tag{341}$$

□

Under PL, the same descent estimate becomes a contraction of objective error. The formula distinguishes a transient geometric term from the error floor: reducing the learning rate suppresses the variance contribution, while a fixed discrepancy budget E^2 can persist. The diminishing-step result specifies when decaying discrepancy also permits the objective error to vanish.

Lemma 11 (Diminishing-step PL recurrence)**.** *If* $t_0 > a > \max\{1, p\}$, $p > 0$, *and*

$$a_{t+1} \le \left(1 - \frac{a}{t + t_0}\right) a_t + \frac{b}{(t + t_0)^2} + \frac{c}{(t + t_0)^{1+p}}, \tag{342}$$

then

$$a_T \le \left(\frac{t_0}{T + t_0}\right)^a a_0 + \frac{b\,2^a}{(T + t_0)^a}\sum_{t<T}(t + t_0)^{a-2} + \frac{c\,2^a}{(T + t_0)^a}\sum_{t<T}(t + t_0)^{a-p-1} \tag{343}$$

$$= O((T + t_0)^{-a} + (T + t_0)^{-1} + (T + t_0)^{-p}). \tag{344}$$

Thus $\eta_t = \eta_0/(t + t_0)$, $\mathbb{E}\|\varepsilon_t\|^2 = O((t + t_0)^{-p})$, *and* $a = \mu m\eta_0 > \max\{1, p\}$ *imply the displayed rate.*

Proof.

$$\prod_{j=s+1}^{T-1}\left(1-\frac{a}{j+t_0}\right)\le\exp\left(-a\sum_{j=s+1}^{T-1}\frac{1}{j+t_0}\right)\le\left(\frac{s+1+t_0}{T+t_0}\right)^a, \tag{345}$$

$$(s+1+t_0)^a\le 2^a(s+t_0)^a,\qquad \sum_{t<T}(t+t_0)^u=O((T+t_0)^{u+1})\quad(u>-1). \tag{346}$$

□

Theorem 7 (Coupled parameter–flag Lyapunov inequality)**.** *Suppose nonnegative sequences $E_t, R_t, \Xi_t, \sigma_t^2$ satisfy*

$$\mathbb{E}F(w_{t+1})\le\mathbb{E}F(w_t)-a\eta\mathbb{E}\|h_t\|^2+c\eta E_t+v\eta^2\sigma_t^2+r\eta R_t, \tag{347}$$

$$E_{t+1}\le\kappa E_t+b\eta^2\mathbb{E}\|h_t\|^2+q\eta^2\sigma_t^2+s\Xi_t,\quad 0\le\kappa<1. \tag{348}$$

Let $a, c>0$, $b\ge 0$,

$$\eta^2\le\frac{a(1-\kappa)}{4bc}\quad(bc=0:\ \textit{no constraint}),\quad \Lambda:=\frac{2c\eta}{1-\kappa},\quad \mathcal{L}_t:=\mathbb{E}F(w_t)-F_{\inf}+\Lambda E_t. \tag{349}$$

Then

$$\mathcal{L}_{t+1}-\mathcal{L}_t\le-\tfrac{a}{2}\eta\mathbb{E}\|h_t\|^2-c\eta E_t+(v+\Lambda q)\eta^2\sigma_t^2+r\eta R_t+\Lambda s\Xi_t, \tag{350}$$

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|h_t\|^2\le\frac{2\mathcal{L}_0}{a\eta T}+\frac{2(v+\Lambda q)\eta}{aT}\sum_{t<T}\sigma_t^2+\frac{2r}{aT}\sum_{t<T}R_t+\frac{4cs}{a(1-\kappa)T}\sum_{t<T}\Xi_t. \tag{351}$$

Proof.

$$\mathcal{L}_{t+1}-\mathcal{L}_t\le-(a\eta-\Lambda b\eta^2)\mathbb{E}\|h_t\|^2+(c\eta-\Lambda(1-\kappa))E_t+(v+\Lambda q)\eta^2\sigma_t^2+r\eta R_t+\Lambda s\Xi_t, \tag{352}$$

$$a\eta-\Lambda b\eta^2\ge a\eta/2,\qquad c\eta-\Lambda(1-\kappa)=-c\eta. \tag{353}$$

Summation and $\mathcal{L}_T\ge 0$ give the conclusion. □

The joint potential accounts for feedback between parameter motion and flag tracking: each can introduce error into the other. Its weight Λ is chosen so that contraction of the flag error absorbs its contribution to parameter descent. The following corollary then connects covariance estimation and tracking to the stationarity bound through explicit intermediate constants.

Corollary 7 (Covariance-to-stationarity error chain)**.** *Suppose a linear block-isotropic surrogate satisfies*

$$\mathbb{E}\|\varepsilon_t\|^2\le 2G_\star^2(\Gamma^a)^2J_P^2E_t+2R_t,\quad d_P(\psi(x),\psi(y))\le J_P\|x-y\|, \tag{354}$$

$$E_{t+1}\le\kappa E_t+B_t. \tag{355}$$

For constant $\eta\le 1/(4LM)$ and $\sup_t\mathbb{E}\mathsf{v}_t\le\overline{v}$,

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|h_t\|^2\le\frac{2\Delta_0}{m\eta T}+\frac{2L\eta\overline{v}}{m} \tag{356}$$

$$+\frac{4}{m}(m^{-1}+L\eta)\left[\frac{G_\star^2(\Gamma^a)^2J_P^2}{T(1-\kappa)}\left(E_0+\sum_{t<T-1}B_t\right)+\frac{1}{T}\sum_{t<T}R_t\right]. \tag{357}$$

For the implemented coordinatewise optimizer, R_t additionally contains the normalization, gauge, momentum, clipping, and decay terms of Definitions 7 and 8 and Lemma 10.

Proof.

$$\|(B(U_t)-B(U_t^\star))g_t\|^2\le G_\star^2(\Gamma^a)^2J_P^2\|x_t-x_t^\star\|^2,\quad \sum_{t<T}E_t\le(E_0+\sum_{t<T-1}B_t)/(1-\kappa). \tag{358}$$

□

A.6. A virtual-iterate theorem for matrix momentum

The virtual iterate subtracts the parameter displacement stored in momentum. Its increment resembles a preconditioned gradient step, with a remainder that records changing coefficients and transport. This algebraic reformulation allows the proof to retain the whole momentum history through variation budgets, rather than discarding it as an unspecified per-step error.

Assumption 3 (Predictable momentum system)**.** *Let $\mathcal{B}_t, B_t \in \mathcal{F}_{t-1}$, with $\|\mathcal{B}_t\|_{\mathrm{op}} \le \beta < 1$, $mI \preceq B_t \preceq MI$, and*

$$m_t = \mathcal{B}_t m_{t-1} + (I - \mathcal{B}_t) g_t + u_t, \quad w_{t+1} = w_t - \eta_t B_t m_t, \quad m_{-1} = 0, \tag{359}$$

$$\mathbb{E}_t g_t = \nabla F(w_t), \quad \mathbb{E}_t \|g_t\|^2 \le G_2^2, \quad \|m_t\| \le G_m, \quad \|\nabla F(z_t)\| \le G_z, \tag{360}$$

$$C_t := \eta_t B_t (I - \mathcal{B}_t)^{-1} \mathcal{B}_t, \quad C_{-1} := 0, \quad z_t := w_t - C_{t-1} m_{t-1}. \tag{361}$$

Lemma 12 (Exact virtual-iterate identity)**.** *Under Assumption 3,*

$$z_{t+1} = z_t - \eta_t B_t g_t + r_t, \tag{362}$$

$$r_t := (C_{t-1} - C_t) m_{t-1} - \eta_t B_t (I - \mathcal{B}_t)^{-1} u_t, \tag{363}$$

$$\|r_t\| \le G_m \|C_t - C_{t-1}\|_{\mathrm{op}} + \frac{\eta_t M}{1-\beta} \|u_t\|, \qquad \|w_t - z_t\| \le \|C_{t-1}\|_{\mathrm{op}} G_m. \tag{364}$$

Proof.

$$\eta_t B_t + C_t = \eta_t B_t (I - \mathcal{B}_t)^{-1}, \tag{365}$$

$$(\eta_t B_t + C_t) m_t = C_t m_{t-1} + \eta_t B_t g_t + \eta_t B_t (I - \mathcal{B}_t)^{-1} u_t, \tag{366}$$

$$z_{t+1} - z_t = -(\eta_t B_t + C_t) m_t + C_{t-1} m_{t-1}. \tag{367}$$

□

Theorem 8 (Stationarity with explicit variation and transport budgets)**.** *Under Assumptions 2 (smoothness and lower bound only) and 3,*

$$\frac{m}{2} \sum_{t<T} \eta_t \mathbb{E} \|\nabla F(w_t)\|^2 \le F(w_0) - F_{\inf} + LM^2 G_2^2 \sum_{t<T} \eta_t^2 \tag{368}$$

$$+ \frac{M^2 L^2 G_m^2}{2m} \sum_{t<T} \eta_t \|C_{t-1}\|_{\mathrm{op}}^2 + G_z \sum_{t<T} \mathbb{E}\|r_t\| + L \sum_{t<T} \mathbb{E}\|r_t\|^2. \tag{369}$$

For constant scalar momentum $\mathcal{B}_t = \beta I$ and constant $\eta = \eta_0/\sqrt{T}$, define $\mathcal{V}_T := \sum_{t<T} \|B_t - B_{t-1}\|_{\mathrm{op}}$, with $B_{-1} = 0$. Then

$$\sum_{t<T} \mathbb{E}\|r_t\| \le \frac{\eta \beta G_m}{1-\beta} \mathbb{E}\mathcal{V}_T + \frac{\eta M}{1-\beta} \sum_{t<T} \mathbb{E}\|u_t\|, \tag{370}$$

$$\mathbb{E}\mathcal{V}_T + \sum_{t<T} \mathbb{E}\|u_t\| = O(\sqrt{T}), \quad \sum_{t<T} \mathbb{E}\|r_t\|^2 = O(1) \Longrightarrow \frac{1}{T} \sum_{t<T} \mathbb{E}\|\nabla F(w_t)\|^2 = O(T^{-1/2}). \tag{371}$$

Proof. Let $h_t = \nabla F(w_t)$, $\delta_t = \nabla F(z_t) - h_t$. Then

$$\|\delta_t\| \le L \|C_{t-1}\|_{\mathrm{op}} G_m, \tag{372}$$

$$-\mathbb{E}_t \langle \nabla F(z_t), B_t g_t \rangle = -\langle h_t + \delta_t, B_t h_t \rangle \tag{373}$$

$$\le -\tfrac{m}{2} \|h_t\|^2 + \tfrac{M^2}{2m} \|\delta_t\|^2, \tag{374}$$

$$F(z_{t+1}) \le F(z_t) - \eta_t \langle \nabla F(z_t), B_t g_t \rangle + \langle \nabla F(z_t), r_t \rangle + \tfrac{L}{2} \| - \eta_t B_t g_t + r_t\|^2, \tag{375}$$

$$\mathbb{E}_t F(z_{t+1}) \le F(z_t) - \tfrac{m}{2} \eta_t \|h_t\|^2 + \frac{\eta_t M^2 L^2 G_m^2}{2m} \|C_{t-1}\|_{\mathrm{op}}^2$$

$$+ G_z \mathbb{E}_t \|r_t\| + LM^2 \eta_t^2 G_2^2 + L \mathbb{E}_t \|r_t\|^2. \tag{376}$$

Summation gives the first assertion. Also $\|C_t\|_{\mathrm{op}} \le \eta M \beta/(1-\beta)$ and $\sum_t \eta \|C_{t-1}\|_{\mathrm{op}}^2 = O(T\eta^3)$. □

A.7. Full-state replace-one stability and generalization

Replacing one training example perturbs more than the current parameter gradient. It also changes covariance sketches, subsequent flags, and both moment states, which can feed back into later parameter updates. We therefore couple two runs with shared randomness and track five distances. The nonnegative transition matrix records how each component can amplify the others over the remaining training horizon.

Definition 9 (Neighboring samples and synchronized coupling)**.**

$$S = (z_1, \dots, z_n) \sim \mathcal{D}^n, \quad z_i' \sim \mathcal{D}, \quad S^{(i)} = (z_1, \dots, z_{i-1}, z_i', z_{i+1}, \dots, z_n), \tag{377}$$

$$F_S(w) := \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i), \quad F_{\mathcal{D}}(w) := \mathbb{E}_{z\sim\mathcal{D}}\ell(w, z), \tag{378}$$

$$A_T(S; \vartheta) = w_T, \quad A_T(S^{(i)}; \vartheta) = w_T', \quad \vartheta = (U_0, \Omega_{1:K}, \textit{batch indices, augmentations, masks}), \tag{379}$$

$$I_t := \mathbf{1}_{i\in\mathcal{I}_t}, \quad |\mathcal{I}_t| = b_t, \quad \mathbb{P}(I_t = 1) = b_t/n. \tag{380}$$

The initial state, probes, schedule, counters, and random seed are shared. The marginal sampling condition permits fresh uniform subsets and uniform random permutations; conditional independence from past states is not required below.

Assumption 4 (Uniform regularity for a coupled pair)**.** *On the trajectories under consideration, assume deterministic envelopes*

$$|\ell(w, z) - \ell(v, z)| \le L_0\|w - v\|, \quad \|\nabla\ell(w, z)\| \le G_\star, \quad \|\nabla\ell(w, z) - \nabla\ell(v, z)\| \le L_g\|w - v\|, \tag{381}$$

$$\Omega_k^\top\Omega_k = I, \quad \mathrm{sym}(\Omega_k^\top Y_k) + \gamma I \succeq \gamma I, \quad \|Y_k\|_{\mathrm{op}} \le Y_\star, \tag{382}$$

$$\|\widetilde{C}_k\|_{\mathrm{op}} \le C_\star, \quad \inf_{s\in[0,1]} \sigma_{\min}(X + s(X' - X)) \ge a_Q > 0, \tag{383}$$

$$X := U + \xi\,\mathrm{grad}\,\Phi(U; \widetilde{C}), \quad \min\{\sigma_{\min}(\overline{U}_k^\top U_k), \sigma_{\min}(\overline{U}_k'^\top U_k')\} \ge a_O > 0, \tag{384}$$

$$0 \le \eta_t\lambda \le 1, \quad a_k \ge 0, \quad a_\perp \ge 0. \tag{385}$$

The gradient passed to the optimizer may additionally undergo the same nonexpansive norm clipping in both runs.

Definition 10 (Five distances and deterministic constants)**.**

$$u_t := \|w_t - w_t'\|, \quad v_t := \|U_t - U_t'\|_F, \quad y_t := \sum_k \omega_k\|Y_{k,t-1} - Y_{k,t-1}'\|_F, \tag{386}$$

$$m_t := \sum_k \|M_{k,t-1}^+ - M_{k,t-1}^{+\prime}\|_F, \quad s_t := \sum_k \|\sqrt{V_{k,t-1}^+} - \sqrt{V_{k,t-1}^{+\prime}}\|_F, \tag{387}$$

$$z_t := (u_t, v_t, y_t, m_t, s_t)^\top \in \mathbb{R}_+^5, \tag{388}$$

$$\bar\rho := \max_k \rho_k, \quad \bar\beta_j := \max_k \beta_{jk}, \quad \underline{\beta}_j := \min_k \beta_{jk}, \tag{389}$$

$$c_Y := 2G_\star\sum_k \omega_k(1 - \rho_k), \quad c_M := \sqrt{K}(1 - \underline{\beta}_1), \quad c_S := \sqrt{K}\sqrt{1 - \underline{\beta}_2}, \tag{390}$$

$$H_\star := 2\sqrt{R}C_\star\sum_k \omega_k, \quad L_U := 2C_\star\sum_k \omega_k + 4H_\star, \quad L_Y := 2\sqrt{R}\max_k L_{N,k}, \tag{391}$$

$$L_Q := \sqrt{2}/a_Q, \quad L_O := 1/a_O, \quad c_O := \sqrt{K}L_O, \tag{392}$$

$$A_M := ((1 - \bar\beta_1)\epsilon)^{-1}, \quad A_S := \frac{G_\star}{(1 - \bar\beta_1)\epsilon^2\sqrt{1 - \bar\beta_2}}, \quad H_0 := G_\star/\epsilon, \quad a_\star := \max_k a_k. \tag{393}$$

For multiple matrices and dense fallback parameters, concatenate their distances; the same construction gives a larger nonnegative matrix, or a five-state bound using valid aggregate envelopes.

Lemma 13 (Gradient, covariance, and horizontal-gradient perturbations)**.** *Under Assumption 4,*

$$\delta G_t := \|G_t - G'_t\|_F \le L_g u_t + (2G_\star/b_t) I_t, \tag{394}$$

$$\|G_t^\top G_t - G_t'^\top G'_t\|_F \le 2G_\star \delta G_t, \tag{395}$$

$$y_t^- := \sum_k \omega_k \|Y_{k,t}^- - Y_{k,t}^{-\prime}\|_F \le \bar{\rho} y_t + c_Y \delta G_t, \tag{396}$$

$$\|\operatorname{grad}\Phi(U;\widetilde{C}) - \operatorname{grad}\Phi(U';\widetilde{C}')\|_F \le L_U \|U - U'\|_F + L_Y \sum_k \omega_k \|Y_k - Y'_k\|_F, \tag{397}$$

$$\sum_k \|Z_k - Z'_k\|_F \le \sqrt{K}(\delta G_t + G_\star v_t). \tag{398}$$

Proof.

$$G_t - G'_t = \frac{1}{b_t} \sum_{j \in \mathcal{I}_t} [\nabla \ell(w_t, z_j) - \nabla \ell(w'_t, z_j)] + \frac{I_t}{b_t} [\nabla \ell(w'_t, z_i) - \nabla \ell(w'_t, z'_i)], \tag{399}$$

$$G^\top G - G'^\top G' = (G - G')^\top G + G'^\top (G - G'), \tag{400}$$

$$\|\mathsf{H}_U(H) - \mathsf{H}_{U'}(H)\|_F \le 4\|H\|_F \|U - U'\|_F, \quad \|\mathsf{H}_U\|_{F\to F} = 1, \tag{401}$$

$$\|H(U;C) - H(U';C')\|_F \le 2C_\star \sum_k \omega_k \|U - U'\|_F + 2\sum_k \omega_k \sqrt{r_k} \|C_k - C'_k\|_{\mathrm{op}}, \tag{402}$$

$$\sum_k \|(G - G')U_k\|_F \le \sqrt{K}\|(G - G')U\|_F \le \sqrt{K}\delta G_t, \tag{403}$$

$$\sum_k \|G'(U_k - U'_k)\|_F \le \sqrt{K} G_\star \|U - U'\|_F. \tag{404}$$

□

Definition 11 (An explicit nonnegative transition matrix)**.** *Let e_w, e_U, e_Y, e_M, e_S be the standard row vectors in $\mathbb{R}^5$. The following rows and scalars define $\mathcal{M}_t \in \mathbb{R}_+^{5\times 5}$ and $d_t \in \mathbb{R}_+^5$ without implicit constants:*

$$r_g := L_g e_w, \qquad d_g := 2G_\star/b_t, \tag{405}$$

$$r_y := \bar{\rho} e_Y + c_Y r_g, \qquad d_y := c_Y d_g, \tag{406}$$

$$r_h := L_Q[(1 + \xi L_U)e_U + \xi L_Y r_y], \qquad d_h := L_Q \xi L_Y d_y, \tag{407}$$

$$r_o := \chi_t c_O (r_h + e_U), \qquad d_o := \chi_t c_O d_h, \tag{408}$$

$$r_U := (1 - \chi_t)e_U + \chi_t r_h + r_o, \qquad d_U := \chi_t d_h + d_o, \tag{409}$$

$$r_m^- := \bar{\beta}_1 e_M + c_M(r_g + G_\star e_U), \qquad d_m^- := c_M d_g, \tag{410}$$

$$r_s^- := \sqrt{\bar{\beta}_2} e_S + c_S(r_g + G_\star e_U), \qquad d_s^- := c_S d_g, \tag{411}$$

$$r_M := r_m^- + G_\star r_o, \qquad d_M := d_m^- + G_\star d_o, \tag{412}$$

$$r_S := (1 - \chi_t) r_s^-, \qquad d_S := (1 - \chi_t) d_s^-, \tag{413}$$

$$\tag{414}$$

$$r_D := a_\star(A_M r_m^- + A_S r_s^- + H_0\sqrt{K} e_U) + (a_\perp/\epsilon)(r_g + 2G_\star e_U), \tag{415}$$

$$d_D := a_\star(A_M d_m^- + A_S d_s^-) + (a_\perp/\epsilon) d_g, \tag{416}$$

$$r_w := (1 - \eta_t \lambda) e_w + \eta_t r_D, \tag{417}$$

$$d_w := \eta_t d_D. \tag{418}$$

$$\mathcal{M}_t := \begin{pmatrix} r_w \\ r_U \\ r_y \\ r_M \\ r_S \end{pmatrix}, \quad d_t := (d_w, d_U, d_y, d_M, d_S)^\top. \tag{419}$$

Theorem 9 (Full-state stability with all feedback paths)**.** *Under Assumption 4, Definition 7 satisfies, componentwise,*

$$z_{t+1} \preceq \mathcal{M}_t z_t + d_t I_t, \qquad z_0 = 0, \tag{420}$$

$$\mathbb{E} z_T \preceq \sum_{t=0}^{T-1} (\mathcal{M}_{T-1} \cdots \mathcal{M}_{t+1}) d_t \frac{b_t}{n}, \tag{421}$$

$$\epsilon_T^{\mathrm{stab}} := \sup_{S \sim S', z} \mathbb{E}_\vartheta |\ell(A_T(S;\vartheta), z) - \ell(A_T(S';\vartheta), z)| \tag{422}$$

$$\leq \frac{L_0}{n} e_w \sum_{t=0}^{T-1} b_t (\mathcal{M}_{T-1} \cdots \mathcal{M}_{t+1}) d_t. \tag{423}$$

Empty products equal I_5. The supremum is over pairs satisfying the stated uniform envelopes; a uniform result over all neighboring samples requires those envelopes uniformly over the sample domain.

Proof. At a refresh, Lemmas 3 and 7 imply

$$\|\overline{U} - \overline{U}'\|_F \leq r_h z_t + d_h I_t, \tag{424}$$

$$\sum_k \|O_k - O_k'\|_F \leq \sqrt{K} L_O (\|\overline{U} - \overline{U}'\|_F + \|U - U'\|_F) \leq r_o z_t + d_o I_t, \tag{425}$$

$$\|\overline{U} O - \overline{U}' O'\|_F \leq \|\overline{U} - \overline{U}'\|_F + \sum_k \|O_k - O_k'\|_F, \tag{426}$$

$$\sum_k \|M_k^- O_k^\top - M_k'^- O_k'^\top\|_F \leq \sum_k \|M_k^- - M_k'^-\|_F + G_\star \sum_k \|O_k - O_k'\|_F. \tag{427}$$

The root-state inequality and the synchronized reset give

$$\sum_k \|M_k^- - M_k'^-\|_F \leq r_m^- z_t + d_m^- I_t, \tag{428}$$

$$\sum_k \|\sqrt{V_k^-} - \sqrt{V_k'^-}\|_F \leq r_s^- z_t + d_s^- I_t, \tag{429}$$

$$\sum_k \|\sqrt{V_k^+} - \sqrt{V_k'^+}\|_F = (1 - \chi_t) \sum_k \|\sqrt{V_k^-} - \sqrt{V_k'^-}\|_F. \tag{430}$$

For the parameter direction,

$$\|HU^\top - H'U'^\top\|_F \leq \|H - H'\|_F + \|H'\|_{\mathrm{op}} \|U - U'\|_F, \tag{431}$$

$$\|G(I - UU^\top) - G'(I - U'U'^\top)\|_F \leq \delta G_t + 2 G_\star v_t, \tag{432}$$

$$\|D_t - D_t'\|_F \leq r_D z_t + d_D I_t, \tag{433}$$

$$u_{t+1} \leq (1 - \eta_t \lambda) u_t + \eta_t \|D_t - D_t'\|_F. \tag{434}$$

Every entry of $\mathcal{M}_t, d_t$ is deterministic and nonnegative. Consequently,

$$\mathbb{E} z_{t+1} \preceq \mathcal{M}_t \mathbb{E} z_t + d_t \mathbb{E} I_t = \mathcal{M}_t \mathbb{E} z_t + d_t b_t / n, \tag{435}$$

$$\mathbb{E}|\ell(w_T, z) - \ell(w_T', z)| \leq L_0 e_w \mathbb{E} z_T. \tag{436}$$

□

The matrix product retains every feedback path after the changed example is sampled. A scalar bound follows if a weighted combination of state distances has controlled growth. Such a weight is an additional property of the coupled dynamics; bounding each one-step map by smoothness alone need not produce contraction.

Corollary 8 (Weighted small-gain and contraction bounds)**.** *Suppose a fixed row $a \in \mathbb{R}^5_{>0}$, $a_w \geq 1$, satisfies*

$$a\mathcal{M}_t \preceq (1+\nu\eta_t)a, \qquad b_t a d_t \leq c\eta_t. \tag{437}$$

Then

$$\epsilon_T^{\mathrm{stab}} \leq \frac{L_0 c}{n} \sum_{t<T} \eta_t \prod_{s=t+1}^{T-1} (1+\nu\eta_s) \tag{438}$$

$$\leq \begin{cases} \dfrac{L_0 c}{n\nu}(e^{\nu S_T} - 1), & \nu > 0, \\ \dfrac{L_0 c}{n} S_T, & \nu = 0. \end{cases} \tag{439}$$

If instead $a\mathcal{M}_t \preceq (1-\nu\eta_t)a$, $0 < \nu\eta_t \leq 1$, then

$$\epsilon_T^{\mathrm{stab}} \leq \frac{L_0 c}{n\nu}\left[1 - \prod_{t<T}(1-\nu\eta_t)\right] \leq \frac{L_0 c}{n\nu}. \tag{440}$$

Proof.

$$a\mathbb{E}z_{t+1} \leq (1 \pm \nu\eta_t)a\mathbb{E}z_t + c\eta_t/n, \qquad \sum_{t<T} \nu\eta_t \prod_{s>t}(1 \pm \nu\eta_s) = \begin{cases} \prod_t(1+\nu\eta_t) - 1, & +, \\ 1 - \prod_t(1-\nu\eta_t), & -. \end{cases} \tag{441}$$

□

Proposition 15 (Regularity failure probability)**.** *If $0 \leq \ell \leq B_\ell$, and the coupled regularity conditions fail before T with probability at most p_T, then the same matrix-product upper bound, denoted $\epsilon_{T,\mathrm{reg}}$, gives*

$$\epsilon_T^{\mathrm{stab}} \leq \epsilon_{T,\mathrm{reg}} + B_\ell p_T. \tag{442}$$

Proof. Let τ_{exit} be the first failure time. Nonnegative killed distances $\widetilde{z}_t := z_t \mathbf{1}_{\tau_{\mathrm{exit}} > t}$ satisfy

$$\widetilde{z}_{t+1} \preceq \mathcal{M}_t \widetilde{z}_t + d_t I_t, \tag{443}$$

$$\mathbb{E}|\ell(w_T, z) - \ell(w'_T, z)| \leq L_0 e_w \mathbb{E}\widetilde{z}_T + B_\ell \mathbb{P}(\tau_{\mathrm{exit}} \leq T). \tag{444}$$

□

Theorem 10 (Expected generalization and concentration)**.** *For iid S, independent algorithm randomness, and uniform replace-one stability $\epsilon_T^{\mathrm{stab}} = \epsilon_T$,*

$$\left|\mathbb{E}_{S,\vartheta}\left[F_{\mathcal{D}}(A_T(S;\vartheta)) - F_S(A_T(S;\vartheta))\right]\right| \leq \epsilon_T. \tag{445}$$

If also $0 \leq \ell \leq B_\ell$, define $f(S) := \mathbb{E}_\vartheta[F_{\mathcal{D}}(A_T(S)) - F_S(A_T(S))]$. Then, with probability at least $1-\delta$ over S,

$$|f(S)| \leq \epsilon_T + (2n\epsilon_T + B_\ell)\sqrt{\frac{\log(2/\delta)}{2n}}. \tag{446}$$

For a bound on a single randomized run, the latter assertion requires the same uniform stability constant for every fixed seed ϑ; it then holds over (S, ϑ).

Proof. Exchangeability and independence imply

$$\mathbb{E}_{S,\vartheta} F_{\mathcal{D}}(A_T(S)) = \frac{1}{n}\sum_i \mathbb{E}_{S,z_i',\vartheta}\ell(A_T(S), z_i'), \tag{447}$$

$$\mathbb{E}_{S,z_i',\vartheta}\ell(A_T(S^{(i)}), z_i') = \mathbb{E}_{S,\vartheta}\ell(A_T(S), z_i), \tag{448}$$

$$\mathbb{E}(F_{\mathcal{D}} - F_S) = \frac{1}{n}\sum_i \mathbb{E}[\ell(A_T(S), z_i') - \ell(A_T(S^{(i)}), z_i')]. \tag{449}$$

For neighboring S, S',

$$|f(S) - f(S')| \le \mathbb{E}_\vartheta |F_{\mathcal{D}}(A_T(S)) - F_{\mathcal{D}}(A_T(S'))| \tag{450}$$

$$+ \frac{1}{n}\sum_{j\neq i} \mathbb{E}_\vartheta |\ell(A_T(S), z_j) - \ell(A_T(S'), z_j)| + B_\ell/n \tag{451}$$

$$\le (2 - 1/n)\epsilon_T + B_\ell/n \le 2\epsilon_T + B_\ell/n. \tag{452}$$

The bounded-differences inequality (McDiarmid, 1989) yields

$$\mathbb{P}(|f(S) - \mathbb{E}f(S)| \ge r) \le 2\exp\left(-\frac{2r^2}{n(2\epsilon_T + B_\ell/n)^2}\right), \quad |\mathbb{E}f(S)| \le \epsilon_T. \tag{453}$$

□

Corollary 9 (Bousquet–Klochkov–Zhivotovskiy concentration)**.** *Under the bounded-loss and uniform expected-stability hypotheses above, there is a numerical constant C such that, for $n \ge 2$, with probability at least $1 - \delta$ over S,*

$$|f(S)| \le C\left[\epsilon_T \log(2n)\log(2/\delta) + B_\ell\sqrt{\frac{\log(2/\delta)}{n}}\right]. \tag{454}$$

If the stability bound holds for every fixed random seed with the same constant, this bound also applies to a single run over (S, ϑ).

Proof. Define the deterministic, sample-dependent loss $\overline{\ell}_S(z) := \mathbb{E}_\vartheta \ell(A_T(S;\vartheta), z)$. It takes values in $[0, B_\ell]$ and satisfies

$$|\overline{\ell}_S(z) - \overline{\ell}_{S'}(z)| \le \mathbb{E}_\vartheta |\ell(A_T(S;\vartheta), z) - \ell(A_T(S';\vartheta), z)| \le \epsilon_T. \tag{455}$$

Apply Corollary 8 of Bousquet et al. (2020) to this uniformly stable loss-valued algorithm. Its generalization gap is exactly $f(S)$. For the single-run assertion condition on the seed and integrate the failure probability. This improves the stability dependence from $\epsilon_T\sqrt{n}$ in the elementary bounded-differences bound to $\epsilon_T \log n$, up to confidence factors and a universal constant; we make no claim that the numerical constant C is optimal. □

A.8. Effective noise dimension, excess risk, and horizon dependence

The next results separate directional noise reduction from generalization and from state storage. The trace identity measures how a predictable preconditioner transforms gradient noise; the excess-risk bound also needs optimization and stability estimates. Compressing optimizer state leaves the parameter domain unchanged, as Proposition 18 makes explicit.

Proposition 16 (Exact directional-noise identities)**.** *For predictable symmetric B_t and conditional noise covariance $\Sigma_t := \mathbb{E}_t[\zeta_t\zeta_t^\top]$,*

$$\mathbb{E}_t\|B_t\zeta_t\|^2 = \operatorname{tr}(B_t^2\Sigma_t), \quad \lambda_{\max}(\Sigma_t)\operatorname{tr}(B_t^2) \ge \operatorname{tr}(B_t^2\Sigma_t), \tag{456}$$

$$B = \sum_{j=1}^{K+1} a_j\Pi_j \quad \Longrightarrow \quad \operatorname{tr}(B^2) = \sum_{j=1}^{K+1} q_j a_j^2. \tag{457}$$

For m rows with independent isotropic noise variance σ^2 per coordinate,

$$\mathbb{E}\|\zeta B\|_F^2 = m\sigma^2\left(\sum_{j=1}^{K} q_j a_j^2 + (d - R)a_\perp^2\right). \tag{458}$$

Proof.

$$\mathbb{E}_t\|B_t\zeta_t\|^2 = \mathbb{E}_t\,\mathrm{tr}(B_t\zeta_t\zeta_t^\top B_t) = \mathrm{tr}(B_t^2\Sigma_t), \tag{459}$$

$$B^2 = \sum_j a_j^2\Pi_j, \quad \mathrm{tr}\,\Pi_j = q_j. \tag{460}$$

□

Corollary 10 (Population excess risk under empirical PL)**.** *Suppose every F_S is μ-PL, a random output A_J satisfies $\mathbb{E}\|\nabla F_S(A_J)\|^2 \le R_T$, and its uniform expected stability is ϵ_T. Then for every fixed population minimizer $w^\star_{\mathcal{D}}$,*

$$\mathbb{E}[F_{\mathcal{D}}(A_J) - F_{\mathcal{D}}(w^\star_{\mathcal{D}})] \le R_T/(2\mu) + \epsilon_T. \tag{461}$$

If, under the additional hypotheses of the stationarity and small-gain results,

$$R_T/(2\mu) \le A/\sqrt{T} + R_{\mathrm{disc}}, \quad \epsilon_T \le B\sqrt{T}/n, \tag{462}$$

then

$$\mathbb{E}[F_{\mathcal{D}}(A_J) - F_{\mathcal{D}}(w^\star_{\mathcal{D}})] \le A/\sqrt{T} + B\sqrt{T}/n + R_{\mathrm{disc}}, \tag{463}$$

$$T^\star_{\mathrm{cont}} = An/B, \quad \inf_{T>0}\{A/\sqrt{T} + B\sqrt{T}/n\} = 2\sqrt{AB/n}. \tag{464}$$

The two assumed rates must hold with constants uniform in the selected horizon.

Proof.

$$F_S(A_J) - \inf_w F_S(w) \le \|\nabla F_S(A_J)\|^2/(2\mu), \tag{465}$$

$$\mathbb{E}\inf_w F_S(w) \le \mathbb{E}F_S(w^\star_{\mathcal{D}}) = F_{\mathcal{D}}(w^\star_{\mathcal{D}}), \tag{466}$$

$$\mathbb{E}[F_{\mathcal{D}}(A_J) - F_{\mathcal{D}}(w^\star_{\mathcal{D}})] \le \mathbb{E}[F_{\mathcal{D}}(A_J) - F_S(A_J)] + \mathbb{E}[F_S(A_J) - \inf_w F_S(w)], \tag{467}$$

$$\frac{\mathrm{d}}{\mathrm{d}T}(AT^{-1/2} + BT^{1/2}/n) = -\tfrac{A}{2}T^{-3/2} + \tfrac{B}{2n}T^{-1/2}. \tag{468}$$

□

The horizon balance expresses a conditional early-stopping tradeoff: more updates reduce optimization error while the stability term can increase. Its minimizing horizon is meaningful only when both assumed rates hold uniformly over the horizons being compared. The movement bound below remains available with fewer state-regularity requirements, though it can be much looser.

Proposition 17 (A movement-only bound)**.** *For a deterministic common initialization w_0, L_0-Lipschitz losses, and $w_{t+1} = w_t - \eta_t d_t$,*

$$|\mathbb{E}[F_{\mathcal{D}}(w_T) - F_S(w_T)]| \le 2L_0\sum_{t<T}\eta_t\mathbb{E}\|d_t\|. \tag{469}$$

Proof.

$$\mathbb{E}[F_{\mathcal{D}}(w_0) - F_S(w_0)] = 0, \quad |(F_{\mathcal{D}} - F_S)(w_T) - (F_{\mathcal{D}} - F_S)(w_0)| \le 2L_0\|w_T - w_0\|, \tag{470}$$

$$\|w_T - w_0\| \le \sum_{t<T}\eta_t\|d_t\|. \tag{471}$$

□

Proposition 18 (State dimension versus parameter domain)**.** *For $m \geq d$, the stored floating-point matrix-state counts are*

$$N_{\mathrm{STRATA}} = 2mR + dR + 2d\sum_{k=1}^{K} \min(d, r_k + p), \qquad N_{\mathrm{Adam}} = 2md, \tag{472}$$

$$\dim \mathfrak{F} = dR - \tfrac{1}{2}(R^2 + \sum_k q_k^2), \qquad \mathcal{W}_{\mathrm{STRATA}} = \mathcal{W}_{\mathrm{Adam}} = \mathbb{R}^{m\times d}, \tag{473}$$

$$U \in \mathrm{St}(d, R) \quad \not\Longrightarrow \quad \mathrm{rank}(W) \leq R. \tag{474}$$

Fixed probes are included in N_{STRATA}; temporary workspaces, parameters, gradients, scalar counters, and fallback states are excluded.

Proof.

$$N_M + N_V = 2m\sum_k q_k = 2mR, \quad N_U = dR, \quad N_Y = N_\Omega = d\sum_k \min(d, r_k + p), \tag{475}$$

$$W_0 = \begin{pmatrix} I_d \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathrm{rank}(W_0) = d > R, \qquad U_0^\top U_0 = I_R. \tag{476}$$

No projection of W onto a rank-constrained set occurs in Definition 7. □

Corollary 11 (Explicit local regularity radii)**.** *At a refresh, let $Z = \xi\,\mathrm{grad}\,\Phi(U; \widetilde{C})$ and $Z' = \xi\,\mathrm{grad}\,\Phi(U'; \widetilde{C}')$. If*

$$\max(\|Z\|_F, \|Z'\|_F) \leq z_\star, \quad e_\star := (1 + z_\star^2)z_\star^2 < 1, \quad \|(U + Z) - (U' + Z')\|_{\mathrm{op}} \leq r_X < 1, \tag{477}$$

then the QR and Procrustes regularity assumptions hold with

$$a_Q = 1 - r_X, \qquad a_O = 1 - e_\star. \tag{478}$$

If $\|\mathrm{D}\psi\| \leq J_\psi$ on a convex chart domain, then also

$$d_P(\psi(x), \psi(y)) \leq \sqrt{2K} J_\psi \|x - y\|. \tag{479}$$

Proof.

$$\sigma_{\min}(U + Z)^2 = \lambda_{\min}(I + Z^\top Z) \geq 1, \tag{480}$$

$$\sigma_{\min}(X + s(X' - X)) \geq \sigma_{\min}(X) - s\|X' - X\|_{\mathrm{op}} \geq 1 - r_X, \tag{481}$$

$$\sigma_{\min}(\overline{U}_j^\top U_j) \geq 1 - \|\overline{U}_j - U_j - Z_j\|_F \geq 1 - e_\star, \tag{482}$$

$$\left\| \frac{\mathrm{d}}{\mathrm{d}s}(P_1, \ldots, P_K)(\psi(y + s(x - y))) \right\|_F \leq \sqrt{2K} J_\psi \|x - y\|. \tag{483}$$

□

A.9. Invariant-block quadratic rates and an exact small-gain criterion

An invariant-block quadratic isolates a setting where differentiated scalar weights can improve conditioning exactly. The result describes the benefit available when the flag aligns with Hessian invariant subspaces. Learning a flag from gradient covariance does not itself ensure this alignment; the perturbation bound quantifies how flag error can erode the ideal rate.

Assumption 5 (Invariant Hessian blocks)**.**

$$F(w) = \tfrac{1}{2}(w - w^\star)^\top H(w - w^\star), \quad H \succ 0, \tag{484}$$

$$H\Pi_j = \Pi_j H, \quad 0 < \ell_j \Pi_j \preceq \Pi_j H \Pi_j \preceq u_j \Pi_j, \quad B = \sum_{j=1}^{K+1} a_j \Pi_j, \quad a_j > 0. \tag{485}$$

The projectors in this assumption act on the parameter vector; a matrix right-action uses $I_m \otimes \Pi_j$ after vectorization.

Theorem 11 (Block conditioning and optimal scalar weights)**.** *Under Assumption 5, let*

$$\ell_B := \min_j a_j \ell_j, \quad u_B := \max_j a_j u_j, \quad \kappa_B := u_B/\ell_B, \quad \eta_\star := 2/(u_B + \ell_B), \tag{486}$$

$$w_{t+1} = w_t - \eta B \nabla F(w_t). \tag{487}$$

Then

$$\|w_t - w^\star\|_{B^{-1}} \le \left[\max_{\lambda \in [\ell_B, u_B]} |1 - \eta\lambda|\right]^t \|w_0 - w^\star\|_{B^{-1}}, \tag{488}$$

$$F(w_t) \le \left(\frac{\kappa_B - 1}{\kappa_B + 1}\right)^{2t} F(w_0) \quad (\eta = \eta_\star), \tag{489}$$

$$\inf_{a_1,\dots,a_{K+1}>0} \frac{\max_j a_j u_j}{\min_j a_j \ell_j} = \max_j \frac{u_j}{\ell_j}, \qquad a_j = 1/\ell_j \text{ attains the infimum}. \tag{490}$$

If $H = \sum_j h_j \Pi_j$ and $a_j = h_j^{-1}$, then $w_1 = w^\star$ for $\eta = 1$.

Proof.

$$x_t := B^{-1/2}(w_t - w^\star), \quad \widetilde{H} := B^{1/2} H B^{1/2} = BH, \tag{491}$$

$$x_{t+1} = (I - \eta\widetilde{H})x_t, \quad \operatorname{spec}(\widetilde{H}) \subset [\ell_B, u_B], \tag{492}$$

$$F(w_t) = \tfrac{1}{2} x_t^\top \widetilde{H} x_t, \quad \widetilde{H}^{1/2} x_t = (I - \eta\widetilde{H})^t \widetilde{H}^{1/2} x_0, \tag{493}$$

$$\min_{\eta>0} \max\{|1 - \eta\ell_B|, |1 - \eta u_B|\} = \frac{u_B - \ell_B}{u_B + \ell_B}, \tag{494}$$

$$\frac{\max_i a_i u_i}{\min_i a_i \ell_i} \ge \frac{a_j u_j}{a_j \ell_j} = u_j/\ell_j \quad (\forall j), \tag{495}$$

$$a_j = 1/\ell_j \Longrightarrow \min_j a_j \ell_j = 1, \quad \max_j a_j u_j = \max_j u_j/\ell_j. \tag{496}$$

□

Proposition 19 (Flag perturbation of quadratic conditioning)**.** *Let $B_\star = B(U^\star)$, $B = B(U)$, and $\delta_B := \Gamma^a d_P(U, U^\star)$. If $\ell_\star I \preceq H^{1/2} B_\star H^{1/2} \preceq u_\star I$, then*

$$\ell_\star - \|H\|_{\mathrm{op}}\delta_B \le \lambda_{\min}(H^{1/2} B H^{1/2}) \le \lambda_{\max}(H^{1/2} B H^{1/2}) \le u_\star + \|H\|_{\mathrm{op}}\delta_B, \tag{497}$$

$$\|H\|_{\mathrm{op}}\delta_B < \ell_\star \Longrightarrow \quad \kappa(H^{1/2} B H^{1/2}) \le \frac{u_\star + \|H\|_{\mathrm{op}}\delta_B}{\ell_\star - \|H\|_{\mathrm{op}}\delta_B}. \tag{498}$$

Proof.

$$\|H^{1/2}(B - B_\star)H^{1/2}\|_{\mathrm{op}} \le \|H\|_{\mathrm{op}} \|B - B_\star\|_{\mathrm{op}} \le \|H\|_{\mathrm{op}}\delta_B. \tag{499}$$

□

The small-gain criterion addresses a different question: when does direct parameter contraction dominate feedback through optimizer state? The resolvent $(I - D)^{-1}$ sums repeated propagation inside the state subsystem. Its coupling cost must remain below the direct contraction margin α, giving an exact criterion for the stated nonnegative envelope.

Theorem 12 (Schur-resolvent small gain)**.** *This is the nonsingular M-matrix/Schur-complement criterion for a nonnegative feedback system (Berman & Plemmons, 1994), specialized to the optimizer's parameter–state partition. Let a nonnegative sensitivity envelope at a fixed step size have the form*

$$\mathcal{A}_\eta := \begin{pmatrix} 1 - \eta\alpha & \eta b^\top \\ c & D \end{pmatrix} \ge 0, \quad b, c \ge 0, \quad D \ge 0, \quad \rho(D) < 1, \quad \eta > 0. \tag{500}$$

Then

$$\rho(\mathcal{A}_\eta) < 1 \quad \Longleftrightarrow \quad \alpha > b^\top (I - D)^{-1} c. \tag{501}$$

If the right side holds, choose $\varepsilon > 0$ so small that

$$v^\top := (b^\top + \varepsilon \mathbf{1}^\top)(I - D)^{-1} > 0, \quad \delta := \alpha - v^\top c > 0, \quad a_\eta := (1, \eta v^\top), \tag{502}$$

$$0 < \gamma \le \delta, \quad \gamma\eta \max_j v_j \le \varepsilon. \tag{503}$$

Then

$$a_\eta \mathcal{A}_\eta \preceq (1 - \gamma\eta) a_\eta. \tag{504}$$

Proof.

$$(I - D)^{-1} = \sum_{k=0}^{\infty} D^k \ge 0, \tag{505}$$

$$v^\top (I - D) = b^\top + \varepsilon \mathbf{1}^\top, \tag{506}$$

$$a_\eta \mathcal{A}_\eta = (1 - \eta(\alpha - v^\top c),\ \eta v^\top - \eta\varepsilon \mathbf{1}^\top) \preceq (1 - \gamma\eta) a_\eta. \tag{507}$$

Thus $\rho(\mathcal{A}_\eta) \le 1 - \gamma\eta < 1$. Conversely, $\rho(\mathcal{A}_\eta) < 1$ implies the existence of $x = (x_0, x_s) > 0$ with $(I - \mathcal{A}_\eta)x > 0$, e.g. $x = (I - \mathcal{A}_\eta)^{-1}\mathbf{1}$. Hence

$$(I - D)x_s > c x_0, \quad x_s > (I - D)^{-1} c x_0, \quad \alpha x_0 > b^\top x_s \ge b^\top (I - D)^{-1} c x_0. \tag{508}$$

□

Corollary 12 (A uniform $1/n$ sensitivity floor)**.** *Suppose a coupled optimizer satisfies*

$$\begin{pmatrix} u_{t+1} \\ s_{t+1} \end{pmatrix} \preceq \mathcal{A}_\eta \begin{pmatrix} u_t \\ s_t \end{pmatrix} + \frac{I_t}{b_t} \begin{pmatrix} \eta d \\ h \end{pmatrix}, \quad \mathbb{P}(I_t = 1) = b_t/n, \quad (u_0, s_0) = 0, \tag{509}$$

with the hypotheses of Theorem 12. Then

$$\mathbb{E}[u_T + \eta v^\top s_T] \le \frac{d + v^\top h}{n\gamma} (1 - (1 - \gamma\eta)^T), \tag{510}$$

$$\epsilon_T^{\mathrm{stab}} \le \frac{L_0 (d + v^\top h)}{n\gamma}. \tag{511}$$

Proof.

$$\mathbb{E}[u_{t+1} + \eta v^\top s_{t+1}] \le (1 - \gamma\eta)\mathbb{E}[u_t + \eta v^\top s_t] + \eta(d + v^\top h)/n. \tag{512}$$

□

Proposition 20 (Local Jacobian envelopes for the actual map)**.** *Let $\mathcal{T}_{t,S}$ denote the complete map of Definition 7, expressed in regular local coordinates. For a convex coupling region $\mathcal{K}$, partition its Jacobian into parameter and state blocks:*

$$\mathrm{D}\mathcal{T}_{t,S}(x) = \begin{pmatrix} J_{ww} & J_{ws} \\ J_{sw} & J_{ss} \end{pmatrix}, \quad \sup_{x \in \mathcal{K}} \|J_{ww}\| \le 1 - \eta\alpha, \quad \sup_{x \in \mathcal{K}} \|J_{ws}\| \le \eta b, \tag{513}$$

$$\sup_{x \in \mathcal{K}} \|J_{sw}\| \le c, \quad \sup_{x \in \mathcal{K}} \|J_{ss}\| \le q < 1. \tag{514}$$

If $\alpha(1 - q) > bc$, the scalar-state version of Theorem 12 applies. Block norms give the matrix-state version. For the broad triangle-inequality envelope of Definition 11, these contraction conditions are additional hypotheses, not consequences of smoothness alone.

Proof.

$$\mathcal{T}_{t,S}(x) - \mathcal{T}_{t,S}(x') = \int_0^1 \mathrm{D}\mathcal{T}_{t,S}(x' + a(x - x'))(x - x')\, \mathrm{d}a. \tag{515}$$

□

A.10. A second generalization route: a full-rank flag posterior

Stability follows perturbations along the entire training trajectory. Here we instead control a distribution around the final parameters, allowing a generalization statement when products of state sensitivities are too large to be informative. The posterior may use the learned flag because the bound holds simultaneously over posteriors; the prior must remain independent of the sample to which the bound is applied.

Definition 12 (Prior, posterior, and binary relative entropy)**.**

$$0 \le \ell(w,z) \le 1, \quad L(Q) := \mathbb{E}_{w\sim Q, z\sim\mathcal{D}}\ell(w,z), \quad \widehat{L}_S(Q) := \mathbb{E}_{w\sim Q}\frac{1}{n}\sum_i \ell(w,z_i), \tag{516}$$

$$\mathrm{kl}(a\|b) := a\log(a/b) + (1-a)\log((1-a)/(1-b)), \tag{517}$$

$$\mathrm{KL}(Q\|P) := \int \log(\mathrm{d}Q/\mathrm{d}P)\,\mathrm{d}Q, \quad Q \ll P, \tag{518}$$

$$\mathrm{kl}_+^{-1}(a,c) := \sup\{b\in[a,1] : \mathrm{kl}(a\|b) \le c\}, \quad c \ge 0. \tag{519}$$

The prior P is independent of S; the posterior may depend on the complete training trajectory.

Lemma 14 (Seeger–Maurer exponential moment)**.** *For iid $X_i \in [0,1]$, $p := \mathbb{E}X_i$, and $\overline{X} = n^{-1}\sum_i X_i$,*

$$\mathbb{E}\exp\{n\,\mathrm{kl}(\overline{X}\|p)\} \le \Xi_n, \qquad \Xi_n := \sum_{k=0}^{n}\binom{n}{k}(k/n)^k(1-k/n)^{n-k}, \tag{520}$$

$$\Xi_n \le n+1 \quad (n\ge 1), \qquad \Xi_n \le 2\sqrt{n} \quad (n \ge 8). \tag{521}$$

Here $0^0 = 1$. The moment bound is attained by every nondegenerate Bernoulli law, so Ξ_n is its exact uniform prefactor (Maurer, 2004).

Proof. For $p \in (0,1)$, the function $f(x) := \exp(n\,\mathrm{kl}(x\|p))$ is convex. For every convex g and $X \in [0,1]$,

$$g(X) \le (1-X)g(0) + Xg(1), \qquad \mathbb{E}g(X) \le (1-p)g(0) + pg(1). \tag{522}$$

Applying this replacement separately to each coordinate yields independent $B_i \sim \mathrm{Bernoulli}(p)$ and

$$\mathbb{E}f(\overline{X}) \le \mathbb{E}f(n^{-1}\textstyle\sum_i B_i) \tag{523}$$

$$= \sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}\left(\frac{k/n}{p}\right)^k\left(\frac{1-k/n}{1-p}\right)^{n-k} \tag{524}$$

$$= \sum_{k=0}^{n}\binom{n}{k}(k/n)^k(1-k/n)^{n-k} = \Xi_n \le n+1. \tag{525}$$

For Bernoulli X_i the comparison is an equality. Maurer's Stirling bound (Maurer, 2004, Theorem 1) gives $\Xi_n \le 2\sqrt{n}$ for $n \ge 8$. At $p \in \{0,1\}$, $X_i = p$ almost surely and the left side equals 1. □

Theorem 13 (Seeger–Maurer PAC–Bayes bound, without Pinsker relaxation)**.** *For any sample-independent P and $\delta \in (0,1)$, with probability at least $1-\delta$ over S, simultaneously for all $Q \ll P$, with $k_Q := \mathrm{KL}(Q\|P) + \log(\Xi_n/\delta)$,*

$$\mathrm{kl}(\widehat{L}_S(Q)\|L(Q)) \le k_Q/n, \tag{526}$$

$$L(Q) \le \mathrm{kl}_+^{-1}(\widehat{L}_S(Q), k_Q/n) \le \min\left\{1, \widehat{L}_S(Q) + \sqrt{\frac{k_Q}{2n}}\right\}. \tag{527}$$

This is the binary-KL form of the PAC–Bayes theorem (Seeger, 2002; Maurer, 2004); the square-root expression is a secondary relaxation rather than the reported certificate.

Proof. Fubini, Lemma 14, and Markov's inequality give

$$\mathbb{E}_S \mathbb{E}_{w\sim P} e^{n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w))} \le \Xi_n, \tag{528}$$

$$\mathbb{P}_S\left(\mathbb{E}_{w\sim P} e^{n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w))} > \frac{\Xi_n}{\delta}\right) \le \delta. \tag{529}$$

On the complementary event, for $f(w) := n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w))$, the Gibbs/Donsker–Varadhan change-of-measure inequality (Donsker & Varadhan, 1975) and joint convexity of binary relative entropy give

$$\mathbb{E}_Q f - \mathrm{KL}(Q\|P) \le \log \mathbb{E}_P e^f, \tag{530}$$

$$n\,\mathrm{kl}(\widehat{L}_S(Q)\|L(Q)) \le \mathbb{E}_Q n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w)) \le \mathrm{KL}(Q\|P) + \log(\Xi_n/\delta), \tag{531}$$

$$\mathrm{kl}(a\|b) \ge 2(a-b)^2. \tag{532}$$

The last inequality follows from $\partial_a^2\,\mathrm{kl}(a\|b) = 1/[a(1-a)] \ge 4$ and Taylor expansion at $a = b$. All assertions extend by continuity at boundary values. □

Corollary 13 (Exact Catoni-type exponential inversion)**.** *For $a \in [0,1]$ and $c \ge 0$,*

$$\mathrm{kl}_+^{-1}(a,c) = \inf_{\gamma>0} \frac{1-\exp(-\gamma a - c)}{1-\exp(-\gamma)}. \tag{533}$$

Thus the preceding theorem can be evaluated by a one-dimensional monotone root or by exponential-temperature optimization (Catoni, 2007). In particular, $\widehat{L}_S(Q) = 0$ gives the exact KL-event consequence

$$L(Q) \le 1 - e^{-k_Q/n} \le k_Q/n, \tag{534}$$

which retains the fast k_Q/n dependence lost by a square-root relaxation.

Proof. For $b \ge a$, Bernoulli entropy duality gives

$$\mathrm{kl}(a\|b) = \sup_{\gamma\ge 0}\{-\gamma a - \log(1-b+be^{-\gamma})\}. \tag{535}$$

The constraint $\mathrm{kl}(a\|b) \le c$ is therefore equivalent to $b \le (1-e^{-\gamma a - c})/(1-e^{-\gamma})$ for every $\gamma > 0$. Taking the largest admissible b proves the identity, including boundary values by limits. This optimization is algebraic on one uniform KL event and incurs no additional selection penalty. □

Proposition 21 (Exact flag-Gaussian complexity)**.** *For $W \in \mathbb{R}^{m\times d}$, let*

$$P = \mathcal{N}(0, \sigma_0^2 I_{md}), \quad Q_{W,U,\sigma} = \mathcal{N}(\mathrm{vec}\,W, I_m \otimes \Sigma_U), \tag{536}$$

$$\Sigma_U := \sum_{j=1}^{K+1} \sigma_j^2 \Pi_j(U), \quad \sigma_j > 0, \quad \sigma_0 > 0. \tag{537}$$

Using rowwise vectorization,

$$\mathrm{KL}(Q_{W,U,\sigma}\|P) = \frac{\|W\|_F^2}{2\sigma_0^2} + \frac{m}{2}\sum_{j=1}^{K+1} q_j\left[\frac{\sigma_j^2}{\sigma_0^2} - 1 - \log\frac{\sigma_j^2}{\sigma_0^2}\right], \tag{538}$$

$$\mathrm{tr}(I_m \otimes \Sigma_U) = m\sum_{j=1}^{K+1} q_j\sigma_j^2, \qquad \det(I_m \otimes \Sigma_U) = \prod_{j=1}^{K+1} (\sigma_j^2)^{mq_j}. \tag{539}$$

For $q_{K+1} = d - R > 0$,

$$\sigma_{K+1} \downarrow 0 \quad \Longrightarrow \quad \mathrm{KL}(Q_{W,U,\sigma}\|P) \longrightarrow \infty. \tag{540}$$

Proof. For positive definite $\Sigma \in \mathbb{R}^{D\times D}$ and $Q = \mathcal{N}(\mu, \Sigma)$,

$$\log \frac{\mathrm{d}Q}{\mathrm{d}P}(x) = \tfrac{1}{2}\log \frac{\det(\sigma_0^2 I_D)}{\det \Sigma} - \tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu) + \frac{\|x\|^2}{2\sigma_0^2}, \tag{541}$$

$$\mathbb{E}_Q (x-\mu)^\top \Sigma^{-1}(x-\mu) = D, \qquad \mathbb{E}_Q \|x\|^2 = \|\mu\|^2 + \operatorname{tr}\Sigma, \tag{542}$$

$$\log\det \Sigma_U = \sum_j q_j \log \sigma_j^2, \quad \operatorname{tr}\Sigma_U = \sum_j q_j \sigma_j^2. \tag{543}$$

□

Positive residual variance keeps the posterior full rank and gives finite complexity relative to the full-rank Gaussian prior. Under this isotropic prior, the displayed KL depends on block dimensions and variances, while posterior orientation can affect the perturbed loss. To obtain a statement about the trained parameter W, the next result controls both empirical and population changes induced by the Gaussian perturbation.

Corollary 14 (A deterministic-center bound)**.** *In addition to $0 \le \ell \le 1$, suppose $\ell(\cdot, z)$ has L_ℓ-Lipschitz gradient for every z. On the event of Theorem 13, simultaneously for all data-dependent $W, U, \sigma_j > 0$, set $v_\sigma := m\sum_j q_j\sigma_j^2$, $a_Q := \min\{1, F_S(W) + L_\ell v_\sigma/2\}$. Then*

$$F_{\mathcal{D}}(W) \le \min\left\{1, \frac{L_\ell v_\sigma}{2} + \mathrm{kl}_+^{-1}(a_Q, k_Q/n)\right\}, \tag{544}$$

where k_Q uses the exact Gaussian KL above. In particular,

$$F_{\mathcal{D}}(W) \le F_S(W) + L_\ell m \sum_j q_j \sigma_j^2 \tag{545}$$

$$+ \sqrt{\frac{\|W\|_F^2/(2\sigma_0^2) + \frac{m}{2}\sum_j q_j[\sigma_j^2/\sigma_0^2 - 1 - \log(\sigma_j^2/\sigma_0^2)] + \log(\Xi_n/\delta)}{2n}}. \tag{546}$$

Proof. For $E \sim \mathcal{N}(0, I_m \otimes \Sigma_U)$,

$$|\ell(W+E,z) - \ell(W,z) - \langle \nabla\ell(W,z), E\rangle| \le \tfrac{1}{2}L_\ell \|E\|_F^2, \tag{547}$$

$$\mathbb{E}E = 0, \quad \mathbb{E}\|E\|_F^2 = m\sum_j q_j\sigma_j^2, \tag{548}$$

$$|L(Q) - F_{\mathcal{D}}(W)| \le \tfrac{1}{2}L_\ell m \sum_j q_j\sigma_j^2, \quad |\widehat{L}_S(Q) - F_S(W)| \le \tfrac{1}{2}L_\ell m\sum_j q_j\sigma_j^2. \tag{549}$$

Insert these two Taylor bounds into the inverse-KL certificate; monotonicity of kl_+^{-1} in its first argument gives the first assertion. Applying Pinsker only afterward gives the displayed analytic relaxation. □

A.11. Convergence of the actual current-gradient-normalized update

The second-moment update includes the current stochastic gradient, so its normalizer and gradient are dependent. We analyze their combined nonlinear field. The proof first establishes monotonicity of that field, then compares the conditional noise distribution with its reflection. This provides a descent term together with a quantitative cost for gradient bias and noise asymmetry, before momentum and refresh errors are introduced.

Definition 13 (Conditional normalization field)**.** *Condition on $\mathcal{F}_{t-1}$, with U_t, the stored moments, and their ages fixed. Use rowwise vectorization, $N = md$, and let $\mathcal{P}_{k,t}$ be the orthogonal parameter-coordinate projector induced by $\Pi_{k,t}$. For each active coordinate i of block k set*

$$\alpha_{k,t} := \frac{1-\beta_{2k}}{1-\beta_{2k}^{c_{V,k,t}}}, \quad c_{i,t} := \frac{\beta_{2k}V_{i,t-1}^+}{1-\beta_{2k}^{c_{V,k,t}}}, \quad h_{i,t}(z) := \frac{a_k}{\sqrt{c_{i,t} + \alpha_{k,t}z^2} + \epsilon}, \tag{550}$$

$$\mathcal{B}_t(g) := \bigoplus_{k=1}^{K} \operatorname{diag}_i h_{i,t}((\operatorname{mat}(g)U_{k,t})_i) \;\oplus\; \frac{a_\perp}{\|\operatorname{mat}(g)\Pi_{K+1,t}\|/\sqrt{N} + \epsilon} I_{m(d-R)}, \tag{551}$$

$$\mathcal{T}_t(g) := \mathcal{B}_t(g)g. \tag{552}$$

The direct sum is expressed in the orthonormal block frame and lifted back to parameter space. In particular, $\mathcal{B}_t(g)$ depends on the current g.

Lemma 15 (Strong monotonicity of the normalization field)**.** *Suppose $\|g_s\| \le G$ for all preceding steps, $a_k, a_\perp > 0$, and*

$$M := \epsilon^{-1} \max\{a_1, \dots, a_K, a_\perp\}, \tag{553}$$

$$\kappa := \min\left\{ \min_k \frac{a_k \epsilon}{(3G+\epsilon)^2}, \frac{a_\perp \epsilon}{(3G/\sqrt{N}+\epsilon)^2} \right\} > 0. \tag{554}$$

Then, on $\|x\|, \|y\| \le 3G$,

$$\mathcal{T}_t(-x) = -\mathcal{T}_t(x), \quad \|\mathcal{T}_t(x) - \mathcal{T}_t(y)\| \le M\|x-y\|, \tag{555}$$

$$\langle x-y, \mathcal{T}_t(x) - \mathcal{T}_t(y)\rangle \ge \kappa \|x-y\|^2. \tag{556}$$

Proof. The exact reset-age recursion gives

$$V^+_{i,t-1} \le (1-\beta_{2k}^{c_{V,k,t}-1})G^2, \tag{557}$$

$$0 \le c_{i,t} \le (1-\alpha_{k,t})G^2, \quad 0 < \alpha_{k,t} \le 1, \tag{558}$$

$$s(z) := \sqrt{c_{i,t} + \alpha_{k,t} z^2}, \quad f(z) := a_k z/(s(z)+\epsilon), \tag{559}$$

$$f'(z) = \frac{a_k(c_{i,t}/s(z)+\epsilon)}{(s(z)+\epsilon)^2} \quad (s(z) > 0), \tag{560}$$

$$|z| \le 3G \Longrightarrow s(z) \le 3G, \quad \frac{a_k\epsilon}{(3G+\epsilon)^2} \le f'(z) \le a_k/\epsilon. \tag{561}$$

For $c_{i,t} = z = 0$, $f'(0) = a_k/\epsilon$ by continuity. On the residual block, Proposition 11 gives the same bounds with $3G$ replaced by $3G/\sqrt{N}$. Thus

$$\kappa I \preceq \mathrm{D}\mathcal{T}_t(x) \preceq MI, \quad \|x\| \le 3G, \tag{562}$$

$$\mathcal{T}_t(x) - \mathcal{T}_t(y) = \int_0^1 \mathrm{D}\mathcal{T}_t(y + s(x-y))(x-y)\,\mathrm{d}s. \tag{563}$$

□

Definition 14 (Bias and symmetry defect)**.**

$$h_t := \nabla F(w_t), \quad \overline{h}_t := \mathbb{E}_t g_t, \quad b_t := \overline{h}_t - h_t, \quad \xi_t := g_t - \overline{h}_t, \tag{564}$$

$$\nu_t := \mathcal{L}(\xi_t \mid \mathcal{F}_{t-1}), \quad \nu_t^- := (-\mathrm{Id})_\# \nu_t, \quad \delta_t := W_1(\nu_t, \nu_t^-), \quad D_t := \|b_t\| + \delta_t/2, \tag{565}$$

$$\mathfrak{a}_t := \|\mathbb{E}_t \mathcal{T}_t(\xi_t)\|, \qquad \widehat{D}_t := \|b_t\| + \mathfrak{a}_t/M, \tag{566}$$

$$W_1(\mu,\nu) := \inf_{(X,Y):\, X\sim\mu, Y\sim\nu} \mathbb{E}\|X-Y\|. \tag{567}$$

Theorem 14 (Odd-field coercivity and Kantorovich–Rubinstein calibration)**.** *If $\|g_t\|, \|h_t\| \le G$, then*

$$\langle h_t, \mathbb{E}_t[\mathcal{B}_t(g_t) g_t]\rangle \ge \kappa\|h_t\|^2 - M\widehat{D}_t\|h_t\| \tag{568}$$

$$\ge \tfrac{\kappa}{2}\|h_t\|^2 - \frac{M^2}{2\kappa}\widehat{D}_t^2. \tag{569}$$

Moreover,

$$\mathfrak{a}_t \le \min\{M\delta_t/2, \mathsf{H}_t\}, \quad \widehat{D}_t \le D_t, \tag{570}$$

$$\mathsf{H}_t^2 := m\sum_{k=1}^K \frac{q_k a_k^2}{\alpha_{k,t}} + N a_\perp^2. \tag{571}$$

If $b_t = 0$ and $\nu_t = \nu_t^-$, the stronger lower bound is $\kappa\|h_t\|^2$; more generally $b_t = 0$ and $\mathfrak{a}_t = 0$ suffice.

Proof. Strong monotonicity applies on the segment from ξ_t to $\xi_t + h_t$, whose norm is at most $3G$. Lipschitz continuity controls the shift b_t:

$$\|\mathbb{E}_t[\mathcal{T}_t(g_t) - \mathcal{T}_t(h_t + \xi_t)]\| \le M\|b_t\|, \tag{572}$$

$$\langle h_t, \mathbb{E}_t[\mathcal{T}_t(h_t + \xi_t) - \mathcal{T}_t(\xi_t)]\rangle \ge \kappa\|h_t\|^2, \tag{573}$$

$$M\widehat{D}_t\|h_t\| \le \tfrac{\kappa}{2}\|h_t\|^2 + \tfrac{M^2}{2\kappa}\widehat{D}_t^2. \tag{574}$$

For a unit vector v, $v^\top\mathcal{T}_t$ is M-Lipschitz and odd. Kantorovich–Rubinstein duality (Villani, 2009) therefore gives

$$2\mathfrak{a}_t = \left\| \int \mathcal{T}_t \,\mathrm{d}\nu_t - \int \mathcal{T}_t \,\mathrm{d}\nu_t^- \right\| \le MW_1(\nu_t, \nu_t^-). \tag{575}$$

Each active normalized coordinate has magnitude at most $a_k/\sqrt{\alpha_{k,t}}$, and the residual norm is at most $a_\perp\sqrt{N}$. Orthogonality gives $\|\mathcal{T}_t(\xi_t)\| \le \mathsf{H}_t$, which supplies the second calibration without a factor ϵ^{-1}. □

Bias and asymmetry enter through different mechanisms. Even an unbiased gradient can acquire a nonzero mean update under nonlinear normalization if its noise is asymmetric. Conversely, conditional symmetry cancels the odd normalization field at zero signal even when the noise has substantial variance. The directly normalized defect $\mathfrak{a}_t$ can be smaller than the Wasserstein envelope $M\delta_t/2$; the latter is a convenient sufficient bound, while the saturation calibration avoids introducing an unnecessary ϵ^{-1} factor into the asymmetry numerator.

Proposition 22 (Exact ambient momentum and refresh defect)**.** *Write $G_t := \mathrm{mat}(g_t)$. Let $\widehat{M}_{k,t}$ denote the bias-corrected first moment before refresh, $A_{k,t} := \widehat{M}_{k,t}U_{k,t}^\top$, and*

$$m_t := \mathrm{vec}\left(\sum_k A_{k,t} + G_t\Pi_{K+1,t}\right), \quad \theta_{k,t} := \frac{\beta_{1k}(1-\beta_{1k}^{t-1})}{1-\beta_{1k}^t}, \quad \Theta_t := \sum_k \theta_{k,t}\mathcal{P}_{k,t}, \tag{576}$$

$$\beta := \max_k \beta_{1k} < 1, \quad G_m := \sqrt{K+1}G. \tag{577}$$

With the first-moment ages $c_{M,k,t} = t$, exactly,

$$m_t = \Theta_t m_{t-1} + (I - \Theta_t)g_t + u_t, \qquad d_t = \mathcal{B}_t(g_t)m_t, \qquad \|m_t\| \le G_m, \tag{578}$$

$$u_t = \mathrm{vec}\sum_k \theta_{k,t}\left[\widehat{M}_{k,t-1}O_{k,t-1}^\top U_{k,t}^\top - \mathrm{mat}(m_{t-1})\Pi_{k,t}\right]. \tag{579}$$

Here $O_{k,t-1} = I$ without a preceding refresh. Furthermore,

$$\|u_t\| \le \beta\sum_k \left[G\|O_{k,t-1}^\top - U_{k,t-1}^\top U_{k,t}\|_{\mathrm{op}} + G_m\|\Pi_{k,t} - \Pi_{k,t-1}\|_{\mathrm{op}}\right], \tag{580}$$

$$\chi_{t-1} = 0 \Longrightarrow u_t = 0. \tag{581}$$

Proof.

$$\widehat{M}_{k,t} = \theta_{k,t}\widehat{M}_{k,t-1}O_{k,t-1}^\top + (1-\theta_{k,t})G_tU_{k,t}, \tag{582}$$

$$\|m_t\|^2 = \sum_k \|\widehat{M}_{k,t}\|_F^2 + \|G_t\Pi_{K+1,t}\|_F^2 \le (K+1)G^2, \tag{583}$$

$$A_{k,t-1}(I - \Pi_{k,t-1}) = 0, \quad (\mathrm{mat}(m_{t-1}) - A_{k,t-1})\Pi_{k,t-1} = 0, \tag{584}$$

$$\widehat{M}_{k,t-1}O^\top U_{k,t}^\top - A_{k,t-1}\Pi_{k,t} = \widehat{M}_{k,t-1}(O^\top - U_{k,t-1}^\top U_{k,t})U_{k,t}^\top, \tag{585}$$

$$(\mathrm{mat}(m_{t-1}) - A_{k,t-1})\Pi_{k,t} = \mathrm{mat}(m_{t-1})(I - \Pi_{k,t-1})(\Pi_{k,t} - \Pi_{k,t-1}). \tag{586}$$

□

Theorem 15 (Actual Adam normalization, momentum, and flag transport)**.** *Assume F is L-smooth, lower bounded, $\|h_t\|, \|g_t\| \le G$, $\lambda = 0$, and the exact map of Definition 7 is used. Define*

$$B_t := \mathcal{B}_t(g_t), \quad C_t := \eta_t B_t (I - \Theta_t)^{-1}\Theta_t, \quad C_0 := 0 \text{ before step } 1, \tag{587}$$

$$z_t := w_t - C_{t-1} m_{t-1}, \quad r_t := (C_{t-1} - C_t) m_{t-1} - \eta_t B_t (I - \Theta_t)^{-1} u_t, \tag{588}$$

$$\overline{\eta} := \sup_t \eta_t, \quad G_z := G + L\overline{\eta} M \beta G_m/(1-\beta), \quad S_T := \sum_{t=1}^{T} \eta_t, \quad \mathsf{J}_t := \min\{MG, \mathsf{H}_t\}. \tag{589}$$

Then

$$\frac{\kappa}{2} \sum_{t=1}^{T} \eta_t \mathbb{E}\|\nabla F(w_t)\|^2 \le \Delta_0 + L \sum_{t=1}^{T} \eta_t^2 \mathbb{E}\mathsf{J}_t^2 + L G_m \sum_{t=1}^{T} \eta_t \mathbb{E}[\mathsf{J}_t \|C_{t-1}\|_{\mathrm{op}}] \tag{590}$$

$$+ \frac{M^2}{2\kappa} \sum_{t=1}^{T} \eta_t \mathbb{E}\widehat{D}_t^2 + G_z \sum_{t=1}^{T} \mathbb{E}\|r_t\| + L \sum_{t=1}^{T} \mathbb{E}\|r_t\|^2, \tag{591}$$

$$\|r_t\| \le G_m \|C_t - C_{t-1}\|_{\mathrm{op}} + \frac{\eta_t M}{1-\beta} \|u_t\|. \tag{592}$$

Here $\Delta_0 := \mathbb{E}F(w_1) - F_{\inf}$. No predictability assumption on B_t is used.

Proof. The virtual-iterate identity is algebraic, hence remains valid for random B_t:

$$z_{t+1} = z_t - \eta_t B_t g_t + r_t, \tag{593}$$

$$\|w_t - z_t\| \le G_m \|C_{t-1}\|_{\mathrm{op}}, \quad \|C_t\|_{\mathrm{op}} \le \eta_t M\beta/(1-\beta), \quad \|B_t g_t\| \le \mathsf{J}_t, \tag{594}$$

$$F(z_{t+1}) \le F(z_t) - \eta_t \langle \nabla F(z_t), B_t g_t \rangle + \langle \nabla F(z_t), r_t \rangle + L\eta_t^2 \|B_t g_t\|^2 + L\|r_t\|^2, \tag{595}$$

$$|\langle \nabla F(z_t) - h_t, B_t g_t \rangle| \le L G_m \mathsf{J}_t \|C_{t-1}\|_{\mathrm{op}}, \tag{596}$$

$$\mathbb{E}_t \langle h_t, B_t g_t \rangle \ge \tfrac{\kappa}{2}\|h_t\|^2 - \tfrac{M^2}{2\kappa}\widehat{D}_t^2, \tag{597}$$

$$\|\nabla F(z_t)\| \le \|h_t\| + L\|w_t - z_t\| \le G_z. \tag{598}$$

Take conditional expectations and sum, using $z_1 = w_1$ and $F(z_{T+1}) \ge F_{\inf}$. □

The bound now separates current-normalization bias, momentum lag, and the variation/transport remainder r_t. The virtual-iterate identity remains exact even though B_t is random and depends on g_t; the descent argument uses coercivity of the combined field. A rate follows only when the accumulated state changes and noise defect satisfy the budgets specified next.

Corollary 15 (Observable variation and batch-size conditions)**.** *Set $\eta_t = \eta_0 T^{-1/2}$ and define*

$$V_T^B := \sum_{t=1}^{T} \|B_t - B_{t-1}\|_{\mathrm{op}}, \quad V_T^\Theta := \sum_{t=1}^{T} \|\Theta_t - \Theta_{t-1}\|_{\mathrm{op}}, \quad U_T := \sum_{t=1}^{T} \|u_t\|, \quad B_0 = \Theta_0 = 0. \tag{599}$$

Then

$$\sum_t \|C_t - C_{t-1}\|_{\mathrm{op}} \le \eta \left[\frac{\beta}{1-\beta} V_T^B + \frac{M}{(1-\beta)^2} V_T^\Theta \right], \tag{600}$$

$$\mathbb{E}[V_T^B + V_T^\Theta + U_T] = O(\sqrt{T}) \Longrightarrow \quad \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\|\nabla F(w_t)\|^2 = O(T^{-1/2}) + \frac{M^2}{\kappa^2 T} \sum_t \mathbb{E}\widehat{D}_t^2. \tag{601}$$

For an average of b independent unbiased sample gradients with conditional variance at most σ^2, without clipping,

$$\widehat{D}_t^2 \le \mathbb{E}_t\|g_t - h_t\|^2 \le \sigma^2/b. \tag{602}$$

Consequently the displayed variation condition and $b \gtrsim \sqrt{T}$ give $T^{-1/2}$ stationarity. With norm clipping, replace the last error by $2\|b_t^{\mathrm{clip}}\|^2 + 2\sigma^2/b$.

Proof.

$$(I-\Theta)^{-1}\Theta = (I-\Theta)^{-1} - I, \tag{603}$$

$$\|(I-\Theta)^{-1} - (I-\Theta')^{-1}\|_{\mathrm{op}} \le (1-\beta)^{-2}\|\Theta - \Theta'\|_{\mathrm{op}}, \tag{604}$$

$$W_1(\nu_t, \nu_t^-) \le 2\mathbb{E}_t\|\xi_t\| \le 2\sqrt{\mathbb{E}_t\|\xi_t\|^2}, \tag{605}$$

$$\|r_t\| \le O(\eta), \quad \sum_t \mathbb{E}\|r_t\| = O(\eta\sqrt{T}) = O(1), \tag{606}$$

$$\sum_t \mathbb{E}\|r_t\|^2 \le (\sup_t \|r_t\|)\sum_t \mathbb{E}\|r_t\| = O(\eta). \tag{607}$$

Clipping is nonexpansive, so for independent copies X, X',

$$\mathrm{Var}(\mathrm{clip}(X)) = \tfrac{1}{2}\mathbb{E}\|\mathrm{clip}(X) - \mathrm{clip}(X')\|^2 \le \tfrac{1}{2}\mathbb{E}\|X - X'\|^2 = \mathrm{Var}(X), \tag{608}$$

$$\widehat{D}_t^2 \le D_t^2 = (\|b_t\| + \delta_t/2)^2 \le 2\|b_t\|^2 + 2\,\mathrm{Var}_t(g_t). \tag{609}$$

□

Proposition 23 (An exact asymmetry-induced stationary bias)**.** *Consider a fixed one-coordinate flag branch with $\beta_1 = \beta_2 = 0$, $a = 1$, $F(w) = w^2/2$, and unbiased additive noise*

$$\mathbb{P}(\xi = 3\sigma) = 1/10, \quad \mathbb{P}(\xi = -\sigma/3) = 9/10, \quad \mathbb{E}\xi = 0, \quad \mathbb{E}\xi^2 = \sigma^2, \tag{610}$$

$$(\xi_t)_{t\ge 1} \textit{ iid}, \quad w_{t+1} = w_t - \eta_t \frac{w_t + \xi_t}{|w_t + \xi_t| + \epsilon}. \tag{611}$$

Its expected update field has a unique root

$$\Psi_\epsilon(w) := \frac{1}{10}\frac{w + 3\sigma}{|w + 3\sigma| + \epsilon} + \frac{9}{10}\frac{w - \sigma/3}{|w - \sigma/3| + \epsilon}, \tag{612}$$

$$w_\epsilon = \tfrac{1}{2}\left[\sqrt{(8\sigma/3 + 5\epsilon/4)^2 + 4\sigma^2} - (8\sigma/3 + 5\epsilon/4)\right] \in (0, \sigma/3), \tag{613}$$

$$\lim_{\epsilon\downarrow 0} w_\epsilon = \sigma/3 > 0. \tag{614}$$

For deterministic $\eta_t > 0$, $\sup_t \eta_t < \infty$, $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$, the iterates satisfy

$$w_t \longrightarrow w_\epsilon \quad \textit{a.s.}, \qquad \|\nabla F(w_t)\|^2 \longrightarrow w_\epsilon^2 > 0 \quad \textit{a.s.} \tag{615}$$

Proof.

$$\Psi_\epsilon'(w) > 0, \quad \Psi_\epsilon(0) < 0, \quad \Psi_\epsilon(\sigma/3) > 0, \tag{616}$$

$$\Psi_\epsilon(w) = 0, \quad 0 < w < \sigma/3 \iff (w + 3\sigma)(w - \sigma/3) + \tfrac{5}{4}\epsilon w = 0. \tag{617}$$

The scalar branch embeds in the exact flag algorithm as

$$W = \begin{pmatrix} w & 0 \\ 0 & 0 \end{pmatrix}, \quad U = e_1 \in \mathrm{St}(2,1), \quad G = \begin{pmatrix} w + \xi & 0 \\ 0 & 0 \end{pmatrix}, \tag{618}$$

$$G\Pi_\perp = 0, \quad C_k = \mathrm{diag}(c_k, 0), \quad \mathrm{grad}\,\Phi(U; C) = 0. \tag{619}$$

If $w > \sigma/3$, every update decreases w; if $w < -3\sigma$, every update increases w. Since every increment has magnitude at most η_t, all iterates lie in a deterministic compact interval $\mathcal{I}$. On $\mathcal{I}$, $\Psi_\epsilon'(w) \ge c_{\mathcal{I}} > 0$, choosing $c_{\mathcal{I}} \le (2\sup_t \eta_t)^{-1}$. Thus

$$\mathbb{E}_t(w_{t+1} - w_\epsilon)^2 \le (w_t - w_\epsilon)^2 - 2\eta_t(w_t - w_\epsilon)\Psi_\epsilon(w_t) + \eta_t^2 \tag{620}$$

$$\le (1 - 2c_{\mathcal{I}}\eta_t)(w_t - w_\epsilon)^2 + \eta_t^2. \tag{621}$$

The almost-supermartingale theorem gives convergence of $(w_t - w_\epsilon)^2$ and $\sum_t \eta_t(w_t - w_\epsilon)^2 < \infty$; hence its limit is zero. □

The counterexample converges to a zero of the expected normalized update field that is separated from the minimizer of F. Its gradient noise is unbiased and has bounded variance, so those properties alone cannot remove this stationary bias. This gives a concrete reason to retain the symmetry-defect term when interpreting the convergence bound for the actual algorithm.

Proposition 24 (Resolving the variation budgets into actual state changes)**.** *For the exact preconditioner let* $S_{k,t} := \sqrt{\widehat{V}_{k,t}}$ *and* $c_t := (\|G_t \Pi_{K+1,t}\|_F/\sqrt{N} + \epsilon)^{-1}$*. With constant* a_k*,*

$$\|B_t - B_{t-1}\|_{\mathrm{op}} \le \sum_k a_k \left[\epsilon^{-2}\|S_{k,t} - S_{k,t-1}\|_{\max} + 2\epsilon^{-1}\|U_{k,t} - U_{k,t-1}\|_{\mathrm{op}}\right] \tag{622}$$

$$+ a_\perp |c_t - c_{t-1}| + (a_\perp/\epsilon)\|P_{K,t} - P_{K,t-1}\|_{\mathrm{op}}, \tag{623}$$

$$|c_t - c_{t-1}| \le \frac{\|G_t - G_{t-1}\|_F + G\|P_{K,t} - P_{K,t-1}\|_{\mathrm{op}}}{\epsilon^2\sqrt{N}}, \tag{624}$$

$$V_T^\Theta \le \sum_k \beta_{1k} + \beta \sum_{t=2}^{T} \sum_k \|\Pi_{k,t} - \Pi_{k,t-1}\|_{\mathrm{op}}. \tag{625}$$

At refresh s *let* $Z_s = \xi_s \operatorname{grad}\Phi(U_s; \widetilde{C}_s)$*,* $z_s = \|Z_s\|_F$*, and* $e_s = (1 + z_s^2)z_s^2 < 1$*. The transport budget obeys*

$$U_T \le \beta\sqrt{K} \sum_{s \le S} \left[G\left(1 + \frac{2}{2 - e_s}\right) e_s + 2G_m \left(z_s + e_s + \frac{2e_s}{2 - e_s}\right)\right]. \tag{626}$$

Consequently $z_s \le H\xi_0/\sqrt{S}$ *and* $S = O(T)$ *imply* $U_T + V_T^\Theta = O(\sqrt{T})$*. This does not by itself control* V_T^B*: second-moment resets and persistent gradient fluctuations remain explicit in* $\|S_{k,t} - S_{k,t-1}\|$ *and* $|c_t - c_{t-1}|$*. A constant refresh step does not imply the required variation rate.*

Proof. Let $\mathcal{U}_{k,t} = I_m \otimes U_{k,t}$. Since $B_{k,t} = a_k \mathcal{U}_{k,t} \operatorname{diag}((S_{k,t} + \epsilon)^{-1})\mathcal{U}_{k,t}^\top$,

$$\|\operatorname{diag}((S + \epsilon)^{-1} - (S' + \epsilon)^{-1})\|_{\mathrm{op}} \le \epsilon^{-2}\|S - S'\|_{\max}, \tag{627}$$

$$\|\mathcal{U}\mathcal{H}\mathcal{U}^\top - \mathcal{U}'\mathcal{H}\mathcal{U}'^\top\|_{\mathrm{op}} \le 2\|\mathcal{H}\|_{\mathrm{op}}\|U - U'\|_{\mathrm{op}}. \tag{628}$$

Also $\theta_{k,t} \uparrow \beta_{1k}$, so its total scalar variation is β_{1k}. Proposition 14 gives

$$\|U^+ - U\|_F \le z_s + e_s + 2e_s/(2 - e_s), \tag{629}$$

$$\sum_k \|\Pi_k^+ - \Pi_k\|_{\mathrm{op}} \le 2\sqrt{K}\|U^+ - U\|_F, \quad \sum_k \|O_k^\top - U_k^\top U_k^+\|_{\mathrm{op}} \le \sqrt{K}(1 + 2/(2 - e_s))e_s. \tag{630}$$

Insert these inequalities into Proposition 22, and use $\sum_s z_s = O(\sqrt{S})$, $\sum_s e_s = O(1)$. □

Corollary 16 (Complete networks and dense fallback tensors)**.** *For finitely many parameter blocks, apply the constructions blockwise and set*

$$B_t = \bigoplus_l B_{l,t}, \quad \Theta_t = \bigoplus_l \Theta_{l,t}, \quad m_t = (m_{1,t}, \ldots, m_{L,t}), \quad u_t = (u_{1,t}, \ldots, u_{L,t}), \tag{631}$$

$$M = \max_l M_l, \quad \kappa = \min_l \kappa_l, \quad \beta = \max_l \beta_l, \quad G^2 = \sum_l G_l^2, \quad G_m^2 = \sum_l G_{m,l}^2. \tag{632}$$

Set $\mathsf{H}_t^2 := \sum_l \mathsf{H}_{l,t}^2$ *and* $\mathsf{J}_t := \min\{MG, \mathsf{H}_t\}$ *for the joint normalized field. A dense fallback tensor is a fixed full-coordinate block with diagonal* $B_{l,t}$*, scalar* $\Theta_{l,t}$ *and* $u_{l,t} = 0$*. Theorem 15 then holds for the joint, coupled network objective, using these aggregate constants and a joint conditional symmetry defect. For decoupled weight decay* $w_{t+1} = (1 - \eta_t\lambda)w_t - \eta_t B_t m_t$*, its algebraic identity becomes*

$$z_{t+1} = z_t - \eta_t B_t g_t + \widetilde{r}_t, \qquad \widetilde{r}_t = r_t - \eta_t\lambda w_t. \tag{633}$$

The finite-time inequality remains valid with r_t *replaced by* $\widetilde{r}_t$*; decay is not silently absorbed into an unbiased data gradient.*

Proof.

$$\|\bigoplus_l B_l\|_{\mathrm{op}} = \max_l \|B_l\|_{\mathrm{op}}, \quad \sum_l \langle h_l, \mathcal{T}_l(h_l + \xi_l^s) - \mathcal{T}_l(\xi_l^s)\rangle \geq (\min_l \kappa_l) \sum_l \|h_l\|^2, \tag{634}$$

$$\|m_t\|^2 = \sum_l \|m_{l,t}\|^2 \leq \sum_l G_{m,l}^2. \tag{635}$$

Apply the same virtual-iterate identity and smoothness inequality to the direct sum. □

Lemma 16 (State-dependent curvature of the normalization field)**.** *In Definition 13, write $a_i = a_{k(i)}$ and $\alpha_{i,t} = \alpha_{k(i),t}$. Restrict active coordinate i to $|z_i| \leq H_i$, and the residual to radius $H_\perp$. Set*

$$s_{i,\max} := \sqrt{c_{i,t} + \alpha_{i,t} H_i^2}, \qquad \kappa_{i,t} := \frac{a_i (c_{i,t}/s_{i,\max} + \epsilon)}{(s_{i,\max} + \epsilon)^2}, \tag{636}$$

$$\kappa_t := \min\left\{ \min_i \kappa_{i,t}, \frac{a_\perp \epsilon}{(H_\perp/\sqrt{N} + \epsilon)^2} \right\}. \tag{637}$$

The value at $s_{i,\max} = 0$ is interpreted as a_i/ϵ. These are exact lower Jacobian bounds on the stated product domain. In particular,

$$c_{i,t} \geq \nu^2 > 0, \quad s_{i,\max} \leq S \quad \Longrightarrow \quad \kappa_{i,t} \geq \frac{a_i \nu^2}{S(S+\epsilon)^2}, \tag{638}$$

$$c_{i,t} = 0 \quad \Longrightarrow \quad \kappa_{i,t} = \frac{a_i \epsilon}{(\sqrt{\alpha_{i,t}} H_i + \epsilon)^2}. \tag{639}$$

Thus a mature variance floor removes the explicit vanishing-ϵ factor in an active block, whereas a reset restores it. The residual can still be the limiting block. If the support domains cover $\xi_t + s h_t$, $0 \leq s \leq 1$, Theorem 14 holds with the predictable κ_t.

Proof. For $s > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{c/s + \epsilon}{(s+\epsilon)^2}\right) = -\frac{c}{s^2(s+\epsilon)^2} - \frac{2(c/s+\epsilon)}{(s+\epsilon)^3} < 0. \tag{640}$$

The active derivative is minimized at the maximal allowed $|z_i|$; the radial residual derivative is minimized at its maximal radius. Apply the integral Jacobian identity on the same convex product domain. □

The state-dependent lower bound explains why reset times deserve separate attention. A positive historical variance can maintain curvature of an active normalization field as ϵ decreases; immediately after resetting it, the lower bound again depends explicitly on ϵ. The residual may still limit the joint constant, so this observation alone does not prescribe an optimal regularization value.

Corollary 17 (PL contraction for the actual normalized momentum map)**.** *Under Theorem 15, additionally suppose, along the actual iterates,*

$$\tfrac{1}{2}\|\nabla F(w_t)\|^2 \geq \mu(F(w_t) - F^\star), \quad 0 < \mu\kappa\eta_t \leq 1, \quad a_t := \mathbb{E}[F(z_t) - F^\star], \tag{641}$$

$$A_t := G G_m \|C_{t-1}\|_{\mathrm{op}} + \tfrac{L}{2} G_m^2 \|C_{t-1}\|_{\mathrm{op}}^2, \tag{642}$$

$$Q_t := \mu\kappa\eta_t \mathbb{E} A_t + L\eta_t^2 \mathbb{E}\mathsf{J}_t^2 + L G_m \eta_t \mathbb{E}[\mathsf{J}_t \|C_{t-1}\|_{\mathrm{op}}] \tag{643}$$

$$+ \frac{M^2 \eta_t}{2\kappa} \mathbb{E}\widehat{D}_t^2 + G_z \mathbb{E}\|r_t\| + L\mathbb{E}\|r_t\|^2. \tag{644}$$

Then, with $q_t := 1 - \mu\kappa\eta_t$,

$$a_{t+1} \leq q_t a_t + Q_t, \tag{645}$$

$$a_{T+1} \leq \left(\prod_{t=1}^T q_t\right) a_1 + \sum_{t=1}^T \left(\prod_{s=t+1}^T q_s\right) Q_t. \tag{646}$$

For constant η and $Q_t \le K_0\eta^2 + Z_0\eta$,

$$a_{T+1} \le (1-\mu\kappa\eta)^T a_1 + \frac{K_0\eta + Z_0}{\mu\kappa}[1-(1-\mu\kappa\eta)^T]. \tag{647}$$

The forcing explicitly retains bias/symmetry, moment normalization, and transport/reset variation; the stated inequality on Q_t is an additional condition on these quantities.

Proof.

$$|F(w_t) - F(z_t)| \le \|\nabla F(w_t)\|\|w_t - z_t\| + \tfrac{L}{2}\|w_t - z_t\|^2 \le A_t, \tag{648}$$

$$-\tfrac{\kappa}{2}\eta_t\|\nabla F(w_t)\|^2 \le -\mu\kappa\eta_t(F(w_t) - F^\star) \tag{649}$$

$$\le -\mu\kappa\eta_t(F(z_t) - F^\star) + \mu\kappa\eta_t A_t. \tag{650}$$

Insert this in the one-step inequality in the proof of Theorem 15 and expand the scalar recurrence. □

A.12. Directional PAC–Bayes complexity and optimal flag variances

The previous Gaussian bound uses one curvature constant for all directions. An anisotropic envelope instead charges perturbations according to the curvature of each flag block. We optimize the variance within every block, then examine the effect of refining the flag and of tracking its oracle inexactly. These are comparisons of posterior bounds under the stated evaluation-loss assumptions.

Assumption 6 (Anisotropic curvature envelope)**.** *For the bounded evaluation loss $0 \le \ell(w,z) \le 1$, assume a sample-independent matrix $H_0 \succeq 0$ such that, globally in w, z,*

$$-H_0 \preceq \nabla_w^2 \ell(w,z) \preceq H_0. \tag{651}$$

For $\mathcal{P}_j := I_m \otimes \Pi_j(U)$, set

$$d_j := mq_j, \quad C_j(U) := \operatorname{tr}(\mathcal{P}_j H_0), \quad \overline{h}_j(U) := C_j(U)/d_j, \quad s_j := \sigma_j^2 > 0, \tag{652}$$

$$\Sigma_U := \sum_j s_j \Pi_j(U), \quad \widetilde{\Sigma}_U := I_m \otimes \Sigma_U = \sum_j s_j \mathcal{P}_j, \quad c_{n,\delta} := \log(\Xi_n/\delta). \tag{653}$$

The curvature assumption concerns the bounded evaluation loss; it does not assert that unbounded training cross-entropy lies in $[0,1]$.

Theorem 16 (Optimized directional posterior)**.** *Under Assumption 6, fix $P = \mathcal{N}(0, \sigma_0^2 I_N)$ independently of the sample. With probability at least $1-\delta$, simultaneously for every learned W, U and every $\lambda > 0$,*

$$F_{\mathcal{D}}(W) \le F_S(W) + \frac{\|W\|_F^2}{2\lambda\sigma_0^2} + \frac{c_{n,\delta}}{\lambda} + \frac{\lambda}{8n} + \mathcal{C}_\lambda(U), \tag{654}$$

$$\mathcal{C}_\lambda(U) := \frac{1}{2\lambda}\sum_{j=1}^{K+1} d_j \log(1 + 2\lambda\sigma_0^2 \overline{h}_j(U)). \tag{655}$$

The optimal positive posterior variances in this bound are exactly

$$s_j^\star(U,\lambda) = \frac{\sigma_0^2}{1 + 2\lambda\sigma_0^2\overline{h}_j(U)}. \tag{656}$$

Proof. For $E \sim \mathcal{N}(0, \widetilde{\Sigma}_U)$, Taylor's formula gives

$$|\ell(W+E,z) - \ell(W,z) - \langle \nabla\ell(W,z), E\rangle| \le \tfrac{1}{2}E^\top H_0 E, \tag{657}$$

$$|L(Q) - F_{\mathcal{D}}(W)|, \quad |\widehat{L}_S(Q) - F_S(W)| \le \tfrac{1}{2}\operatorname{tr}(H_0\widetilde{\Sigma}_U) = \tfrac{1}{2}\sum_j C_j s_j. \tag{658}$$

On the single event of Theorem 13,

$$F_{\mathcal{D}}(W) \le F_S(W) + \sum_j C_j s_j + \sqrt{\frac{\|W\|_F^2/(2\sigma_0^2) + \frac{1}{2}\sum_j d_j[s_j/\sigma_0^2 - 1 - \log(s_j/\sigma_0^2)] + c_{n,\delta}}{2n}}, \tag{659}$$

$$\sqrt{a/(2n)} \le a/\lambda + \lambda/(8n) \quad (a \ge 0, \lambda > 0). \tag{660}$$

For each j, define the strictly convex scalar objective

$$J_j(s) := C_j s + \frac{d_j}{2\lambda}[s/\sigma_0^2 - 1 - \log(s/\sigma_0^2)], \tag{661}$$

$$J_j'(s) = C_j + \frac{d_j}{2\lambda}(\sigma_0^{-2} - s^{-1}), \quad J_j''(s) = \frac{d_j}{2\lambda s^2} > 0, \tag{662}$$

$$J_j'(s_j^\star) = 0, \quad J_j(s_j^\star) = \frac{d_j}{2\lambda}\log(1 + 2\lambda\sigma_0^2 C_j/d_j). \tag{663}$$

The event is uniform over all posteriors; optimization over data-dependent U, s_j, λ therefore requires no additional union bound. □

The optimal variance is smaller in blocks with greater average curvature: perturbations there have a larger loss cost. The logarithmic complexity results from balancing that cost against the KL penalty for shrinking variance. Here λ is a bound-optimization parameter; it is separate from the optimizer's learning rate and from weight-decay notation used earlier. The closed form optimizes this analytic envelope; an evaluated certificate can be tighter by using the inverse-KL bound before its scalar relaxation.

Proposition 25 (A strict benefit of posterior flag refinement)**.** *Suppose a block $\mathcal{P}$ of dimension d is refined into orthogonal blocks $\mathcal{P}_1, \dots, \mathcal{P}_J$, with*

$$\mathcal{P} = \sum_{j=1}^{J} \mathcal{P}_j, \quad d = \sum_j d_j, \quad h_j := \operatorname{tr}(\mathcal{P}_j H_0)/d_j, \quad \overline{h} := \sum_j (d_j/d) h_j. \tag{664}$$

Then

$$\sum_j \frac{d_j}{2\lambda}\log(1 + 2\lambda\sigma_0^2 h_j) \le \frac{d}{2\lambda}\log(1 + 2\lambda\sigma_0^2\overline{h}). \tag{665}$$

For $d_j > 0$, equality holds if and only if $h_1 = \cdots = h_J$. Thus refining a fixed flag can strictly improve the optimized posterior bound when the block curvatures differ. This concerns the posterior family, not an ordering of the trained optimizers' test errors. More quantitatively, let $h_- := \min_j h_j$, $h_+ := \max_j h_j$, $V_h := \sum_j (d_j/d)(h_j - \overline{h})^2$, and let $\mathcal{G}_\lambda$ be the coarse cost minus the refined cost. Then

$$\frac{d\lambda\sigma_0^4 V_h}{(1 + 2\lambda\sigma_0^2 h_+)^2} \le \mathcal{G}_\lambda \le \frac{d\lambda\sigma_0^4 V_h}{(1 + 2\lambda\sigma_0^2 h_-)^2}. \tag{666}$$

The two sides agree to leading order as the block curvatures coalesce.

Proof.

$$f(h) := \log(1 + 2\lambda\sigma_0^2 h), \qquad f''(h) = -\frac{(2\lambda\sigma_0^2)^2}{(1 + 2\lambda\sigma_0^2 h)^2} < 0, \tag{667}$$

$$\sum_j (d_j/d) f(h_j) \le f(\sum_j (d_j/d) h_j). \tag{668}$$

For the quantitative statement apply Taylor's formula at $\overline{h}$. The weighted linear term cancels, and $-f''$ lies between its values at h_+ and h_-. Multiplication by $d/(2\lambda)$ gives the two-sided strong-concavity remainder. Continuity of f'' proves the local sharpness. □

Refinement permits different variances where a coarser posterior required a shared one. Strict concavity makes this flexibility useful precisely when the subblocks have unequal average curvature. Whether a covariance-trained flag discovers such directions is a separate alignment question. The next result controls the change in complexity caused by tracking a chosen oracle inexactly, without assuming that oracle minimizes the curvature cost.

Proposition 26 (Tracking error in the directional complexity)**.** *Partition $H_0 \in \mathbb{R}^{md\times md}$ into $d \times d$ row-coordinate blocks H_{ab}, and define $A_0 := \sum_{a=1}^{m} H_{aa} \succeq 0$. Then Set $s_A := \lambda_{\max}(A_0) - \lambda_{\min}(A_0)$. Then*

$$C_j(U) = \operatorname{tr}(\Pi_j(U)A_0), \tag{669}$$

$$|\mathcal{C}_\lambda(U) - \mathcal{C}_\lambda(V)| \le \sigma_0^2 s_A \sqrt{d}\, d_P(U,V). \tag{670}$$

In the regular oracle chart, let $\widehat{x}_t^\star$ and $x_t^\star$ be the oracles for $\widehat{C}_t$ and C_t, respectively, and $U_t = \psi(x_t)$. Then

$$|\mathcal{C}_\lambda(U_t) - \mathcal{C}_\lambda(\psi(x_t^\star))| \le \sigma_0^2 s_A \sqrt{d}\, J_P \left(\|x_t - \widehat{x}_t^\star\| + \frac{2J_\psi}{\mu_\phi} \sum_k \omega_k \sqrt{r_k}\epsilon_{k,t} \right). \tag{671}$$

Proof. Centering by $c_A := (\lambda_{\max}(A_0) + \lambda_{\min}(A_0))/2$ costs nothing because every projector difference is traceless. The trace/nuclear norm duality and the exact overlap identity give For $g_j(c) := d_j(2\lambda)^{-1}\log(1 + 2\lambda\sigma_0^2 c/d_j)$,

$$0 \le g_j'(c) = \frac{\sigma_0^2}{1 + 2\lambda\sigma_0^2 c/d_j} \le \sigma_0^2, \tag{672}$$

$$|C_j(U) - C_j(V)| \le \|A_0 - c_A I\|_{\mathrm{op}} \|\Pi_j(U) - \Pi_j(V)\|_* \tag{673}$$

$$\le \tfrac{s_A}{2}\sqrt{2q_j}\|\Pi_j(U) - \Pi_j(V)\|_F, \tag{674}$$

$$\sum_{j=1}^{K+1} \|\Pi_j(U) - \Pi_j(V)\|_F^2 \le 2 d_P(U,V)^2, \tag{675}$$

$$\sum_j \sqrt{2q_j}\|\Pi_j(U) - \Pi_j(V)\|_F \le \sqrt{2\sum_j q_j}\sqrt{2d_P(U,V)^2} = 2\sqrt{d}\, d_P(U,V). \tag{676}$$

Apply the triangle inequality and Lemma 4 to the two oracle branches. In particular $A_0 = cI$ gives zero orientation cost exactly. The bound depends on spectral spread, removing an isotropic component that cannot distinguish flags. □

Proposition 27 (A sample-independent anisotropic flag prior)**.** *Let a fixed prior flag V have projectors Π_l^0 and ranks q_l^0, and let*

$$\Lambda_0 := \sum_l \tau_l^2 \Pi_l^0 \succ 0, \quad P_V := \mathcal{N}(0, I_m \otimes \Lambda_0), \quad Q_U := \mathcal{N}(\operatorname{vec} W, I_m \otimes \Sigma_U), \tag{677}$$

$$T_{lj} := \operatorname{tr}(\Pi_l^0 \Pi_j(U)), \quad T_{lj} \ge 0, \quad \sum_j T_{lj} = q_l^0, \quad \sum_l T_{lj} = q_j. \tag{678}$$

Then the exact orientation-dependent cost is

$$\mathrm{KL}(Q_U \| P_V) = \tfrac{1}{2}\operatorname{tr}(W\Lambda_0^{-1}W^\top) + \tfrac{m}{2}\left[\sum_{l,j} \frac{\sigma_j^2}{\tau_l^2} T_{lj} - d + \sum_l q_l^0 \log \tau_l^2 - \sum_j q_j \log \sigma_j^2 \right]. \tag{679}$$

A data-dependent V is admissible if it is learned from an independent pilot sample S_0 and the bound is evaluated on a disjoint iid sample S_1 of size n_1:

$$S_0 \perp S_1, \quad V = V(S_0), \tag{680}$$

$$\mathcal{E}(S_0) := \left\{ \begin{array}{l} \forall Q \ll P_{V(S_0)} : \\ \mathrm{kl}(\widehat{L}_{S_1}(Q) \| L(Q)) \le \dfrac{\mathrm{KL}(Q\|P_{V(S_0)}) + \log(\Xi_{n_1}/\delta)}{n_1} \end{array} \right\}, \tag{681}$$

$$\mathbb{P}_{S_1|S_0}(\mathcal{E}(S_0)) \ge 1 - \delta. \tag{682}$$

The posterior may use both samples; the empirical term in this bound uses S_1.

Proof.

$$\operatorname{tr}(\Lambda_0^{-1}\Sigma_U) = \sum_{l,j} \tau_l^{-2}\sigma_j^2 \operatorname{tr}(\Pi_l^0\Pi_j), \tag{683}$$

$$T_{lj} = \operatorname{tr}(\Pi_j\Pi_l^0\Pi_j) = \|\Pi_l^0\Pi_j\|_F^2 \geq 0, \tag{684}$$

$$\log\det\Lambda_0 - \log\det\Sigma_U = \sum_l q_l^0 \log\tau_l^2 - \sum_j q_j \log\sigma_j^2. \tag{685}$$

Substitution in the Gaussian density-ratio identity proves the first assertion. Conditioning on S_0 makes the prior independent of S_1; apply Theorem 13 conditionally, then integrate over S_0. □

B. Expanded Convergence and Generalization Bridge

All definitions, hypotheses, and proofs are in Appendix A. The central question is how accurately learned gradient subspaces affect an optimizer whose moments and normalizers also evolve. We follow the chain from covariance estimation to flag tracking, adaptive-state perturbations, and parameter descent. For generalization, we examine both sensitivity to one training example and the complexity of a distribution around the learned parameters. These routes expose different assumptions and complementary terms.

Write $P_k = U_{:r_k}U_{:r_k}^\top$, $\Pi_k = P_k - P_{k-1}$, $\Phi(U) = \sum_k \omega_k \operatorname{tr}(P_kC_k)$, and $d_P^2(U,V) = \sum_k \|P_k(U) - P_k(V)\|_F^2$. The key geometric reduction is the overlap coupling $T_{ja} = \operatorname{tr}(\Pi_j(U)\Pi_a(V))$: Lemma 1 gives $d_P^2(U,V) = \sum_{j,a} |j-a| T_{ja}$ exactly. A misplaced direction is charged only for the prefix boundaries it crosses. This replaces repeated triangle inequalities and yields the sharp factor K in the assignment-distance comparison. Hoffman–Kruskal integrality then resolves the commuting optimum.

$$[C_k, C_l] = 0, \quad C_k = Q\operatorname{diag}(\lambda_{ki})Q^\top, \quad b_{ji} = \sum_{k\geq j} \omega_k\lambda_{ki}, \tag{686}$$

$$\max_{[U]} \Phi(U) = \max_{\substack{x\geq 0,\ \sum_j x_{ji}=1 \\ \sum_i x_{ji}=q_j}} \sum_{j,i} b_{ji}x_{ji} = \max_{\substack{x\in\{0,1\}^{(K+1)\times d} \\ \sum_j x_{ji}=1,\ \sum_i x_{ji}=q_j}} \sum_{j,i} b_{ji}x_{ji}. \tag{687}$$

Theorem 1 gives (687); Proposition 6 gives

$$\exists\{C_k\}: \quad [C_k,C_l]=0, \quad \operatorname{grad}\Phi(U_0)=0, \quad \operatorname{Hess}\Phi(U_0) \prec 0, \quad \Phi(U_0) < \max_U \Phi(U). \tag{688}$$

The assignment formula characterizes the global optimum, while the counterexample shows why local ascent requires a basin assumption even when the covariance matrices commute. The following tracking bounds therefore describe a regular local oracle branch. Under the regular-chart hypotheses of Assumption 1, Lemma 4 and Theorem 3 imply

$$-L_\phi I \preceq \nabla^2 f_C \preceq -\mu_\phi I, \quad \|\widehat{C}_k - C_k\|_{\mathrm{op}} \leq \epsilon_k \Longrightarrow \|x^\star_{\widehat{C}} - x^\star_C\| \leq \frac{2J_\psi}{\mu_\phi}\sum_k \omega_k\sqrt{r_k}\epsilon_k, \tag{689}$$

$$\sqrt{E_{s+1}} \leq q\sqrt{E_s} + b_s, \quad q<1 \Longrightarrow \sqrt{E_s} \leq q^s\sqrt{E_0} + \sum_{j<s} q^{s-1-j}b_j. \tag{690}$$

Here b_s is the L^2 norm of the combined forcing. The optimal quadratic contraction is $q_\star = (L_\phi - \mu_\phi)/(L_\phi + \mu_\phi)$; centered noise has variance gain $\xi^2/(1-q^2)$, while persistent drift has RMS gain $1/(1-q)$. We first isolate a predictable positive preconditioner and retain the exact algorithmic discrepancy as an error term. This yields a transparent descent bound; the analysis of current-gradient normalization below removes that predictability requirement under explicit noise and state-variation controls. For L-smooth F, let $h_t = \nabla F(w_t)$, $g_t = h_t + b_t + \zeta_t$, $\mathbb{E}_t\zeta_t = 0$, $\mathbb{E}_t\|\zeta_t\|^2 \leq \sigma_t^2$, and

$$w_{t+1} = w_t - \eta_t(B_tg_t + e_t), \quad B_t \in \mathcal{F}_{t-1}, \quad mI \preceq B_t \preceq MI, \tag{691}$$

$$\varepsilon_t = e_t + B_tb_t, \quad S_T = \sum_{t<T}\eta_t, \quad \mathsf{v}_t = \operatorname{tr}(B_t^2\Sigma_t). \tag{692}$$

Theorems 5 and 6 yield

$$\eta_t \le 1/(4LM), \quad \Delta_0 := \mathbb{E}F(w_0) - F_{\inf}, \tag{693}$$

$$\tfrac{1}{2}\|\nabla F\|^2 \ge \mu(F - F^\star) \quad \text{(PL case)}. \tag{694}$$

$$\frac{\sum_{t<T}\eta_t\mathbb{E}\|h_t\|^2}{S_T} \le \frac{2\Delta_0}{mS_T} + \frac{2L\sum_{t<T}\eta_t^2\mathbb{E}\mathsf{v}_t}{mS_T} \tag{695}$$

$$+ \frac{2\sum_{t<T}\left[\eta_t\mathbb{E}\|\varepsilon_t\|^2_{B_t^{-1}} + L\eta_t^2\mathbb{E}\|\varepsilon_t\|^2\right]}{mS_T}, \tag{696}$$

$$\mathbb{E}(F(w_T) - F^\star) \le (1-\mu m\eta)^T\Delta_0 + \frac{L\eta\overline{v} + (m^{-1}+L\eta)E^2}{\mu m}. \tag{697}$$

Here $\Sigma_t = \mathbb{E}_t[\zeta_t\zeta_t^\top]$, $\sup_t \mathbb{E}\mathsf{v}_t \le \overline{v} \le M^2\sigma^2$, $\mathbb{E}\|\varepsilon_t\|^2 \le E^2$, and the PL step-size restrictions apply. The exact implemented discrepancy is expanded in Definition 8 and Proposition 10.

For neighboring samples $S \sim S^{(i)}$, let $z_t = (\delta w_t, \delta U_t, \delta Y_t, \delta M_t, \delta\sqrt{V_t})^\top$. Definition 11 and Theorems 9 and 10 give

$$z_{t+1} \preceq \mathcal{M}_t z_t + d_t I_t, \quad z_0 = 0, \quad \mathbb{P}(I_t = 1) = b_t/n, \tag{698}$$

$$|\mathbb{E}(F_{\mathcal{D}}(w_T) - F_S(w_T))| \le \epsilon_T^{\mathrm{stab}} \le \frac{L_0}{n}e_w\sum_{t<T} b_t(\mathcal{M}_{T-1}\cdots\mathcal{M}_{t+1})d_t. \tag{699}$$

For $0 \le \ell \le 1$, a sample-independent prior P, and any posterior $Q \ll P$, Theorem 13 also gives, simultaneously in Q with probability at least $1-\delta$,

$$L(Q) \le \mathrm{kl}_+^{-1}\left(\widehat{L}_S(Q), \frac{\mathrm{KL}(Q\|P) + \log(\Xi_n/\delta)}{n}\right). \tag{700}$$

Here Ξ_n is the exact Seeger–Maurer moment constant, at most $2\sqrt{n}$ for $n \ge 8$; retaining inverse KL avoids the Pinsker relaxation. Proposition 21 evaluates this complexity exactly for a full-rank Gaussian posterior aligned with the learned flag.

For the implemented current-gradient normalizer $\mathcal{T}_t(g) = B_t(g)g$, Theorem 14 replaces predictability of B_t by an explicit conditional symmetry defect:

$$b_t = \mathbb{E}_t g_t - \nabla F(w_t), \quad \delta_t = W_1(\mathcal{L}(\xi_t \mid \mathcal{F}_{t-1}), \mathcal{L}(-\xi_t \mid \mathcal{F}_{t-1})), \tag{701}$$

$$\widehat{D}_t = \|b_t\| + \|\mathbb{E}_t\mathcal{T}_t(\xi_t)\|/M \le \|b_t\| + \delta_t/2, \tag{702}$$

$$\langle\nabla F(w_t), \mathbb{E}_t\mathcal{T}_t(g_t)\rangle \ge \tfrac{\kappa}{2}\|\nabla F(w_t)\|^2 - \tfrac{M^2}{2\kappa}\widehat{D}_t^2. \tag{703}$$

Theorem 15 then controls the actual moment/reset/transport map by its observable variation budgets. Proposition 23 gives a closed-form nonzero stationary bias for asymmetric, unbiased noise.

Under the bounded-loss anisotropic-curvature hypothesis of Assumption 6, Theorem 16 gives

$$\mathcal{C}_\lambda(U) = \frac{1}{2\lambda}\sum_j mq_j\log(1 + 2\lambda\sigma_0^2\overline{h}_j(U)), \qquad (\sigma_j^2)^\star = \frac{\sigma_0^2}{1 + 2\lambda\sigma_0^2\overline{h}_j(U)}. \tag{704}$$

Proposition 25 proves a strict decrease in this optimized posterior cost when a refined block separates unequal average curvatures. It also bounds the improvement above and below by the weighted variance of those curvatures. The tracking sensitivity depends on the spectral spread of the partial trace of H_0, so isotropic curvature cancels exactly (Proposition 26). This last comparison identifies when additional posterior directions improve the bound. Translating it into an optimizer advantage also requires the learned flag and the resulting parameter trajectory to have the appropriate geometry.

C. Complete Matrices and Reproducibility Notes

The full PINN STRATA grid with $\epsilon = 10^{-6}$ gives Burgers $(0.066, 0.027, 0.428, 0.187, 0.454, 0.440)$ over lr/rank cells $(5e{-}3/32, 5e{-}3/64, 1e{-}3/32, 1e{-}3/64, 5e{-}4/32, 5e{-}4/64)$; Burgers-g2 $(0.049, 0.026, 0.445, 0.393, 0.472, 0.443)$;

and Poisson $(1.768, 2.932, 0.177, 0.187, 0.060, 0.718)$. The corresponding Single-Grassmann grid is Burgers $(1.947, 2.777, 0.084, 0.052, 0.335, 0.156)$, Burgers-g2 $(0.046, 0.026, 0.140, 0.144, 0.426, 0.411)$, and Poisson $(\mathrm{NaN}, \mathrm{NaN}, 0.973, \mathrm{NaN}, \mathrm{NaN}, 1.496)$. The CNN grid gives CIFAR-10 STRATA $(0.9328, 0.9401)$ and Single-Grassmann $(0.9349, 0.9403)$ for lr $(3e{-}4, 1e{-}3)$; CIFAR-100 STRATA $(0.6943, 0.7140)$ and Single-Grassmann $(0.7125, 0.7257)$. The MLP STRATA grid gives MNIST validation accuracies $(0.982, 0.960, 0.552)$ and Fashion $(0.901, 0.899, 0.897)$ over lr $(3e{-}4, 1e{-}3, 3e{-}3)$.

All formal training used seed 42. Raw data are in the project 'experiments/' and 'analysis/' directories, with independent verification scripts for the epoch-axis δ_{nest} analysis and state-budget calculations. The paper uses the epoch-axis interpretation as primary; the learn-progress axis is retained only as a caveat because it can overweight late bins for tasks that reach near-convergence early.

D. Additional Numerical Checks and Protocol Audit

Synthetic checks of the corrected theory. All synthetic random starts use a fixed generator initialized with seed 42; multiple starts here are landscape diagnostics, not additional benchmark seeds. For the explicit commuting example in Proposition 6, 100 random orthogonal starts followed by 2000 projected-ascent steps at step size 0.05 give 88 global solutions (objective 6) and 12 strict nonglobal local maxima (objective 5). Finite-difference second derivatives at the latter are approximately $(-1, -2, -1)$ in the three plane-rotation coordinates. This validates the counterexample, not a universal basin probability. For the basis-dependence example, with $\epsilon = 10^{-6}$ the identity-basis update is approximately $(0.999999, 0)$ and the rotated-basis update is $(1.414212, 0)$, a difference of 0.414213. A deterministic diagonal quadratic check with Hessian $\mathrm{diag}(1, 2, 4)$ and preconditioner $\mathrm{diag}(0.5, 1, 1.5)$ satisfies the descent inequality at every one of 1000 steps; squared gradient decreases from 21 to 8.98×10^{-7}. These checks exercise algebra and local behavior, not the stochastic assumptions of real neural training.

Recomputed selection audit. We independently recomputed terminal-100-epoch means from raw logs. Classification configurations are selected only by validation accuracy; the resulting Fashion-MNIST Single-Grassmann test accuracy is 0.9033, correcting the previous 0.9025. For PINNs, pure terminal PDE-loss selection chooses Poisson STRATA lr 10^{-3}, rank 64, with relative L_2 error 0.1874. The previously highlighted 0.0603 belongs to lr 5×10^{-4}, rank 32 and is not the minimum-PDE-loss configuration. It remains a diagnostic configuration, not our primary validation-selected endpoint. The selected Single-Grassmann Poisson configuration gives 2.0028; the grid also contains nonfinite outcomes. The new result supports a limited robustness observation rather than a general optimization advantage. The archived full grids and audited endpoint tables use the corrected selection.

The corrected endpoints are reported in Table 25.

A near-full-rank implementation boundary. The previous initialization allocated Y_k at width $r_k + p$ even when an orthonormal probe could have at most d columns. For $R < d < R + p$, this produced incompatible sketch shapes. We now cap both allocations at $\min(d, r_k + p)$ in STRATA and the Grassmann baselines. Finite one-step checks pass for an 80×65 matrix with $R = 64$, $p = 8$, including the one-block independent baseline. This boundary correction does not change archived configurations whose sketch widths already fit.

Independent-baseline residual audit. The archived Independent-Grassmann code accepted $a_\perp$ but omitted the residual update. New runs add the normalized residual on the true orthogonal complement of the independent subspaces' union, obtained from a rank-revealing SVD at refresh times. Its cached basis is included in state accounting. With one block, the corrected baseline matches STRATA for 60 float64 steps to maximum absolute error 8.9×10^{-16}. Archived independent-baseline results are not silently relabeled as results of the corrected algorithm; they cannot isolate nesting under matched residual treatment.

Proposal coverage. The original six broad proposal families must not be confused with the eight existing dataset rows. Existing completed results cover portions of vision, graph learning and scientific ML; the original archive has no completed language-pretraining, continual-learning or artificial-distribution-shift family. The November revision launches an isolated training server queue for full-data Rotated-MNIST continual learning and shifts, Split-CIFAR-100 replay, CIFAR distribution shifts and WikiText-103 GPT-2 experiments. New queue entries, smoke tests and partial trajectories are not counted as completed results. Generated Darcy/FNO and additional citation-graph adapters are separate from existing archived evidence.

Completion of the full proposal additionally requires the remaining image, continual-learning, corpus and ablation matrix; no missing cell is replaced with an inferred result.

E. New Long-Horizon Runs: Completion Ledger

Only runs that completed their prescribed budget are listed. All configurations use seed 42; no statistical significance is inferred. Endpoints are means over the terminal 100 epochs or evaluation units. These are individual configurations, not test-selected optima. GPT-2 language units are predefined ten-step units, not corpus epochs; the separate WikiText-2 sequence runs use complete corpus epochs.

Table 8. Completed configurations. Full model, rank, base method, ablation, runtime and source paths are retained in the accompanying completion matrix. Accuracy is higher-is-better; PPL and relative L2 are lower-is-better.

ID	Task	Method	Metric	lr	Units	Val	Test
0	rotated_mnist_cl	strata	acc	3e-04	1000	0.57360	0.57655
1	rotated_mnist_cl	single	acc	3e-04	1000	0.56580	0.57109
70	wikitext103	strata	PPL	3e-04	1000	143.79571	144.54240
2	rotated_mnist_cl	adamw	acc	3e-04	1000	0.62196	0.62193
5	rotated_mnist_cl	strata	acc	1e-03	1000	0.56372	0.56741
6	rotated_mnist_cl	single	acc	1e-03	1000	0.54107	0.54505
172	Cora	strata	acc	3e-04	1000	0.68296	0.70120
173	Cora	single	acc	3e-04	1000	0.67532	0.69317
174	Cora	adamw	acc	3e-04	1000	0.67128	0.71734
3	rotated_mnist_cl	galore	acc	3e-04	1000	0.59439	0.59691
292	tinyimagenet	strata	acc	3e-04	1000	0.40812	0.39914
238	imagenet100	single	acc	3e-04	1000	0.75966	0.73214
294	tinyimagenet	single	acc	3e-04	1000	0.41744	0.41126
175	Cora	galore	acc	3e-04	1000	0.76050	0.77180
176	Cora	independent	acc	3e-04	1000	0.66796	0.69555
177	Cora	strata	acc	1e-03	1000	0.67186	0.69636
178	Cora	single	acc	1e-03	1000	0.68538	0.71643
179	Cora	adamw	acc	1e-03	1000	0.67248	0.71538
180	Cora	galore	acc	1e-03	1000	0.71760	0.74274
181	Cora	independent	acc	1e-03	1000	0.65686	0.67054
182	CiteSeer	strata	acc	3e-04	1000	0.57568	0.56758
222	rotated_mnist_cl	strata	acc	3e-04	1000	0.56998	0.57354
408	wine_red	strata	acc	3e-04	1000	0.74786	0.70776
488	wikitext2_words	strata	PPL	3e-04	1000	115.81714	108.71847
183	CiteSeer	single	acc	3e-04	1000	0.51166	0.50525
409	wine_red	single	acc	3e-04	1000	0.74714	0.70175
184	CiteSeer	adamw	acc	3e-04	1000	0.58050	0.57365
410	wine_red	adamw	acc	3e-04	1000	0.75709	0.71022
411	wine_red	galore	acc	3e-04	1000	0.73915	0.71851
185	CiteSeer	galore	acc	3e-04	1000	0.62548	0.62643
412	wine_red	independent	acc	3e-04	1000	0.74573	0.70614
186	CiteSeer	independent	acc	3e-04	1000	0.60210	0.57178
413	wine_red	strata	acc	1e-03	1000	0.76068	0.68421
187	CiteSeer	strata	acc	1e-03	1000	0.56290	0.54614
414	wine_red	single	acc	1e-03	1000	0.76940	0.68829

ID	Task	Method	Metric	lr	Units	Val	Test
188	CiteSeer	single	acc	1e-03	1000	0.58402	0.59175
415	wine_red	adamw	acc	1e-03	1000	0.75641	0.69803
416	wine_red	galore	acc	1e-03	1000	0.73504	0.69298
417	wine_red	independent	acc	1e-03	1000	0.76923	0.69096
189	CiteSeer	adamw	acc	1e-03	1000	0.55422	0.55906
428	adult	strata	acc	3e-04	1000	0.81558	0.81486
190	CiteSeer	galore	acc	1e-03	1000	0.60380	0.59303
191	CiteSeer	independent	acc	1e-03	1000	0.60204	0.59399
518	wine_red	strata	acc	3e-04	1000	0.75214	0.70175
192	PubMed	strata	acc	3e-04	1000	0.69348	0.67515
519	wine_red	strata	acc	1e-03	1000	0.75214	0.69298
520	wine_red	strata	acc	3e-04	1000	0.75641	0.71491
521	wine_red	strata	acc	1e-03	1000	0.75214	0.69311
193	PubMed	single	acc	3e-04	1000	0.70666	0.67902
522	wine_red	strata	acc	3e-04	1000	0.76496	0.70175
194	PubMed	adamw	acc	3e-04	1000	0.67232	0.66846
523	wine_red	strata	acc	1e-03	1000	0.77778	0.70175
195	PubMed	galore	acc	3e-04	1000	0.75898	0.75801
524	wine_red	strata	acc	3e-04	1000	0.73671	0.71930
525	wine_red	strata	acc	1e-03	1000	0.78632	0.70175
526	wine_red	strata	acc	3e-04	1000	0.74786	0.71053
196	PubMed	independent	acc	3e-04	1000	0.57778	0.56541
527	wine_red	strata	acc	1e-03	1000	0.74786	0.68833
528	wine_red	strata	acc	3e-04	1000	0.73932	0.70614
197	PubMed	strata	acc	1e-03	1000	0.71792	0.71623
529	wine_red	strata	acc	1e-03	1000	0.75214	0.68127
530	wine_red	strata	acc	3e-04	1000	0.74786	0.71053
198	PubMed	single	acc	1e-03	1000	0.66728	0.65871
531	wine_red	strata	acc	1e-03	1000	0.73944	0.68421
532	wine_red	strata	acc	3e-04	1000	0.74684	0.71491
199	PubMed	adamw	acc	1e-03	1000	0.70412	0.69061
533	wine_red	strata	acc	1e-03	1000	0.75534	0.70711
200	PubMed	galore	acc	1e-03	1000	0.72228	0.70575
534	wine_red	strata	acc	3e-04	1000	0.74786	0.71491
535	wine_red	strata	acc	1e-03	1000	0.76068	0.70175
201	PubMed	independent	acc	1e-03	1000	0.74190	0.74151
536	wine_red	strata	acc	3e-04	1000	0.73932	0.71298
537	wine_red	strata	acc	1e-03	1000	0.76329	0.69096
202	Cora	strata	acc	3e-04	1000	0.65480	0.67123
538	adult	strata	acc	3e-04	1000	0.82061	0.81588
203	Cora	single	acc	3e-04	1000	0.68356	0.70138
204	Cora	adamw	acc	3e-04	1000	0.66464	0.67840
205	Cora	galore	acc	3e-04	1000	0.74424	0.76033
206	Cora	independent	acc	3e-04	1000	0.67734	0.69344
207	Cora	strata	acc	1e-03	1000	0.67370	0.68975
539	adult	strata	acc	1e-03	1000	0.80477	0.81382
208	Cora	single	acc	1e-03	1000	0.68036	0.69609

ID	Task	Method	Metric	lr	Units	Val	Test
209	Cora	adamw	acc	1e-03	1000	0.65556	0.67565
210	Cora	galore	acc	1e-03	1000	0.67564	0.70496
429	adult	single	acc	3e-04	1000	0.81353	0.81487
211	Cora	independent	acc	1e-03	1000	0.63766	0.65550
212	Darcy-generated	strata	relL2	3e-04	2000	0.18722	0.19899
540	adult	strata	acc	3e-04	1000	0.81325	0.81136
430	adult	adamw	acc	3e-04	1000	0.81828	0.81723
541	adult	strata	acc	1e-03	1000	0.80801	0.81149
431	adult	galore	acc	3e-04	1000	0.82256	0.82160
542	adult	strata	acc	3e-04	1000	0.81189	0.81374
213	Darcy-generated	single	relL2	3e-04	2000	0.16107	0.18230
432	adult	independent	acc	3e-04	1000	0.81152	0.81171
433	adult	strata	acc	1e-03	1000	0.81400	0.81111
543	adult	strata	acc	1e-03	1000	0.80778	0.81480
214	Darcy-generated	adamw	relL2	3e-04	2000	0.14744	0.15944
215	Darcy-generated	galore	relL2	3e-04	2000	0.18957	0.20337
544	adult	strata	acc	3e-04	1000	0.82029	0.81986
434	adult	single	acc	1e-03	1000	0.80007	0.81104
216	Darcy-generated	independent	relL2	3e-04	2000	0.16984	0.18200
217	Darcy-generated	strata	relL2	1e-03	2000	0.12678	0.13637
218	Darcy-generated	single	relL2	1e-03	2000	0.11738	0.12830
219	Darcy-generated	adamw	relL2	1e-03	2000	0.10621	0.11464
220	Darcy-generated	galore	relL2	1e-03	2000	0.12789	0.13988
221	Darcy-generated	independent	relL2	1e-03	2000	0.11664	0.12661
223	rotated_mnist_cl	strata	acc	1e-03	1000	0.57962	0.58171
224	rotated_mnist_cl	strata	acc	3e-04	1000	0.57108	0.57635
225	rotated_mnist_cl	strata	acc	1e-03	1000	0.56744	0.56916
226	rotated_mnist_cl	strata	acc	3e-04	1000	0.59567	0.59354
227	rotated_mnist_cl	strata	acc	1e-03	1000	0.58941	0.59460
228	rotated_mnist_cl	strata	acc	3e-04	1000	0.52993	0.53637
229	rotated_mnist_cl	strata	acc	1e-03	1000	0.52214	0.52569
230	rotated_mnist_cl	strata	acc	3e-04	1000	0.56862	0.57415
231	rotated_mnist_cl	strata	acc	1e-03	1000	0.55904	0.56346
232	rotated_mnist_cl	strata	acc	3e-04	1000	0.54933	0.55402
435	adult	adamw	acc	1e-03	1000	0.80611	0.81447
436	adult	galore	acc	1e-03	1000	0.81108	0.81365
437	adult	independent	acc	1e-03	1000	0.80609	0.81304
448	wine_white	strata	acc	3e-04	1000	0.76511	0.72423
449	wine_white	single	acc	3e-04	1000	0.73595	0.73171
450	wine_white	adamw	acc	3e-04	1000	0.75858	0.72037
451	wine_white	galore	acc	3e-04	1000	0.75425	0.72597
452	wine_white	independent	acc	3e-04	1000	0.74510	0.73892
453	wine_white	strata	acc	1e-03	1000	0.73983	0.70879
454	wine_white	single	acc	1e-03	1000	0.72550	0.72482
455	wine_white	adamw	acc	1e-03	1000	0.75162	0.70362
456	wine_white	galore	acc	1e-03	1000	0.73979	0.72023
457	wine_white	independent	acc	1e-03	1000	0.73475	0.71593

ID	Task	Method	Metric	lr	Units	Val	Test
468	covertype	strata	acc	3e-04	1000	0.94863	0.94766
469	covertype	single	acc	3e-04	1000	0.95147	0.95204
470	covertype	adamw	acc	3e-04	1000	0.95556	0.95387
471	covertype	galore	acc	3e-04	1000	0.95282	0.95273
472	covertype	independent	acc	3e-04	1000	0.95002	0.94874
489	wikitext2_words	single	PPL	3e-04	1000	117.79782	110.28693
545	adult	strata	acc	1e-03	1000	0.80988	0.81318
546	adult	strata	acc	3e-04	1000	0.81490	0.81471
547	adult	strata	acc	1e-03	1000	0.80696	0.81131
548	adult	strata	acc	3e-04	1000	0.81109	0.81579
549	adult	strata	acc	1e-03	1000	0.80963	0.81092
550	adult	strata	acc	3e-04	1000	0.81698	0.81576
551	adult	strata	acc	1e-03	1000	0.81956	0.81710
552	adult	strata	acc	3e-04	1000	0.80891	0.81459
553	adult	strata	acc	1e-03	1000	0.80542	0.81182
554	adult	strata	acc	3e-04	1000	0.81613	0.82046
555	adult	strata	acc	1e-03	1000	0.80512	0.81190
556	adult	strata	acc	3e-04	1000	0.81252	0.81545
557	adult	strata	acc	1e-03	1000	0.80587	0.81024
608	PubMed	strata	acc	3e-04	1000	0.66432	0.63141
609	PubMed	single	acc	3e-04	1000	0.67700	0.65567
610	PubMed	adamw	acc	3e-04	1000	0.72004	0.70140
611	PubMed	galore	acc	3e-04	1000	0.76750	0.75915
612	PubMed	independent	acc	3e-04	1000	0.67412	0.65001
613	PubMed	strata	acc	1e-03	1000	0.73198	0.71742
614	PubMed	single	acc	1e-03	1000	0.72020	0.71208
615	PubMed	adamw	acc	1e-03	1000	0.72312	0.70445
616	PubMed	galore	acc	1e-03	1000	0.76286	0.75407
617	PubMed	independent	acc	1e-03	1000	0.72084	0.68782
618	PubMed	strata	acc	3e-04	1000	0.72014	0.69601
619	PubMed	single	acc	3e-04	1000	0.73342	0.71160
620	PubMed	adamw	acc	3e-04	1000	0.73420	0.71654
621	PubMed	galore	acc	3e-04	1000	0.77330	0.76626
622	PubMed	independent	acc	3e-04	1000	0.74344	0.73025
623	PubMed	strata	acc	1e-03	1000	0.74076	0.71174
638	PubMed	strata	acc	3e-04	1000	0.69376	0.67773
639	PubMed	strata	acc	1e-03	1000	0.71302	0.70196
640	PubMed	strata	acc	3e-04	1000	0.69346	0.67923
641	PubMed	strata	acc	1e-03	1000	0.68728	0.68762
642	PubMed	strata	acc	3e-04	1000	0.70624	0.70345
643	PubMed	strata	acc	1e-03	1000	0.71646	0.71188
644	PubMed	strata	acc	3e-04	1000	0.64888	0.63121
645	PubMed	strata	acc	1e-03	1000	0.70986	0.69363
646	PubMed	strata	acc	3e-04	1000	0.69820	0.67599
647	PubMed	strata	acc	1e-03	1000	0.69942	0.70164
648	PubMed	strata	acc	3e-04	1000	0.69988	0.67823
649	PubMed	strata	acc	1e-03	1000	0.67162	0.67233

ID	Task	Method	Metric	lr	Units	Val	Test
650	PubMed	strata	acc	3e-04	1000	0.69730	0.67639
651	PubMed	strata	acc	1e-03	1000	0.69448	0.67971
652	PubMed	strata	acc	3e-04	1000	0.66102	0.64627
653	PubMed	strata	acc	1e-03	1000	0.71310	0.70829
654	PubMed	strata	acc	3e-04	1000	0.68724	0.66675
655	PubMed	strata	acc	1e-03	1000	0.70572	0.69553
656	PubMed	strata	acc	3e-04	1000	0.68404	0.67246
657	PubMed	strata	acc	1e-03	1000	0.70994	0.70900
658	PubMed	strata	acc	3e-04	1000	0.69918	0.67856
659	PubMed	single	acc	3e-04	1000	0.70280	0.68041
660	PubMed	independent	acc	3e-04	1000	0.69246	0.67143
661	PubMed	strata	acc	1e-03	1000	0.65368	0.65201
662	PubMed	single	acc	1e-03	1000	0.70510	0.68944
663	PubMed	independent	acc	1e-03	1000	0.72100	0.72332
664	PubMed	strata	acc	3e-04	1000	0.68556	0.66021
665	PubMed	single	acc	3e-04	1000	0.69188	0.68123
666	PubMed	independent	acc	3e-04	1000	0.66610	0.65012
667	PubMed	strata	acc	1e-03	1000	0.75018	0.73383
668	PubMed	single	acc	1e-03	1000	0.62142	0.61951
669	PubMed	independent	acc	1e-03	1000	0.67426	0.65143
71	wikitext103	single	PPL	3e-04	1000	139.27219	139.67201
72	wikitext103	adamw	PPL	3e-04	1000	80.10692	79.99655
73	wikitext103	galore	PPL	3e-04	1000	239.92938	239.15308
74	wikitext103	independent	PPL	3e-04	1000	140.89543	140.80677
75	wikitext103	strata	PPL	1e-03	1000	312.79283	313.84985
76	wikitext103	single	PPL	1e-03	1000	279.10226	275.62732
77	wikitext103	adamw	PPL	1e-03	1000	301.44893	295.99329
78	wikitext103	galore	PPL	1e-03	1000	193.51594	192.19865
79	wikitext103	independent	PPL	1e-03	1000	252.92951	251.62580
378	core50	strata	acc	3e-04	1000	0.11000	0.08769
380	core50	single	acc	3e-04	1000	0.10997	0.08582
10	rotated_mnist_shift	strata	acc	3e-04	1000	0.97642	0.97570
11	rotated_mnist_shift	single	acc	3e-04	1000	0.97686	0.97896
12	rotated_mnist_shift	adamw	acc	3e-04	1000	0.97908	0.97986
13	rotated_mnist_shift	galore	acc	3e-04	1000	0.97713	0.97617
14	rotated_mnist_shift	independent	acc	3e-04	1000	0.97564	0.97674
15	rotated_mnist_shift	strata	acc	1e-03	1000	0.97718	0.97615
16	rotated_mnist_shift	single	acc	1e-03	1000	0.97214	0.97358
17	rotated_mnist_shift	adamw	acc	1e-03	1000	0.98314	0.98057
18	rotated_mnist_shift	galore	acc	1e-03	1000	0.98003	0.97958
19	rotated_mnist_shift	independent	acc	1e-03	1000	0.97449	0.97422
50	cifar100_class_shift	strata	acc	3e-04	1000	0.53488	0.53486
51	cifar100_class_shift	single	acc	3e-04	1000	0.53675	0.54195
52	cifar100_class_shift	adamw	acc	3e-04	1000	0.57668	0.55842
53	cifar100_class_shift	galore	acc	3e-04	1000	0.55069	0.53056
54	cifar100_class_shift	independent	acc	3e-04	1000	0.53668	0.53372
55	cifar100_class_shift	strata	acc	1e-03	1000	0.53702	0.53092

ID	Task	Method	Metric	lr	Units	Val	Test
56	cifar100_class_shift	single	acc	1e-03	1000	0.57694	0.57370
57	cifar100_class_shift	adamw	acc	1e-03	1000	0.47818	0.46563
58	cifar100_class_shift	galore	acc	1e-03	1000	0.54574	0.56323
59	cifar100_class_shift	independent	acc	1e-03	1000	0.51951	0.52140
60	cifar10_label_shift	strata	acc	3e-04	1000	0.72311	0.69377
61	cifar10_label_shift	single	acc	3e-04	1000	0.73559	0.71973
62	cifar10_label_shift	adamw	acc	3e-04	1000	0.74820	0.74254
63	cifar10_label_shift	galore	acc	3e-04	1000	0.71905	0.72506
64	cifar10_label_shift	independent	acc	3e-04	1000	0.72782	0.72025
65	cifar10_label_shift	strata	acc	1e-03	1000	0.76913	0.75012
66	cifar10_label_shift	single	acc	1e-03	1000	0.79489	0.76650
67	cifar10_label_shift	adamw	acc	1e-03	1000	0.81155	0.80212
68	cifar10_label_shift	galore	acc	1e-03	1000	0.81096	0.79996
69	cifar10_label_shift	independent	acc	1e-03	1000	0.78062	0.76964
233	rotated_mnist_cl	strata	acc	1e-03	1000	0.56065	0.56390
234	rotated_mnist_cl	strata	acc	3e-04	1000	0.58203	0.58595
235	rotated_mnist_cl	strata	acc	1e-03	1000	0.58910	0.59227
418	wine_red	strata	acc	3e-04	1000	0.78205	0.71500
419	wine_red	single	acc	3e-04	1000	0.76795	0.69702
420	wine_red	adamw	acc	3e-04	1000	0.75214	0.71491
421	wine_red	galore	acc	3e-04	1000	0.75214	0.71053
422	wine_red	independent	acc	3e-04	1000	0.73944	0.69298
423	wine_red	strata	acc	1e-03	1000	0.74423	0.71539
424	wine_red	single	acc	1e-03	1000	0.74786	0.69298
425	wine_red	adamw	acc	1e-03	1000	0.75214	0.72232
426	wine_red	galore	acc	1e-03	1000	0.75214	0.73246
427	wine_red	independent	acc	1e-03	1000	0.75641	0.70175
438	adult	strata	acc	3e-04	1000	0.81177	0.81615
439	adult	single	acc	3e-04	1000	0.80934	0.81281
440	adult	adamw	acc	3e-04	1000	0.81446	0.81909
441	adult	galore	acc	3e-04	1000	0.81655	0.81838
442	adult	independent	acc	3e-04	1000	0.81440	0.81506
443	adult	strata	acc	1e-03	1000	0.81208	0.81509
444	adult	single	acc	1e-03	1000	0.80320	0.81324
445	adult	adamw	acc	1e-03	1000	0.80935	0.81613
446	adult	galore	acc	1e-03	1000	0.80524	0.81637
447	adult	independent	acc	1e-03	1000	0.80916	0.81197
458	wine_white	strata	acc	3e-04	1000	0.74254	0.70875
459	wine_white	single	acc	3e-04	1000	0.74254	0.72310
460	wine_white	adamw	acc	3e-04	1000	0.72826	0.71736
461	wine_white	galore	acc	3e-04	1000	0.72698	0.71004
462	wine_white	independent	acc	3e-04	1000	0.75641	0.73888
463	wine_white	strata	acc	1e-03	1000	0.71243	0.71189
464	wine_white	single	acc	1e-03	1000	0.73427	0.72904
465	wine_white	adamw	acc	1e-03	1000	0.71893	0.71019
466	wine_white	galore	acc	1e-03	1000	0.71658	0.72146
467	wine_white	independent	acc	1e-03	1000	0.74545	0.72732

ID	Task	Method	Metric	lr	Units	Val	Test
473	covertype	strata	acc	1e-03	1000	0.96251	0.96224
474	covertype	single	acc	1e-03	1000	0.96334	0.96302
475	covertype	adamw	acc	1e-03	1000	0.96361	0.96350
476	covertype	galore	acc	1e-03	1000	0.96271	0.96271
477	covertype	independent	acc	1e-03	1000	0.96099	0.96147
478	covertype	strata	acc	3e-04	1000	0.96115	0.96145
479	covertype	single	acc	3e-04	1000	0.96317	0.96257
480	covertype	adamw	acc	3e-04	1000	0.96497	0.96448
481	covertype	galore	acc	3e-04	1000	0.96235	0.96187
482	covertype	independent	acc	3e-04	1000	0.96093	0.96071
483	covertype	strata	acc	1e-03	1000	0.96704	0.96644
484	covertype	single	acc	1e-03	1000	0.96763	0.96642
485	covertype	adamw	acc	1e-03	1000	0.96482	0.96494
486	covertype	galore	acc	1e-03	1000	0.96671	0.96597
487	covertype	independent	acc	1e-03	1000	0.96711	0.96755
490	wikitext2_words	adamw	PPL	3e-04	1000	181.51046	169.37738
491	wikitext2_words	galore	PPL	3e-04	1000	128.70924	121.75232
492	wikitext2_words	independent	PPL	3e-04	1000	117.97227	111.10367
493	wikitext2_words	strata	PPL	1e-03	1000	137.43132	130.28771
494	wikitext2_words	single	PPL	1e-03	1000	146.58642	136.84381
496	wikitext2_words	galore	PPL	1e-03	1000	141.68800	133.12894
497	wikitext2_words	independent	PPL	1e-03	1000	142.34695	133.91073
558	Cora	strata	acc	3e-04	1000	0.58672	0.55909
559	Cora	single	acc	3e-04	1000	0.58500	0.55900
560	Cora	adamw	acc	3e-04	1000	0.69592	0.70995
561	Cora	galore	acc	3e-04	1000	0.78062	0.78155
562	Cora	independent	acc	3e-04	1000	0.58212	0.55522
563	Cora	strata	acc	1e-03	1000	0.69102	0.69798
564	Cora	single	acc	1e-03	1000	0.70684	0.71909
565	Cora	adamw	acc	1e-03	1000	0.69450	0.72322
566	Cora	galore	acc	1e-03	1000	0.75262	0.76573
567	Cora	independent	acc	1e-03	1000	0.73068	0.73650
568	Cora	strata	acc	3e-04	1000	0.41448	0.40589
569	Cora	single	acc	3e-04	1000	0.41818	0.40877
570	Cora	adamw	acc	3e-04	1000	0.60468	0.62115
571	Cora	galore	acc	3e-04	1000	0.74026	0.75034
572	Cora	independent	acc	3e-04	1000	0.41246	0.41750
573	Cora	strata	acc	1e-03	1000	0.46380	0.46532
574	Cora	single	acc	1e-03	1000	0.65010	0.65061
575	Cora	adamw	acc	1e-03	1000	0.69744	0.70599
576	Cora	galore	acc	1e-03	1000	0.72298	0.74158
577	Cora	independent	acc	1e-03	1000	0.58934	0.56409
578	CiteSeer	strata	acc	3e-04	1000	0.48006	0.46952
579	CiteSeer	single	acc	3e-04	1000	0.43156	0.44258
580	CiteSeer	adamw	acc	3e-04	1000	0.51832	0.49720
581	CiteSeer	galore	acc	3e-04	1000	0.63558	0.61661
582	CiteSeer	independent	acc	3e-04	1000	0.43744	0.44473

ID	Task	Method	Metric	lr	Units	Val	Test
583	CiteSeer	strata	acc	1e-03	1000	0.53562	0.49199
584	CiteSeer	single	acc	1e-03	1000	0.54720	0.51555
585	CiteSeer	adamw	acc	1e-03	1000	0.60566	0.59270
586	CiteSeer	galore	acc	1e-03	1000	0.61904	0.59259
587	CiteSeer	independent	acc	1e-03	1000	0.54792	0.50296
588	CiteSeer	strata	acc	3e-04	1000	0.42874	0.39781
589	CiteSeer	single	acc	3e-04	1000	0.47720	0.45770
590	CiteSeer	adamw	acc	3e-04	1000	0.50564	0.50128
591	CiteSeer	galore	acc	3e-04	1000	0.65014	0.65124
592	CiteSeer	independent	acc	3e-04	1000	0.53174	0.52278
593	CiteSeer	strata	acc	1e-03	1000	0.56726	0.55747
594	CiteSeer	single	acc	1e-03	1000	0.56434	0.53980
595	CiteSeer	adamw	acc	1e-03	1000	0.57294	0.55302
596	CiteSeer	galore	acc	1e-03	1000	0.55940	0.53048
597	CiteSeer	independent	acc	1e-03	1000	0.59654	0.55836
598	CiteSeer	strata	acc	3e-04	1000	0.47550	0.45184
599	CiteSeer	single	acc	3e-04	1000	0.42926	0.43050
600	CiteSeer	adamw	acc	3e-04	1000	0.52690	0.50331
601	CiteSeer	galore	acc	3e-04	1000	0.63712	0.61499
602	CiteSeer	independent	acc	3e-04	1000	0.42150	0.43925
603	CiteSeer	strata	acc	1e-03	1000	0.51716	0.47508
604	CiteSeer	single	acc	1e-03	1000	0.53544	0.49874
605	CiteSeer	adamw	acc	1e-03	1000	0.60938	0.60848
606	CiteSeer	galore	acc	1e-03	1000	0.63520	0.61041
607	CiteSeer	independent	acc	1e-03	1000	0.55070	0.50420
624	PubMed	single	acc	1e-03	1000	0.75320	0.74308
625	PubMed	adamw	acc	1e-03	1000	0.75052	0.72323
626	PubMed	galore	acc	1e-03	1000	0.76438	0.75346
627	PubMed	independent	acc	1e-03	1000	0.75774	0.73659
628	PubMed	strata	acc	3e-04	1000	0.66512	0.63999
629	PubMed	single	acc	3e-04	1000	0.67972	0.64939
630	PubMed	adamw	acc	3e-04	1000	0.70236	0.68710
631	PubMed	galore	acc	3e-04	1000	0.75988	0.74449
632	PubMed	independent	acc	3e-04	1000	0.66546	0.64440
633	PubMed	strata	acc	1e-03	1000	0.68362	0.66341
634	PubMed	single	acc	1e-03	1000	0.71634	0.71067
635	PubMed	adamw	acc	1e-03	1000	0.73228	0.70880
636	PubMed	galore	acc	1e-03	1000	0.76184	0.74985
637	PubMed	independent	acc	1e-03	1000	0.69912	0.66932
670	PubMed	single	acc	3e-04	1000	0.71554	0.71640
671	PubMed	single	acc	1e-03	1000	0.73948	0.72923
672	PubMed	single	acc	3e-04	1000	0.69382	0.69613
673	PubMed	single	acc	1e-03	1000	0.62916	0.61564
674	PubMed	single	acc	3e-04	1000	0.69020	0.69950
675	PubMed	single	acc	1e-03	1000	0.72560	0.72434
676	ETTh1	strata	MSE	3e-04	1000	0.91637	0.89922
677	ETTh1	single	MSE	3e-04	1000	1.06275	0.88655

ID	Task	Method	Metric	lr	Units	Val	Test
678	ETTh1	adamw	MSE	1e-04	1000	1.28210	1.26977
679	ETTh1	galore	MSE	1e-04	1000	0.73812	0.76942
680	ETTh1	independent	MSE	3e-04	1000	1.04993	0.89897
681	ETTh1	strata	MSE	1e-03	1000	0.98299	0.88613
682	ETTh1	single	MSE	1e-03	1000	1.11589	0.87013
683	ETTh1	adamw	MSE	3e-04	1000	1.46292	1.31620
684	ETTh1	galore	MSE	3e-04	1000	1.19797	1.14226
685	ETTh1	independent	MSE	1e-03	1000	1.10118	0.83715
686	ETTh1	strata	MSE	3e-04	1000	0.94795	0.91651
687	ETTh1	single	MSE	3e-04	1000	1.07601	0.89658
688	ETTh1	adamw	MSE	1e-03	1000	1.24838	1.10378
689	ETTh1	galore	MSE	1e-03	1000	1.39158	1.75306
690	ETTh1	independent	MSE	3e-04	1000	1.04792	0.90163
691	ETTh1	strata	MSE	1e-03	1000	0.98292	0.87753
692	ETTh1	single	MSE	1e-03	1000	1.12002	0.88601
693	ETTh1	adamw	MSE	3e-03	1000	1.11725	0.92354
694	ETTh1	galore	MSE	3e-03	1000	1.43248	0.99045
695	ETTh1	independent	MSE	1e-03	1000	1.11775	0.80900
696	ETTh1	strata	MSE	3e-04	1000	1.15296	0.95096
697	ETTh1	single	MSE	3e-04	1000	1.16195	0.95721
698	ETTh1	adamw	MSE	1e-02	1000	0.94111	0.86679
699	ETTh1	galore	MSE	1e-02	1000	1.22093	0.84632
700	ETTh1	independent	MSE	3e-04	1000	1.22655	1.05103
701	ETTh1	strata	MSE	1e-03	1000	1.19479	1.05471
702	ETTh1	single	MSE	1e-03	1000	1.32485	1.02055
703	ETTh1	adamw	MSE	3e-02	1000	1.48635	1.11085
704	ETTh1	galore	MSE	3e-02	1000	1.32699	2.64401
705	ETTh1	independent	MSE	1e-03	1000	1.31802	0.88625
706	ETTh2	strata	MSE	3e-04	1000	1.07510	2.34749
707	ETTh2	single	MSE	3e-04	1000	0.82090	2.31532
708	ETTh2	adamw	MSE	1e-04	1000	1.50903	1.42992
709	ETTh2	galore	MSE	1e-04	1000	1.40733	2.52593
710	ETTh2	independent	MSE	3e-04	1000	1.44955	1.71429
711	ETTh2	strata	MSE	1e-03	1000	1.17938	3.03325
712	ETTh2	single	MSE	1e-03	1000	1.59519	5.85606
713	ETTh2	adamw	MSE	3e-04	1000	1.82006	3.47191
714	ETTh2	galore	MSE	3e-04	1000	1.61952	3.06158
715	ETTh2	independent	MSE	1e-03	1000	1.36697	2.28023
716	ETTh2	strata	MSE	3e-04	1000	0.95426	2.06459
717	ETTh2	single	MSE	3e-04	1000	0.80694	2.37970
718	ETTh2	adamw	MSE	1e-03	1000	2.29168	2.14275
719	ETTh2	galore	MSE	1e-03	1000	4.03384	2.89176
720	ETTh2	independent	MSE	3e-04	1000	1.57753	1.76863
721	ETTh2	strata	MSE	1e-03	1000	1.25061	3.24254
722	ETTh2	single	MSE	1e-03	1000	1.57979	5.53129
723	ETTh2	adamw	MSE	3e-03	1000	1.25122	2.16884
724	ETTh2	galore	MSE	3e-03	1000	3.55828	6.28914

ID	Task	Method	Metric	lr	Units	Val	Test
725	ETTh2	independent	MSE	1e-03	1000	1.31254	2.18874
726	ETTh2	strata	MSE	3e-04	1000	0.85177	1.74404
727	ETTh2	single	MSE	3e-04	1000	1.04042	2.14237
728	ETTh2	adamw	MSE	1e-02	1000	1.54196	3.15369
729	ETTh2	galore	MSE	1e-02	1000	1.00745	1.79170
730	ETTh2	independent	MSE	3e-04	1000	2.43952	1.86961
731	ETTh2	strata	MSE	1e-03	1000	1.18671	2.34007
732	ETTh2	single	MSE	1e-03	1000	1.27554	3.93957
733	ETTh2	adamw	MSE	3e-02	1000	1.54203	3.15372
734	ETTh2	galore	MSE	3e-02	1000	1.03937	3.25346
735	ETTh2	independent	MSE	1e-03	1000	2.60794	2.20963
736	ml100k	strata	RMSE	3e-04	1000	0.96273	0.96967
737	ml100k	single	RMSE	3e-04	1000	1.00523	1.00532
738	ml100k	adamw	RMSE	1e-04	1000	0.91533	0.93295
739	ml100k	galore	RMSE	1e-04	1000	0.94025	0.95778
740	ml100k	independent	RMSE	3e-04	1000	1.00694	1.01763
741	ml100k	strata	RMSE	1e-03	1000	1.09648	1.08972
742	ml100k	single	RMSE	1e-03	1000	1.15610	1.14820
743	ml100k	adamw	RMSE	3e-04	1000	1.00674	1.01382
744	ml100k	galore	RMSE	3e-04	1000	0.96247	0.97599
745	ml100k	independent	RMSE	1e-03	1000	1.16351	1.17252
746	ml100k	strata	RMSE	3e-04	1000	0.96273	0.96967
747	ml100k	single	RMSE	3e-04	1000	1.00523	1.00532
748	ml100k	adamw	RMSE	1e-03	1000	1.04344	1.04229
749	ml100k	galore	RMSE	1e-03	1000	1.15211	1.17608
750	ml100k	independent	RMSE	3e-04	1000	1.00694	1.01763
751	ml100k	strata	RMSE	1e-03	1000	1.09647	1.08972
752	ml100k	single	RMSE	1e-03	1000	1.15609	1.14820
753	ml100k	adamw	RMSE	3e-03	1000	1.04296	1.04202
754	ml100k	galore	RMSE	3e-03	1000	1.43743	1.44852
755	ml100k	independent	RMSE	1e-03	1000	1.16351	1.17252
756	ml100k	strata	RMSE	3e-04	1000	0.96257	0.96949
757	ml100k	single	RMSE	3e-04	1000	1.00521	1.00529
758	ml100k	adamw	RMSE	1e-02	1000	1.01296	1.01644
759	ml100k	galore	RMSE	1e-02	1000	1.85762	1.90548
760	ml100k	independent	RMSE	3e-04	1000	1.00684	1.01755
761	ml100k	strata	RMSE	1e-03	1000	1.09626	1.08954
762	ml100k	single	RMSE	1e-03	1000	1.15597	1.14808
763	ml100k	adamw	RMSE	3e-02	1000	1.00569	1.00898
764	ml100k	galore	RMSE	3e-02	1000	2.00236	2.06601
765	ml100k	independent	RMSE	1e-03	1000	1.16331	1.17230
766	ml1m	strata	RMSE	3e-04	1000	1.03092	1.05052
767	ml1m	single	RMSE	3e-04	1000	1.07528	1.09444
768	ml1m	adamw	RMSE	1e-04	1000	0.96096	0.98066
769	ml1m	galore	RMSE	1e-04	1000	0.89748	0.91712
770	ml1m	independent	RMSE	3e-04	1000	1.05630	1.08335
771	ml1m	strata	RMSE	1e-03	1000	1.24435	1.25975

ID	Task	Method	Metric	lr	Units	Val	Test
772	ml1m	single	RMSE	1e-03	1000	1.29115	1.30582
773	ml1m	adamw	RMSE	3e-04	1000	1.17433	1.19114
774	ml1m	galore	RMSE	3e-04	1000	0.98716	1.01147
775	ml1m	independent	RMSE	1e-03	1000	1.28009	1.30368
776	ml1m	strata	RMSE	3e-04	1000	1.03094	1.05054
777	ml1m	single	RMSE	3e-04	1000	1.07530	1.09446
778	ml1m	adamw	RMSE	1e-03	1000	1.52068	1.53384
779	ml1m	galore	RMSE	1e-03	1000	1.14201	1.18503
780	ml1m	independent	RMSE	3e-04	1000	1.05629	1.08335
781	ml1m	strata	RMSE	1e-03	1000	1.24435	1.25976
782	ml1m	single	RMSE	1e-03	1000	1.29116	1.30582
783	ml1m	adamw	RMSE	3e-03	1000	1.81062	1.82567
784	ml1m	galore	RMSE	3e-03	1000	1.37212	1.44711
785	ml1m	independent	RMSE	1e-03	1000	1.28009	1.30367
786	ml1m	strata	RMSE	3e-04	1000	1.03278	1.05213
787	ml1m	single	RMSE	3e-04	1000	1.07662	1.09574
788	ml1m	adamw	RMSE	1e-02	1000	1.67616	1.67385
789	ml1m	galore	RMSE	1e-02	1000	1.57440	1.68607
790	ml1m	independent	RMSE	3e-04	1000	1.05453	1.08146
791	ml1m	strata	RMSE	1e-03	1000	1.24467	1.26003
792	ml1m	single	RMSE	1e-03	1000	1.29148	1.30613
793	ml1m	adamw	RMSE	3e-02	1000	1.50056	1.50141
794	ml1m	galore	RMSE	3e-02	1000	1.56793	1.63850
795	ml1m	independent	RMSE	1e-03	1000	1.27905	1.30269
796	gas_drift	strata	acc	3e-04	1000	0.82265	0.81767
797	gas_drift	single	acc	3e-04	1000	0.85400	0.85375
798	gas_drift	adamw	acc	1e-04	1000	0.83539	0.83036
799	gas_drift	galore	acc	1e-04	1000	0.83006	0.81090
800	gas_drift	independent	acc	3e-04	1000	0.82289	0.81550
801	gas_drift	strata	acc	1e-03	1000	0.79489	0.80248
802	gas_drift	single	acc	1e-03	1000	0.85311	0.83380
803	gas_drift	adamw	acc	3e-04	1000	0.84425	0.82420
804	gas_drift	galore	acc	3e-04	1000	0.90191	0.90977
805	gas_drift	independent	acc	1e-03	1000	0.72418	0.70191
806	gas_drift	strata	acc	3e-04	1000	0.84972	0.84624
807	gas_drift	single	acc	3e-04	1000	0.84865	0.84787
808	gas_drift	adamw	acc	1e-03	1000	0.82752	0.81969
809	gas_drift	galore	acc	1e-03	1000	0.90496	0.90656
810	gas_drift	independent	acc	3e-04	1000	0.81555	0.80662
811	gas_drift	strata	acc	1e-03	1000	0.86518	0.85453
812	gas_drift	single	acc	1e-03	1000	0.84680	0.83885
813	gas_drift	adamw	acc	3e-03	1000	0.85438	0.82335
814	gas_drift	galore	acc	3e-03	1000	0.86965	0.86761
815	gas_drift	independent	acc	1e-03	1000	0.76659	0.75160
816	gas_drift	strata	acc	3e-04	1000	0.82474	0.81789
817	gas_drift	single	acc	3e-04	1000	0.84389	0.84120
818	gas_drift	adamw	acc	1e-02	1000	0.78638	0.74923

ID	Task	Method	Metric	lr	Units	Val	Test
819	gas_drift	galore	acc	1e-02	1000	0.78677	0.75995
820	gas_drift	independent	acc	3e-04	1000	0.82691	0.80659
821	gas_drift	strata	acc	1e-03	1000	0.81552	0.77587
822	gas_drift	single	acc	1e-03	1000	0.75290	0.75997
823	gas_drift	adamw	acc	3e-02	1000	0.20228	0.21588
824	gas_drift	galore	acc	3e-02	1000	0.77278	0.75453
825	gas_drift	independent	acc	1e-03	1000	0.72306	0.72491
826	gas_drift	strata	acc	3e-04	1000	0.95166	0.94171
827	gas_drift	single	acc	3e-04	1000	0.95459	0.94800
828	gas_drift	adamw	acc	1e-04	1000	0.93413	0.93237
829	gas_drift	galore	acc	1e-04	1000	0.92632	0.92101
830	gas_drift	independent	acc	3e-04	1000	0.96087	0.95010
831	gas_drift	strata	acc	1e-03	1000	0.96972	0.95956
832	gas_drift	single	acc	1e-03	1000	0.96936	0.96409
833	gas_drift	adamw	acc	3e-04	1000	0.96731	0.95725
834	gas_drift	galore	acc	3e-04	1000	0.96211	0.95326
835	gas_drift	independent	acc	1e-03	1000	0.96976	0.96059
836	gas_drift	strata	acc	3e-04	1000	0.95536	0.94627
837	gas_drift	single	acc	3e-04	1000	0.95500	0.94745
838	gas_drift	adamw	acc	1e-03	1000	0.97093	0.96001
839	gas_drift	galore	acc	1e-03	1000	0.97444	0.96232
840	gas_drift	independent	acc	3e-04	1000	0.96117	0.95059
841	gas_drift	strata	acc	1e-03	1000	0.97321	0.96721
842	gas_drift	single	acc	1e-03	1000	0.96649	0.96462
843	gas_drift	adamw	acc	3e-03	1000	0.97452	0.96536
844	gas_drift	galore	acc	3e-03	1000	0.97095	0.96146
845	gas_drift	independent	acc	1e-03	1000	0.97133	0.96156
846	gas_drift	strata	acc	3e-04	1000	0.96291	0.95493
847	gas_drift	single	acc	3e-04	1000	0.96230	0.95461
848	gas_drift	adamw	acc	1e-02	1000	0.96929	0.95992
849	gas_drift	galore	acc	1e-02	1000	0.95600	0.95324
850	gas_drift	independent	acc	3e-04	1000	0.96466	0.95530
851	gas_drift	strata	acc	1e-03	1000	0.97487	0.96506
852	gas_drift	single	acc	1e-03	1000	0.96116	0.95267
853	gas_drift	adamw	acc	3e-02	1000	0.45878	0.42244
854	gas_drift	galore	acc	3e-02	1000	0.95290	0.93774
855	gas_drift	independent	acc	1e-03	1000	0.97214	0.96343
4	rotated_mnist_cl	independent	acc	3e-04	1000	0.56890	0.57278
7	rotated_mnist_cl	adamw	acc	1e-03	1000	0.59213	0.59452
8	rotated_mnist_cl	galore	acc	1e-03	1000	0.57700	0.58166
9	rotated_mnist_cl	independent	acc	1e-03	1000	0.54958	0.55253
20	rotated_mnist_cl	strata	acc	3e-04	1000	0.85091	0.86012
21	rotated_mnist_cl	single	acc	3e-04	1000	0.86009	0.87170
22	rotated_mnist_cl	adamw	acc	3e-04	1000	0.88943	0.89805
23	rotated_mnist_cl	galore	acc	3e-04	1000	0.86630	0.87717
24	rotated_mnist_cl	independent	acc	3e-04	1000	0.85136	0.86388
25	rotated_mnist_cl	strata	acc	1e-03	1000	0.87586	0.88582

ID	Task	Method	Metric	lr	Units	Val	Test
26	rotated_mnist_cl	single	acc	1e-03	1000	0.85666	0.86967
27	rotated_mnist_cl	adamw	acc	1e-03	1000	0.89949	0.90720
28	rotated_mnist_cl	galore	acc	1e-03	1000	0.88049	0.88880
29	rotated_mnist_cl	independent	acc	1e-03	1000	0.85804	0.87140

F. Recovered Image Data and Extended Protocol

This extension fills previously missing image and continual-learning datasets. Acquired data, successful smoke tests, queued configurations, and completed formal results are distinct states; only completed full-budget runs enter Appendix E.

Recovered classification data. We recovered the public ImageNet-100 subset associated with contrastive multiview coding (Tian et al., 2020) from an existing source Arrow cache and copied it to training server. The reused mirror contains 126689 training images and 5000 official validation images across 100 classes, with shorter-side-160-pixel preprocessing already applied by the mirror. A fixed image-index/label hash allocates 114129/12560 of the original training images to training/internal validation; the 5000 official validation images are reserved as test. Direct Arrow reads preserve the cached image bytes. Tiny-ImageNet is recovered from an existing training server copy: the corresponding fixed split contains 90067 training, 9933 internal-validation and 10000 official-validation-as-test examples. The official unlabeled Tiny-ImageNet test partition is not used to invent test labels.

Reacquired continual-learning data. For DomainNet (Peng et al., 2019), we use the cleaned official six-domain release, in the fixed order clipart, infograph, painting, quickdraw, real, sketch. Ten percent by a predeclared path hash of each official training split forms internal validation; official test lists remain separate. For CORe50 (Lomonaco & Maltoni, 2017), we use the official RGB 128×128 data and the NC_inc/run0 filelists, comprising nine class-incremental batches. The official fixed test sessions are retained; labels and their batch assignment are recovered from the filelists. Thirty-frame clip groups within the training data are hash-selected for internal validation. The validated CORe50 manifest contains 107869 training, 12025 internal-validation and 44972 fixed-test images; its 30 configurations have been activated, but no completed CORe50 endpoint is claimed. The validated DomainNet manifest contains 368689 training, 41143 internal-validation and 176743 official-test images across 345 classes and six domains; formal training has started, while complete comparisons remain pending. These preparation rules are specified before seeing new optimizer results. Downloads are CRC-checked and their hashes retained; acquisition does not itself establish an experimental result.

Matched training and search. All new configurations use seed 42, 1000 total epochs and training from scratch. Classification covers ResNet-18, ResNet-50, ViT-S/16 and ConvNeXt-T; inputs are 224 pixels for ImageNet-100 and the native 64 pixels for Tiny-ImageNet. Continual experiments use ResNet-18 and 128-pixel inputs, with fine-tuning, replay or EWC. We compare five adaptive/projection optimizers at maximum rank 64 where applicable, with learning rates $\{3 \times 10^{-4}, 10^{-3}\}$. Classification additionally includes SGD-momentum at $\{0.03, 0.1\}$ and official single-device Muon with auxiliary Adam at $\{0.01, 0.02\}$ (auxiliary rate 0.03 times the Muon rate). Thus each method receives two predefined trials per setting; rates are method-scaled rather than forced to share numerically inappropriate units. The Muon source commit is pinned. Forward computation uses BF16 autocast with FP32 weights/state for all methods, and adaptive branches use $\epsilon = 10^{-6}$. The cosine learning-rate schedule restarts with warmup at each task boundary while flags and moments carry across tasks.

Replay uses a deterministic 2000-example buffer selected independently of outcomes, and EWC uses a diagonal batch-gradient-square Fisher proxy with coefficient 10. Reports retain per-task accuracy, forgetting, validation forward transfer, gradient norms, memory allocation and shared-server wall time. Timing under concurrent GPU use is descriptive rather than an isolated throughput comparison. The new image matrix contains 172 configurations; all require a validated manifest and a successful real-data smoke test before activation.

G. Full Loss Trajectories and Expanded Experimental Scope

Frozen plotting and selection protocol. Every plotted value comes from a retained JSON trajectory. The figure audit records source paths, SHA-256 digests, the selected metric, point counts, and the last observed abscissa. Classification selections use terminal-100 validation accuracy among completed 1000-epoch trials. The scientific panels restrict both

methods to rank 64 and choose within that rank using terminal-100 PDE residual; exact-solution errors never select the curve. Seed 42 is used throughout. Archived epoch indices and new one-based epochs are retained separately. Curves are not stretched to a common horizon, and unfinished traces do not enter endpoint or AUC comparisons. No confidence intervals are claimed.

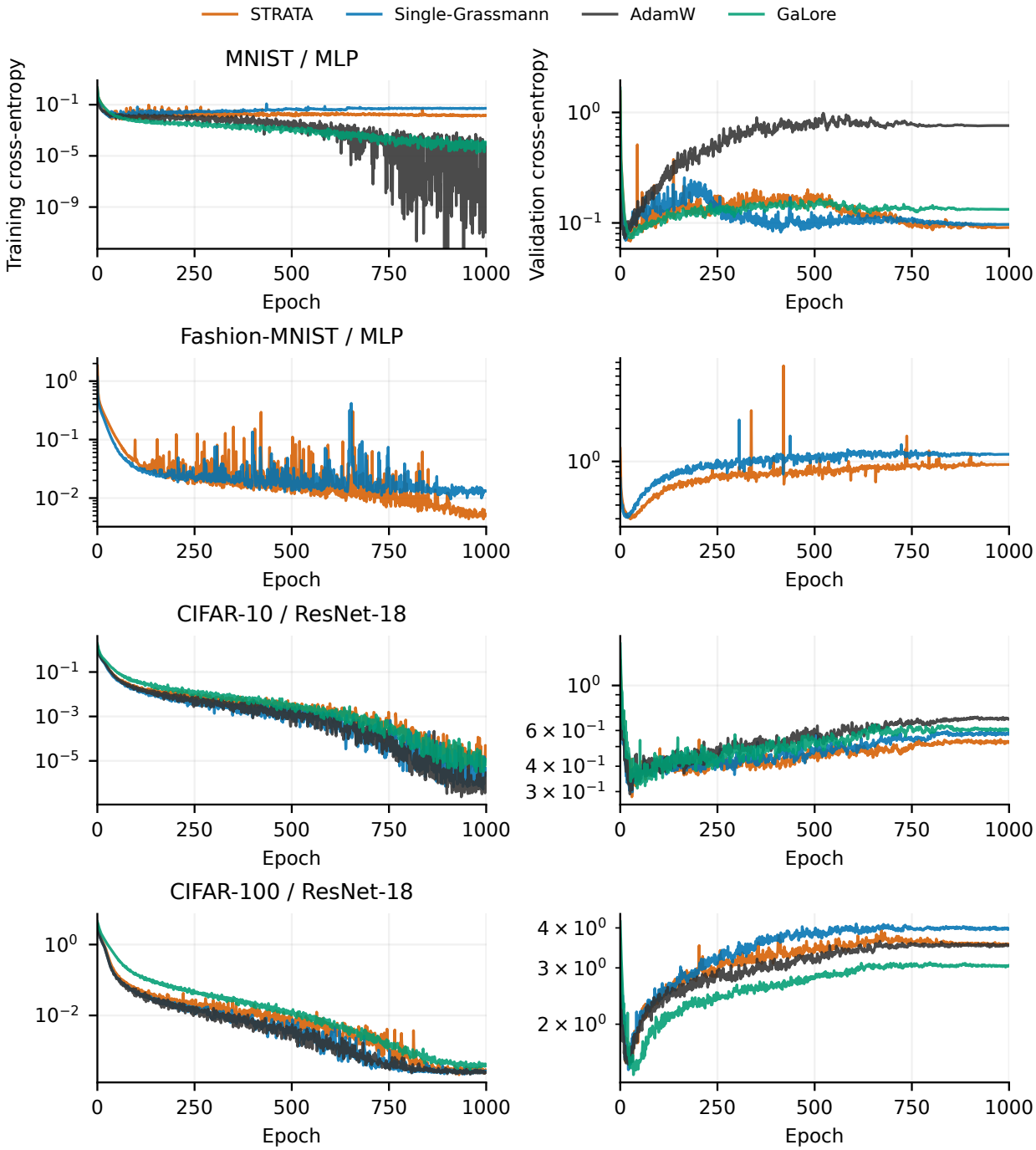


Figure 2. **Training and validation cross-entropy for the selected classifiers.** Each row is one dataset; left: training loss; right: held-out validation loss. The same selected run supplies both panels. A declining training loss with rising validation loss is retained. Missing Fashion-MNIST baselines are left absent. These are single-seed archived trajectories with unequal baseline search sizes.

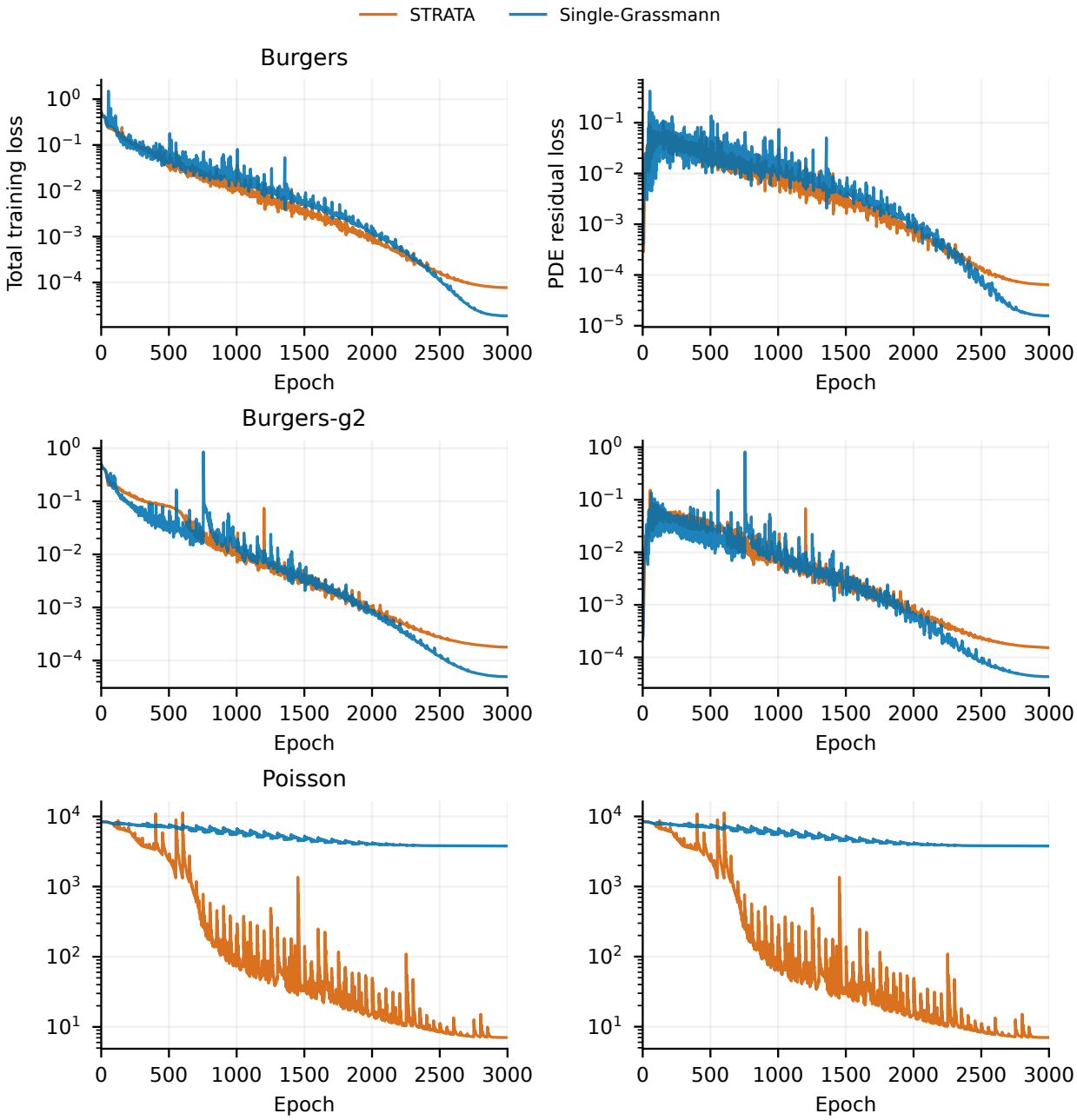


Figure 3. **Scientific-ML convergence at a strictly matched maximum rank of 64.** Left: total training loss; right: PDE residual loss. Exact-solution endpoint errors are tabulated separately. Rows: Burgers, Burgers-g2, Poisson. Each method's learning rate is selected by terminal PDE loss within rank 64. For Poisson this gives Single-Grassmann 2.04357 and STRATA 0.18741. The former differs from 2.00280, the rank-32 winner of the pooled-rank Single-Grassmann grid. Neither exact-solution error nor a favorable intermediate point selects a run.

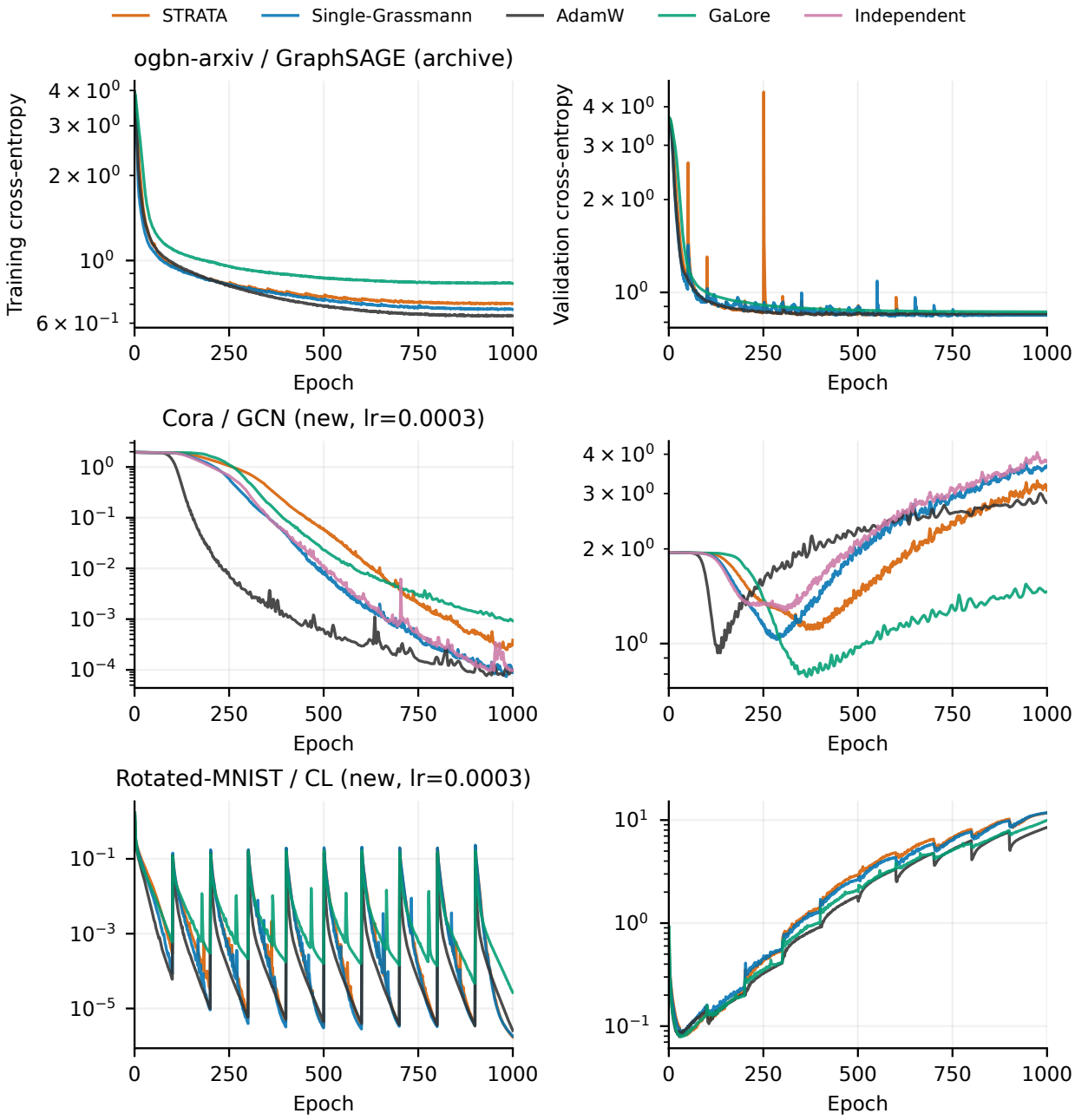


Figure 4. **Graph and continual-learning loss trajectories.** Left: training cross-entropy; right: validation cross-entropy. Rows show archived ogbn-arxiv, newly completed Cora/GCN configurations at the fixed rate 3×10^{-4}, and newly completed Rotated-MNIST fine-tuning configurations at that same fixed rate. In continual learning the held-out loss averages seen tasks, whose set grows at task boundaries; it is not the loss on an unchanging distribution. Only completed configurations are included in these panels.

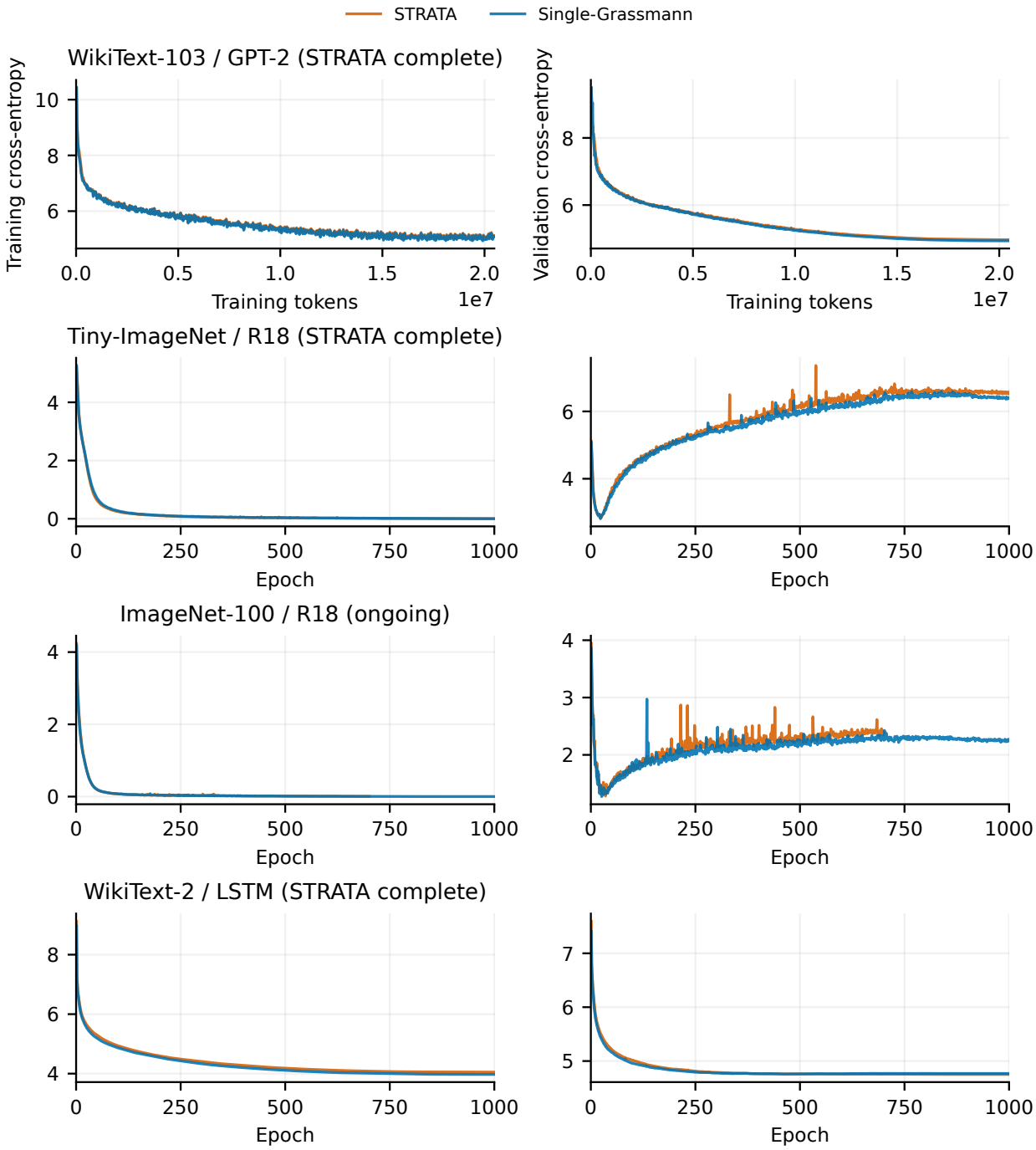


Figure 5. **Large-task trajectories and configuration completion status.** Left: training loss; right: validation loss. Rows show WikiText-103/GPT-2, Tiny-ImageNet/ResNet-18, ImageNet-100/ResNet-18, and WikiText-2/LSTM at rate 3×10^{-4}. GPT-2 uses actual training tokens; the other rows use observed epochs. Each WikiText-2 epoch covers the complete next-token training corpus. A title marks any completed STRATA configuration; missing methods and unfinished searches do not become completed comparison cohorts. Other traces stop at their last observed epoch. A missing method has no available trace at the snapshot. The different language tokenizations also prevent interpreting perplexity across the two language-model rows as a matched comparison. These traces do not establish a family-level advantage.

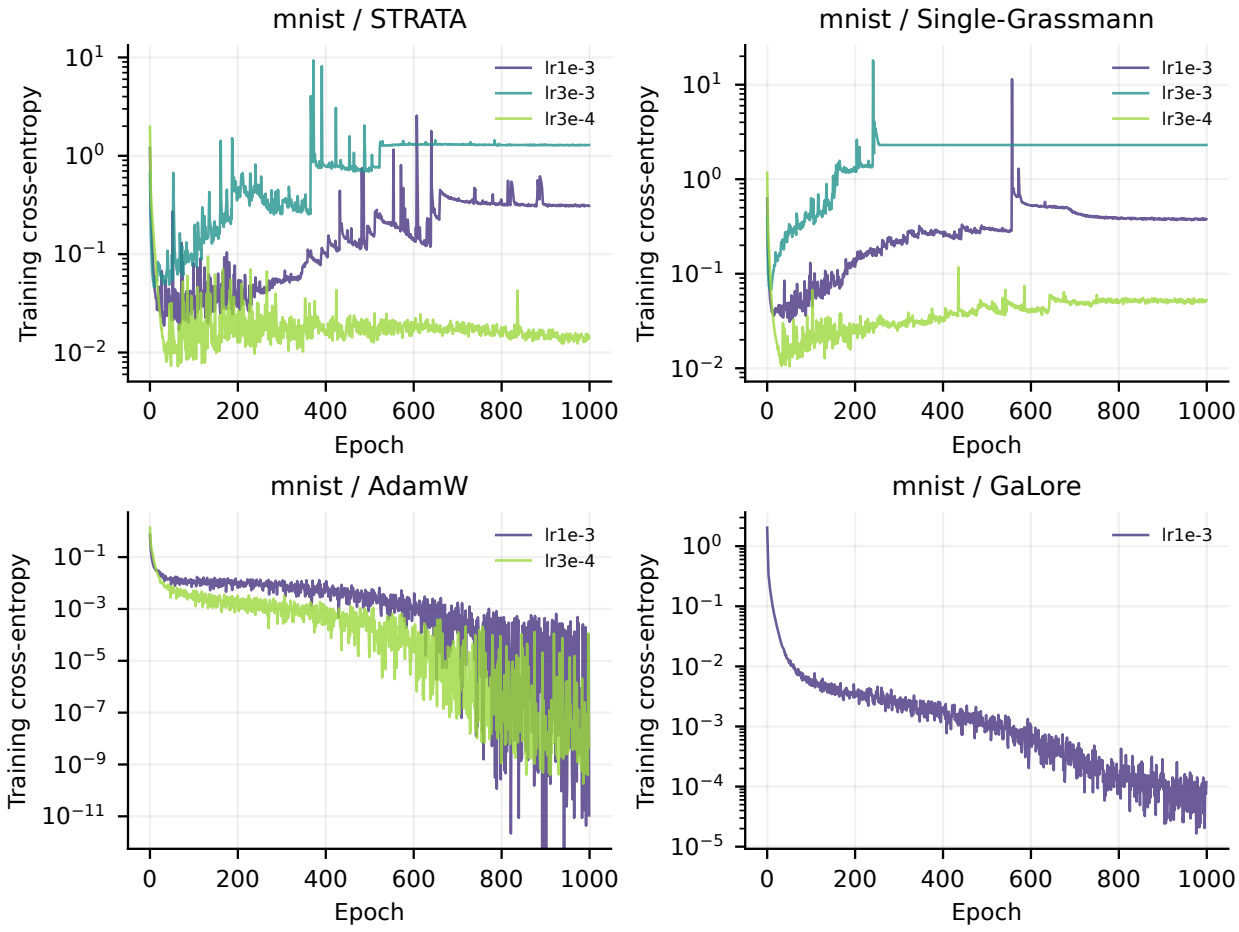


Figure 6. **All completed MNIST classifier-grid training losses.** Separate panels show methods; lines show all available completed learning-rate trials. Poor configurations are retained. A 397-epoch partial GaLore trial is excluded from the complete-grid view.

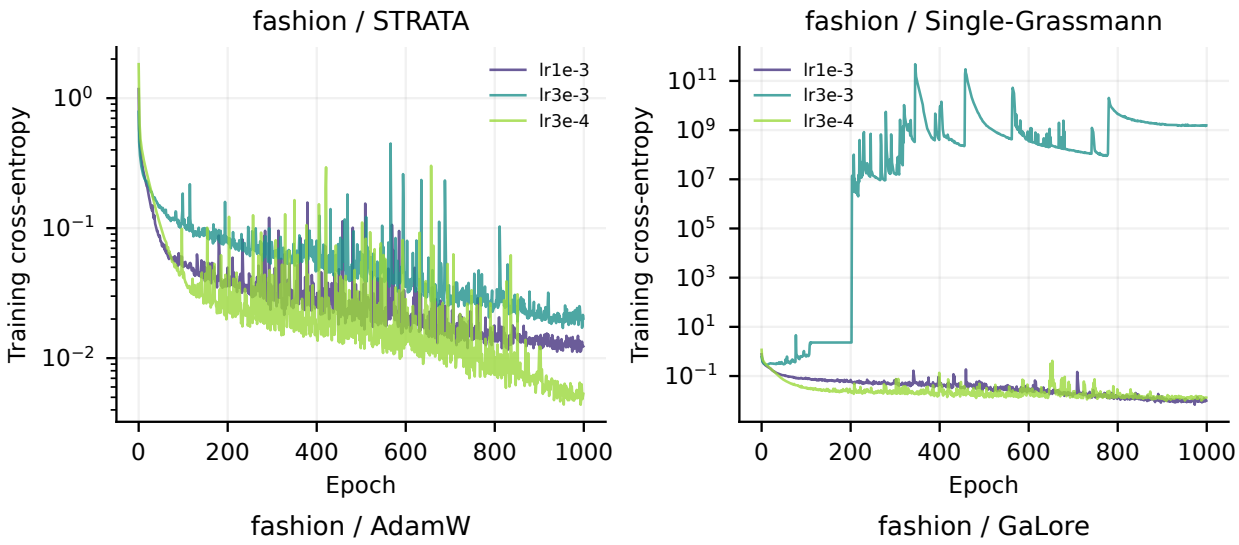


Figure 7. **All completed Fashion-MNIST classifier-grid training losses.** Missing archived AdamW/GaLore grids are absent. The unstable Single-Grassmann trial remains visible rather than being removed from the learning-rate matrix.

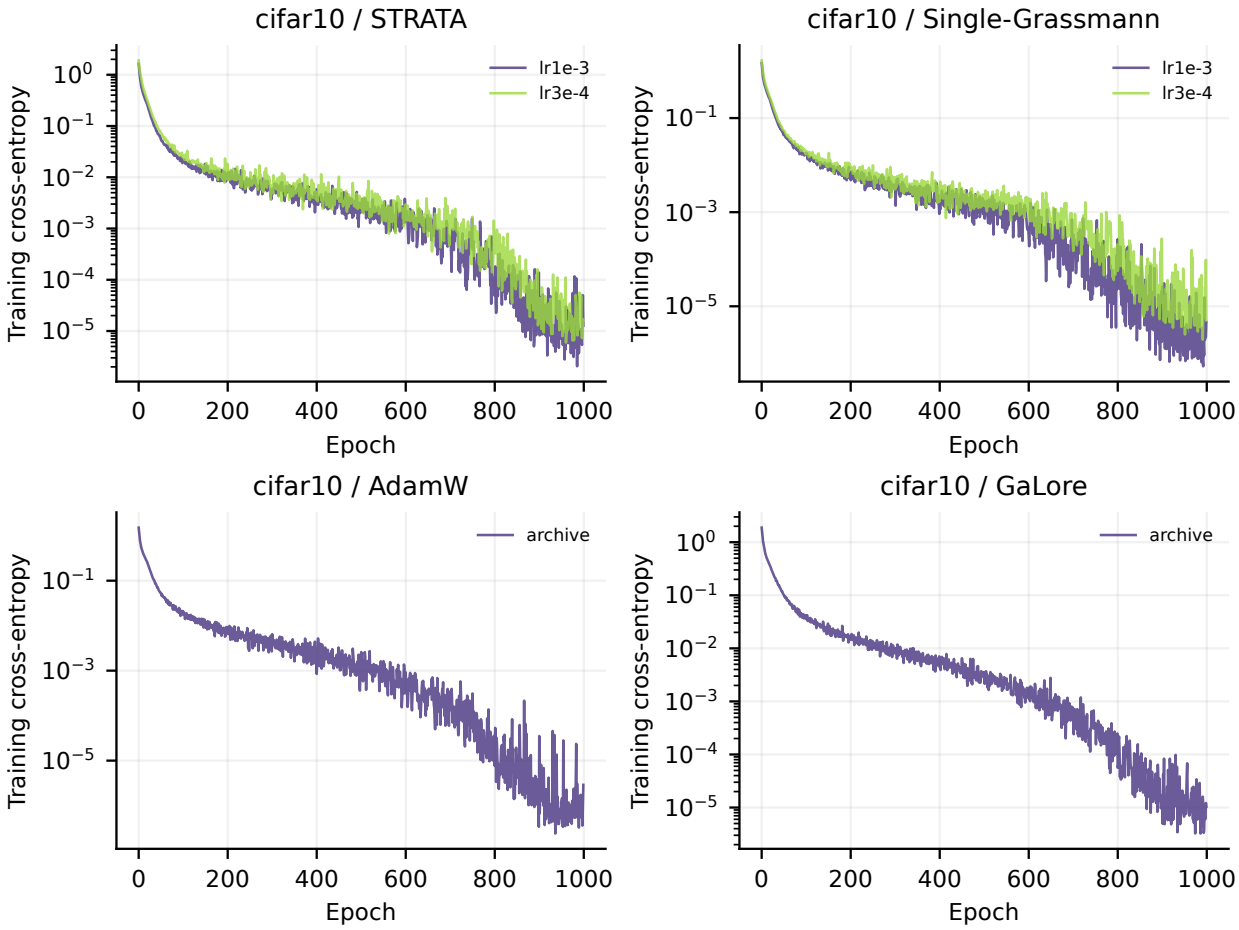


Figure 8. **All completed CIFAR-10 classifier-grid training losses.** STRATA/Single-Grassmann have two available rates; each archived AdamW/GaLore reference has one. These unequal grids are disclosed rather than represented as equal hyperparameter budgets.

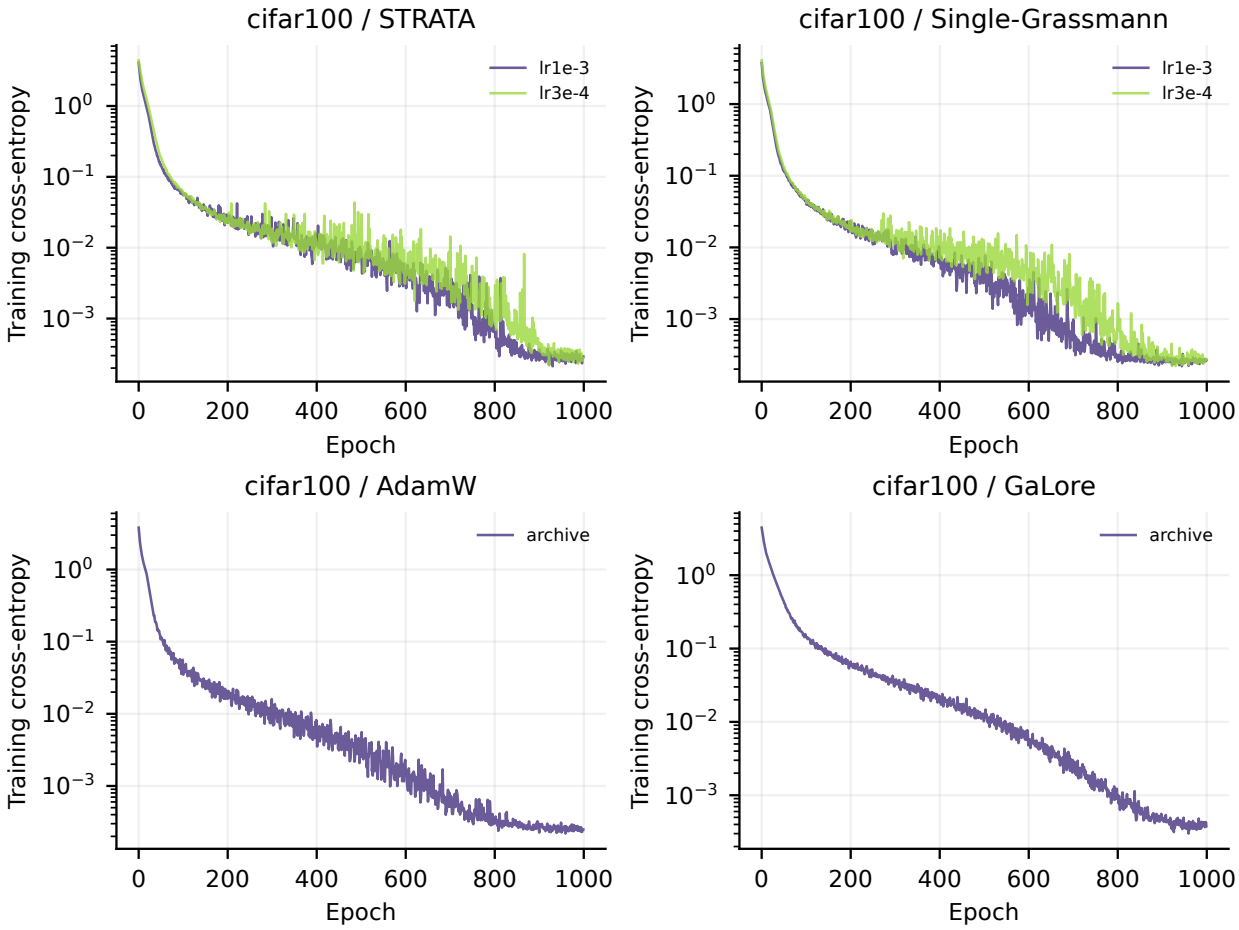


Figure 9. **All completed CIFAR-100 classifier-grid training losses.** The full training paths support optimization diagnostics while the held-out panels prevent interpreting a small terminal training loss as evidence of superior generalization.

Table 9. Selected classifier convergence diagnostics. AUC integrates raw training loss over the observed archived epoch indices (0–999); $g_{\log}$ is the mean logged gradient norm, not a claim to the exact full-objective gradient. Missing gradient norms were not recorded in those archives and are left absent. Configurations use the selection rule of Figure 35.

Dataset	Method	Loss AUC	Train loss	Val loss	$g_{\log}$
mnist	strata	21.9	0.0144	0.0917	4.92
mnist	single	40.8	0.0513	0.0969	15.8
mnist	adamw	6.13	7.5e-05	0.759	–
mnist	galore	7.21	7.76e-05	0.133	0.0133
fashion	strata	35.2	0.0059	0.939	0.639
fashion	single	32.4	0.0131	1.16	1.99
cifar10	strata	19.4	1.77e-05	0.527	0.00133
cifar10	single	16	4.69e-06	0.578	0.000361
cifar10	adamw	17.4	3.19e-06	0.688	–
cifar10	galore	27.3	1.81e-05	0.607	–
cifar100	strata	59	0.000286	3.56	0.00343
cifar100	single	49.2	0.000261	3.98	0.00111
cifar100	adamw	48.7	0.000259	3.53	–
cifar100	galore	101	0.000407	3.05	–

Breadth relative to the ManifoldFlow evaluation. The graph additions use custom 256-wide models: attention is single-head, SAGE-style aggregation uses the stated symmetric adjacency, and GCNII-style initial/identity mixing is fixed. These are controlled architecture variants, not reproductions of canonical published endpoints. The reference evaluation combines recurrent language models, tabular and image MLPs, a small Transformer, convolutional heads, component ablations, and scaling diagnostics. Our expanded matrix adds UCI Adult (Becker & Kohavi, 1996), Covertype (Blackard, 1998), red/white Wine Quality (Cortez et al., 2009) with 256- and 512-wide two-hidden-layer MLPs, and WikiText-2 (Merity et al., 2016) with two-layer LSTM, two-layer GRU and a two-block Transformer. All five adaptive/projection methods receive rates 3×10^{-4} and 10^{-3}. Every formal tabular and sequence trial is prescribed 1000 complete epochs. These 110 additional configurations enlarge planned coverage; only completed entries in Appendix E count as results.

Preprocessing and language budgets. Adult retains the official test split and hashes 10% of official training for validation. Wine uses separate red/white tasks with target quality ≥ 6; identical full rows share a fixed 70/15/15 hash split. Covertype uses all 581012 records with the same hash split rule. All feature means, scales, and categorical dictionaries are fitted on training only. The WikiText-2 vocabulary is fitted on its official training split, with minimum frequency two and at most 30000 words. Its frozen splits contain 2088628/217646/245569 tokens. Every new sequence epoch visits the complete next-token training corpus; recurrent state is reset between contiguous blocks. This differs explicitly from the separate GPT-2 fixed-ten-step evaluation units. All preprocessing hashes and source revisions are retained.

Mechanistic ablations. A further 40 trials compare ten STRATA variants on Adult and red Wine Quality at both rates, keeping the same data, architecture, and full horizon. The full method, K=1, independent subspaces and full AdamW controls are reused from the matched tabular cohorts. Equal versus heterogeneous (β_1, β_2) is a meaningful memory comparison: the heterogeneous variant uses $(.99, .9, .5)$ and $(.999, .99, .9)$, while the default has equal decays. The learned flag freezes after 200 updates in the freeze variant; this implementation also stops refresh-time variance resets, so it is a refresh-policy intervention rather than an isolated basis-motion change. The periodic-SVD variant refreshes a nested eigenspace chain of the pooled multi-timescale covariance; pooling changes its flag objective, so it is a baseline rather than a pure retraction-only intervention. Least-squares transport uses $M^+ = MU_{\text{old}}^\top U_{\text{new}}$ and is distinguished from both the implemented rotation and omission of rotation. Remaining variants equalize/reverse covariance memories, equalize coefficients, remove the residual, fix a random flag, or omit transport. Each new code path is smoke-tested before formal activation; smoke runs never count as completed evidence.

G.1. New completed breadth-cohort loss curves

Every displayed cohort contains five completed full-budget methods. Learning rates are selected on validation only after both predeclared rates have completed for every method; otherwise a completed common-rate cohort is labeled fixed-rate. Each curve is traced to its complete log in the accompanying audit.

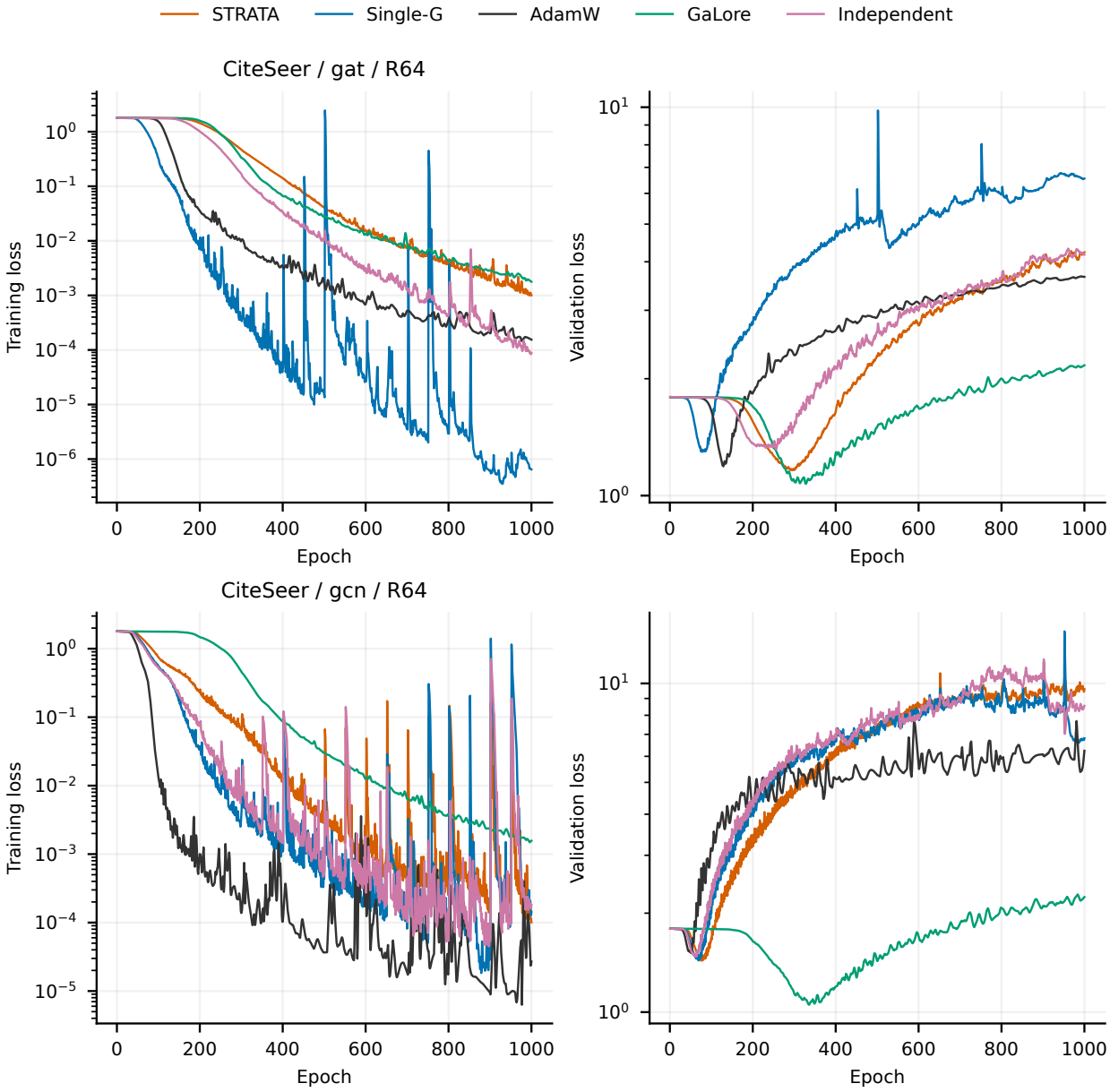


Figure 10. **Completed breadth-cohort training and validation losses.** CiteSeer/gat: two-rate validation selection; CiteSeer/gcn: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

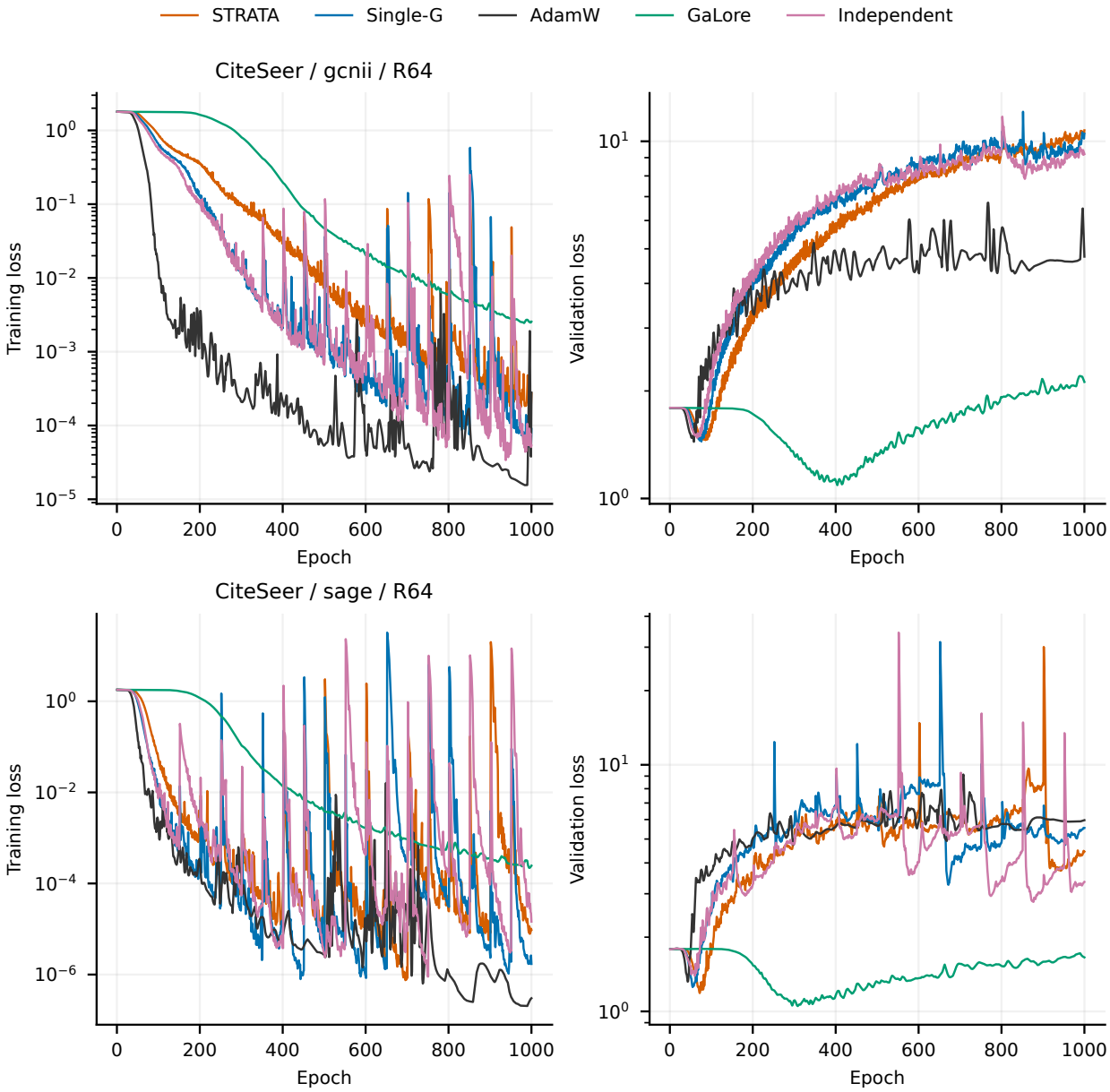


Figure 11. **Completed breadth-cohort training and validation losses.** CiteSeer/gcnii: two-rate validation selection; CiteSeer/sage: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

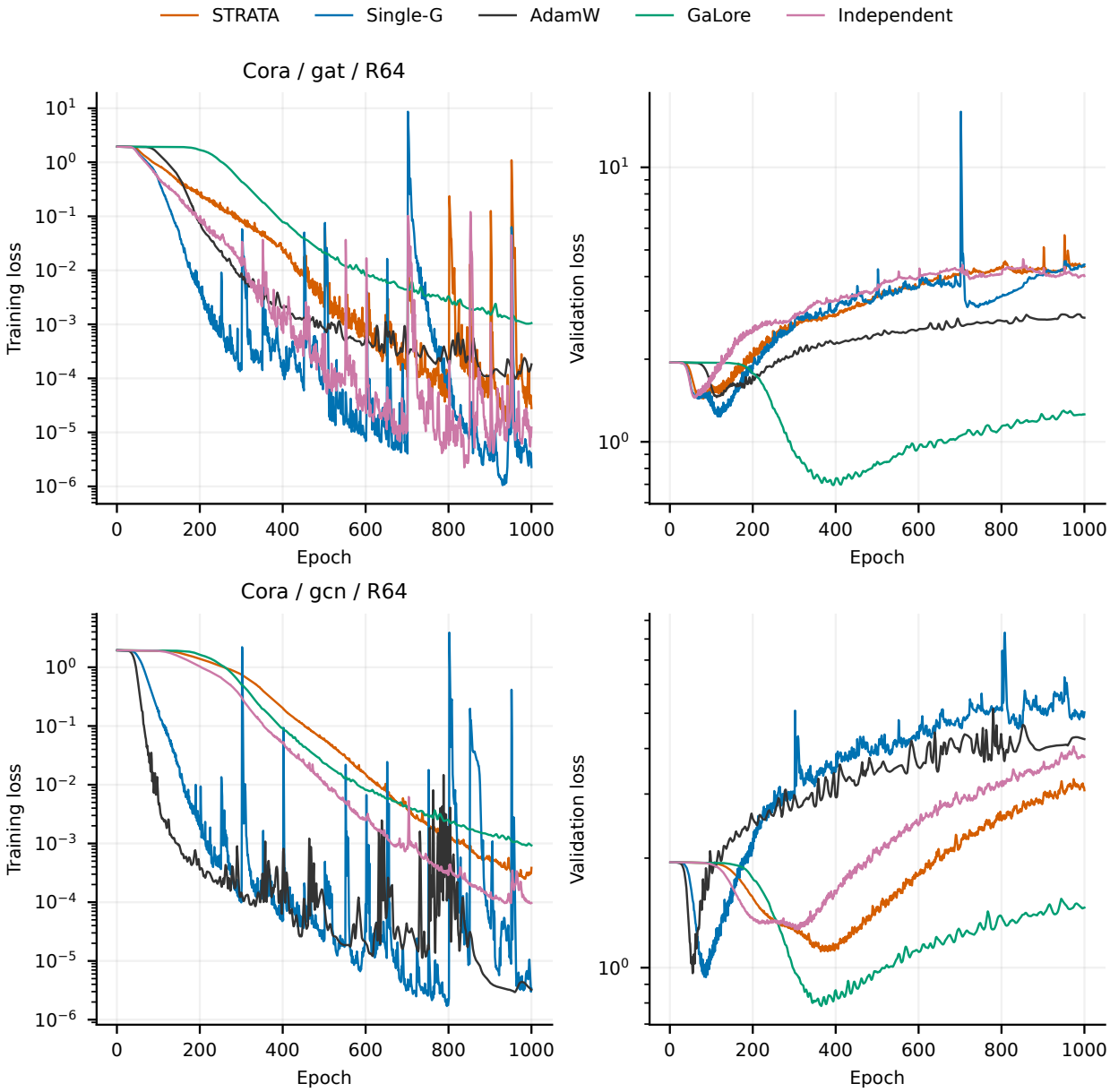


Figure 12. **Completed breadth-cohort training and validation losses.** Cora/gat: two-rate validation selection; Cora/gcn: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

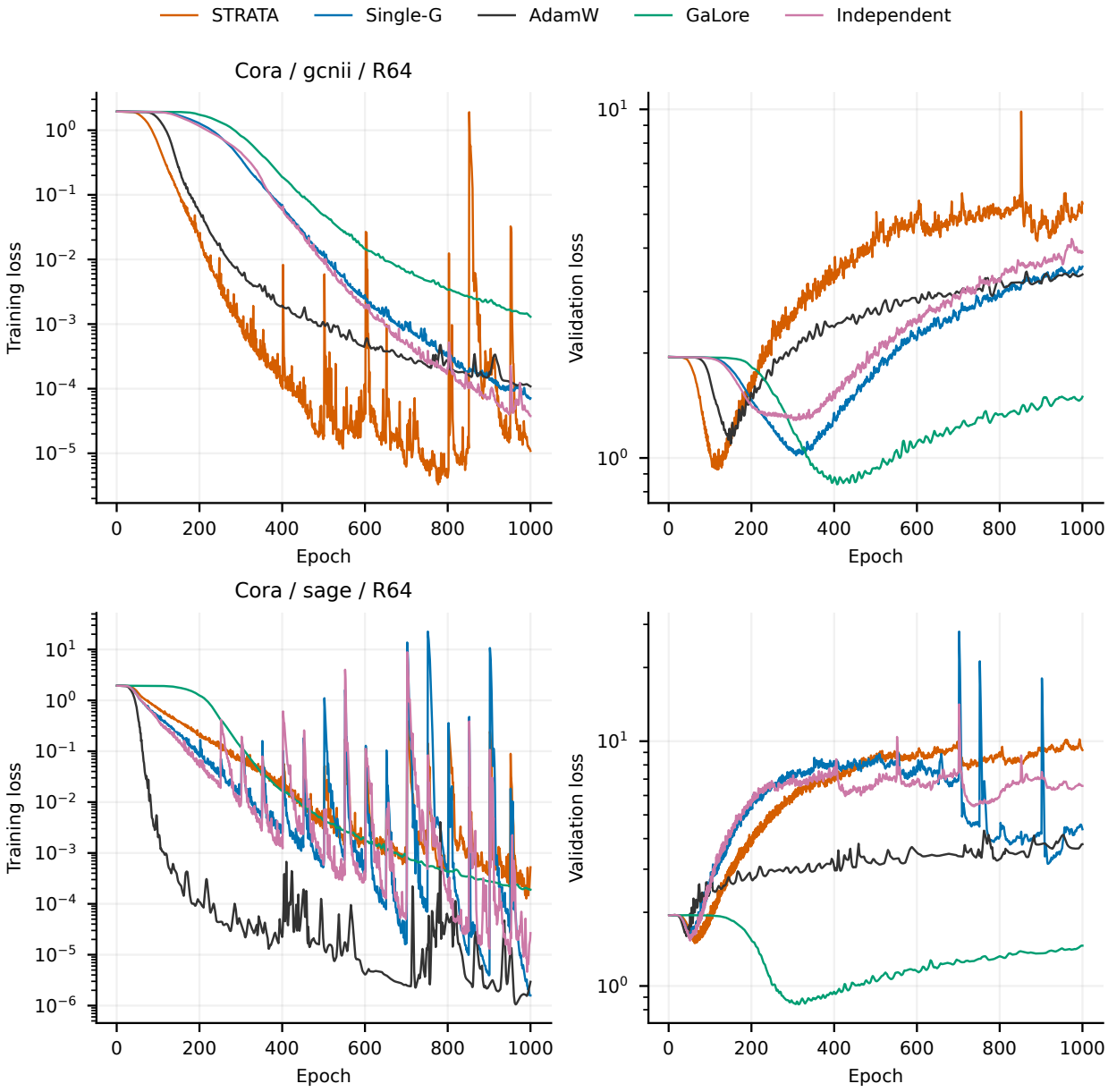


Figure 13. **Completed breadth-cohort training and validation losses.** Cora/gcnii: two-rate validation selection; Cora/sage: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

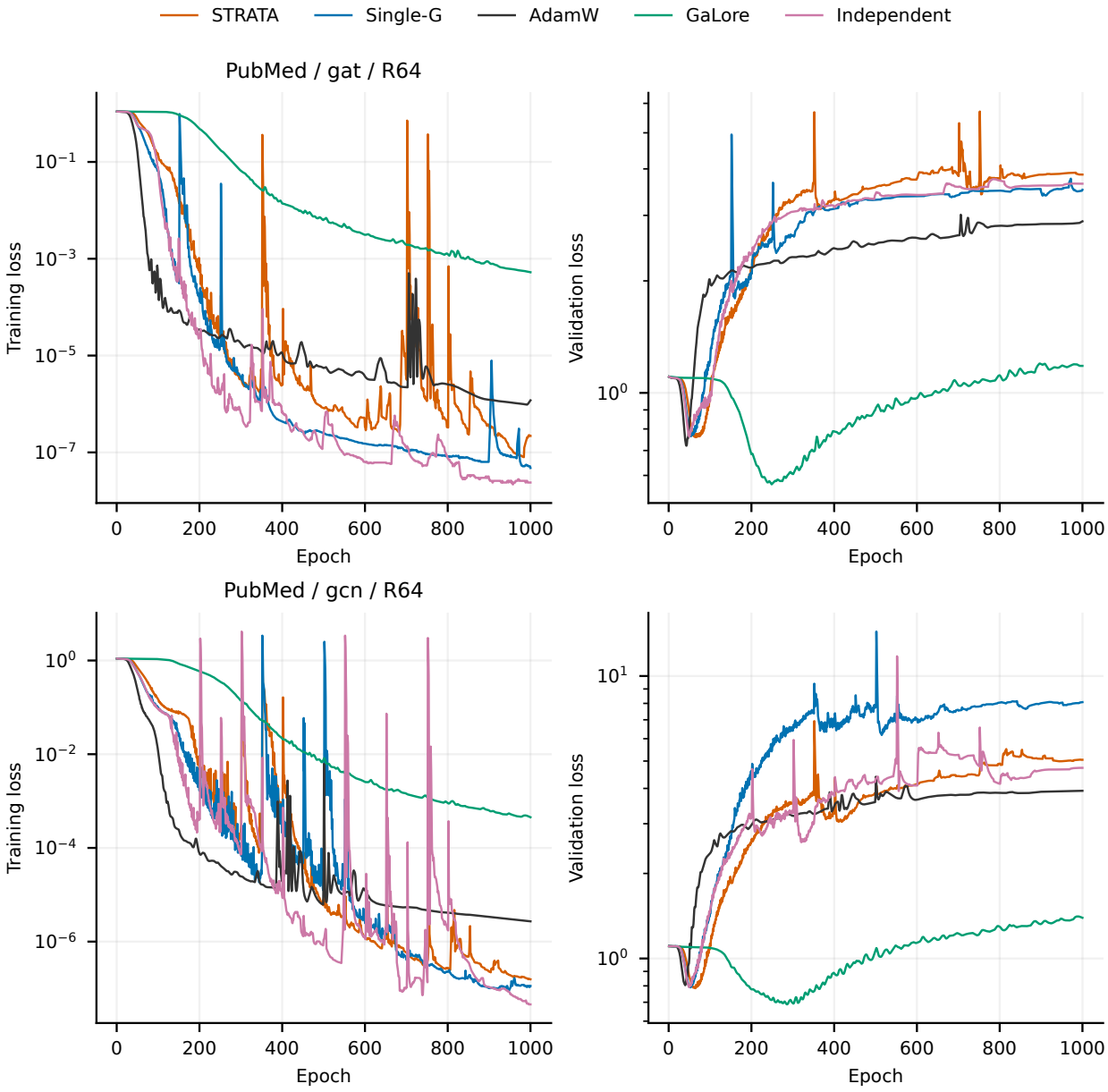


Figure 14. **Completed breadth-cohort training and validation losses.** PubMed/gat: two-rate validation selection; PubMed/gcn: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

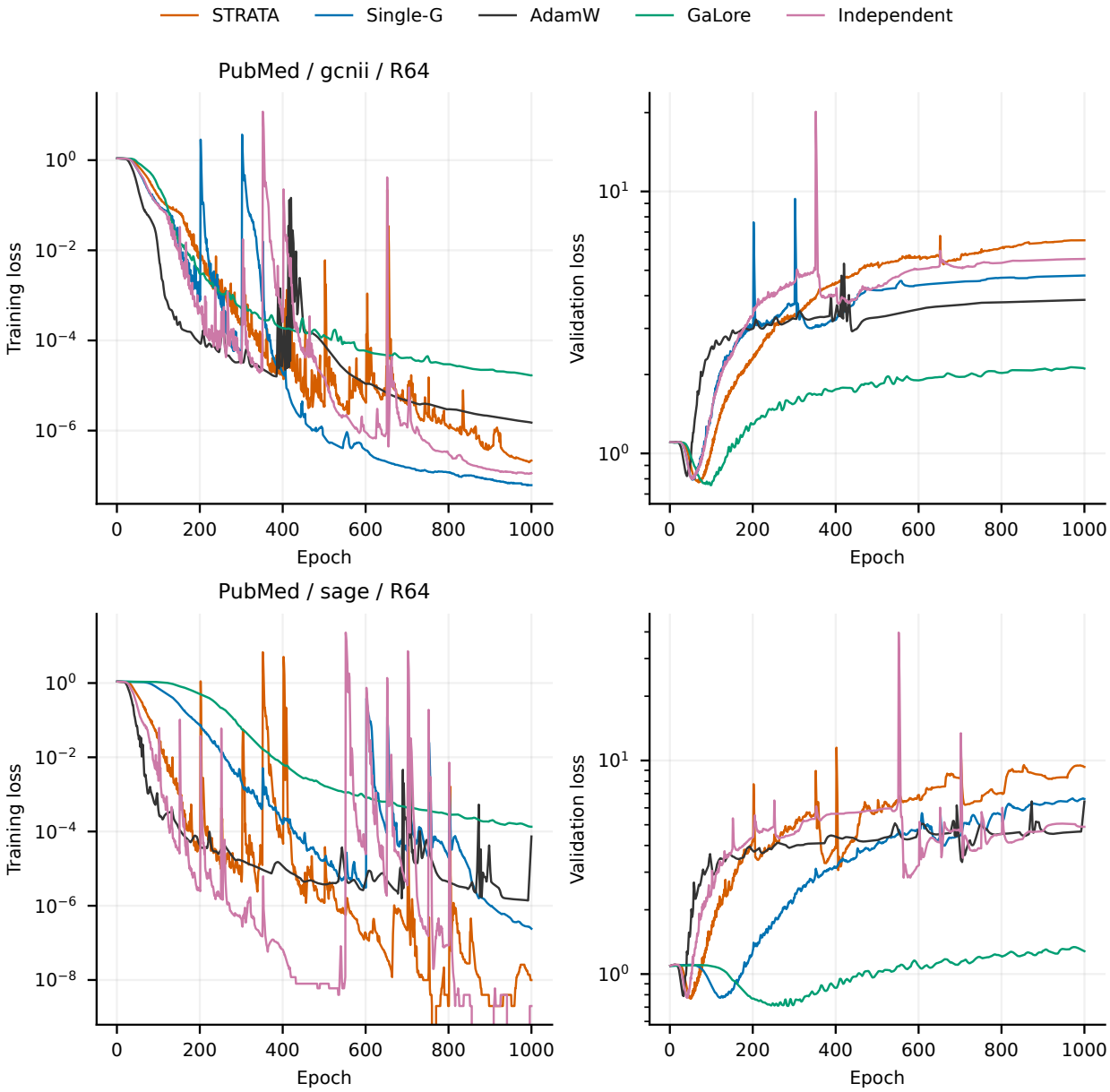


Figure 15. **Completed breadth-cohort training and validation losses.** PubMed/gcnii: two-rate validation selection; PubMed/sage: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

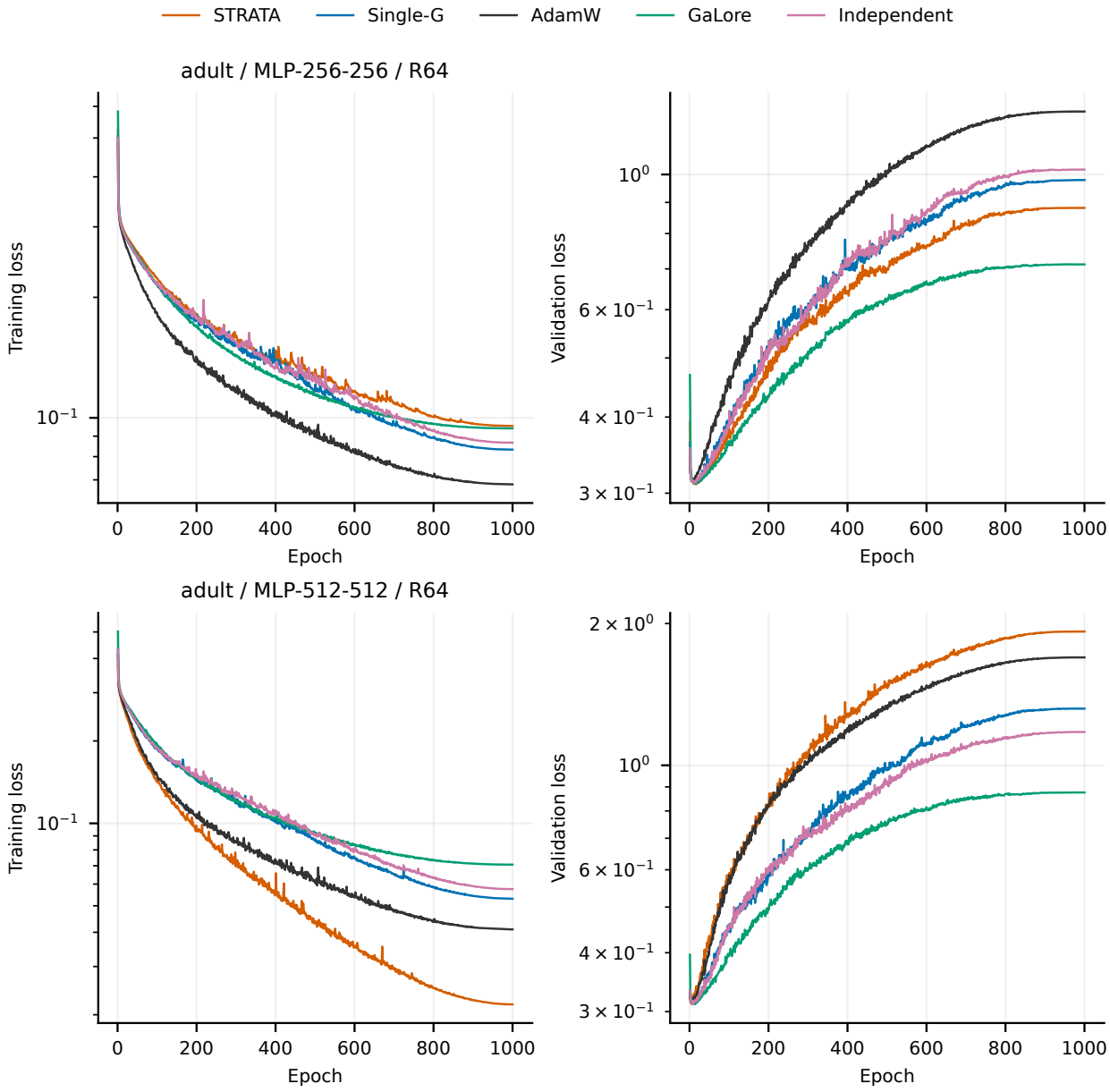


Figure 16. **Completed breadth-cohort training and validation losses.** adult/MLP-256-256: two-rate validation selection; adult/MLP-512-512: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

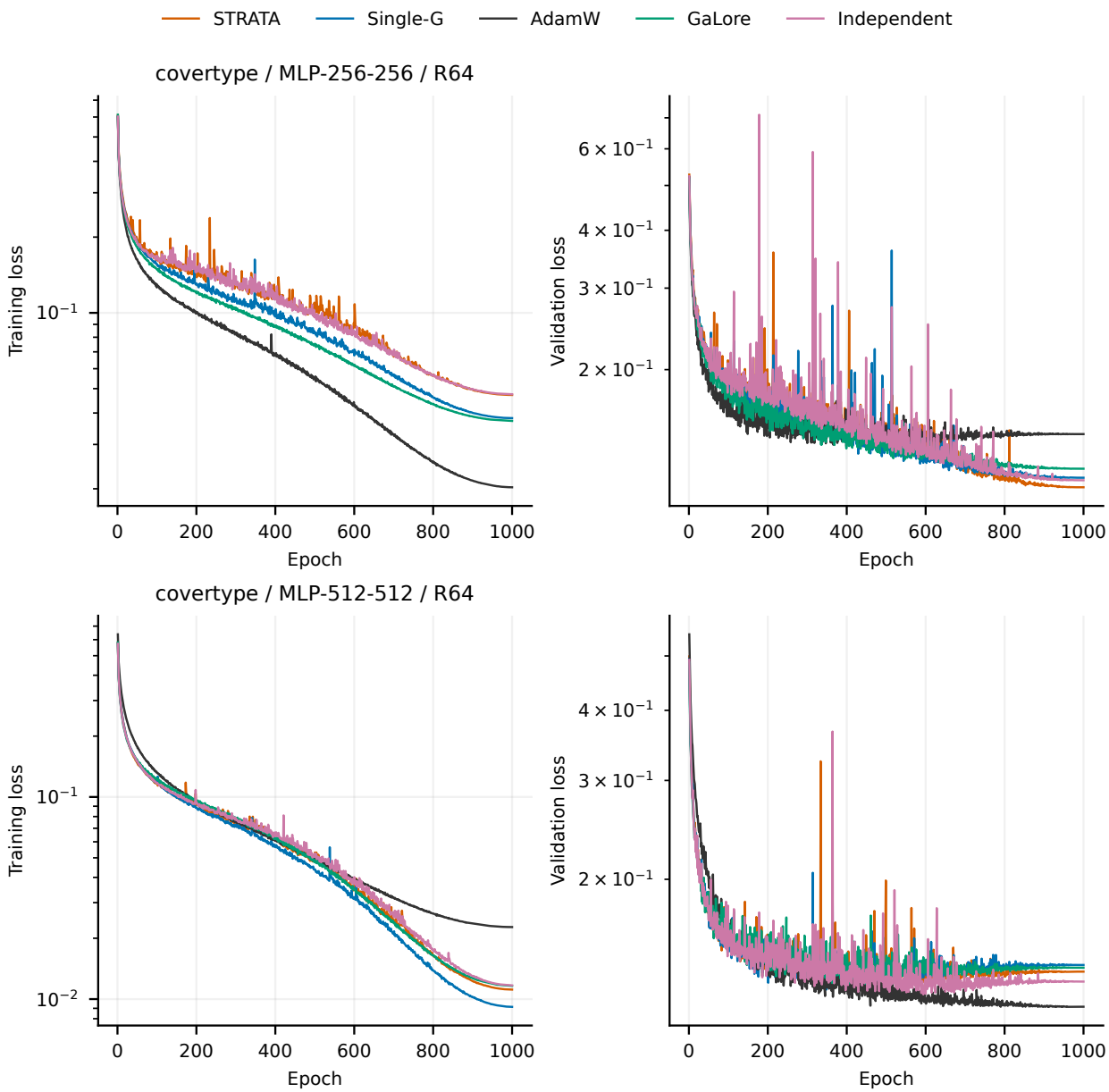


Figure 17. **Completed breadth-cohort training and validation losses.** covertype/MLP-256-256: two-rate validation selection; covertype/MLP-512-512: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

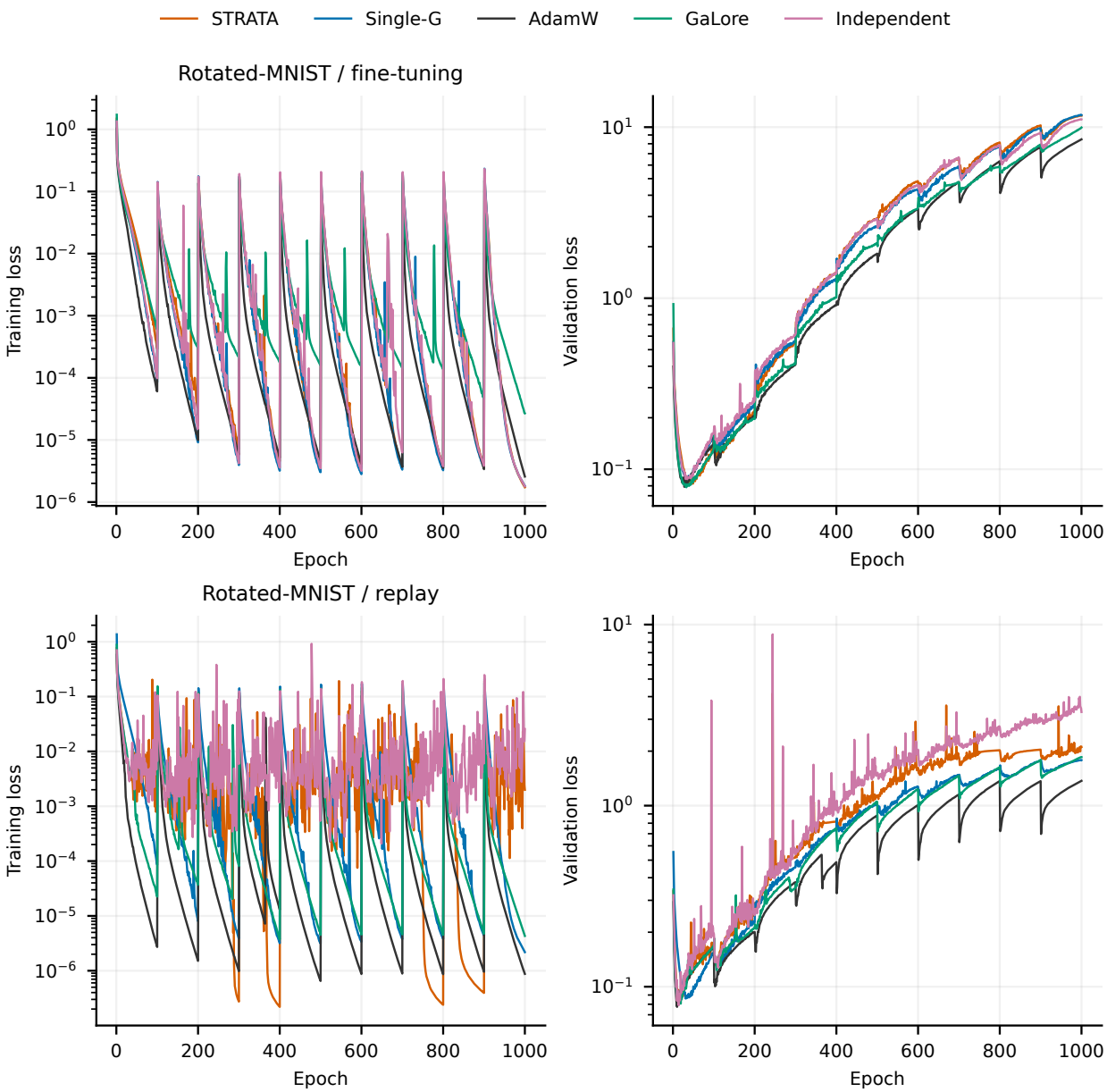


Figure 18. **Completed breadth-cohort training and validation losses.** rotated_mnist_cl/MLP-256-256/finetune: two-rate validation selection; rotated_mnist_cl/MLP-256-256/replay: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. Continual validation averages seen tasks, so its evaluation distribution changes at task boundaries. No smoothing or extrapolation.

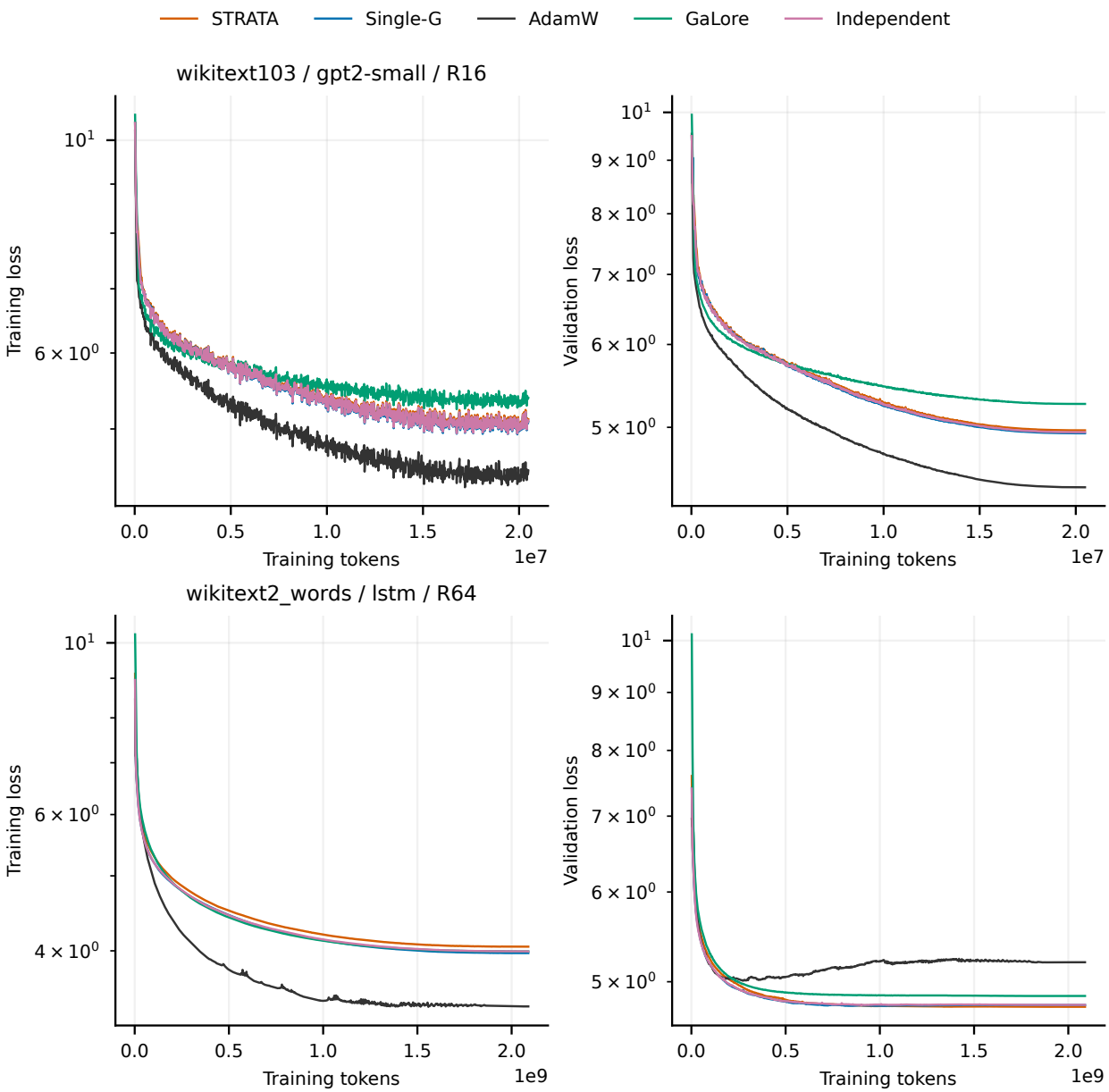


Figure 19. **Completed breadth-cohort training and validation losses.** wikitext103/gpt2-small: two-rate validation selection; wikitext2_words/lstm: fixed rate 0.0003; search incomplete. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

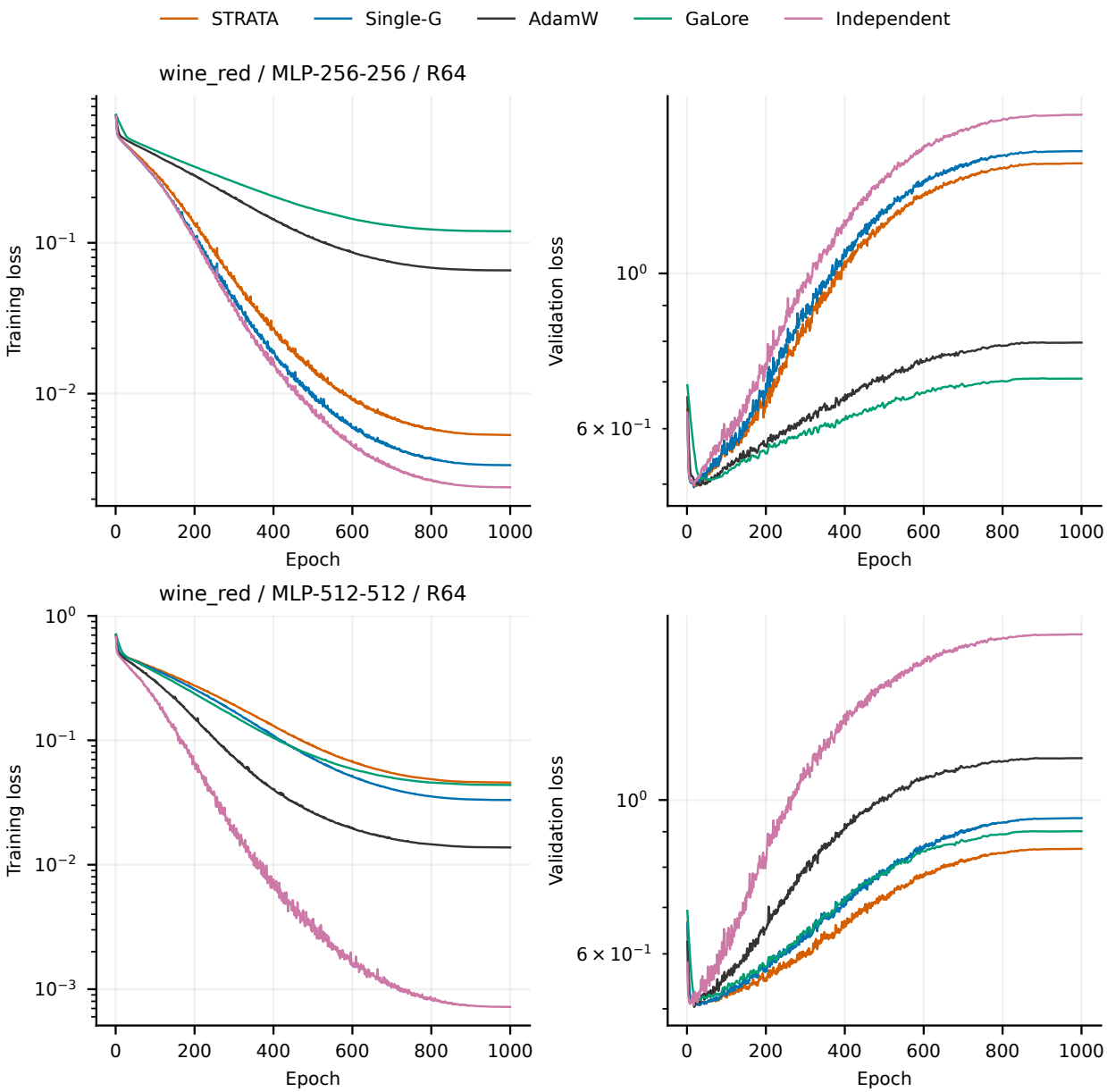


Figure 20. **Completed breadth-cohort training and validation losses.** wine_red/MLP-256-256: two-rate validation selection; wine_red/MLP-512-512: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

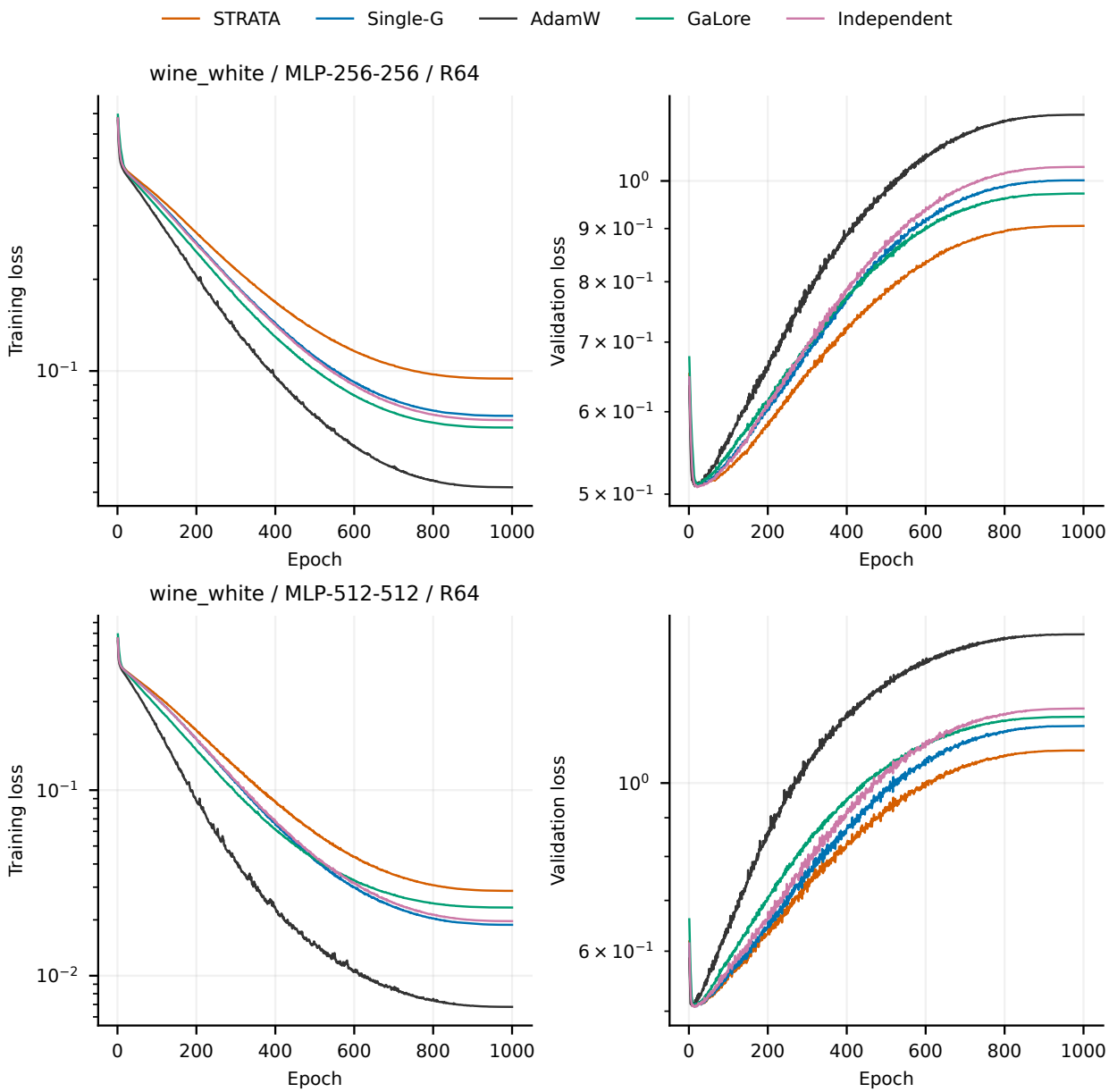


Figure 21. **Completed breadth-cohort training and validation losses.** wine_white/MLP-256-256: two-rate validation selection; wine_white/MLP-512-512: two-rate validation selection. Left: training; right: validation. Raw complete trajectories, seed 42. No smoothing or extrapolation.

G.2. Completed ablation loss trajectories

Each panel uses one predeclared fixed learning rate and only full-budget completed runs. The ablation matrix may still be incomplete; missing variants are not assigned values. All completed variants, including adverse ones, are retained.

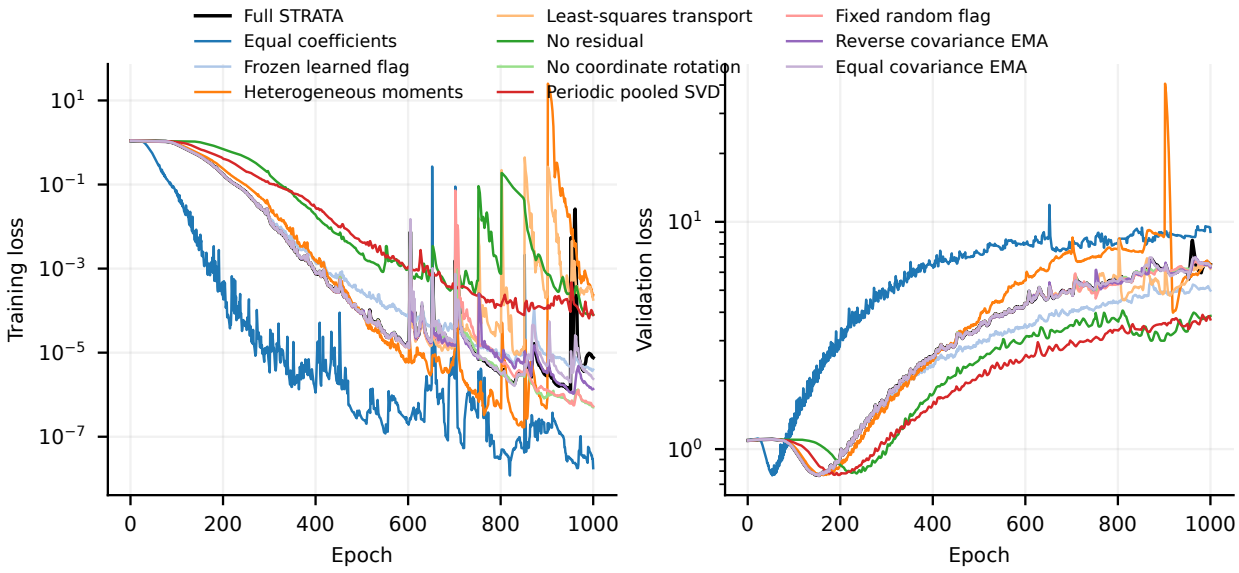


Figure 22. **Completed fixed-rate ablation trajectories.** PubMed/sage/64/0.0003. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

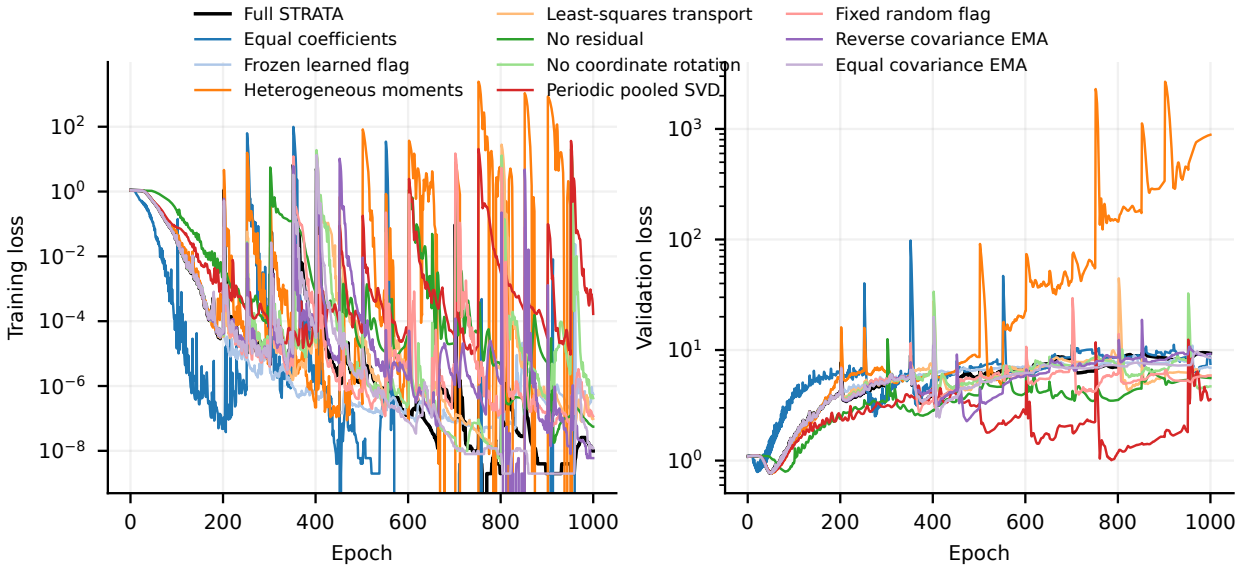


Figure 23. **Completed fixed-rate ablation trajectories.** PubMed/sage/64/0.001. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

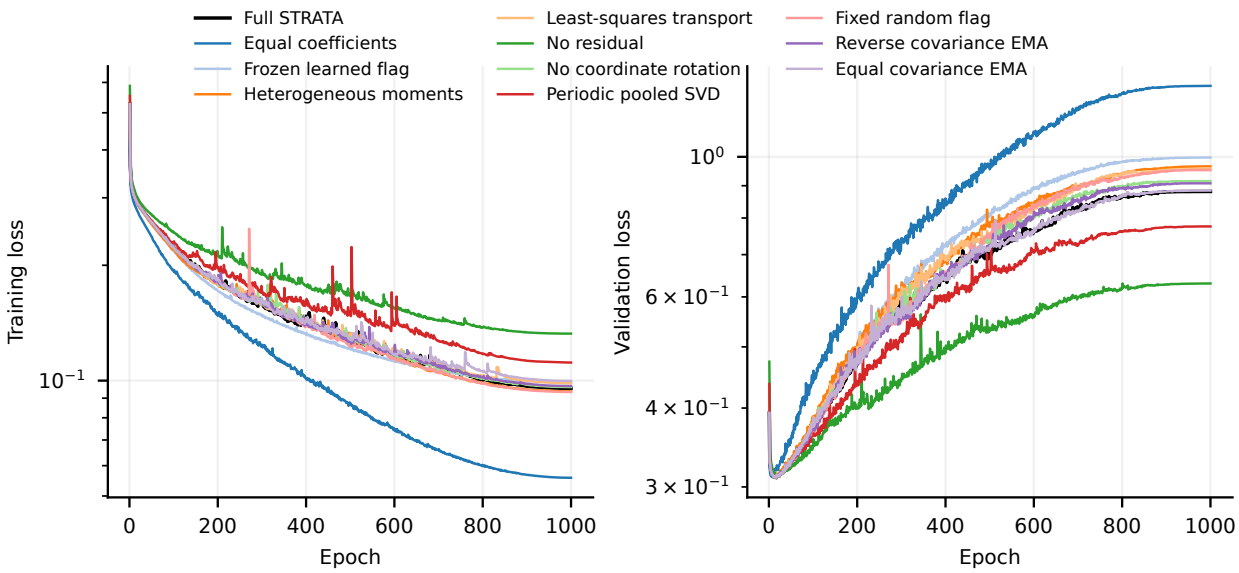


Figure 24. **Completed fixed-rate ablation trajectories.** adult/MLP-256-256/64/0.0003. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

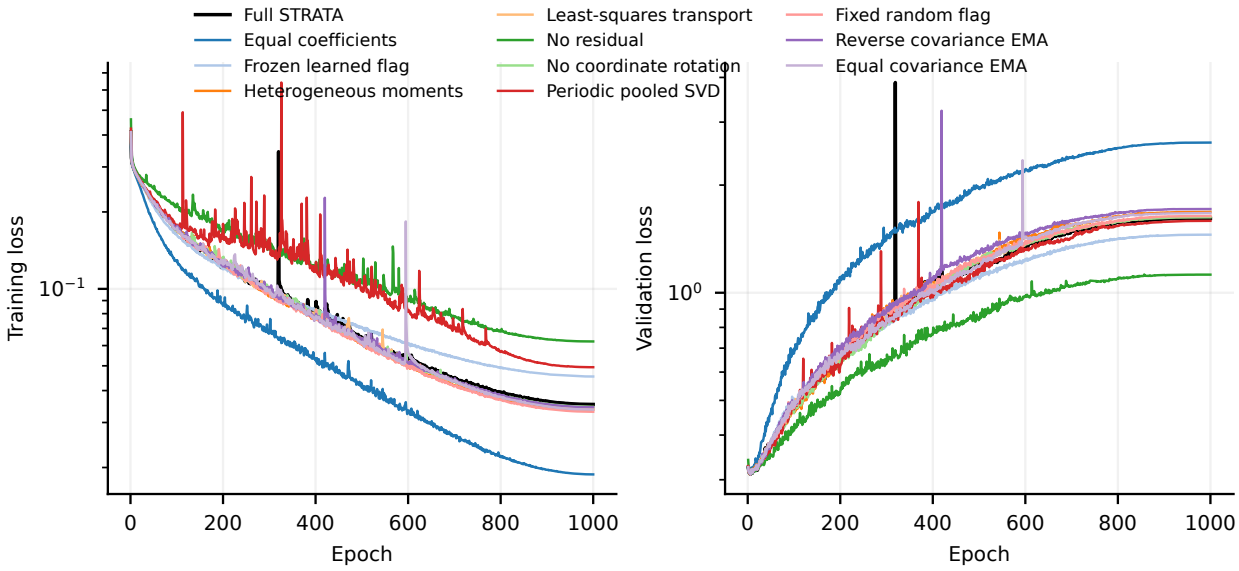


Figure 25. **Completed fixed-rate ablation trajectories.** adult/MLP-256-256/64/0.001. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

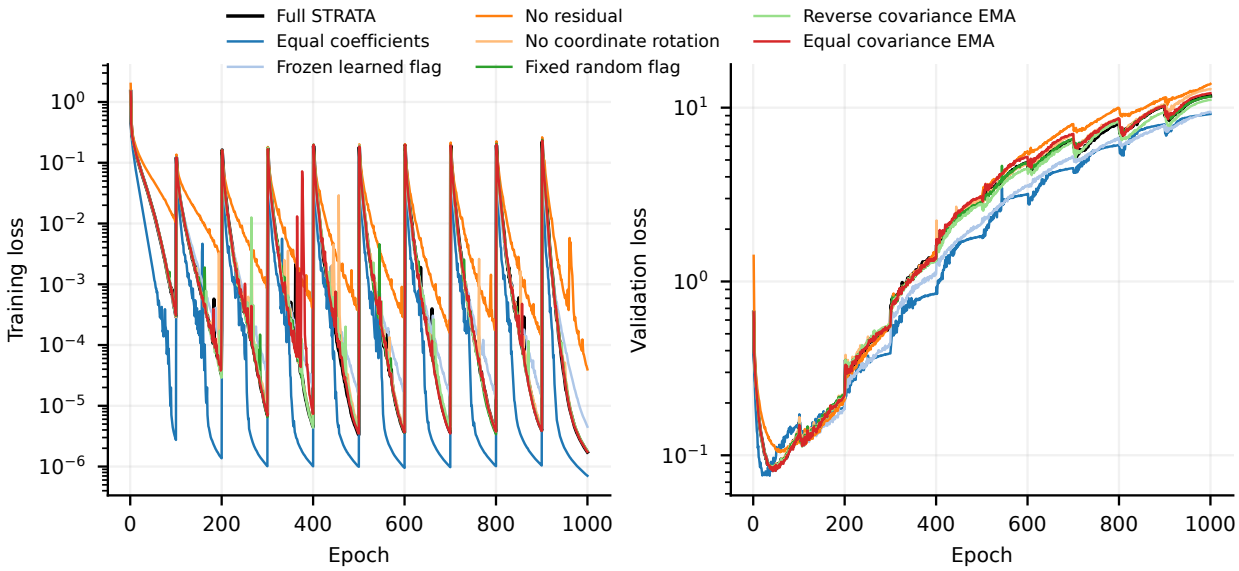


Figure 26. **Completed fixed-rate ablation trajectories.** rotated_mnist_cl/MLP-256-256/64/0.0003. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

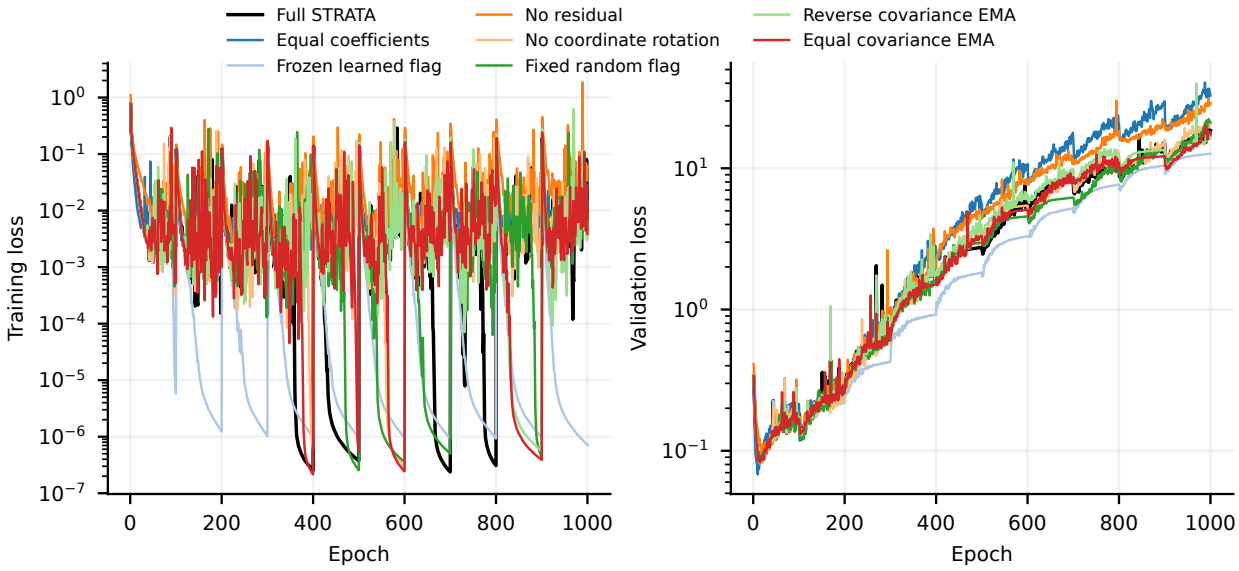


Figure 27. **Completed fixed-rate ablation trajectories.** rotated_mnist_cl/MLP-256-256/64/0.001. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

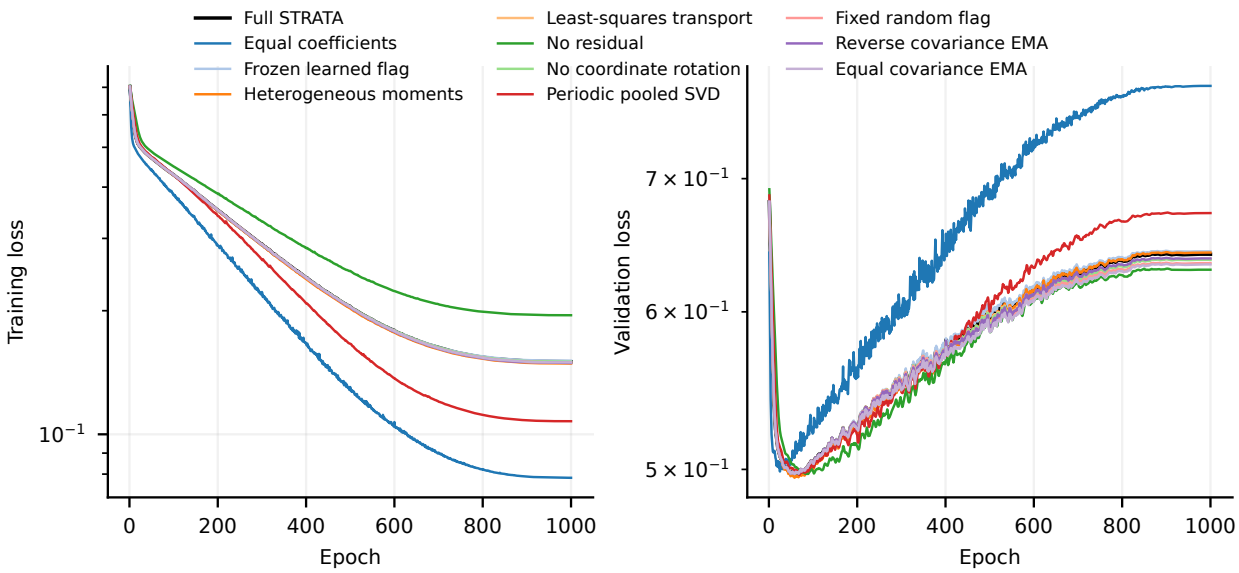


Figure 28. **Completed fixed-rate ablation trajectories.** wine_red/MLP-256-256/64/0.0003. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

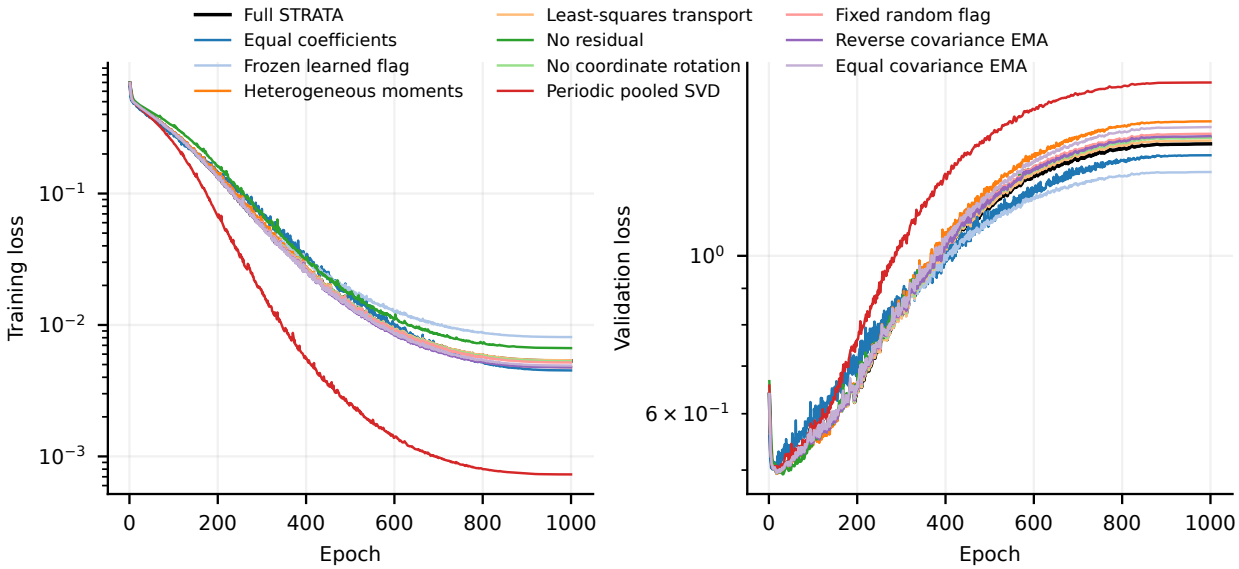


Figure 29. **Completed fixed-rate ablation trajectories.** wine_red/MLP-256-256/64/0.001. Left: training; right: validation. Raw losses for the full method and every currently completed variant, seed 42. No smoothing, test selection or extrapolation. In continual learning the validation set averages seen tasks.

Table 10. **Exact normalization-bias counterexample.** Symmetric and asymmetric zero-mean noises have variance σ^2, $\sigma = .3$. The true optimum is zero. The asymmetric stationary root is given by Proposition 23.

ϵ	Symmetric root	Asymmetric root
0.0001	0	0.09998750
0.001	0	0.09987514
0.01	0	0.09876392
0.1	0	0.08877693
1	0	0.04300047
10	0	0.00676348

Table 11. **Finite-trajectory diagnostics at $\epsilon = .01$.** Means use the final 5000 of 30000 seed-42 stochastic-approximation iterates. These are finite-run observations, not asymptotic estimates.

Update / noise	Final-5000 mean w
Normalized / skew	0.09878739
Normalized / symmetric	0.06416802
SGD / skew	0.00508841
SGD / symmetric	0.00119654

H. Diagnostics of the Actual Adaptive-Update Theory

Implementation-level identity checks. A float64 test calls the unmodified STRATA matrix update for 100 steps with 20 flag refreshes on an 8×6 matrix, ranks $(1, 2, 3)$ and nonidentical moment decays. It reconstructs the ambient moment, exact adaptive direction, and virtual iterate using the actual stored states. Maximum identity errors are 2.18×10^{-16}, 2.20×10^{-15}, and 3.56×10^{-17}, respectively. These tests exercise the exact identities in Proposition 22 and Theorem 15, including refresh steps; they do not assert that the variation budgets vanish on neural benchmarks. The analytic asymmetric root has update-field residual 6.3×10^{-15}. One hundred scalar posterior-variance checks give maximum first-order residual 5.4×10^{-15} and nonnegative refinement gains, checking Theorem 16 and Proposition 25. Source hashes and complete numerical outputs are retained.

Table 12. **Implementation-level variation-budget checks.** The float64 trajectory has 100 steps and 20 flag refreshes. Maxima summarize each sequence separately; inequalities are checked step by step. A violation means slack below -10^{-10}.

Quantity	Steps	Max observed	Max bound	Min slack	Violations
Preconditioner	99	7.231e+01	1.443e+03	+1.049e+02	0
Momentum coefficient	99	4.737e-01	1.302e+00	+1.019e-11	0
Transport defect	100	4.782e-03	8.445e-02	-1.242e-16	0

Table 13. **Exact adaptive-update identities.** Maximum residual over the same 100-step trajectory.

Identity	Maximum residual
ambient momentum	2.173e-16
exact direction	2.199e-15
virtual iterate	3.560e-17

I. Completed Evidence, Resource Audits, and Controlled Follow-ups

All comparisons below use seed 42. A two-rate result requires both rates for every method, and all prescribed epochs must finish. Rates and exploratory flag-step settings are selected on validation only. Incomplete configurations remain in the execution counts, without extrapolated endpoints.

Proposal coverage and remaining work. Configuration counts measure execution coverage, not positive results. The graph and tabular grids are complete; image and continual-learning runs continue; the LSTM grid has nine completed candidates and one recorded failure. WikiText-2 epochs traverse its full corpus; GPT-2 uses fixed ten-step evaluation units, explicitly not corpus epochs or the 1000-corpus-epoch-equivalent proposal budget. OpenWebText/C4 subsets and corpus-switch experiments are not established completed benchmarks in this matrix. Core50 and DomainNet acquisition is complete, but readiness does not imply completion of their long-horizon runs.

Table 14. Execution snapshot by disjoint queue family. No pending, running, deferred or failed configuration is a completed endpoint.

Family	Complete	Running	Pending	Deferred	Data wait	Failed
ablation	54	0	0	0	0	0
continual	22	5	5	68	0	0
forecasting	60	0	0	0	0	0
graph	120	0	0	0	0	0
graph mechanism/sweep	38	0	0	0	0	0
language	10	0	0	92	0	0
nonstationary	30	0	0	0	0	0
recommendation	60	0	0	0	0	0
scientific	10	0	0	0	0	0
sensor_drift	60	0	0	0	0	0
sequence	9	0	0	20	0	1
tabular	80	0	0	0	0	0
vision	3	2	9	98	0	0

Compute-scope revision (2026-11-04). After observing the earlier exploratory comparisons, we reduced the remaining computational scope. The original 856-configuration ledger is preserved: 278 unstarted configurations are explicitly deferred, not counted as completed or removed according to their eventual result. All seven jobs already running at the decision remain unchanged; no trajectory is shortened, and all retained formal runs still require 1000 epochs. Thirty pending configurations remain eligible, leaving 37 unfinished runs at the decision. This is an adaptive exploratory scope revision, not completion of the original full proposal grid.

The priority is to complete the existing Rotated-MNIST fine-tuning grid and its replay extension, motivated by the earlier comparison-specific continual-learning gain and the multi-timescale memory hypothesis. Both retain five methods and both original rates. For ResNet-18 on ImageNet-100, Tiny-ImageNet, DomainNet, and CORe50, the retained comparison uses one rate per method: the first-listed rate in the original queue, fixed without selecting among their test outcomes. All original methods are retained within each setting, including SGD with momentum and Muon in the two image-classification settings. These four groups support fixed-rate comparisons only, not validation-selected two-rate claims. Additional already-running rates remain individual configurations and do not alter that rule.

For the five adaptive/projection methods, the fixed rate is 3×10^{-4}; image SGD and Muon use their original first-listed rates 0.03 and 0.01, respectively. The Rotated-MNIST search retains $\{3 \times 10^{-4}, 10^{-3}\}$ per method. Deferred work includes

Table 15. **Retained scope after compute reduction.** Counts below are full comparison candidates, including already completed or running members. Priority determines execution order, not inclusion of favorable outcomes. Every completed comparator, including stronger baselines, remains reportable.

Setting	Model/base	Methods	Candidates	Rate protocol
Rotated-MNIST	MLP/fine-tuning	5	10	Two original rates
Rotated-MNIST	MLP/replay	5	10	Two original rates
ImageNet-100	ResNet-18	7	7	Fixed first-listed rate
Tiny-ImageNet	ResNet-18	7	7	Fixed first-listed rate
DomainNet	ResNet-18/fine-tuning	5	5	Fixed first-listed rate
CORe50	ResNet-18/fine-tuning	5	5	Fixed first-listed rate

larger vision architectures, additional GPT-2 model/rank grids, GRU/Transformer sequence variants, and the unstarted EWC and other replay expansions. Deferral is a scope decision and supplies no evidence for or against STRATA. The LSTM candidate already in progress at the scope revision has since completed; its failed AdamW candidate remains visible below.

Recorded numerical failure. The WikiText-2/LSTM AdamW candidate at learning rate 10^{-3} stopped at epoch 276 after training, validation, and test losses became non-finite. Its log contains 275 finite epochs of the prescribed 1000; it supplies no completed endpoint. Nine of the ten planned LSTM configurations are complete, including the final Independent candidate. The two-rate comparison remains incomplete because of this failed AdamW candidate. A completed common-rate subset, when displayed, is labeled fixed-rate rather than validation-selected.

Measured state–quality tradeoffs. State includes every unique optimizer tensor storage in a final checkpoint, including fixed probes; weights, gradients and transient workspaces are separate. Peak allocation and shared-server timing are reported independently, so a reduction in stored state is not asserted to be a peak-memory or speed reduction.

Table 16. Complete-cohort resource audit. Ratios compare STRATA with full AdamW after each method independently selects its learning rate on validation. Accuracy differences are STRATA minus Single-G; perplexity and relative-L2 differences retain their lower-is-better sign.

Task	Model	State ratio	Peak ratio	Test	Δ Single-G
CiteSeer	gat	0.402	0.983	0.56758	-0.02417
CiteSeer	gcn	0.402	0.964	0.49199	-0.02356
CiteSeer	gcnii	0.402	0.967	0.47508	-0.02366
CiteSeer	sage	0.379	0.961	0.55747	+0.01767
Cora	gat	0.559	0.992	0.69798	-0.02111
Cora	gcn	0.558	0.977	0.70120	-0.01523
Cora	gcnii	0.558	0.978	0.68975	-0.01163
Cora	sage	0.478	0.968	0.46532	-0.18529
Darcy-generated	FNO	1.027	1.003	0.13637	+0.00807
PubMed	gat	0.782	1.000	0.71174	-0.03134
PubMed	gcn	0.781	0.998	0.71742	+0.00534
PubMed	gcnii	0.781	0.998	0.66341	-0.04726
PubMed	sage	0.581	0.996	0.71623	+0.03721
adult	MLP-256-256	1.000	1.000	0.81486	-0.00001
adult	MLP-512-512	0.535	0.986	0.81509	+0.00227
cifar100_class_shift	ResNet18-CIFAR	0.244	0.979	0.53092	-0.04278
cifar10_label_shift	ResNet18-CIFAR	0.241	0.979	0.75012	-0.01638
covertype	MLP-256-256	0.925	1.000	0.96224	-0.00077
covertype	MLP-512-512	0.513	0.994	0.96644	+0.00002
rotated_mnist_shift	MLP-256-256	0.578	0.997	0.97615	-0.00281
wikitext103	gpt2-small	0.040	0.822	144.54240	+4.87039
wine_red	MLP-256-256	0.911	1.017	0.68421	-0.00408
wine_red	MLP-512-512	0.469	1.019	0.71500	+0.01798
wine_white	MLP-256-256	0.911	0.998	0.72423	-0.00747
wine_white	MLP-512-512	0.469	0.957	0.70875	-0.01435

Completed Darcy/FNO comparison. All ten two-rate configurations finish 2000 epochs on the retained generated 16×16 operator dataset. Validation relative L_2 selects the rate; PDE residual and spectral errors are secondary test metrics. Frequency errors are mean squared FFT magnitudes of errors in training-standard-deviation target units. This is not an external standard Darcy benchmark. Spectral real/imaginary tensors retain dense Adam state; eligible pointwise matrices use projected state.

Table 17. Completed generated-Darcy/FNO results, terminal-100 test metrics.

Method	lr	Rel-L_2	PDE residual	Low-freq	High-freq
STRATA	1e-03	0.1364	3.45	274	0.115
Single-G	1e-03	0.1283	3.06	257	0.101
AdamW	1e-03	0.1146	2.17	206	0.1
GaLore	1e-03	0.1399	3.8	290	0.128
Independent	1e-03	0.1266	2.83	257	0.104

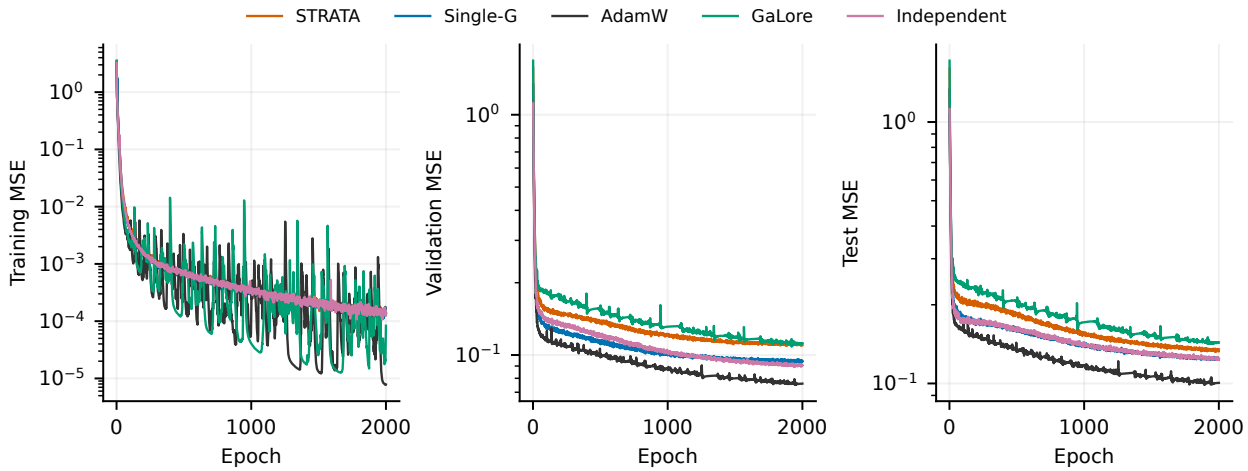


Figure 30. Complete 2000-epoch Darcy/FNO loss trajectories: training, validation, and test MSE. Rates are validation-selected from complete two-rate grids; test residual and spectrum never select a curve. All methods and adverse outcomes are retained.

PubMed mechanism follow-up. The ten variants and both rates are frozen before follow-up outcomes. Their controls reuse the original two full-method configurations. Freezing after 200 updates also stops variance resets; periodic pooled SVD changes the flag objective. Thus these are explicitly identified interventions, not pure isolated geometry controls.

Exploratory flag-step sensitivity. The declared grid is $\xi \in \{10^{-3}, 10^{-1}, 10\}$ and $\eta \in \{3 \times 10^{-4}, 10^{-3}\}$ for each of STRATA, Single-G and Independent: six equal-budget candidates per method. Defaults remain unchanged. Larger raw-objective steps test whether weak flag motion limits the default protocol. Baseline numerical paths were checked unchanged for all five optimizers over 55 paired updates; all 20 new graph-path smoke tests passed. New runs save float32 bases and evaluate span motion with float64 QR/SVD; old float16 flag snapshots are not used to infer tiny principal angles. The span diagnostic is $D_{\text{span}} = \sqrt{1 - \|Q_{10}^{\top} Q_{1000}\|_F^2 / r}$, with fresh orthonormal QR factors and represented rank r. Independent uses the concatenated subspace union when it has full column rank; excluded rank-deficient matrices are not assigned zero motion.

State-matched capacity control. Same-rank comparisons allocate more optimizer state to STRATA than to Single-Grassmann. We therefore choose the closest integer Single-G rank using initialized tensor storage only, before observing any new accuracy; the target includes probes. Both methods receive the same six-candidate flag-step/rate grid. This is an approximately equal-state comparison, separate from the equal-rank default protocol. Single-G rank 82 uses 2423128 bytes, versus 2433048 for rank-64 STRATA, a signed difference of -0.408%. The choice uses exact shape-derived state counts over

Table 18. Complete PubMed/SAGE ablation grid, validation-selected rates, terminal-100 accuracies. Differences are relative to full STRATA. No variant is promoted to the default based on test results.

Variant	lr	Val	Test	Δ test (pp)
full	1e-03	0.71792	0.71623	+0.000
same_rho	1e-03	0.71302	0.70196	-1.427
reverse_rho	3e-04	0.69346	0.67923	-3.700
equal_a	1e-03	0.71646	0.71188	-0.435
no_residual	1e-03	0.70986	0.69363	-2.260
random_flag	1e-03	0.69942	0.70164	-1.459
no_transport	3e-04	0.69988	0.67823	-3.800
frozen_after_warmup	3e-04	0.69730	0.67639	-3.984
heterogeneous_memory	1e-03	0.71310	0.70829	-0.794
periodic_svd	1e-03	0.70572	0.69553	-2.070
least_squares_transport	1e-03	0.70994	0.70900	-0.723

Table 19. Matched six-candidate exploratory search per geometric optimizer, selected on validation. Span motion compares the total represented space at checkpoints 10 and 1000, using float64 QR/SVD and averaging eligible matrices. It is not an oracle tracking error.

Method	ξ	lr	Val	Test	Span motion
STRATA	10	1e-03	0.75018	0.73383	0.129
Single-G	0.001	3e-04	0.70666	0.67902	8.39e-06
Independent	0.001	1e-03	0.74190	0.74151	1.25e-05

Table 20. **Complete flag-step sensitivity profiles.** Each point selects its learning rate on validation after both candidates finish. Test values are reporting only.

Method	ξ	lr	Val acc	Test acc
STRATA	0.001	1e-03	0.71792	0.71623
STRATA	0.1	3e-04	0.69918	0.67856
STRATA	10	1e-03	0.75018	0.73383
Single-G	0.001	3e-04	0.70666	0.67902
Single-G	0.1	1e-03	0.70510	0.68944
Single-G	10	3e-04	0.69188	0.68123
Independent	0.001	1e-03	0.74190	0.74151
Independent	0.1	1e-03	0.72100	0.72332
Independent	10	1e-03	0.67426	0.65143

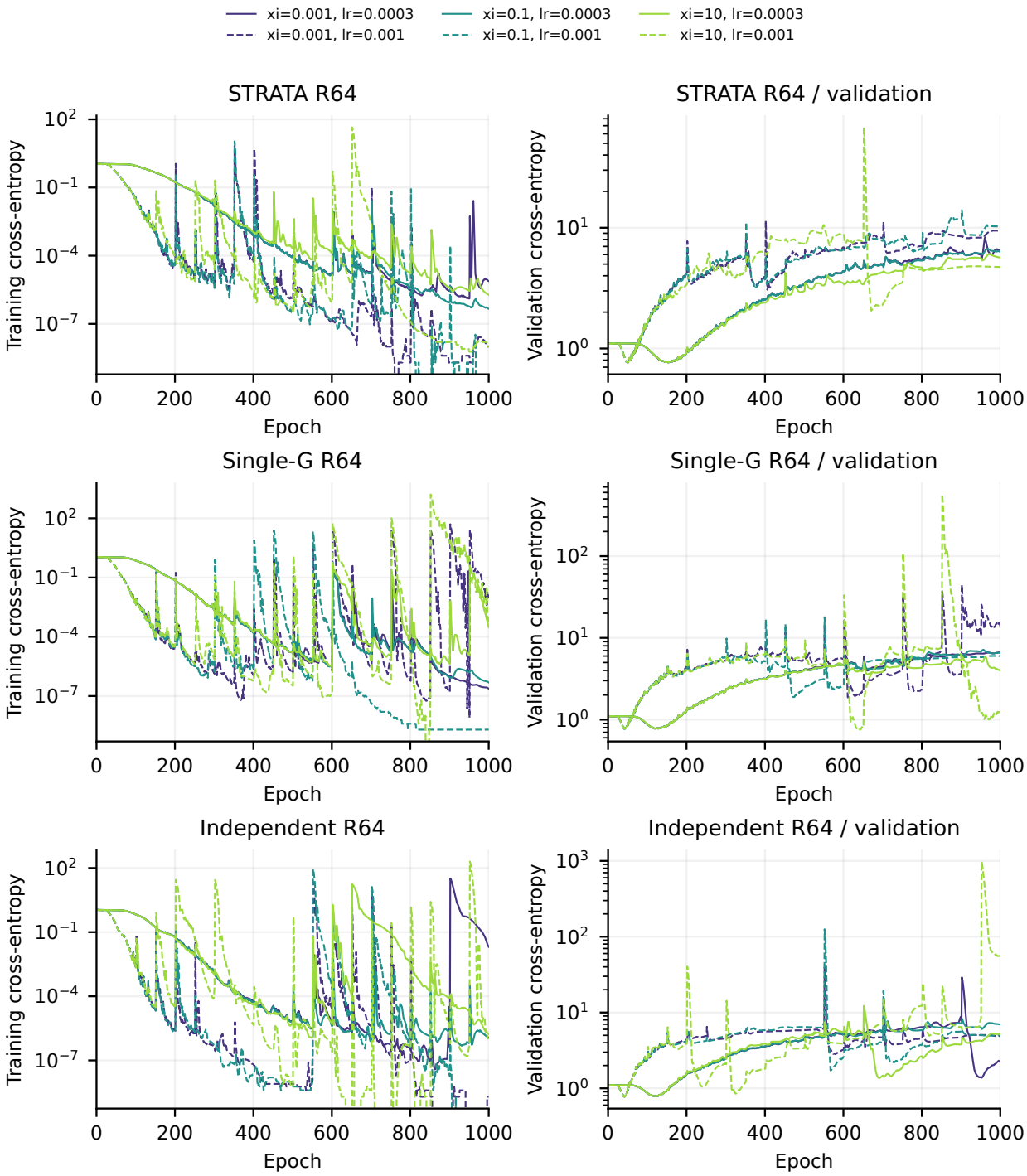


Figure 31. All 18 completed equal-rank step/rate trajectories, without selection or smoothing. Rows show STRATA, Single-G and Independent; columns show training and validation cross-entropy. Colors denote flag step and line styles distinguish the two learning rates. Every candidate runs 1000 epochs.

integer ranks 4–255, including probe-rank saturation, checked against actual initialized storage at four ranks and an archived STRATA final checkpoint; no validation or test score chooses rank.

Table 21. Completed state-budget control. Rank is fixed from storage alone; every row finishes two rates at each of three flag steps. Default test selects the rate at $\xi = 10^{-3}$; six-grid test selects both hyperparameters on validation. Independent uses blocks of dimensions $(16, 16, 32)$, whose total rank budget is 64. State includes probes and the cached union basis when present.

Method	State MiB	ξ_6	lr_6	Test default	Test six-grid
STRATA R64	2.320	10	1e-03	0.71623	0.73383
Single-G R64	1.820	0.001	3e-04	0.67902	0.67902
Single-G R82	2.311	0.001	1e-03	0.72923	0.72923
Independent total R64	2.195	0.001	1e-03	0.74151	0.74151

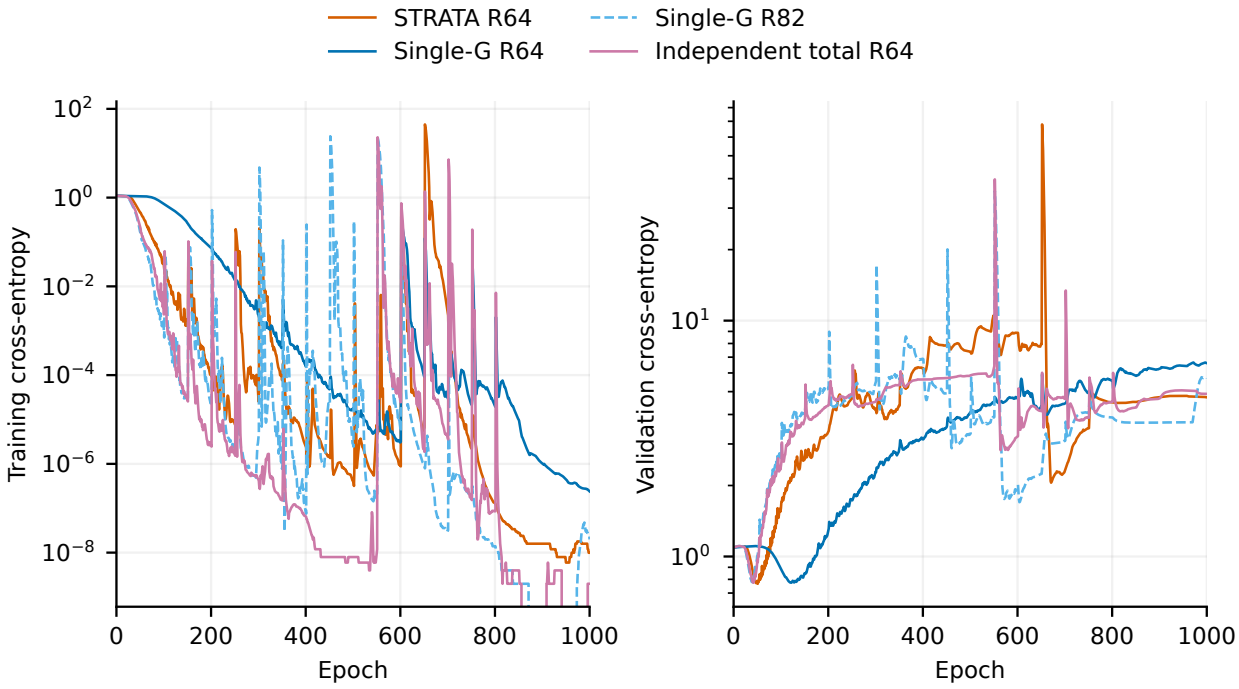


Figure 32. Completed capacity-control training and validation losses after equal six-candidate validation selection. The additional Single-G rank is determined using state storage alone; all six candidates are shown separately below.

Table 22. All six completed Single-G state-control candidates. The rank is chosen using storage alone; validation, not test, selects the reported optimum.

ξ	lr	Val acc	Test acc
0.001	3e-04	0.71554	0.71640
0.001	1e-03	0.73948	0.72923
0.1	3e-04	0.69382	0.69613
0.1	1e-03	0.62916	0.61564
10	3e-04	0.69020	0.69950
10	1e-03	0.72560	0.72434

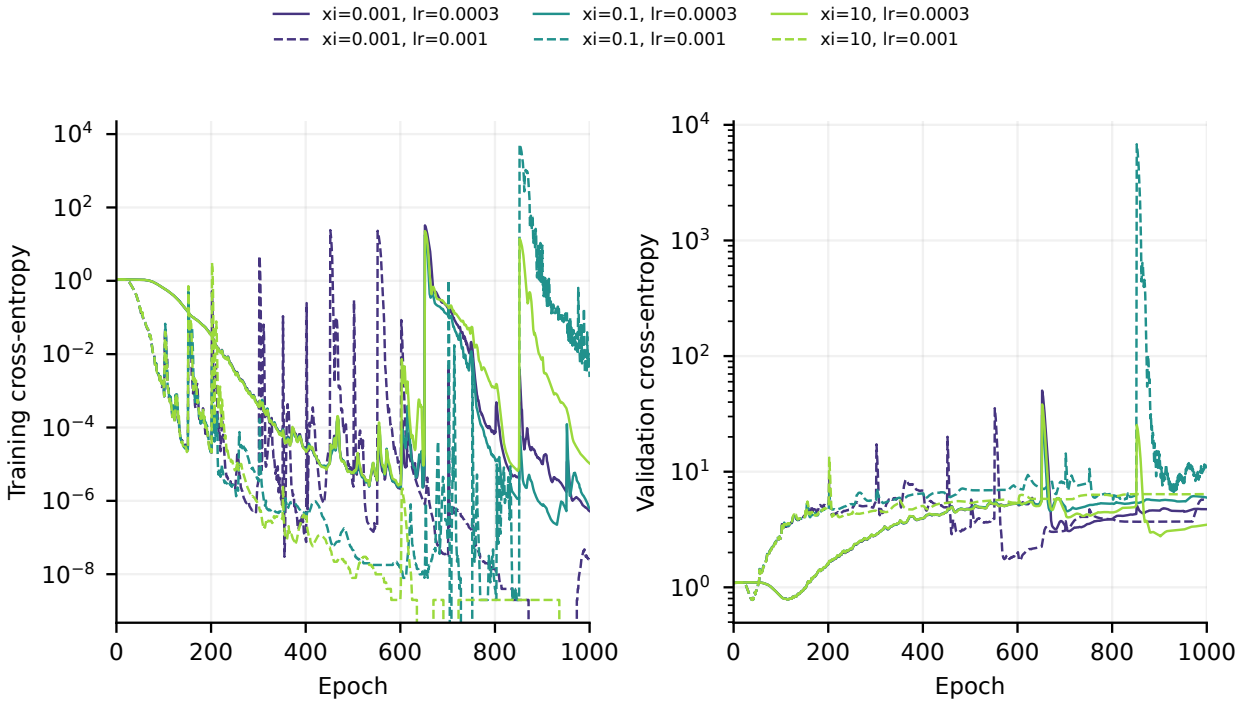


Figure 33. All six completed Single-G capacity-control training and validation trajectories. No candidate is omitted, selected or smoothed in this grid view.

I.1. Completed language and label-shift comparisons

Language protocol and endpoint. All ten WikiText-103/GPT-2-small candidates finish their declared budget before selection. The model has 123849984 parameters and trains from scratch; projected methods use maximum rank 16. Each run uses 1000 units of ten updates, batch size eight and context length 256, totaling 10000 updates and 20480000 training tokens. These are fixed-token-budget comparisons, not 1000-corpus-epoch runs. Each method independently selects between 3×10^{-4} and 10^{-3} using mean validation cross-entropy over units 901–1000. Reported perplexity is $\exp(100^{-1} \sum_{u=901}^{1000} \ell_u)$, not the arithmetic mean of per-unit perplexities; the same selected configuration supplies its test endpoint. All candidates use seed 42 and identical frozen token splits. Complete candidate endpoints remain in Appendix E.

Table 23. **Completed WikiText-103/GPT-2-small comparison.** Both rates finish for every method. Lower perplexity is better; bold marks the lowest validation and test perplexity. State counts unique final-checkpoint optimizer tensor storage, including probes. Peak is maximum recorded CUDA allocation, a separate quantity.

Method	Selected lr	Val PPL	Test PPL	State MiB	Peak MiB
STRATA	3e-04	143.80	144.54	38.00	4179.2
Single-G	3e-04	139.27	139.67	29.90	4170.2
AdamW	3e-04	**80.11**	**80.00**	944.90	5081.8
GaLore	1e-03	193.52	192.20	22.97	4164.0
Independent	3e-04	140.90	140.81	36.84	4178.1

STRATA's test perplexity is 24.8% lower than GaLore's, but higher than AdamW, Single-Grassmann, and Independent. Its optimizer state is 4.0% of AdamW's, while its measured peak allocation is 82.2% of AdamW's. This separates stored-state savings from the weaker held-out result and from peak memory; it does not establish a language-family ranking.

CIFAR-10 label shift. All ten ResNet-18 fine-tuning configurations finish 1000 epochs. Four consecutive 250-epoch phases upweight classes with label modulo four equal to the phase index by a factor of ten. Validation and test accuracy use the corresponding current-phase weights; the terminal 100 epochs are all in the last phase. Each method independently selects its rate on terminal-100 validation accuracy. STRATA attains 75.012%, below Single-Grassmann's 76.650% and the other three methods; the negative comparison is retained. Table 28 includes it in the complete classifier matrix.

Table 24. **Completed CIFAR-10 label-shift comparison.** Validation-selected rates, seed 42, terminal-100 weighted accuracy (percent). Higher accuracy is better. State and peak use the same separate definitions as Table 23.

Method	Selected lr	Val acc.	Test acc.	State MiB	Peak MiB
STRATA	1e-03	76.913	75.012	20.56	3151.9
Single-G	1e-03	79.489	76.650	18.62	3154.7
AdamW	1e-03	**81.155**	**80.212**	85.25	3219.2
GaLore	1e-03	81.096	79.996	16.44	3152.3
Independent	1e-03	78.062	76.964	20.07	3151.4

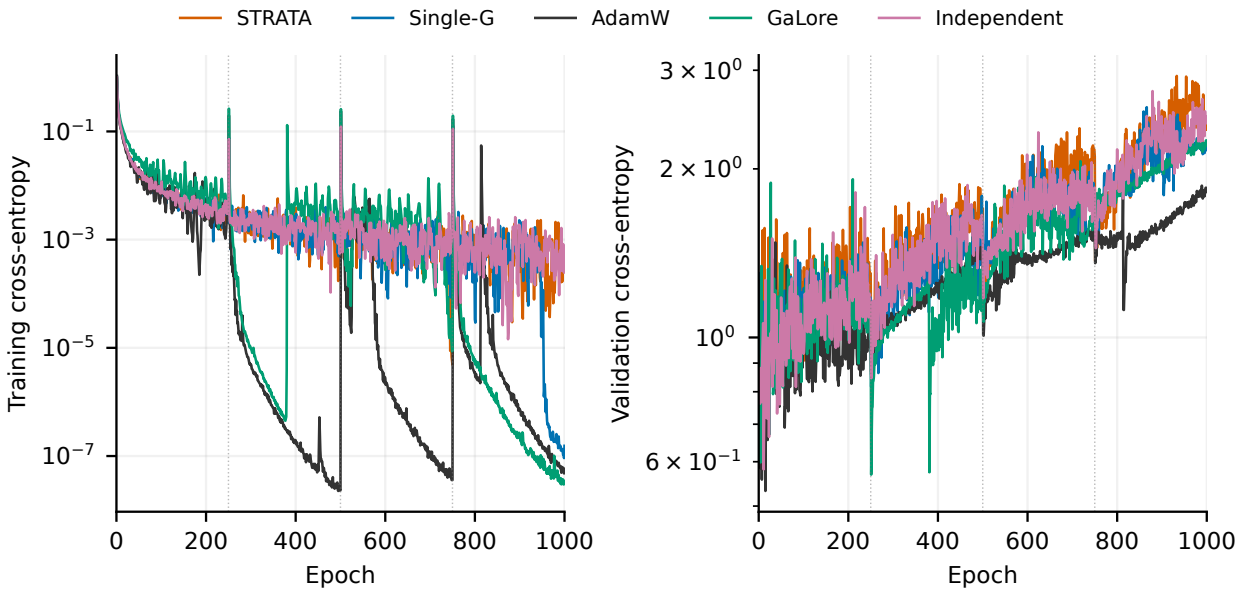


Figure 34. **Completed CIFAR-10 label-shift losses.** Raw training and validation cross-entropy for all five validation-selected configurations. Vertical guides mark phase boundaries at epochs 250, 500, and 750; the validation weighting changes across phases. Every trajectory finishes 1000 epochs. No smoothing or extrapolation.

J. Complete Comparison Matrices and Archived Diagnostics

The main-text cases illustrate restricted positive comparisons. This appendix retains the full comparison matrix, archived non-improvements and losses; no result is removed according to its sign. The standard graph matrix contains twelve dataset–architecture settings, of which three improve on Single-Grassmann after the complete two-rate validation search. The other nine remain part of the empirical scope.

Loss trajectories and generalization. Figure 35 replaces endpoint-only inspection with complete recorded optimization paths. Low training cross-entropy does not establish a test-accuracy advantage. The accompanying held-out cross-entropies expose late overfitting and confidence effects that accuracy alone can conceal. Appendix G retains both curves and adverse hyperparameter outcomes. All observed epochs are plotted; an archived index of zero is the first logged trained epoch, not a reconstructed common initialization.

Table 25. **Validation-audited head-to-head endpoints.** PINNs restrict both methods to rank 64 and select by terminal PDE loss. Classifiers select by validation accuracy. Bold compares the two methods; all values use seed 42.

Task	Metric	Single-G	STRATA
burgers	Rel-L_2 ↓	**0.02607**	0.02666
burgersg2	Rel-L_2 ↓	0.02617	**0.02615**
poisson	Rel-L_2 ↓	2.04357	**0.18741**
mnist	Accuracy ↑	0.98011	**0.98273**
fashion	Accuracy ↑	**0.90328**	0.89907
cifar10	Accuracy ↑	**0.94034**	0.94014
cifar100	Accuracy ↑	**0.72574**	0.71402

Archived comparison. Table 26 and Table 25 support a restrained conclusion. STRATA becomes stable after the numerical fixes, but it is not a universal improvement over Single-Grassmann. In the archived Burgers, Burgers-g2, MNIST, Fashion, CIFAR-10, and ogbn-arxiv comparisons, the practical conclusion is a tie or near-tie. In CIFAR-100, Single-Grassmann is

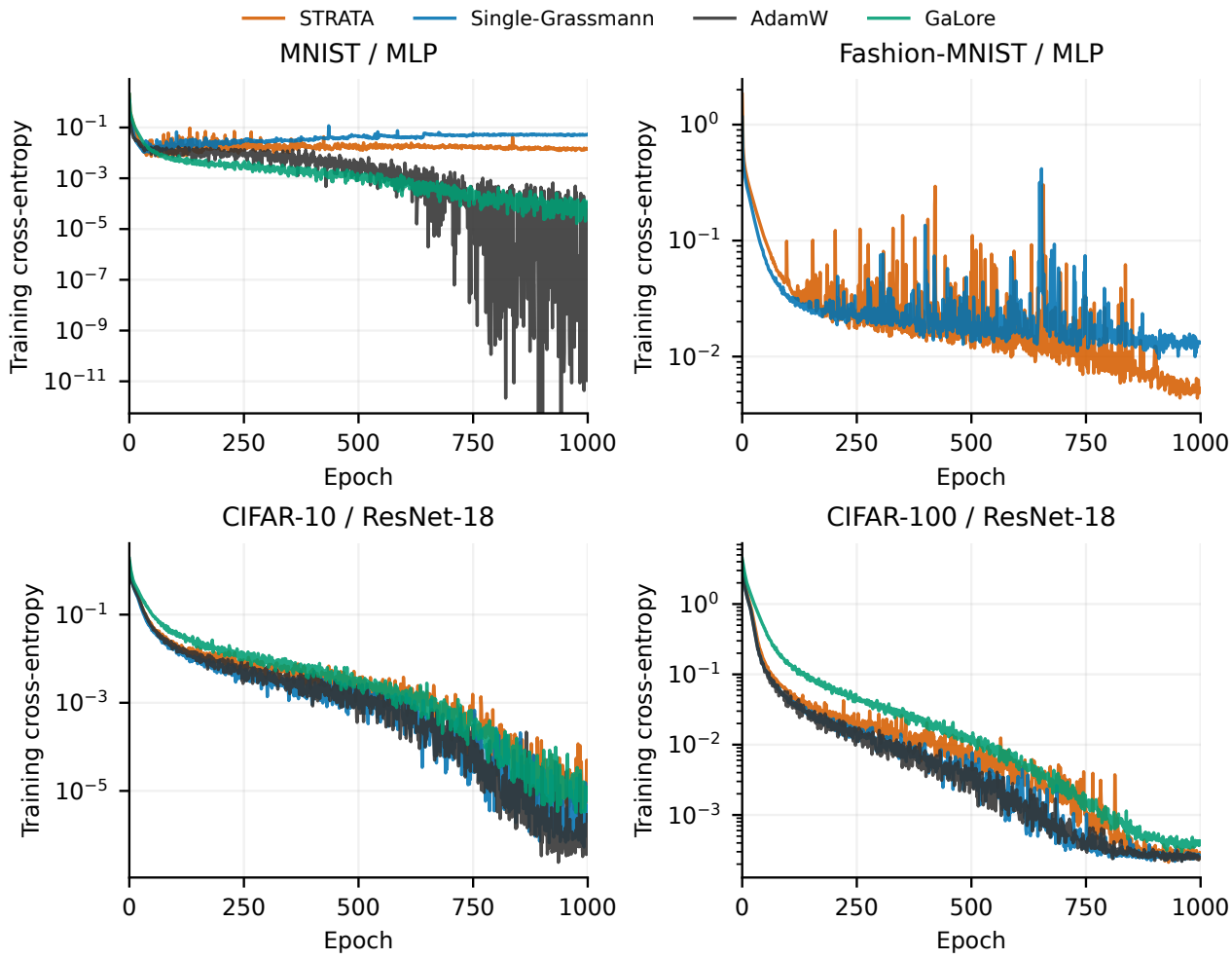


Figure 35. **Full 1000-epoch training-loss trajectories.** Four archived classification settings, seed 42. Within each available method grid we select a completed run by terminal-100 validation accuracy and plot its raw training cross-entropy. Baseline grids have unequal sizes; these panels are descriptive convergence comparisons, not an equal-search causal experiment. No smoothing, extrapolation, seed averaging, or test-based selection is used. Training and validation curves for the same runs, every completed classifier-grid loss, fixed-rank PINN curves, graph/continual curves, and separately labeled ongoing runs appear in Appendix G.

ahead by 0.0117. In Poisson, STRATA is much better than Single-Grassmann in a grid where the latter is unstable, but full AdamW is still substantially better at 0.0079 versus 0.1874.

δ_{nest} is descriptive. The epoch-axis rejudgment is crucial. Graph is unique only in its low starting point, approximately 0.05, followed by collapse to about 0.49. It is not unique in rising: Burgers and Poisson also rise strongly, with Spearman correlations 0.901 and 0.822, and Burgers after its short initial dip has Spearman 0.993. Because rising, flat, and falling diagnostics all map mostly to head-to-head ties, δ_{nest} is useful for describing gradient structure but not for predicting STRATA's performance.

A5 mechanism ablation. Setting $a_1 = \cdots = a_K = a_\perp$ makes $\Gamma^a = 0$ in the linear surrogate, while coordinatewise Adam remains basis dependent (Appendix A). Equalizing scalar layer weights does not substantially degrade the tested cells: in Burgers at lr 5×10^{-3} and rank 64, full STRATA gives 0.0267 and A5 gives 0.0270; in Poisson at lr 5×10^{-4} and rank 32, full STRATA gives 0.0603 and A5 gives 0.0673. These archived cells alone do not establish a benefit from distinct layer weights. The new PubMed controls below test a different empirical regime, with every intervention and adverse outcome retained.

Table 26. Archived benchmark endpoints. Bold marks the better of STRATA and Single-Grassmann only; these are single-seed endpoints, not multi-seed estimates.

Task	AdamW	GaLore	Single-G	STRATA	Verdict
Burgers rel-$L_2 \downarrow$	0.0286	0.436	**0.0261**	0.0267	tie
Burgers-g2 rel-$L_2 \downarrow$	0.0486	–	0.0262	**0.0262**	tie
Poisson rel-$L_2 \downarrow$	0.0079	–	2.0436	**0.1874**	SG unstable
MNIST acc. ↑	0.987	0.985	0.9801	**0.9827**	tie
Fashion acc. ↑	–	–	**0.9033**	0.8991	tie
CIFAR-10 acc. ↑	0.9419	0.9412	**0.9403**	0.9401	tie
CIFAR-100 acc. ↑	0.7358	0.7051	**0.7257**	0.7140	SG slight
ogbn-arxiv val acc. ↑	0.7351	0.7234	**0.7349**	0.7337	tie

Table 27. **Epoch-axis nesting diagnostics.** First/last values and early/late-half means use the complete retained diagnostic trajectory. Spearman ρ uses recorded epochs. Diagnostic trends do not establish a performance ranking.

Task	First	Last	Early mean	Late mean	ρ
ogbn-arxiv	0.047	0.493	0.330	0.477	+0.998
MNIST	0.309	0.204	0.249	0.204	-0.766
Fashion	0.322	0.345	0.344	0.335	-0.065
Burgers	0.880	0.690	0.416	0.618	+0.901
Poisson	0.573	0.415	0.287	0.407	+0.822
CIFAR	0.512	0.601	0.397	0.434	+0.224

New complete comparison cohorts. Table 28 includes a cohort only when all five methods have completed the stated budget. Two-rate rows require all ten trials before validation-based learning-rate selection; fixed-rate rows use 3×10^{-4} for every method. An incomplete search is not reported as a validation-selected winner. After completing both rates on Cora/GCN, validation selects STRATA 3×10^{-4} and Single-Grassmann 10^{-3}. Their test accuracies are 0.70120 and 0.71643; the earlier positive fixed-rate difference therefore does not survive the completed search. GaLore reaches 0.77180 in this cohort. This adverse comparison is retained. On the completed PubMed/SAGE two-rate cohort, STRATA reaches 0.71623 versus Single-Grassmann 0.67902, a descriptive gain of 3.721 percentage points. The highest test accuracy among the validation-selected methods is 0.75801 (galore); this single-seed observation does not establish a graph-family advantage.

Completed Rotated-MNIST replay protocol. Both fine-tuning and replay use the same seed-42 split (55000 training, 5000 validation and 10000 test examples), two 256-wide hidden layers, rank 64, and ten rotations from 0° to 90° in 10° increments. Each rotation receives 100 epochs. Replay stores 200 examples after each task and adds one recalled example per four current-task examples in each minibatch (current batch size 512); all five methods use this same replay rule. Endpoints average epochs 901–1000, evaluating all ten seen rotations. Every method completes both original rates before terminal validation accuracy selects the rate: 10^{-3} for STRATA, AdamW, GaLore and Independent, and 3×10^{-4} for Single-Grassmann. STRATA improves on Single-Grassmann by 1.412 percentage points and on Independent by 1.443 points; AdamW and GaLore remain higher. These single-seed comparisons do not establish statistical significance. The loss curves retain all selected methods without smoothing; the mean validation loss covers an increasing set of seen rotations at task boundaries.

Table 28. New full-budget comparison cohorts, seed 42, terminal-100 test accuracy. A two-rate search selects each method using terminal validation accuracy. Fixed-rate rows use no rate selection. Only complete five-method cohorts are shown; the full ledger retains every completed trial.

Task/model	Protocol	STRATA	Single-G	AdamW	GaLore	Indep.
CiteSeer/gat	2-rate val	0.5676	0.5918	0.5737	**0.6264**	0.5718
CiteSeer/gcn	2-rate val	0.4920	0.5156	0.5927	**0.6166**	0.5030
CiteSeer/gcnii	2-rate val	0.4751	0.4987	0.6085	**0.6150**	0.5042
CiteSeer/sage	2-rate val	0.5575	0.5398	0.5530	**0.6512**	0.5584
Cora/gat	2-rate val	0.6980	0.7191	0.7100	**0.7816**	0.7365
Cora/gcn	2-rate val	0.7012	0.7164	0.7154	**0.7718**	0.6956
Cora/gcnii	2-rate val	0.6898	0.7014	0.6784	**0.7603**	0.6934
Cora/sage	2-rate val	0.4653	0.6506	0.7060	**0.7503**	0.5641
PubMed/gat	2-rate val	0.7117	0.7431	0.7232	**0.7663**	0.7366
PubMed/gcn	2-rate val	0.7174	0.7121	0.7045	**0.7592**	0.6878
PubMed/gcnii	2-rate val	0.6634	0.7107	0.7088	**0.7499**	0.6693
PubMed/sage	2-rate val	0.7162	0.6790	0.6906	**0.7580**	0.7415
adult/MLP-256-256	2-rate val	0.8149	0.8149	0.8172	**0.8216**	0.8117
adult/MLP-512-512	2-rate val	0.8151	0.8128	**0.8191**	0.8184	0.8151
cifar100_class_shift/ResNet18-CIFAR	2-rate val	0.5309	**0.5737**	0.5584	0.5306	0.5337
cifar10_label_shift/ResNet18-CIFAR	2-rate val	0.7501	0.7665	**0.8021**	0.8000	0.7696
covertype/MLP-256-256	2-rate val	0.9622	0.9630	**0.9635**	0.9627	0.9615
covertype/MLP-512-512	2-rate val	0.9664	0.9664	0.9645	0.9660	**0.9675**
Rotated-MNIST/MLP (fine-tuning)	2-rate val	0.5765	0.5711	**0.6219**	0.5969	0.5728
Rotated-MNIST/MLP (replay)	2-rate val	0.8858	0.8717	**0.9072**	0.8888	0.8714
rotated_mnist_shift/MLP-256-256	2-rate val	0.9761	0.9790	**0.9806**	0.9796	0.9767
wine_red/MLP-256-256	2-rate val	0.6842	0.6883	0.7102	**0.7185**	0.6910
wine_red/MLP-512-512	2-rate val	**0.7150**	0.6970	0.7149	0.7105	0.7018
wine_white/MLP-256-256	2-rate val	0.7242	0.7317	0.7204	0.7260	**0.7389**
wine_white/MLP-512-512	2-rate val	0.7088	0.7231	0.7174	0.7100	**0.7389**

K. Prospective Real-Data Tasks Beyond the Current Cases

Purpose and frozen scope. The additional study comprises six settings in three directions: ETTh1 and ETTh2 forecasting, MovieLens100K and MovieLens1M explicit-rating prediction, and gas-sensor drift with fine-tuning or replay. The task list, data digests, splits, models, six candidates per method, metrics and failure policy are frozen before formal outcomes in the retained plan. The goal is to test structured-gradient and adaptation hypotheses beyond citation graphs; neither dataset readiness nor a queued configuration is evidence of improvement. We retain all declared settings and candidate failures, irrespective of the sign of the result.

ETT forecasting. We use the original hourly transformer series (Zhou et al., 2021), each with seven numeric channels. The first 8640 observations define training; validation and test targets lie in $[8640, 11520)$ and $[11520, 14400)$, respectively. The remaining observations do not enter these trials. With lookback $L = 96$ and horizon $H = 24$, a split $[a, b)$ contains precisely

$$t \in \{\max(a, L), \dots, b - H\}, \qquad x_t = \mathrm{vec}(z_{t-L:t}), \quad y_t = \mathrm{vec}(z_{t:t+H}).$$

Thus an input uses past observations only, and every target horizon is contained in its own split. Channel means and standard deviations use the training prefix only. The common forecaster is a $672 \to 256 \to 256 \to 168$ MLP with GELU hidden activations. Its primary metric is MSE in training-standard-deviation units; MAE is secondary. These are controlled optimizer comparisons on ETT data, not claims to reproduce the Informer architecture or a forecasting state-of-the-art endpoint.

Explicit-rating recommendation. MovieLens100K and MovieLens1M come from the official GroupLens releases (Harper & Konstan, 2015). MovieLens100K preserves the official `u1.test`; a SHA-256 user/item hash assigns ten percent of `u1.base` to validation, yielding $71978/8022/20000$ ratings. MovieLens1M uses chronological $70/15/15$ splits within each user, with item ID breaking timestamp ties, yielding $697378/149931/152900$ ratings. User/item pairs do not overlap

across splits. Catalog IDs may be known without using held-out ratings. All methods fit the same biased matrix-factorization model,

$$\widehat{r}_{ui} = \mu_{\text{train}} + b_u + c_i + \frac{p_u^\top q_i}{\sqrt{128}}, \qquad p_u, q_i \in \mathbb{R}^{128}.$$

Training minimizes observed-rating MSE. The primary evaluation is raw-rating RMSE; these tasks do not support a top-k retrieval or NDCG claim. Embeddings use dense gradients, so every optimizer sees the same parameterization and examples. Complete epochs traverse all training ratings.

Real sensor drift. The UCI Gas Sensor Array Drift data (Vergara, 2012) contain 13910 records, 128 features, six labels and ten ordered batches. A global SHA-256 hash of the float32 feature row and label fixes $70/15/15$ training/validation/test roles; identical records keep the same role across batches. Only batch-one training rows determine feature normalization. The common classifier is a $128 \to 256 \to 256 \to 6$ ReLU MLP. Training spends 100 complete epochs in each successive domain, for 1000 total epochs, and never trains on future domains. Fine-tuning uses current-domain training rows. Replay additionally samples at most 128 old examples per current minibatch from a deterministic 512-example sample of past training domains, refreshed at domain boundaries. This is the replay sample size, not a claim that the resident benchmark arrays consume constant storage. Flags and moments carry across domain switches. Evaluation at domain k averages the per-domain held-out accuracies equally,

$$A_k(\theta) = \frac{1}{k} \sum_{j=1}^{k} \frac{1}{|D_j|} \sum_{(x,y)\in D_j} \mathbf{1}\{\arg\max f_\theta(x) = y\}.$$

Per-domain accuracies are retained. Since the set of seen domains grows, these curves are not measurements of convergence to one fixed population objective.

Equal search counts and computational budgets. STRATA, Single-Grassmann and Independent use maximum/total rank 32 and six candidates,

$$\eta \in \{3 \times 10^{-4}, 10^{-3}\}, \qquad \xi \in \{10^{-3}, 10^{-1}, 10\}.$$

Independent's actual block dimensions are $(8, 8, 16)$, totaling 32. AdamW and GaLore each use six distinct learning rates,

$$\eta \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}.$$

Thus the search counts are equal; the hyperparameter spaces are explicitly different. These are comparisons of the full algorithms, not a pure isolated nesting intervention. All use seed42, float32 computation, clipping at norm10 and zero weight decay. A cosine schedule with five-epoch warmup spans the full static task, or restarts within each 100-epoch drift domain. Each prescribed epoch traverses the complete relevant training set. Six CPU workers use two intra-op threads each after checking capacity; CPU wall time is shared-server timing, and a zero CUDA-allocation field means not applicable, not zero total memory. State counts include fixed probes and persistent buffers.

Selection and complete reporting. For every method and setting, the selected candidate optimizes the mean primary validation metric over the terminal 100 epochs: MSE/RMSE is minimized and accuracy maximized. Test metrics never select a configuration. The same reporting rule applies to favorable and unfavorable results. A setting is summarized only after its 30 candidates complete or record a failure; a failed candidate has no fabricated endpoint, and a method with no complete candidate has no selected score. Raw losses, validation/test metrics, flag snapshots, ten-epoch checkpoints, source hashes and data hashes remain available. Small smoke checks are stored separately and excluded from the formal queue. The single-seed protocol does not justify statistical-significance claims or a guaranteed winner in any task family.

Relation to the conditional theory. Replay makes persistent past-domain corrections available alongside current-domain gradients while preserving the model architecture and optimizer tensor-state sizes. The fine-tuning/replay contrast therefore probes a data-regime interaction; it does not isolate a geometric component of the optimizer. ETT windows overlap, chronological ratings share users and sensor domains are dependent and shifted. These data do not by themselves establish the sampling assumptions of the generalization theorems, and unclipped MSE is not a bounded loss. The task comparisons are empirical tests of optimization and state–quality tradeoffs, not numerical certificates of the conditional convergence or generalization bounds.

K.1. Prospective new-task execution and complete comparisons

This extension was declared before its outcomes: six settings, five methods and six candidates per method. Every formal run uses seed 42 and 1000 complete epochs. Configuration selection uses terminal-100 validation metrics. A comparison appears only after all 30 declared candidates have completed or recorded failure; failures have no fabricated endpoint. A method with no complete candidate has no selected score. Negative outcomes and every declared setting remain in this record.

Table 29. Execution coverage of every declared new setting. Counts are configurations, not quality wins; failed candidates remain visible.

Family	Setting	Complete	Running	Pending	Failed
forecasting	ETTh1 / H24	30	0	0	0
forecasting	ETTh2 / H24	30	0	0	0
recommendation	MovieLens100K / ratings	30	0	0	0
recommendation	MovieLens1M / ratings	30	0	0	0
sensor_drift	Gas sensors / fine-tuning	30	0	0	0
sensor_drift	Gas sensors / replay	30	0	0	0

Table 30. All fully searched new settings. Each displayed method independently selects among six candidates on validation. Values are terminal-100 test metrics; MSE and RMSE are lower-is-better, accuracy is higher-is-better. No setting is omitted according to outcome.

Setting	Metric	STRATA	Single-G	AdamW	GaLore	Indep.
ETTh1 / H24	MSE	0.89922	0.88655	0.86679	**0.76942**	0.90163
ETTh2 / H24	MSE	**1.74404**	2.37970	2.16884	1.79170	2.18874
MovieLens100K / ratings	RMSE	0.96949	1.00529	**0.93295**	0.95778	1.01755
MovieLens1M / ratings	RMSE	1.05052	1.09444	0.98066	**0.91712**	1.08146
Gas sensors / fine-tuning	acc	0.85453	0.85375	0.82335	**0.90656**	0.80659
Gas sensors / replay	acc	0.96506	0.96409	**0.96536**	0.96232	0.96343

ETTh1 / H24 complete search. Primary metric: MSE. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 31. **ETTh1 / H24: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	3e-04	0.001	0.91637	0.89922	0.792	✗
Single-G	3e-04	0.001	1.06275	0.88655	0.585	✗
AdamW	1e-02	–	0.94111	0.86679	2.146	✗
GaLore	1e-04	–	**0.73812**	**0.76942**	0.377	✓
Independent	3e-04	0.1	1.04792	0.90163	0.751	✗

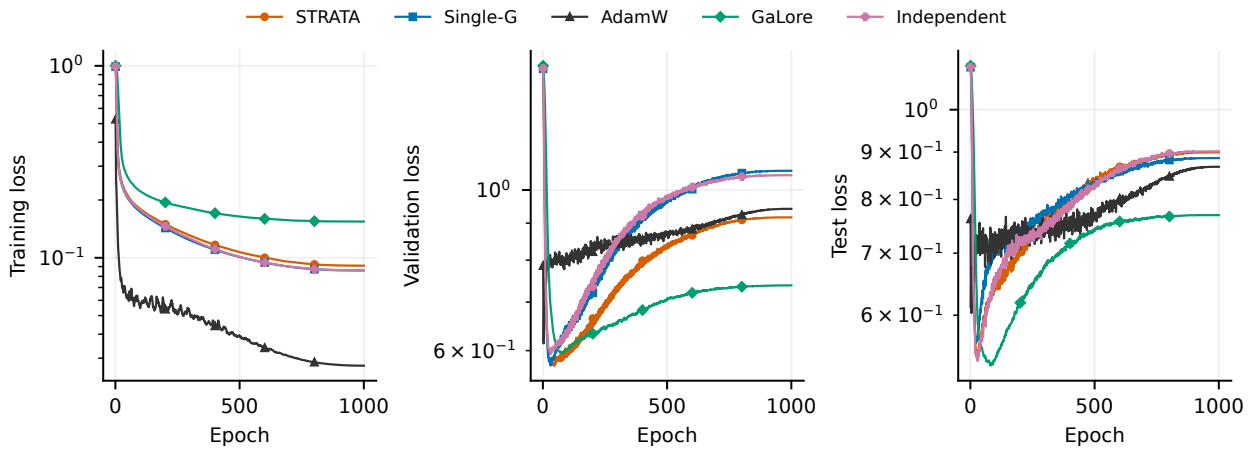


Figure 36. Complete ETTh1 / H24 loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed.

ETTh2 / H24 complete search. Primary metric: MSE. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 32. **ETTh2 / H24: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	3e-04	10	0.85177	**1.74404**	0.792	✓
Single-G	3e-04	0.1	**0.80694**	2.37970	0.585	✗
AdamW	3e-03	–	1.25122	2.16884	2.146	✗
GaLore	1e-02	–	1.00745	1.79170	0.377	✓
Independent	1e-03	0.1	1.31254	2.18874	0.751	✗

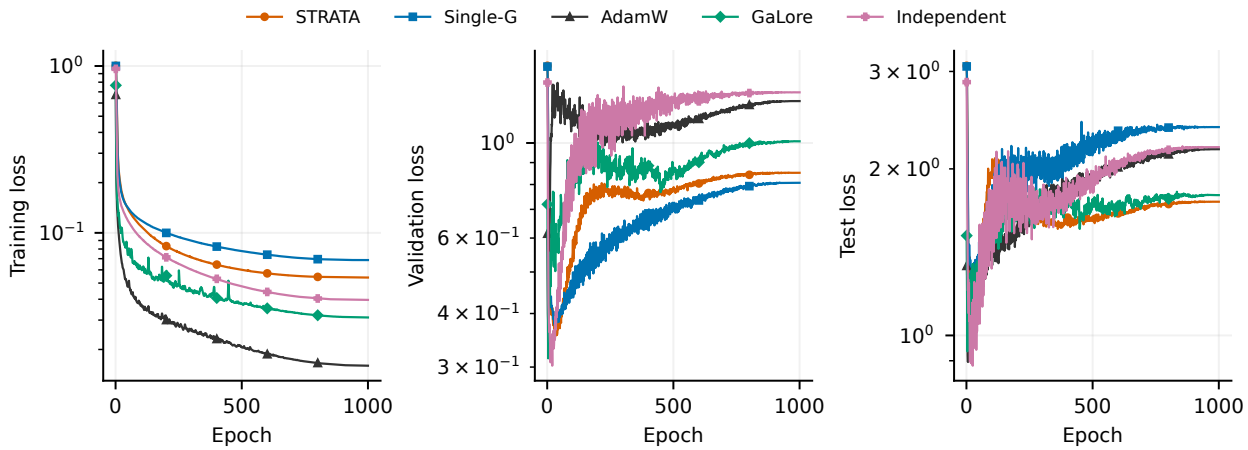


Figure 37. Complete ETTh2 / H24 loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed.

MovieLens100K / ratings complete search. Primary metric: RMSE. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 33. **MovieLens100K / ratings: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	3e-04	10	0.96257	0.96949	0.848	✗
Single-G	3e-04	10	1.00521	1.00529	0.770	✗
AdamW	1e-04	–	**0.91533**	**0.93295**	2.584	✓
GaLore	1e-04	–	0.94025	0.95778	0.692	✓
Independent	3e-04	10	1.00684	1.01755	0.833	✗

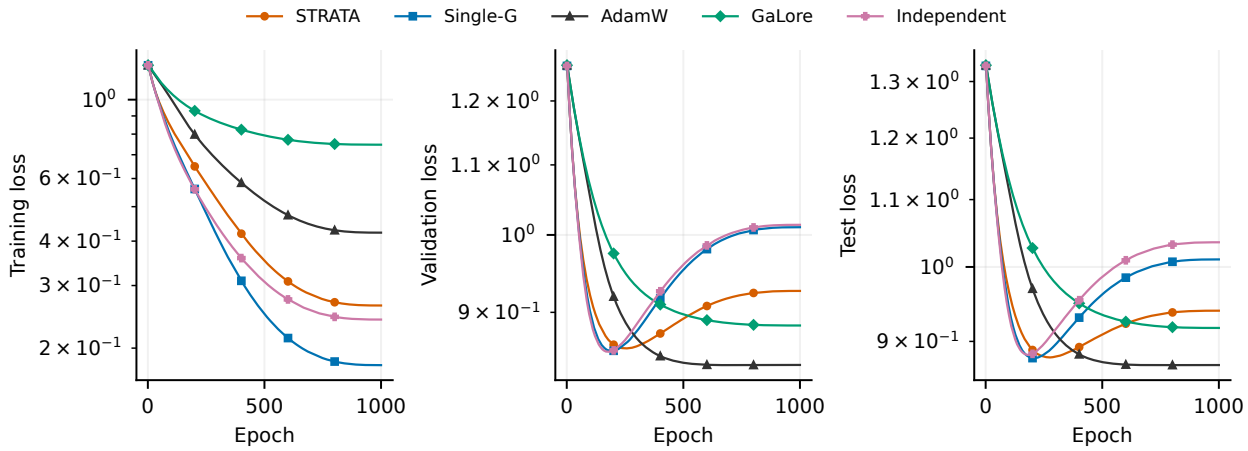


Figure 38. Complete MovieLens100K / ratings loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed.

MovieLens1M / ratings complete search. Primary metric: RMSE. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 34. **MovieLens1M / ratings: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	3e-04	0.001	1.03092	1.05052	2.703	✗
Single-G	3e-04	0.001	1.07528	1.09444	2.625	✗
AdamW	1e-04	–	0.96096	0.98066	9.834	✗
GaLore	1e-04	–	**0.89748**	**0.91712**	2.547	✓
Independent	3e-04	10	1.05453	1.08146	2.688	✗

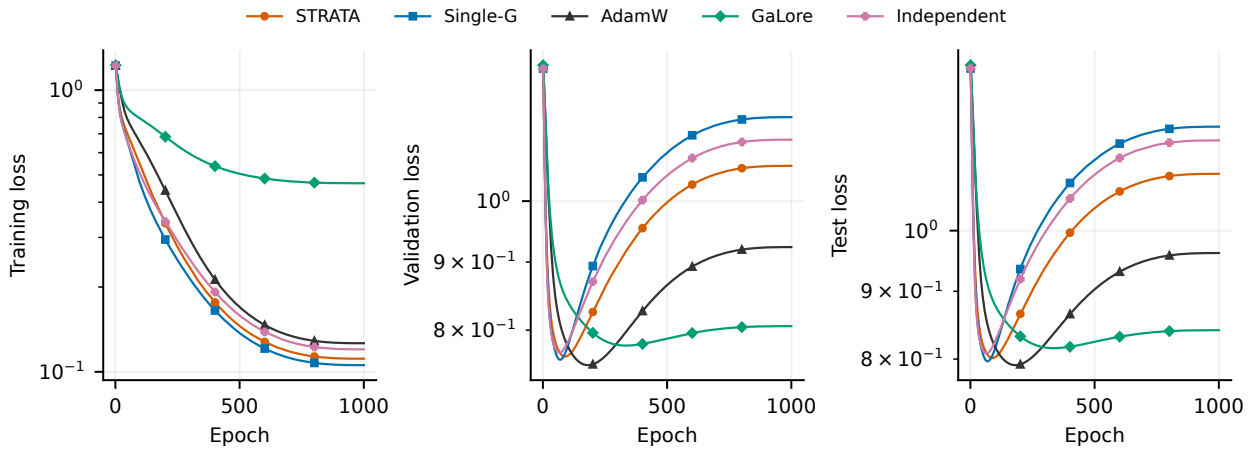


Figure 39. Complete MovieLens1M / ratings loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed.

Gas sensors / fine-tuning complete search. Primary metric: acc. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 35. **Gas sensors / fine-tuning: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	1e-03	0.1	0.86518	0.85453	0.422	✗
Single-G	3e-04	0.001	0.85400	0.85375	0.305	✗
AdamW	3e-03	–	0.85438	0.82335	0.766	✗
GaLore	1e-03	–	**0.90496**	**0.90656**	0.188	✓
Independent	3e-04	10	0.82691	0.80659	0.398	✗

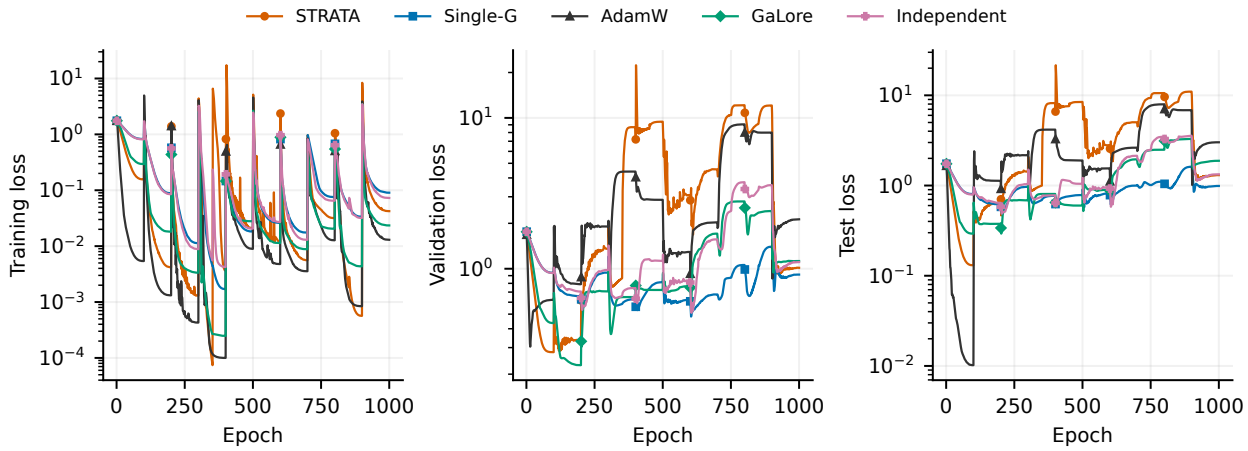


Figure 40. Complete Gas sensors / fine-tuning loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed; sensor-drift evaluation averages domains seen so far and changes at domain boundaries.

Gas sensors / replay complete search. Primary metric: acc. Recorded candidate failures: STRATA 0, Single-G 0, AdamW 0, GaLore 0, Independent 0. Shared-CPU wall time is not isolated throughput; tensor-state size is distinct from peak memory.

Table 36. **Gas sensors / replay: selected configurations and stored state.** Each method selects among six candidates on validation. Bold marks the best validation/test value; the green row identifies STRATA. Pareto status uses test quality and stored state, not peak memory.

Method	lr	ξ	Val	Test	State MiB	Pareto
STRATA	1e-03	10	**0.97487**	0.96506	0.422	✓
Single-G	1e-03	0.001	0.96936	0.96409	0.305	✓
AdamW	3e-03	–	0.97452	**0.96536**	0.766	✓
GaLore	1e-03	–	0.97444	0.96232	0.188	✓
Independent	1e-03	10	0.97214	0.96343	0.398	✗

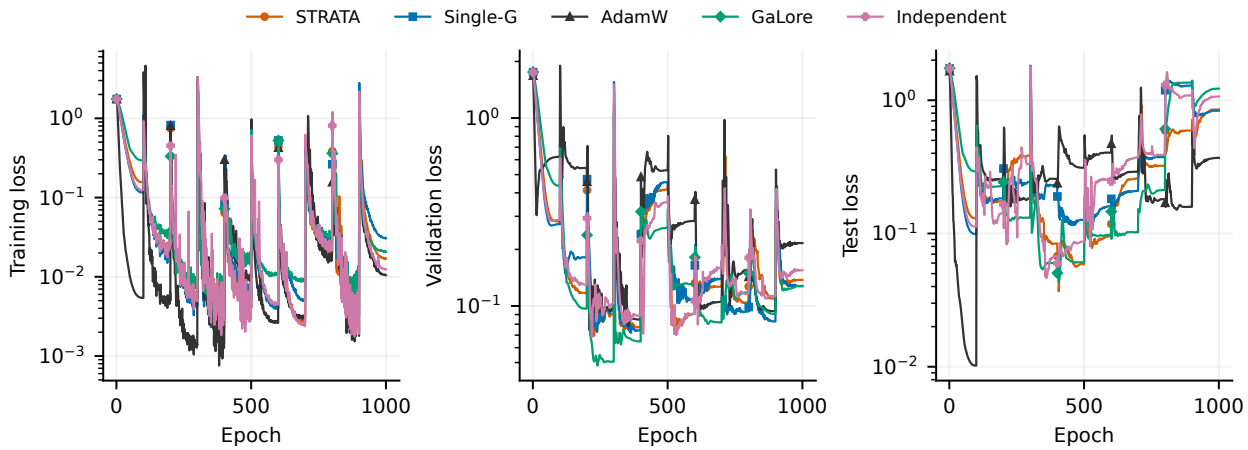


Figure 41. Complete Gas sensors / replay loss trajectories. Left to right: training, validation, and test loss. Selection uses validation only after all candidates finish or record failure. All five methods and adverse outcomes remain visible. Curves are raw and unsmoothed; sensor-drift evaluation averages domains seen so far and changes at domain boundaries.